

Topological Energy Condensate Theory (TECT): 15

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I. FINITE-AUDIT REFORMULATION OF THE REMAINING TECT MICROSCOPIC PROGRAM

A. Goal

The correct next step is not to introduce a new qualitative mechanism, but to rewrite the remaining open microscopic questions as a finite operator audit for the actual linearized Bloch operator of the refined locked TECT branch.

Let

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger, \quad \chi \gg 1,$$

and let the full microscopic linearization around the chosen defect/background be denoted by

$$\mathcal{L}_{\text{mic}}(p).$$

Fix the selected shell pair $\pm G_*$, and let

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \quad \mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

where $\mathcal{H}_D$ is the shell-relevant doubled local block and $\mathcal{H}_R$ is the remote sector.

The purpose of this section is to reduce all remaining open tasks to finite checks on

$$\Pi_D \mathcal{L}_{\text{mic}}(p) \Pi_D, \quad \Pi_D \mathcal{L}_{\text{mic}}(0) \Pi_R, \quad \Pi_R \mathcal{L}_{\text{mic}}(p) \Pi_R,$$

where Π_D, Π_R are the corresponding spectral/projected block projectors.

B. Step A. Actual witness carrier audit

The first question is whether the actual microscopic TECT linearization contains a nontrivial shell-relevant (i, a) -carrier channel.

Let the shell-relevant enriched tensor space be

$$V_{\text{enr}} = \{N_{ij}^a \text{ surviving the actual microscopic linearization}\}.$$

For each transverse shell direction e_α , $\alpha = 1, 2$, define

$$C_j^\alpha(N) = e_{\alpha i} N_{ij}^{a_\alpha}, \quad L_\alpha(N) = G_{*j} C_j^\alpha(N) = G_{*j} e_{\alpha i} N_{ij}^{a_\alpha}.$$

Then the witness condition is

$$L_1(N) \neq 0, \quad L_2(N) \neq 0.$$

The key theorem-level fact is the following dichotomy: either $L_\alpha \equiv 0$ on V_{enr} by an exact symmetry prohibition, or else $L_\alpha \neq 0$ on an open dense full-measure subset of V_{enr} . Therefore the actual carrier audit reduces to constructing a single admissible witness

$$N_{\text{seed}}^{(\alpha)} \in V_{\text{enr}}$$

such that

$$L_\alpha(N_{\text{seed}}^{(\alpha)}) \neq 0.$$

Hence the completion criterion for Step A is:

$$\boxed{\exists N_{\text{seed}}^{(1)}, N_{\text{seed}}^{(2)} \in V_{\text{enr}} \text{ with } L_1(N_{\text{seed}}^{(1)}) \neq 0, \quad L_2(N_{\text{seed}}^{(2)}) \neq 0.}$$

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C. Step B. Opposite-valley survival and matching

Let

$$W^\pm = (W_\parallel^\pm, W_I^\pm, W_J^\pm)$$

denote the shell witness triples extracted from the projected $+$ and $-$ valleys in matched opposite-valley frames.
The actual Dirac pairing condition is

$$\boxed{W^- = W^+}.$$

This is not a conceptual problem any more; it is an explicit finite comparison problem: one must project the actual microscopic operator to both shell blocks and compare the three coefficients.

The completion criterion for Step B is the explicit equality

$$\boxed{W_\parallel^- = W_\parallel^+, \quad W_I^- = W_I^+, \quad W_J^- = W_J^+}.$$

D. Step C. Zero constant mixing as a finite label audit

At $p = 0$, the full operator has block form

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}.$$

If the residual symmetry group at the crossing is H_* , then equivariance implies

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore a sufficient and exact theorem-level criterion is

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0}.$$

In the present TECT setting, the correct practical version is not an abstract “ordinary cubic 2D irrep” statement. Rather, $\mathcal{H}_D$ is the shell-relevant carrier

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

and each remote sector $K_a \subset \mathcal{H}_R$ must differ from $\mathcal{H}_D$ in at least one shell-relevant label:

$\pm G_*$, transverse/non-volumetric character, presence of enriched internal doublet, grading/parity/chirality, additional co

Thus Step C reduces to the finite table audit: for every remote sector K_a , check whether it matches $\mathcal{H}_D$ in all labels. If not, then

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

If this holds for all a , then

$$V(0) = 0.$$

E. Step D. Remote-gap audit

The remote gap is an independent spectral property of the remote block:

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset \quad \text{for } |p| \leq p_*.$$

This is not implied by the shell determinant. Hence the correct theorem-level route is either:

a. Route D1: direct finite lower bound Split

$$H_R(p) = D_R(p) + R_R(p),$$

where D_R is the diagonal or benchmark part and R_R the residual mixing. Define

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(D_R(p))), \quad \eta_{R,\rho} := \sup_{|p| \leq \rho} \|R_R(p)\|.$$

Then

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

b. Route D2: bounded perturbation from a gapped benchmark If

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset$$

and

$$\sup_{|p| \leq p_*} \|\delta H_R(p)\| < \Delta_0,$$

then

$$\boxed{\Delta = \Delta_0 - \sup \|\delta H_R\| > 0.}$$

Hence Step D is a finite Bloch-spectral inequality problem.

F. Step E. Mass-protection audit

The doubled local Dirac block has the form

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i).$$

The space of genuine constant Dirac masses is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Therefore mass protection is exactly the finite operator-algebra condition

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\},}$$

where A_{res} is the algebra of residual-symmetry-allowed constant Hermitian operators on the isolated doubled block.

A sufficient mechanism is an exact residual valley symmetry. If

$$Q_V = \tau_z \otimes \mathbf{1}_2$$

generates an exact $U(1)_V$, then

$$[Q_V, H_D(p)] = 0,$$

while

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2$$

transform nontrivially and are therefore forbidden. The weaker discrete remnant $\mathbb{Z}_2^V$ also suffices.

Hence Step E reduces to computing the residual constant operator algebra A_{res} from the actual microscopic symmetry realization and checking its intersection with $\mathcal{M}_D$.

G. Step F. Combined finite-audit theorem

Assume the actual microscopic realization satisfies:

1. there exist two admissible enriched witnesses with

$$L_1 \neq 0, \quad L_2 \neq 0;$$

2. the opposite-valley witness triples match:

$$W^- = W^+;$$

3. the remote sectors satisfy the shell-relevant mismatch audit, hence

$$V(0) = 0;$$

4. the remote block satisfies one of the gap inequalities above, hence

$$\Delta > 0;$$

5. the residual constant operator algebra satisfies

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Then, for $|E| < \Delta/2$,

$$H_{\text{eff}}(E, p) = H_D(p) - \Sigma(E, p), \quad \Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger,$$

with

$$\Sigma(E, p) = O\left(\frac{|p|^2}{\Delta}\right),$$

and therefore

$$H_{\text{eff}}(E, p) = \sum_{i=1}^3 v_i p_i \Gamma_i + O\left(\frac{|p|^2}{\Delta}\right),$$

with no allowed constant Dirac mass term.

Thus the complete remaining TECT microscopic frontier is reduced to a finite, explicitly checkable operator audit on the actual projected Bloch linearization.

H. Normalization constants and what they do not affect

The overlap constants $\nu_{\parallel}, \nu_{\perp}$ enter the dictionary between the microscopic Class II parameters and the reduced witness coefficients:

$$g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J.$$

They are needed for absolute normalization and quantitative matching, but not for the purely existential Weyl/Dirac statement as long as they are finite and nonzero. Therefore their determination is a calibration problem, not an existence obstruction.

II. FINAL SUMMARY

The strongest honest current status is:

the TECT local Dirac problem is not yet fully unconditionally closed,

but

it has been reduced to a finite explicit microscopic audit problem.

The remaining finite audits are:

(I) actual witness carrier realization,

(II) opposite-valley survival and matching,

(III) zero constant mixing $V(0) = 0$,

(IV) positive remote gap $\Delta > 0$,

(V) mass protection $A_{\text{res}} \cap \mathcal{M}_D = \{0\}$.

More explicitly:

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0,$$

$$\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho > 0,$$

$$A_{\text{res}} \cap \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\} = \{0\} \implies \text{no constant Dirac mass.}$$

Therefore, once the actual microscopic linearized Bloch operator passes these finite audits,

$$H_{\text{eff}}(E, p) = \sum_{i=1}^3 v_i p_i \Gamma_i + O(|p|^2/\Delta),$$

so the theory contains a spectrally isolated massless local Dirac cone.

Finally, the constants $\nu_{\parallel}, \nu_{\perp}$ affect only the absolute normalization of the microscopic-to-reduced dictionary, not the existential form of the Dirac/Weyl theorem.

III. ACTUAL MICROSCOPIC CARRIER AUDIT

A. Goal

The remaining microscopic question after the explicit Class II witness theorem is not whether the shell-level witness logic is correct. That part is already closed inside the concrete Class II truncation.

The actual remaining question is:

Does the true microscopic TECT linearization contain a nontrivial shell-relevant (i, a) -carrier sector? (1)

Equivalently, the issue is whether the full linearized transport operator, after projection to the relevant defect/shell sector, contains a nonzero component in the explicit seed space that generates the reduced witness coefficients

$$(\lambda_{\parallel}, \alpha, \beta).$$

B. What the witness actually needs

The shell-level Weyl/Dirac witness is controlled by the enriched tensor

$$N_{ij}^a,$$

through the shell-adapted coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta,$$

or equivalently

$$A, \quad B, \quad C.$$

The minimal shell-adapted tensor family is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (2)$$

Therefore the microscopic carrier must simultaneously supply:

1. one spatial derivative index i or j ,
2. one reduced internal / shell-tangent index a .

This is the structural meaning of an (i, a) -carrier.

C. Finite-dimensional witness functionals

Let V_{mic} be the finite-dimensional vector space of symmetry-allowed microscopic enriched tensors surviving the linearization around the chosen defect background.

The shell witness functionals are:

$$W_{\parallel}(N) := g_c(N), \quad W_E(N) := g_E(N), \quad W_O(N) := g_O(N). \quad (3)$$

Equivalently,

$$\lambda_{\parallel}(N) = c_{\parallel} W_{\parallel}(N), \quad \alpha(N) = c_{\perp} W_E(N), \quad \beta(N) = c_{\perp} W_O(N), \quad (4)$$

with nonzero defect-overlap constants $c_{\parallel}, c_{\perp}$.

Define the longitudinal and transverse witness conditions by

$$\mathfrak{W}_{\parallel}(N) := \lambda_{\parallel}(N) \neq 0, \quad \det T_{\perp}(N) := \alpha(N)^2 + \beta(N)^2 \neq 0. \quad (5)$$

Since $W_{\parallel}$ is linear and $W_E^2 + W_O^2$ is polynomial on V_{mic} , the witness problem is finite-dimensional from the start.

D. Exact dichotomy: symmetry-forbidden versus generic nonvanishing

Let $V \subset W$ be the symmetry-allowed microscopic tensor space inside the ambient tensor space W . For each shell channel α , define the target covector

$$T_{ija}^{(\alpha)} = e_i^{\alpha} G_*^j u_a^{(\alpha)}.$$

Then the corresponding shell contraction is

$$L_{\alpha}(N) = \langle T^{(\alpha)}, N \rangle. \quad (6)$$

If $P_V : W \rightarrow V$ denotes the orthogonal projection onto the allowed tensor space, then

$$L_{\alpha}|_V \equiv 0 \quad \Longleftrightarrow \quad P_V T^{(\alpha)} = 0. \quad (7)$$

Hence exactly one of the following holds:

1. $P_V T^{(\alpha)} = 0$: the channel is symmetry-forbidden;
2. $P_V T^{(\alpha)} \neq 0$: the channel is generically nonvanishing on an open dense full-measure subset of V .

Therefore, to prove that a witness channel survives, it is enough to exhibit one admissible seed tensor with nonzero projection.

E. Minimal UV classification of admissible carriers

At lowest derivative order, any local microscopic extension capable of producing the reduced shell tensor must effectively introduce an (i, a) -carrier. The minimal viable classes are:

1. Class I: direct shell–projector derivative couplings,
2. Class II: heavy auxiliary internal vector mediators X_i^a ,
3. Class III: connection-like mediators A_i^a .

Among these, Class II is the strongest current explicit class, because it dynamically generates the carrier rather than inserting it by hand.

F. Explicit Class II carrier mechanism

The explicit Class II UV structure is

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi]. \quad (8)$$

After elimination of the heavy mediator X_i^a , the shell-relevant quadratic mixing takes the form

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.}, \quad (9)$$

hence

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (10)$$

Define

$$\Gamma := \frac{\lambda g_X m}{M_X^2}. \quad (11)$$

For the shell-adapted seed family

$$m^a = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

one gets

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J. \quad (12)$$

Thus the reduced witness coordinates satisfy

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J. \quad (13)$$

This shows that the three Class II mediator directions correspond exactly to the three shell-reduced invariant coordinates:

$$X_i^3 \leftrightarrow \text{longitudinal seed}, \quad X_i^1 \leftrightarrow \text{transverse-even seed}, \quad X_i^2 \leftrightarrow \text{transverse-odd seed}.$$

G. Seed subspace and embedding criterion

Define the explicit seed space

$$V_{\text{seed}} = \text{span} \left\{ N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)} \right\}. \quad (14)$$

The microscopic carrier audit succeeds if the actual microscopic linearization contains a nontrivial shell-relevant tensor subspace

$$V_{\text{car}} \subset V_{\text{mic}}$$

whose defect/shell projection has nonzero overlap with V_{seed} .

Equivalently, it is sufficient to find

$$N_{\parallel} \in V_{\text{mic}} \quad \text{such that} \quad W_{\parallel}(N_{\parallel}) \neq 0, \quad (15)$$

and

$$N_{\perp} \in V_{\text{mic}} \quad \text{such that} \quad (W_E(N_{\perp}), W_O(N_{\perp})) \neq (0, 0). \quad (16)$$

Once these two seed conditions hold, finite-dimensional genericity implies that the good set

$$W_{\parallel} \neq 0, \quad W_E^2 + W_O^2 \neq 0$$

is open, dense, and full measure inside the allowed carrier space.

H. Operational microscopic audit

The actual microscopic audit is therefore finite and explicit:

a. (A) Linearize the microscopic theory. Linearize the complete refined locked TECT equations around the relevant defect background

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger.$$

b. (B) Extract the first-order transport kernel. Write the shell-relevant first-order operator in the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

c. (C) Project to the low-mode sector. Choose the low-mode basis $\{u_a^\pm\}$ at the opposite shell pair $\pm G_*$, and compute

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

d. (D) Match to the shell tensor basis. Read off the induced tensor coefficients in the basis

$$(N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}),$$

equivalently extract

$$(A, B, C) \quad \text{or} \quad (\lambda_{\parallel}, \alpha, \beta).$$

e. (E) Perform the projection test. Verify

$$P_{V_{\text{seed}}} N_{\text{mic}} \neq 0, \quad (17)$$

and more explicitly confirm the existence of:

1. at least one longitudinal seed component,

$$\lambda_{\parallel} \neq 0;$$

2. at least one transverse seed component,

$$(\alpha, \beta) \neq (0, 0).$$

I. Strongest honest theorem-level statement

We can now state the precise theorem-level status.

[Actual microscopic carrier audit] Assume:

1. the refined locked TECT linearization admits a finite shell-relevant enriched tensor space V_{mic} ;
2. the microscopic carrier sector contains an effective (i, a) -carrier channel;
3. after shell/defect projection, V_{mic} contains one admissible longitudinal seed and one admissible transverse seed.

Then the local shell witness survives:

$$W_{\parallel} \neq 0, \quad (W_E, W_O) \neq (0, 0) \quad (18)$$

on an open dense full-measure subset of the allowed carrier sector.

Equivalently, the shell-level local Weyl nondegeneracy is generically realized.

The longitudinal statement follows because $W_{\parallel}$ is a linear functional on the allowed carrier space. If it is nonzero on one admissible seed, then its zero set is a proper hyperplane.

The transverse statement follows because $W_E^2 + W_O^2$ is a nonzero polynomial once one admissible transverse seed exists. Its zero set is then a proper algebraic subset.

Intersecting the two complementary good sets gives an open dense full-measure subset on which both conditions hold simultaneously.

J. Sharp honest frontier

Thus the actual microscopic carrier problem is no longer conceptually open. It is reduced to a finite yes/no projection test:

Does the complete microscopic TECT linearization project nontrivially onto the explicit shell-relevant seed space V_{seed} ?
(19)

Inside the concrete Class II truncation, the answer is already yes. What remains open is only whether the final complete microscopic TECT realization actually contains that same nontrivial carrier sector in the physically relevant defect background.

IV. FINAL SUMMARY OF THE ACTUAL MICROSCOPIC CARRIER AUDIT

The actual microscopic carrier problem is not the infinite-dimensional question “does some abstract witness exist?” Rather, it is the finite projection problem:

Does the true microscopic TECT linearization contain a shell-relevant (i, a) -carrier sector?

The witness requires the enriched tensor

$$N_{ij}^a,$$

whose shell-adapted form is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}.$$

Thus the microscopic carrier must supply both:

one spatial derivative index
and
one reduced internal index.

At lowest derivative order, the admissible minimal carrier classes are:

Class I, Class II, Class III.

Among them, Class II is the strongest current explicit class because it realizes all three seed coordinates simultaneously. Its quadratic shell-relevant reduction yields

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab},$$

hence

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J, \quad \Gamma = \frac{\lambda g_X m}{M_X^2},$$

and therefore

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

So the explicit seed space

$$V_{\text{seed}} = \text{span} \left\{ N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)} \right\}$$

is exactly the shell-reduced realization of the Class II carrier decomposition.

Hence the strongest honest conclusion is:

the actual microscopic carrier problem is fully reduced to a finite projection / seed-embedding audit.

What is already closed:

inside the explicit Class II truncation, the witness problem is generically solved.

What remains open:

whether the final complete microscopic TECT realization actually contains that nontrivial carrier sector.

V. CONSTRUCTION OF THE FULL BLOCH LINEARIZATION PROGRAM FOR THE REFINED LOCKED + CLASS II MICROSCOPIC PDE

A. Goal and no-loop rule

The final target is the actual microscopic projected valley operator extracted from the full periodic TECT linearization. From this point onward, we do *not* re-prove the reduced witness theorem. That part is already closed once the coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are obtained and satisfy

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

The only correct next calculation is therefore:

full periodic microscopic operator $\rightarrow$ Bloch fiber $\rightarrow$ shell truncation $\rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta)$.

B. Refined locked + Class II microscopic starting point

Let the microscopic variables be:

$$\Psi(x, t) \in \mathbb{C}^3, \quad X_i^a(x, t),$$

with a labeling the reduced internal carrier directions and i a spatial index.

The working microscopic branch is the refined locked TECT sector together with the explicit Class II mediator:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi].$$

We do not assume that this is already the final complete closed TECT action. Rather, this is the strongest explicit microscopic branch currently available for the actual PDE derivation.

C. Periodic BCC background

Let the locked periodic condensate be

$$\Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger,$$

where

$$\phi_0(x) = \sum_{G \in S_{\text{BCC}}} A_G e^{iG \cdot x}$$

is a periodic BCC background built from the selected reciprocal shell S_{BCC} . For the microscopic carrier construction we take the static mediator background

$$X_{0i}^a = 0$$

unless a nonzero saddle requires otherwise.

We then expand

$$\Psi = \Psi_0 + \delta\Psi, \quad X_i^a = \delta X_i^a.$$

D. Linearization around the periodic background

The full linearized microscopic system takes the schematic form

$$\mathcal{L}_{\text{full}} \begin{pmatrix} \delta\Psi \\ \delta X \end{pmatrix} = 0,$$

with block operator

$$\mathcal{L}_{\text{full}} = \begin{pmatrix} \mathcal{L}_{\Psi\Psi} & \mathcal{L}_{\Psi X} \\ \mathcal{L}_{X\Psi} & \mathcal{L}_{XX} \end{pmatrix}.$$

Here:

1. $\mathcal{L}_{\Psi\Psi}$ is the linearization of the refined locked TECT operator around (Ψ_0, P_0) ;
2. $\mathcal{L}_{XX}$ contains the heavy mediator mass term and any derivative corrections if present;
3. $\mathcal{L}_{\Psi X}, \mathcal{L}_{X\Psi}$ come from the linearization of the Class II couplings.

The crucial point is that this linearized operator is periodic with the BCC periodicity of $\phi_0(x)$. Hence it admits a Floquet–Bloch decomposition.

E. Bloch fiber decomposition

For a Bloch momentum k , write

$$\delta\Psi_k(x) = e^{ik \cdot x} u_k(x), \quad \delta X_{i,k}^a(x) = e^{ik \cdot x} v_{i,k}^a(x),$$

with u_k, v_k periodic on the BCC unit cell.

Then the full linearized operator decomposes fiberwise:

$$\mathcal{L}_{\text{full}} \simeq \int_{\text{BZ}}^{\oplus} \mathcal{H}_{\text{Bloch}}(k) dk,$$

where

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}.$$

Expand the periodic Bloch amplitudes in reciprocal harmonics:

$$u_k(x) = \sum_{G \in \mathcal{G}} u_G e^{iG \cdot x}, \quad v_{i,k}^a(x) = \sum_{G \in \mathcal{G}} v_{i,G}^a e^{iG \cdot x}.$$

Then $\mathcal{H}_{\text{Bloch}}(k)$ becomes an infinite matrix in the reciprocal-harmonic basis:

$$(\mathcal{H}_{\text{Bloch}}(k))_{G,G'}.$$

This is the actual operator-theoretic starting point of the full PDE program.

F. Heavy-mediator elimination at the Bloch-fiber level

Because the mediator is heavy, the first practical reduction is to eliminate the X -sector inside each Bloch fiber. Formally,

$$H_{XX}(k) = M_X^2 \mathbf{1} + \delta H_{XX}(k), \quad M_X^2 \gg |k|^2, |G|^2.$$

Hence, to leading order in M_X^{-2} ,

$$v_k = -H_{XX}(k)^{-1} H_{X\Psi}(k) u_k.$$

Substituting back gives the effective Bloch operator for the Ψ -sector:

$$\boxed{H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k).}$$

This is the periodic enriched microscopic operator whose shell-valley block must be analyzed.

G. Selection of the shell-relevant harmonic sector

Choose a shell/valley point k_* associated with the opposite shell pair $\pm G_*$. We write

$$k = k_* + p, \quad |p| \ll |G_*|.$$

Let $S_* \subset \mathcal{G}$ denote the finite set of near-resonant reciprocal harmonics around k_* . Define the shell projector

$$\Pi_{S_*} : \mathcal{H}_{\text{Bloch}}(k) \rightarrow \mathcal{H}_{S_*}(k).$$

Then the shell-projected effective Bloch operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}.$$

This is the first finite-dimensional operator that belongs to the actual full-PDE derivation. Everything before this point is still exact operator setup; everything after this point is finite matrix analysis.

H. Valley projection inside the shell block

Assume the shell block contains an isolated two-dimensional complex low-energy sector. Let

$$P_* : \mathcal{H}_{S_*}(0) \rightarrow \mathcal{V}_*$$

be the spectral projector onto that valley subspace:

$$\dim_{\mathbb{C}} \mathcal{V}_* = 2.$$

Define the first derivatives

$$K_i := \partial_{p_i} H_{\text{shell}}(p)|_{p=0}.$$

Then the projected linear valley operator is

$$H_{\text{val}}^{(1)}(p) = p_i P_* K_i P_*.$$

This is the exact place where the actual microscopic coefficients are to be extracted. Equivalently,

$$\boxed{H_{\text{shell}}(p) \rightarrow K_i \rightarrow P_* K_i P_*}$$

is the true computational bottleneck.

I. Extraction of the Pauli-symbol coefficients

Choose the shell-adapted frame

$$\{\hat{n}, e_1, e_2\}, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2.$$

Inside the valley space $\mathcal{V}_*$, decompose

$$P_* K_i P_*$$

in the Pauli basis.

The target form is

$$\boxed{H_{\text{val}}^{(1)}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2) + \gamma_i p_i \mathbf{1}_2.}$$

The identity term $\gamma_i p_i \mathbf{1}_2$ is irrelevant for Weyl nondegeneracy. The actual witness coefficients are therefore extracted by the Pauli projections:

$$\lambda_{\parallel} = \frac{1}{2} \text{tr}[\sigma_3 (\hat{n}_i P_* K_i P_*)],$$

$$\alpha = \frac{1}{2} \text{tr}[\sigma_1 (e_{1i} P_* K_i P_*) + \sigma_2 (e_{2i} P_* K_i P_*)],$$

$$\beta = \frac{1}{2} \text{tr}[\sigma_2 (e_{1i} P_* K_i P_*) - \sigma_1 (e_{2i} P_* K_i P_*)].$$

Equivalently, in enriched-tensor language this is the extraction of the shell tensor

$$N_{ij}^a$$

from the actual projected Bloch derivative operator.

J. Why enriched structure is mandatory

A plain scalar pair truncation is not enough. The minimal scalar shell-pair Bloch problem yields at most a rank-deficient linear traceless symbol and therefore does not generically produce a full Weyl cone.

Hence the full-PDE program must retain enriched internal structure:

multi-component field, projector sector, or Class II-mediated carrier block.

This is exactly why the refined locked + Class II branch is the correct working microscopic branch for the complete derivation.

K. Actual completion criterion

The full-PDE derivation of the shell witness is complete once the following finite tasks are carried out:

1. construct the periodic fiber operator $H_{\text{eff}}(k)$;
2. choose the near-resonant harmonic set S_* and build $H_{\text{shell}}(p)$;
3. isolate the two-dimensional valley projector P_* ;
4. compute

$$K_i = \partial_{p_i} H_{\text{shell}}(0);$$

5. evaluate the three Pauli projections giving

$$\lambda_{\parallel}, \quad \alpha, \quad \beta.$$

After that, the witness theorem is not to be reproved. One simply checks

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

L. Sharp honest status

At the present stage, what remains open is not the abstract logic of Weyl/Dirac reduction. It is the explicit derivation of the microscopic objects

$$H_{\text{eff}}(k), \quad H_{\text{shell}}(p), \quad P_* K_i P_*,$$

from the final periodic refined locked TECT/Class II operator.

Therefore the next concrete theorem target is:

construct the shell-projected enriched Bloch operator of the refined locked + Class II microscopic PDE and extract $(\lambda_{\parallel}, \alpha, \beta)$.

(20)

VI. FINAL SUMMARY

The project goal is fixed:

derive the actual full microscopic PDE / Bloch linearization and extract the local Dirac data from it.
--

Hence the correct no-loop progression is:

periodic refined locked microscopic operator $\rightarrow$ Bloch fiber $\rightarrow$ shell truncation $\rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta)$.
--

The key facts are:

1. the minimal locked isotropic PDE does not by itself produce the required enriched shell tensor;
2. the correct next calculation is the periodic BCC Bloch linearization, not another abstract witness proof;
3. a plain scalar shell-pair truncation is insufficient, so enriched internal structure must be kept;
4. the refined locked + Class II branch is therefore the correct current microscopic branch for the full-PDE derivation.

The actual completion criterion is finite:

$$\boxed{H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).}$$

Once these are computed, one only checks

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

and then moves on immediately to

$$V(0) = 0, \quad \Delta > 0, \quad \text{mass protection.}$$

So the repetitive phase is over. The next stage is an actual operator-construction stage.

VII. FULL PERIODIC BLOCH-FIBER CONSTRUCTION FOR THE REFINED LOCKED + CLASS II MICROSCOPIC PDE

A. Conventions and the no-loop rule

The final target is the actual microscopic projected valley operator extracted from the full periodic TECT linearization. From this point onward, we do *not* re-prove the witness theorem. The only legitimate next task is to construct the periodic Bloch operator and extract its projected first derivatives.

To remove the remaining notation loop, we now fix the *folded Bragg convention*. Instead of working with the unfolded shell variable

$$q = G_* + p,$$

we work at the Bragg quasimomentum

$$\boxed{k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.}$$

This is equivalent to the unfolded description after Brillouin-zone folding, but it is the correct convention for the finite-dimensional Bloch matrix calculation.

B. The explicit working microscopic branch

The strongest explicit microscopic branch currently available is the refined locked TECT sector coupled to the Class II heavy mediator:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (21)$$

We do not claim that (21) is already the final complete TECT action. Rather, it is the strongest explicit branch in which the full-PDE derivation can now be pushed concretely.

C. The 12-mode BCC background

We choose the first BCC reciprocal shell

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |\mathcal{S}_{12}| = 12. \quad (22)$$

Equivalently, choosing one representative from each opposite pair,

$$\phi_0(x) = 2A \sum_{\alpha=1}^6 \cos(Q_\alpha \cdot x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}. \quad (23)$$

The locked background is

$$\Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger. \quad (24)$$

Let

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs},$$

be an orthonormal internal basis adapted to the locked decomposition

$$\mathbb{C}^3 = \mathbb{C}z_0 \oplus Q_{z_0}.$$

D. Bloch-harmonic expansion of the microscopic fluctuations

Let Γ^* denote the reciprocal lattice generated by the BCC periodic background. For each Bloch momentum k , expand

$$\delta\Psi_k(x) = e^{ik \cdot x} \sum_{G \in \Gamma^*} \sum_{r=0}^2 \psi_{G,r}(k) u_r e^{iG \cdot x}, \quad (25)$$

and

$$\delta X_{i,k}^a(x) = e^{ik \cdot x} \sum_{G \in \Gamma^*} X_{i,G}^a(k) e^{iG \cdot x}. \quad (26)$$

Thus the full linearized periodic problem is an infinite-dimensional Bloch-fiber eigenproblem on the coefficient space

$$\ell^2(\Gamma^*) \otimes \mathbb{C}_{\text{int}}^3 \oplus \ell^2(\Gamma^*) \otimes \mathbb{C}_{\text{space}}^3 \otimes \mathbb{C}_{\text{med}}^{N_a}.$$

E. Full Bloch block operator

In the harmonic basis, the linearized Bloch operator has the block form

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}. \quad (27)$$

1. A. The $\Psi\Psi$ -block

Write

$$(H_{\Psi\Psi}(k))_{(G,r),(G',s)} = \delta_{GG'} \delta_{rs} h_r(k+G) + U_{rs}^{(\text{per})}(G-G') + U_{rs}^{(\text{lock})}(G-G'). \quad (28)$$

Here:

1. $h_r(q)$ is the translation-invariant bare kernel in the u_r -channel;

2. $U^{(\text{per})}$ comes from the periodic BCC background ϕ_0 ;
3. $U^{(\text{lock})}$ contains the projector-locking and internal-background Hessian terms.

For the isotropic Brazovskii-type bare part,

$$h_0(q) = r_0 + Z|q|^2 + Y|q|^4, \quad (29)$$

and similarly for the locked transverse channels, up to channel-dependent mass shifts.

2. B. The ΨX - and $X\Psi$ -blocks

Define the background Fourier coefficients

$$\Phi_Q := A, \quad Q \in \mathcal{S}_{12}, \quad (30)$$

so that

$$\phi_0(x) = \sum_{Q \in \mathcal{S}_{12}} \Phi_Q e^{iQ \cdot x}.$$

Also define the internal overlap tensors

$$\Xi_{rs}^a := u_r^\dagger \lambda^a u_s. \quad (31)$$

The linearized J -vertex has the universal harmonic selection rule

$$(\mathcal{J}_i^a(k))_{(G,r),(G',s)} = \sum_{Q \in \mathcal{S}_{12}} \delta_{G'-G,Q} \Phi_Q \Xi_{rs}^a \mathcal{D}_i(k; G, G'; Q), \quad (32)$$

where $\mathcal{D}_i$ is linear in the Bloch-shifted momentum and in the background wavevector:

$$\mathcal{D}_i(k; G, G'; Q) = c_1 (k_i + G'_i) + c_2 Q_i. \quad (33)$$

The constants c_1, c_2 depend only on the normalization convention for the microscopic current J_i^a .

Likewise, the K -vertex has the same harmonic support pattern:

$$(\mathcal{K}_i^a(k))_{(G,r),(G',s)} = \sum_{Q \in \Gamma^*} \delta_{G'-G,Q} \Pi_Q^{a;rs} \tilde{\mathcal{D}}_i(k; G, G'; Q), \quad (34)$$

where $\Pi_Q^{a;rs}$ are the Fourier coefficients of the projector/background-induced transport kernel, and $\tilde{\mathcal{D}}_i$ is again linear in the shifted momentum.

Hence

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad H_{X\Psi}(k) = H_{\Psi X}(k)^\dagger. \quad (35)$$

3. C. The XX -block

To leading order in the heavy-mediator expansion,

$$(H_{XX}(k))_{(G,i,a),(G',j,b)} = \delta_{GG'} \delta_{ij} \delta_{ab} M_X^2 + \Delta_{(G,i,a),(G',j,b)}^{XX}(k), \quad (36)$$

with

$$\|\Delta^{XX}(k)\| \ll M_X^2$$

in the heavy regime.

F. Elimination of the heavy mediator inside each Bloch fiber

Define

$$\mathcal{C}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k). \quad (37)$$

Then the exact fiberwise Schur complement is

$$\boxed{H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k) H_{XX}(k)^{-1} \mathcal{C}(k)^\dagger.} \quad (38)$$

In the heavy-mass expansion,

$$H_{XX}(k)^{-1} = \frac{1}{M_X^2} \mathbf{1} - \frac{1}{M_X^4} \Delta^{XX}(k) + O(M_X^{-6}), \quad (39)$$

hence

$$\boxed{H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}).} \quad (40)$$

At the harmonic-matrix level,

$$(H_{\text{eff}}(k))_{(G,r),(G',s)} = (H_{\Psi\Psi}(k))_{(G,r),(G',s)} - \frac{1}{M_X^2} \sum_{G'',i,a} \mathcal{C}_{(G,r),(G'',i,a)}(k) \overline{\mathcal{C}_{(G',s),(G'',i,a)}(k)} + O(M_X^{-4}). \quad (41)$$

This is the first explicit microscopic formula that one can directly use in the full PDE program.

G. Choice of the near-Bragg shell sector

Let $G_* \in \mathcal{S}_{12}$ be the selected shell vector, and let

$$k_* = \frac{G_*}{2}.$$

The near-Bragg resonant harmonics are those $G \in \Gamma^*$ for which

$$|k_* + G|$$

lies near the shell minimum q_* .

Choose a finite harmonic set $S_* \subset \Gamma^*$ containing the resonant pair and the nearest background-coupled harmonics. Define the shell projector

$$\Pi_{S_*} : \ell^2(\Gamma^*) \otimes \mathbb{C}_{\text{int}}^3 \rightarrow \mathcal{H}_{S_*}. \quad (42)$$

Then

$$\boxed{H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}.} \quad (43)$$

This is now a *finite* matrix. Its size is

$$3 |S_*|$$

before any internal or spectral reduction.

H. Linear expansion of the shell block

Expand

$$H_{\text{shell}}(p) = H_{\text{shell}}(0) + p_i K_i + O(|p|^2), \quad K_i := \partial_{p_i} H_{\text{shell}}(p)|_{p=0}. \quad (44)$$

Using (40),

$$K_i = K_i^{(\Psi\Psi)} + K_i^{(\text{II})} + O(M_X^{-4}), \quad (45)$$

with

$$K_i^{(\Psi\Psi)} = \Pi_{S_*} \partial_{k_i} H_{\Psi\Psi}(k)|_{k=k_*} \Pi_{S_*}, \quad (46)$$

and

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \partial_{k_i} (\mathcal{C}(k) \mathcal{C}(k)^\dagger)|_{k=k_*} \Pi_{S_*}. \quad (47)$$

Equivalently,

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}. \quad (48)$$

I. Why the Class II derivative term is the correct leading target

For the isotropic bare kernel (29),

$$\partial_{q_i} h_0(q) = 2(Z + 2Y|q|^2) q_i. \quad (49)$$

At the Brazovskii shell,

$$|q| = q_*, \quad q_*^2 = -\frac{Z}{2Y}, \quad (50)$$

one gets

$$\partial_{q_i} h_0(q)|_{|q|=q_*} = 0. \quad (51)$$

Thus the isotropic first-order shell transport vanishes identically on-shell. So the first genuinely nontrivial linear shell tensor must come from either:

1. the full 12-mode periodic mixing beyond the locked single-mode cancellation,
2. or the explicit Class II self-energy contribution (48),
3. or both.

This is exactly why the present full-PDE program must focus on (48) instead of recycling the minimal isotropic calculation.

J. Valley isolation and final projected coefficient formulas

Assume the shell block contains a spectrally isolated two-dimensional complex valley subspace

$$\mathcal{V}_* \subset \mathcal{H}_{S_*}, \quad \dim_{\mathbb{C}} \mathcal{V}_* = 2,$$

with spectral projector P_* .

Then the linear valley Hamiltonian is

$$h_+^{\text{lin}}(p) = P_*(p_i K_i) P_*. \quad (52)$$

Choose the shell-adapted frame

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2, \quad \hat{n} = \frac{G_*}{|G_*|}.$$

The target reduced form is

$$h_+^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2) + \gamma_i p_i \mathbf{1}_2. \quad (53)$$

Hence the actual microscopic coefficients are extracted by

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{V}_*} [\sigma_3 P_* (\hat{n}_i K_i) P_*], \quad (54)$$

$$\alpha = \frac{1}{2} \text{Tr}_{\mathcal{V}_*} [\sigma_1 P_* (e_{1i} K_i) P_* + \sigma_2 P_* (e_{2i} K_i) P_*], \quad (55)$$

$$\beta = \frac{1}{2} \text{Tr}_{\mathcal{V}_*} [\sigma_2 P_* (e_{1i} K_i) P_* - \sigma_1 P_* (e_{2i} K_i) P_*]. \quad (56)$$

K. What has actually been achieved in this step

This step does *not* yet diagonalize the full final microscopic TECT operator. But it does remove the remaining conceptual ambiguity in the full-PDE program.

The actual microscopic task is now reduced to the following finite construction:

1. build the harmonic matrices $H_{\Psi\Psi}(k)$, $\mathcal{J}(k)$, $\mathcal{K}(k)$;
2. form the mediator-eliminated effective Bloch operator (41);
3. choose the near-Bragg shell set S_* and construct $H_{\text{shell}}(p)$;
4. isolate P_* and compute K_i ;
5. evaluate (54)–(56).

So the full microscopic frontier is no longer vague. It is exactly the explicit construction of the finite shell matrix and its projected derivatives.

VIII. FINAL SUMMARY

The no-loop full-PDE program is now fixed in the folded Bragg convention

$$k_* = G_*/2.$$

The 12-mode BCC background is

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

Expanding the refined locked + Class II microscopic theory in Bloch harmonics gives the block fiber operator

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}.$$

After eliminating the heavy mediator,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k) H_{XX}(k)^{-1} \mathcal{C}(k)^\dagger, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k),$$

and in the heavy-mass regime,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}).$$

Projecting to the near-Bragg harmonic shell S_* yields the finite shell matrix

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*},$$

with first derivatives

$$K_i = \partial_{p_i} H_{\text{shell}}(p)|_{p=0}.$$

The leading Class II contribution is

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}.$$

Since the isotropic Brazovskii derivative vanishes exactly on the shell,

$$\partial_{q_i} h_0(q)|_{|q|=q_*} = 0,$$

the first genuinely nontrivial shell transport data must come from full periodic mixing and/or the Class II self-energy derivative.

Finally, once the two-dimensional valley projector P_* is isolated, the actual microscopic coefficients are

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}[\sigma_3 P_* (\hat{n}_i K_i) P_*],$$

$$\alpha = \frac{1}{2} \text{Tr}[\sigma_1 P_* (e_{1i} K_i) P_* + \sigma_2 P_* (e_{2i} K_i) P_*],$$

$$\beta = \frac{1}{2} \text{Tr}[\sigma_2 P_* (e_{1i} K_i) P_* - \sigma_1 P_* (e_{2i} K_i) P_*].$$

Therefore the full microscopic frontier is now completely localized to one finite construction pipeline:

$$(H_{\Psi\Psi}, \mathcal{J}, \mathcal{K}) \rightarrow H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

IX. MINIMAL FINITE SHELL MATRIX AT THE BRAGG POINT AND THE EMERGENT ENRICHED PAIR SPACE

A. Step B goal

The goal of this step is to pass from the infinite Bloch fiber operator

$$H_{\text{eff}}(k)$$

to the smallest finite shell matrix that still retains the actual Dirac/Weyl data.

We work in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

The core question is:

What is the minimal finite harmonic sector that contains the actual shell-pair valley data?

B. Core resonant harmonic pair

At the Bragg point $k_* = G_*/2$, the two harmonics

$$G = 0, \quad G = -G_*$$

are exactly degenerate at the bare shell level because

$$|k_* + 0| = \left| \frac{G_*}{2} \right|, \quad |k_* - G_*| = \left| -\frac{G_*}{2} \right|.$$

Therefore the minimal resonant harmonic pair is

$$\boxed{S_{\text{core}} := \{0, -G_*\}}. \quad (57)$$

This is the folded version of the opposite shell pair $\pm G_*$ in the unfolded picture.

C. Why the core pair is not yet the whole story

Because the BCC background has Fourier support on the 12-mode shell

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\},$$

the periodic background scattering couples harmonics by shifts

$$G \mapsto G + Q, \quad Q \in \mathcal{S}_{12}.$$

Hence the core pair S_{core} is not closed under one scattering step.

The one-step closure generated from S_{core} is

$$S_{\text{cl}}^{(1)} = S_{\text{core}} \cup (0 + \mathcal{S}_{12}) \cup (-G_* + \mathcal{S}_{12}). \quad (58)$$

For a generic choice of $G_* \in \mathcal{S}_{12}$, this contains 20 distinct harmonics after removing duplicates.

Thus the correct finite shell program has two natural levels:

1. the minimal core pair S_{core} , which captures the principal Bragg degeneracy;
2. the one-step closed shell $S_{\text{cl}}^{(1)}$, whose spectators renormalize the core pair through Schur complement.

D. Core/spectator decomposition

Let

$$S_* := S_{\text{cl}}^{(1)}.$$

Then decompose the shell Hilbert space as

$$\mathcal{H}_{S_*} = \mathcal{H}_{\text{core}} \oplus \mathcal{H}_{\text{sp}}, \quad (59)$$

where

$$\mathcal{H}_{\text{core}} := \text{span}\{|0, r\rangle, |-G_*, r\rangle\},$$

and r runs over the shell-relevant internal doublet basis of Q_{z_0} . Hence

$$\boxed{\mathcal{H}_{\text{core}} \cong \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2, \quad \dim_{\mathbb{C}} \mathcal{H}_{\text{core}} = 4.} \quad (60)$$

The remainder

$$\mathcal{H}_{\text{sp}} = \mathcal{H}_{S_*} \ominus \mathcal{H}_{\text{core}}$$

is the spectator shell sector.

E. Block form of the shell matrix

The finite shell matrix has the block form

$$H_{\text{shell}}(p) = \begin{pmatrix} H_{cc}(p) & H_{cs}(p) \\ H_{sc}(p) & H_{ss}(p) \end{pmatrix}, \quad (61)$$

where:

- $H_{cc}(p)$ acts on $\mathcal{H}_{\text{core}} \cong \mathbb{C}^2 \otimes \mathbb{C}^2$,
- $H_{ss}(p)$ acts on the spectator sector,
- H_{cs}, H_{sc} are the background-induced and mediator-induced couplings between the two.

F. Core effective Hamiltonian by Schur complement

Assume the spectator block is gapped away from the target shell energy E_* :

$$\text{dist}(E_*, \text{spec } H_{ss}(0)) \geq \Delta_{\text{sp}} > 0. \quad (62)$$

Then the exact finite core effective Hamiltonian is

$$H_{\text{core}}^{\text{eff}}(E, p) = H_{cc}(p) - H_{cs}(p) (H_{ss}(p) - E)^{-1} H_{sc}(p). \quad (63)$$

For the low-energy shell program one expands this at $E \approx E_*$, $p \approx 0$:

$$H_{\text{core}}^{\text{eff}}(p) = H_{\text{core}}^{\text{eff}}(E_*, 0) + p_i \mathcal{K}_i^{\text{core}} + O(|p|^2). \quad (64)$$

This is the actual finite matrix object from which the projected coefficients must be extracted.

G. Canonical pair $\otimes$ internal decomposition

Let τ^a act on the pair basis

$$|+\rangle := |0\rangle, \quad |-\rangle := |-G_*\rangle,$$

and let s^a act on the internal doublet basis of Q_{z_0} .

Then any Hermitian 4×4 core operator can be expanded in

$$\tau^\mu \otimes s^\nu, \quad \mu, \nu = 0, 1, 2, 3.$$

The *minimal* symmetry-compatible shell-core expansion near $p = 0$ is

$$\begin{aligned} H_{\text{core}}^{\text{eff}}(p) = & E_*(p) \mathbf{1}_4 + m_1 \tau^1 \otimes \mathbf{1}_2 + m_2 \tau^2 \otimes \mathbf{1}_2 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 \\ & + \tau^1 \otimes (p_1 W^{(1)} + p_2 W^{(2)}) + \tau^2 \otimes (p_1 \widetilde{W}^{(1)} + p_2 \widetilde{W}^{(2)}) + O(|p|^2), \end{aligned} \quad (65)$$

where $W^{(a)}, \widetilde{W}^{(a)} \in \text{Herm}(2)$ act on the internal doublet.

The constant terms

$$m_1 \tau^1 \otimes \mathbf{1}_2, \quad m_2 \tau^2 \otimes \mathbf{1}_2$$

are the intervalley constant mixing terms. If they are nonzero, they generate a Bragg mass at $p = 0$.

H. The plain scalar pair versus the enriched pair

For a plain periodic scalar pair problem, the minimal two-harmonic Bragg truncation gives only

$$E_0(p)\mathbf{1}_2 + v_{\parallel}p_{\parallel}\tau^3 + m\tau^1 + O(|p|^2),$$

so its linear traceless symbol is only one-dimensional. Therefore the plain scalar pair does not generically produce a full Weyl cone.

In contrast, the enriched shell core

$$\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$$

is large enough to support the Dirac generators

$$\Gamma_1 = \tau^1 \otimes s^1, \quad \Gamma_2 = \tau^1 \otimes s^2, \quad \Gamma_3 = \tau^3 \otimes \mathbf{1}_2,$$

which satisfy

$$\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}\mathbf{1}_4.$$

So the full-PDE shell target is not a plain 2×2 scalar pair Hamiltonian, but an enriched 4×4 pair $\otimes$ internal Hamiltonian.

I. Canonical gauge choice for the off-diagonal pair block

By a pair-space $U(1)$ rephasing, one may absorb one overall complex phase in the intervalley block. Hence, after a convenient basis choice, the linear transverse off-diagonal block can be written using τ^1 only:

$$H_{\text{core}}^{\text{can}}(p) = E_*(p)\mathbf{1}_4 + m\tau^1 \otimes \mathbf{1}_2 + v_{\parallel}p_{\parallel}\tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1W^1 + p_2W^2) + O(|p|^2), \quad (66)$$

where $W^1, W^2 \in \text{Herm}(2)$.

The full shell-core problem is now compressed to determining:

$$m, \quad v_{\parallel}, \quad W^1, \quad W^2.$$

J. Minimal conditions for an actual Weyl/Dirac shell block

The folded enriched shell Hamiltonian (66) becomes a massless Dirac principal operator if:

a. (D1) *No constant intervalley mass*

$$\boxed{m = 0.} \quad (67)$$

b. (D2) *Nonzero longitudinal splitting*

$$\boxed{v_{\parallel} \neq 0.} \quad (68)$$

c. (D3) *Full-rank transverse internal symbol*

$$\boxed{W^1, W^2 \text{ are traceless Hermitian, nonzero, and linearly independent;}} \quad (69)$$

and, after internal basis reduction,

$$\boxed{W^1 = \nu_1 s^1, \quad W^2 = \nu_2 s^2, \quad \nu_1 \nu_2 \neq 0.} \quad (70)$$

Under (67)–(70),

$$\boxed{H_{\text{core}}^{\text{can}}(p) = E_*(p)\mathbf{1}_4 + v_{\parallel}p_{\parallel}\tau^3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2 + O(|p|^2),} \quad (71)$$

which is an exact massless Dirac principal symbol on the enriched shell space.

K. How the 2-dimensional valley sector appears

There are two equivalent descriptions.

a. Folded description. One keeps the full 4-dimensional core space

$$\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

and (71) is already the local massless Dirac operator.

b. Unfolded description. One unfolds back to the opposite shell pair $\pm G_*$. Then the same 4×4 folded Dirac block decomposes into two opposite-chirality 2×2 Weyl valleys. Equivalently, there exist rank-two projectors

$$P_+, P_-, \quad \dim_{\mathbb{C}} P_{\pm} \mathcal{H}_{\text{core}} = 2,$$

selecting the two valley sectors.

Therefore the practical existence conditions for a 2-dimensional valley subspace are:

1. the spectator sector is separated by a finite gap $\Delta_{\text{sp}} > 0$;
2. the core effective Hamiltonian has $m = 0$ at $p = 0$;
3. the longitudinal derivative coefficient satisfies $v_{\parallel} \neq 0$;
4. the transverse internal matrices W^1, W^2 are full-rank in the Pauli sense.

L. Actual projector-derivative data to compute next

Once $H_{\text{core}}^{\text{eff}}(p)$ is obtained, one computes

$$\mathcal{K}_i^{\text{core}} = \partial_{p_i} H_{\text{core}}^{\text{eff}}(p) \Big|_{p=0}.$$

In the canonical basis (66),

$$\mathcal{K}_{\parallel}^{\text{core}} = v_{\parallel} \tau^3 \otimes \mathbf{1}_2, \quad \mathcal{K}_1^{\text{core}} = \tau^1 \otimes W^1, \quad \mathcal{K}_2^{\text{core}} = \tau^1 \otimes W^2.$$

So the remaining actual microscopic task is now finite and concrete: extract $m, v_{\parallel}, W^1, W^2$ from the Schur-complemented shell matrix built from the full periodic refined locked + Class II Bloch operator.

X. FINAL SUMMARY

At the folded Bragg point

$$k_* = \frac{G_*}{2},$$

the minimal resonant harmonic pair is

$$S_{\text{core}} = \{0, -G_*\}.$$

Because the periodic BCC background scatters harmonics by

$$G \mapsto G + Q, \quad Q \in \mathcal{S}_{12},$$

the core pair must be supplemented by spectator harmonics. The smallest systematic choice is the one-step closure

$$S_{\text{cl}}^{(1)} = S_{\text{core}} \cup (0 + \mathcal{S}_{12}) \cup (-G_* + \mathcal{S}_{12}).$$

After splitting

$$\mathcal{H}_{S_*} = \mathcal{H}_{\text{core}} \oplus \mathcal{H}_{\text{sp}},$$

the exact finite core Hamiltonian is obtained by Schur complement:

$$H_{\text{core}}^{\text{eff}}(E, p) = H_{cc}(p) - H_{cs}(p)(H_{ss}(p) - E)^{-1}H_{sc}(p).$$

The shell-relevant core space is

$$\boxed{\mathcal{H}_{\text{core}} \cong \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2.}$$

This is the minimal enriched shell space. A plain scalar 2×2 Bragg pair is not enough, because its linear traceless symbol is only one-dimensional. The enriched 4×4 shell core is the smallest space capable of carrying the three Dirac generators

$$\Gamma_1 = \tau^1 \otimes s^1, \quad \Gamma_2 = \tau^1 \otimes s^2, \quad \Gamma_3 = \tau^3 \otimes \mathbf{1}_2.$$

After basis reduction, the shell-core Hamiltonian takes the canonical form

$$H_{\text{core}}^{\text{can}}(p) = E_*(p)\mathbf{1}_4 + m\tau^1 \otimes \mathbf{1}_2 + v_{\parallel}p_{\parallel}\tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1W^1 + p_2W^2) + O(|p|^2).$$

The shell block is massless Dirac at principal order if

$$m = 0, \quad v_{\parallel} \neq 0, \quad W^1 = \nu_1 s^1, \quad W^2 = \nu_2 s^2, \quad \nu_1 \nu_2 \neq 0.$$

Equivalently,

$$H_{\text{core}}^{\text{can}}(p) = E_*(p)\mathbf{1}_4 + v_{\parallel}p_{\parallel}\tau^3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau^1 \otimes s^1 + \nu_2 p_2 \tau^1 \otimes s^2 + O(|p|^2).$$

Thus the remaining full-PDE frontier is now a completely finite computation:

$$\boxed{(H_{cc}, H_{cs}, H_{ss}) \rightarrow H_{\text{core}}^{\text{eff}} \rightarrow (m, v_{\parallel}, W^1, W^2).}$$

XI. EXPLICIT BLOCK-OVERLAP FORMULAE FOR THE CORE SHELL MATRIX

A. Purpose

The goal of this step is to stop the structural loop completely and replace it by an explicit finite matrix-element computation.

At the Bragg point

$$k_* = \frac{G_*}{2},$$

the enriched shell basis is taken as

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2, \quad (72)$$

where $u_a(x)$ are periodic internal-envelope functions spanning the shell doublet. The projected shell Hamiltonian has the 2×2 block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}, \quad (73)$$

with $E_{\pm}(p)$ and $B(p)$ acting on the internal doublet. This finite block contains all shell-level Dirac data.

B. Core and spectator basis

Fix the folded core pair

$$S_{\text{core}} = \{0, -G_*\}.$$

Let the internal shell-doublet index be $a = 1, 2$, and let spectator harmonics be labeled by

$$\eta \in S_{\text{sp}} := S_* \setminus S_{\text{core}}.$$

Define the core basis vectors

$$|+, a\rangle := |0, a\rangle, \quad |-, a\rangle := |-G_*, a\rangle,$$

and spectator basis vectors

$$|\eta, b\rangle.$$

After eliminating the heavy mediator X , the effective Bloch operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k)H_{XX}(k)^{-1}\mathcal{C}(k)^\dagger, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

The shell matrix is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}.$$

C. Exact block definitions

Define the shell-core blocks by matrix elements:

$$(E_+(p))_{ab} = \langle +, a | H_{\text{shell}}(p) | +, b \rangle, \quad (E_-(p))_{ab} = \langle -, a | H_{\text{shell}}(p) | -, b \rangle, \quad (74)$$

$$(B(p))_{ab} = \langle +, a | H_{\text{shell}}(p) | -, b \rangle. \quad (75)$$

Likewise, define the core-spectator coupling blocks

$$(C_+(p))_{a,\eta b} = \langle +, a | H_{\text{shell}}(p) | \eta, b \rangle, \quad (C_-(p))_{a,\eta b} = \langle -, a | H_{\text{shell}}(p) | \eta, b \rangle, \quad (76)$$

and the spectator block

$$(H_{ss}(p))_{\eta a, \zeta b} = \langle \eta, a | H_{\text{shell}}(p) | \zeta, b \rangle. \quad (77)$$

Then the finite shell matrix decomposes as

$$H_{\text{shell}}(p) = \begin{pmatrix} E_+(p) & B(p) & C_+(p) \\ B(p)^\dagger & E_-(p) & C_-(p) \\ C_+(p)^\dagger & C_-(p)^\dagger & H_{ss}(p) \end{pmatrix}. \quad (78)$$

D. Exact core effective block after spectator elimination

Assume the spectator block remains spectrally separated from the target shell energy E_* . Then the exact core effective Hamiltonian is

$$H_{\text{core}}^{\text{eff}}(E, p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix} - \begin{pmatrix} C_+(p) \\ C_-(p) \end{pmatrix} (H_{ss}(p) - E)^{-1} (C_+(p)^\dagger \ C_-(p)^\dagger). \quad (79)$$

Therefore the renormalized pair-diagonal and pair-off-diagonal blocks are

$$E_+^{\text{eff}}(E, p) = E_+(p) - C_+(p) R_{\text{sp}}(E, p) C_+(p)^\dagger, \quad (80)$$

$$E_-^{\text{eff}}(E, p) = E_-(p) - C_-(p) R_{\text{sp}}(E, p) C_-(p)^\dagger, \quad (81)$$

$$B^{\text{eff}}(E, p) = B(p) - C_+(p) R_{\text{sp}}(E, p) C_-(p)^\dagger, \quad R_{\text{sp}}(E, p) := (H_{ss}(p) - E)^{-1}. \quad (82)$$

This is the first fully explicit finite formula for the actual microscopic shell core.

E. What each object means physically

The block $E_{\pm}(p)$ is the direct core self-energy in the $+$ and $-$ pair sectors. The block $B(p)$ is the direct pair-mixing matrix. The blocks $C_{\pm}(p)$ encode virtual scattering from the core pair into the one-step spectator shell. The Schur term

$$C_+(p)R_{\text{sp}}(E, p)C_-(p)^{\dagger}$$

is the virtual core $\rightarrow$ spectator $\rightarrow$ core intervalley process that renormalizes the pair-off-diagonal shell mixing.

Thus the actual microscopic constants are not read from $B(p)$ alone, but from the renormalized matrix $B^{\text{eff}}(E, p)$.

F. Selection rules from the 12-mode BCC background

Because the BCC background Fourier support is exactly the first shell

$$\mathcal{S}_{12},$$

every one-step background scattering changes the harmonic label by

$$G \mapsto G + Q, \quad Q \in \mathcal{S}_{12}.$$

Hence the core-core pair-mixing channel is directly allowed because

$$0 - (-G_*) = G_* \in \mathcal{S}_{12}.$$

Therefore the direct Bragg off-diagonal block $B(0)$ is generically nonzero already at the periodic background level.

Likewise, a spectator harmonic η couples to the $+$ core iff

$$\eta - 0 \in \mathcal{S}_{12},$$

and to the $-$ core iff

$$\eta + G_* \in \mathcal{S}_{12}.$$

These are the exact one-step selection rules defining the nonzero entries of C_+ and C_- .

So the entire finite shell block is sparse and combinatorial: every nonzero element is dictated by one of these shell-difference rules, dressed by the corresponding internal overlap matrix.

G. Direct microscopic origin of the constant mass matrix

At $p = 0$, define the renormalized constant pair-mixing matrix

$$\boxed{M_0 := B^{\text{eff}}(E_*, 0) = B(0) - C_+(0) R_{\text{sp}}(E_*, 0) C_-(0)^{\dagger}.} \quad (83)$$

This is the actual Bragg mass matrix on the internal doublet. Because M_0 is a general complex 2×2 matrix, the constant pair-off-diagonal part of the full 4×4 core Hamiltonian is

$$\begin{pmatrix} 0 & M_0 \\ M_0^{\dagger} & 0 \end{pmatrix} = \tau^1 \otimes \Re M_0 - \tau^2 \otimes \Im M_0. \quad (84)$$

Therefore the complete massless-crossing condition is exactly

$$\boxed{M_0 = 0.} \quad (85)$$

This is the precise microscopic version of the statement that all constant pair-off-diagonal mass terms must vanish on the projected shell block.

H. Direct microscopic origin of the longitudinal slope

The shell-level longitudinal coefficient is extracted from the renormalized pair-diagonal difference. Define

$$D^{\text{eff}}(E, p) := \frac{1}{2} (E_+^{\text{eff}}(E, p) - E_-^{\text{eff}}(E, p)). \quad (86)$$

Then the scalar longitudinal slope is

$$v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} D^{\text{eff}}(E_*, p) \Big|_{p=0}. \quad (87)$$

Equivalently,

$$v_{\parallel} = \frac{1}{4} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[E_+(p) - E_-(p) - C_+(p) R_{\text{sp}}(E_*, p) C_+(p)^{\dagger} + C_-(p) R_{\text{sp}}(E_*, p) C_-(p)^{\dagger} \right]_{p=0}. \quad (88)$$

This is the exact finite overlap formula for the longitudinal shell splitting.

I. Direct microscopic origin of the transverse Dirac matrices

Now define the renormalized off-diagonal derivative matrices

$$W^i := \partial_{p_i} B^{\text{eff}}(E_*, p) \Big|_{p=0}, \quad i = 1, 2. \quad (89)$$

Using (82),

$$\begin{aligned} W^i &= \partial_{p_i} B(p) \Big|_{p=0} - \partial_{p_i} \left[C_+(p) R_{\text{sp}}(E_*, p) C_-(p)^{\dagger} \right]_{p=0} \\ &= \partial_{p_i} B(p) \Big|_{p=0} - (\partial_{p_i} C_+) R_0 C_-^{\dagger} - C_+ R_0 (\partial_{p_i} C_-^{\dagger}) + C_+ R_0 (\partial_{p_i} H_{ss}) R_0 C_-^{\dagger} \Big|_{p=0}, \end{aligned} \quad (90)$$

where

$$R_0 := (H_{ss}(0) - E_*)^{-1},$$

and we have suppressed the explicit (0) on the undifferentiated blocks for readability.

Equation (90) is the main actual formula of this step. It shows that the transverse Dirac matrices come from two sources:

1. direct core pair-mixing derivatives $\partial_{p_i} B$;
2. virtual spectator renormalization through differentiated Schur complement terms.

J. Traceless Dirac part of the transverse matrices

Since an arbitrary W^i may contain an irrelevant scalar internal part, define

$$\widetilde{W}^i := W^i - \frac{1}{2} \text{tr}_{\text{int}}(W^i) \mathbf{1}_2. \quad (91)$$

The actual transverse Dirac data are carried by $\widetilde{W}^1, \widetilde{W}^2$.

Write

$$\widetilde{W}^i = w_a^i s^a, \quad w^i \in \mathbb{R}^3.$$

Then the projected shell block is Dirac at principal order precisely when the two vectors w^1, w^2 are linearly independent, and in the canonical anticommuting case one may choose an internal basis in which

$$\widetilde{W}^1 = \nu_1 s^1, \quad \widetilde{W}^2 = \nu_2 s^2, \quad \nu_1 \nu_2 \neq 0.$$

K. What the plain scalar sector does and does not contribute

The plain scalar BCC Bragg pair already gives a direct constant pair-mixing term $B(0)$, i.e. a Bragg mass channel. But it does not generically provide two independent traceless transverse internal matrices at linear order.

Therefore the actual role of the enriched microscopic structure is sharply separated:

1. the periodic BCC background naturally supplies the Bragg pair structure and hence the potential constant mass channel M_0 ;
2. the enriched internal structure plus the Class II-induced transport terms are what can generate nontrivial $\widetilde{W}^1, \widetilde{W}^2$.

So the true full-PDE task is not vague any more: one must compute $M_0, v_{\parallel}, W^1, W^2$ from the microscopic overlap blocks above.

L. Final finite extraction dictionary

The full 4×4 core Hamiltonian near $p = 0$ is therefore

$$H_{\text{core}}^{\text{eff}}(p) = E_*(p)\mathbf{1}_4 + \tau^3 \otimes D_{\text{tr}}^{\text{eff}}(p) + \tau^1 \otimes \Re B^{\text{eff}}(p) - \tau^2 \otimes \Im B^{\text{eff}}(p), \quad (92)$$

and, in the minimal Dirac reduction regime,

$$H_{\text{core}}^{\text{eff}}(p) = E_0(p)\mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1 \widetilde{W}^1 + p_2 \widetilde{W}^2) + O(|p|^2), \quad (93)$$

provided

$$M_0 = 0$$

and the non-identity internal corrections in the pair-diagonal block are negligible or removable.

Thus the actual microscopic quantities to compute next are exactly

$$\boxed{M_0, \quad v_{\parallel}, \quad \widetilde{W}^1, \quad \widetilde{W}^2.} \quad (94)$$

XII. FINAL SUMMARY

At the Bragg point

$$k_* = \frac{G_*}{2},$$

the enriched shell-projected Hamiltonian has the finite block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^{\dagger} & E_-(p) \end{pmatrix},$$

where $E_{\pm}(p)$ and $B(p)$ are 2×2 matrices on the internal doublet.

Including the one-step spectator shell and integrating it out gives the exact renormalized core blocks

$$E_{\pm}^{\text{eff}}(E, p) = E_{\pm}(p) - C_{\pm}(p)(H_{ss}(p) - E)^{-1}C_{\pm}(p)^{\dagger},$$

$$B^{\text{eff}}(E, p) = B(p) - C_+(p)(H_{ss}(p) - E)^{-1}C_-(p)^{\dagger}.$$

The exact microscopic shell quantities are therefore:

$$\boxed{M_0 := B^{\text{eff}}(E_*, 0),}$$

the constant Bragg mass matrix,

$$v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left. \frac{E_{+}^{\text{eff}}(E_*, p) - E_{-}^{\text{eff}}(E_*, p)}{2} \right|_{p=0},$$

the longitudinal shell slope,
and

$$W^i = \partial_{p_i} B^{\text{eff}}(E_*, p) \Big|_{p=0}, \quad i = 1, 2,$$

the transverse pair-mixing matrices.

Explicitly,

$$W^i = \partial_{p_i} B - (\partial_{p_i} C_+) R_0 C_-^{\dagger} - C_+ R_0 (\partial_{p_i} C_-^{\dagger}) + C_+ R_0 (\partial_{p_i} H_{ss}) R_0 C_-^{\dagger} \Big|_{p=0}, \quad R_0 = (H_{ss}(0) - E_*)^{-1}.$$

The traceless Dirac data are carried by

$$\widetilde{W}^i = W^i - \frac{1}{2} \text{tr}(W^i) \mathbf{1}_2.$$

Hence the full-PDE shell problem is no longer abstract. It is exactly the finite overlap computation

$$(H_{cc}, H_{cs}, H_{ss}) \longrightarrow M_0, v_{\parallel}, \widetilde{W}^1, \widetilde{W}^2.$$

The minimal massless Dirac regime is obtained when

$$M_0 = 0, \quad v_{\parallel} \neq 0, \quad \widetilde{W}^1, \widetilde{W}^2 \text{ are linearly independent traceless Hermitian matrices.}$$

XIII. ACTUAL HARMONIC MATRIX ELEMENTS AND THE EXPLICIT EXTRACTION OF $M_0, v_{\parallel}, W^1, W^2$

A. Step D goal

The shell-level Dirac problem has already been reduced to the finite projected quantities

$$B_0, \quad v_{\parallel}, \quad W^1, \quad W^2.$$

The only correct next step is therefore to write these objects as actual harmonic matrix elements of the periodic microscopic operator.

We work in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

B. Constant-envelope shell basis

To obtain explicit primitive-cell selection rules, adopt the simplest enriched shell ansatz:

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad a = 1, 2, \quad \xi_a^{\dagger} \xi_b = \delta_{ab}.$$

Define the shell basis states

$$\Psi_{+,a}^{(p)}(x) = \frac{1}{\sqrt{V_{\text{cell}}}} e^{i(k_*+p) \cdot x} \xi_a, \quad \Psi_{-,a}^{(p)}(x) = \frac{1}{\sqrt{V_{\text{cell}}}} e^{i(k_*+p-G_*) \cdot x} \xi_a. \quad (95)$$

Then the shell-projected Hamiltonian has block form

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^{\dagger} & E_-(p) \end{pmatrix},$$

with $E_{\pm}(p), B(p) \in \text{Mat}_{2 \times 2}(\mathbb{C})$.

C. Effective enriched periodic operator

After eliminating the heavy Class II mediator X , the effective periodic operator may be organized as

$$\mathcal{H}_{\text{eff}} = h_0(-i\nabla) \otimes \mathbf{1}_2 + U_{\text{eff}}(x) \otimes \mathbf{1}_2 + \sum_{\alpha=1}^2 T_{\text{eff}}^\alpha(x) (-i\partial_\alpha) \otimes s^\alpha + \mathcal{R}_{\text{eff}}, \quad (96)$$

where:

1. h_0 is the scalar bare shell kernel;
2. $U_{\text{eff}}(x)$ is the scalar periodic pair-mixing potential;
3. $T_{\text{eff}}^\alpha(x)$ are the effective enriched derivative coefficients;
4. $\mathcal{R}_{\text{eff}}$ denotes higher-order, pair-diagonal internal, or subleading terms.

Expand the periodic coefficient fields in primitive-cell Fourier modes:

$$U_{\text{eff}}(x) = \sum_{Q \in \Gamma^*} \hat{U}_{\text{eff}}(Q) e^{iQ \cdot x}, \quad T_{\text{eff}}^\alpha(x) = \sum_{Q \in \Gamma^*} \hat{T}_{\text{eff}}^\alpha(Q) e^{iQ \cdot x}. \quad (97)$$

D. Exact Bragg selection rule for core off-diagonal elements

Consider any periodic coefficient $F(x) = \sum_Q \hat{F}(Q) e^{iQ \cdot x}$. Then

$$\begin{aligned} \langle \Psi_{+,a}^{(p)}, F(x) \Psi_{-,b}^{(p)} \rangle &= \frac{1}{V_{\text{cell}}} \sum_Q \hat{F}(Q) \int_{\text{cell}} d^3x e^{-i(k_*+p) \cdot x} e^{iQ \cdot x} e^{i(k_*+p-G_*) \cdot x} \xi_a^\dagger \xi_b \\ &= \delta_{ab} \hat{F}(G_*). \end{aligned} \quad (98)$$

Thus the off-diagonal shell-core channel sees only the $Q = G_*$ Fourier coefficient.

Therefore the constant pair-mixing mass matrix is

$$\boxed{M_0 = B^{\text{eff}}(0) = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2 + M_0^{(\mathcal{R})}}. \quad (99)$$

In the minimal constant-envelope reduction, $M_0^{(\mathcal{R})}$ is absent and one simply has

$$M_0 = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2.$$

E. Off-diagonal derivative matrix elements

Now consider one enriched derivative term

$$T_{\text{eff}}^\alpha(x) (-i\partial_\alpha) \otimes s^\alpha.$$

Acting on the negative-pair basis state gives

$$(-i\partial_\alpha) \Psi_{-,b}^{(p)} = (k_* + p - G_*)_\alpha \Psi_{-,b}^{(p)}.$$

Because $k_* = G_*/2$ is parallel to the shell normal $\hat{n}$, one has

$$(k_* - G_*)_\alpha = 0, \quad \alpha = 1, 2$$

in the transverse frame. Hence

$$(-i\partial_\alpha) \Psi_{-,b}^{(p)} = p_\alpha \Psi_{-,b}^{(p)} \quad (\alpha = 1, 2).$$

Therefore

$$\langle \Psi_{+,a}^{(p)}, T_{\text{eff}}^\alpha(x) (-i\partial_\alpha) \otimes s^\alpha \Psi_{-,b}^{(p)} \rangle = p_\alpha \hat{T}_{\text{eff}}^\alpha(G_*) (s^\alpha)_{ab}. \quad (100)$$

Thus the exact constant-envelope extraction formula is

$$\boxed{W^\alpha = \hat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2.} \quad (101)$$

This is the explicit reason the entire shell-level transverse Dirac problem reduces to first-shell Fourier coefficients.

F. Direct Class II origin of the enriched coefficients

The minimal Class II mediated UV structure is

$$\mathcal{L}_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^A X_i^A + \alpha_X X_i^A J_i^A[\Psi] + \beta_X X_i^A K_i^A[P, \psi].$$

Integrating out X gives the mixed effective term

$$-\frac{\alpha_X \beta_X}{M_X^2} J_i^A[\Psi] K_i^A[P, \psi].$$

Take the minimal current choices

$$J_i^A[\Psi] = (\partial_i \Psi^\dagger) T_A \Psi + \text{h.c.}, \quad K_i^A[P, \psi] = \text{Tr}(T_A \partial_i P) \mathcal{O}[\psi],$$

or, after shell reduction,

$$K_i^A \sim \partial_i \delta P^A \psi.$$

Around the periodic locked background

$$\Psi_0(x) = \sum_{Q \in \mathcal{S}_{12}} A_Q e^{iQ \cdot x} z_0,$$

the shell/background current has first-shell Fourier coefficients of the form

$$\hat{J}_i^A(Q) = iQ_i A_Q j^A \quad (Q \in \mathcal{S}_{12}), \quad (102)$$

for some reduced internal coefficient j^A determined by the background internal overlaps.

Likewise, the shell-relevant part of the projector/test current may be decomposed as

$$K_i^A[P, \psi] = \sum_{\alpha=1}^2 K_{i,\alpha}^A(x) (-i\partial_\alpha) \otimes s^\alpha + \dots, \quad (103)$$

with Fourier coefficients $\hat{K}_{i,\alpha}^A(Q)$.

Therefore the effective enriched coefficient field generated by Class II is

$$T_{(II)}^\alpha(x) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A J_i^A[\Psi_0](x) K_{i,\alpha}^A(x) + \dots, \quad (104)$$

and its G_* -Fourier coefficient is the convolution

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q')}. \quad (105)$$

In the minimal first-shell-dominant route, the leading contribution is

$$\hat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \hat{J}_i^A(G_*) \hat{K}_{i,\alpha}^A(0), \quad (106)$$

hence, using (102),

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) \propto iA_{G_*} (C_\alpha \cdot G_*)}, \quad (107)$$

with a proportionality constant depending on the microscopic reduced internal overlaps.

Therefore

$$\boxed{W_{(II)}^\alpha = \hat{T}_{(II)}^\alpha(G_*) s^\alpha \propto iA_{G_*} (C_\alpha \cdot G_*) s^\alpha}. \quad (108)$$

This is the direct Class II microscopic realization of the already-known shell-level criterion

$$W^\alpha \neq 0 \iff A_{G_*} \neq 0, \quad C_\alpha \cdot G_* \neq 0.$$

G. Bare/Class II/spectator decomposition

The actual effective coefficients split naturally into three parts:

$$\hat{U}_{\text{eff}}(G_*) = \hat{U}_{\text{bare}}(G_*) + \hat{U}_{(II)}(G_*) + \hat{U}_{(\text{sp})}(G_*), \quad (109)$$

$$\hat{T}_{\text{eff}}^\alpha(G_*) = \hat{T}_{\text{bare}}^\alpha(G_*) + \hat{T}_{(II)}^\alpha(G_*) + \hat{T}_{(\text{sp})}^\alpha(G_*). \quad (110)$$

Accordingly,

$$M_0 = M_0^{\text{bare}} + M_0^{(II)} + M_0^{(\text{sp})}, \quad (111)$$

$$W^\alpha = W_{\text{bare}}^\alpha + W_{(II)}^\alpha + W_{(\text{sp})}^\alpha. \quad (112)$$

The meaning of the three pieces is:

a. Bare periodic scalar piece.

$$M_0^{\text{bare}} = \hat{U}_{\text{bare}}(G_*) \mathbf{1}_2.$$

This is the ordinary Bragg mass channel already present in the periodic scalar background.

b. Direct Class II self-energy piece.

$$W_{(II)}^\alpha = \hat{T}_{(II)}^\alpha(G_*) s^\alpha.$$

This is the first genuinely enriched transverse shell transport generated by the mixed $J \cdot K$ mechanism.

c. Spectator Schur correction. The one-step spectator shell produces

$$M_0^{(\text{sp})} = - \left[C_+(0) R_0 C_-(0)^\dagger \right]_{Q=G_*},$$

and

$$W_{(\text{sp})}^\alpha = -\partial_{p_\alpha} \left[C_+(p) R_{\text{sp}}(E_*, p) C_-(p)^\dagger \right]_{p=0}. \quad (113)$$

This term is algebraically explicit and finite, but depends on the actual chosen one-step spectator closure $S_*^{(1)}$.

H. Longitudinal slope

The longitudinal shell slope comes from the pair-diagonal difference:

$$D^{\text{eff}}(p) = \frac{1}{2} (E_+^{\text{eff}}(p) - E_-^{\text{eff}}(p)).$$

Its scalar coefficient is

$$v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} D^{\text{eff}}(p) \Big|_{p=0}. \quad (114)$$

This splits as

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}. \quad (115)$$

Here:

a. Bare Bragg slope.

$$v_{\parallel}^{\text{bare}} = \frac{1}{4} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \left[h_0(k_* + p) - h_0(k_* + p - G_*) \right]_{p=0}. \quad (116)$$

This is the standard folded Bragg-pair longitudinal splitting.

b. Class II diagonal correction.

$$\delta v_{\parallel}^{(II)}$$

comes from the pair-diagonal part of

$$-\frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger$$

after projection.

c. Spectator correction.

$$\delta v_{\parallel}^{(\text{sp})}$$

comes from the $p_{\parallel}$ -derivative of the diagonal Schur-complement renormalization.

I. What vanishes and what does not in the current working branch

For the current refined locked isotropic projector-stiffness model alone, the direct quadratic projector-mixing source vanishes:

$$J = 0 \quad \implies \quad \delta N_{ij}^a = 0$$

at that order. Therefore, in the present working branch,

$$\boxed{\hat{T}_{\text{bare}}^\alpha(G_*) = 0 \quad \text{for the pure locked isotropic/projector-stiffness route alone.}} \quad (117)$$

Hence the first nontrivial transverse matrices must come from

$$\hat{T}_{(II)}^\alpha(G_*) \quad \text{and/or} \quad \hat{T}_{(\text{sp})}^\alpha(G_*).$$

By contrast, the constant Bragg channel M_0 and the longitudinal Bragg slope $v_{\parallel}$ already have direct periodic core contributions.

J. Final finite extraction dictionary

At this point the full microscopic shell problem has been reduced to the following finite Fourier-overlap dictionary:

$$\boxed{M_0 = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2 = \left[\hat{U}_{\text{bare}}(G_*) + \hat{U}_{(II)}(G_*) + \hat{U}_{(\text{sp})}(G_*) \right] \mathbf{1}_2,} \quad (118)$$

$$\boxed{W^\alpha = \hat{T}_{\text{eff}}^\alpha(G_*) s^\alpha = \left[\hat{T}_{\text{bare}}^\alpha(G_*) + \hat{T}_{(II)}^\alpha(G_*) + \hat{T}_{(\text{sp})}^\alpha(G_*) \right] s^\alpha, \quad \alpha = 1, 2,} \quad (119)$$

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}.} \quad (120)$$

So the actual remaining full-PDE work is no longer conceptual. It is to compute the first-shell Fourier coefficients

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*),$$

together with the single scalar slope $v_{\parallel}$.

XIV. FINAL SUMMARY

Using the constant-envelope shell basis at the folded Bragg point

$$k_* = \frac{G_*}{2},$$

the shell-projected Bloch Hamiltonian reduces to a finite block problem whose off-diagonal core matrix elements are controlled by the $Q = G_*$ primitive-cell Fourier coefficients.

The exact core extraction formulas are:

$$M_0 = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2,$$

for the constant pair-mixing Bragg mass,

$$W^\alpha = \hat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2,$$

for the transverse Dirac matrices, and

$$v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \frac{E_+^{\text{eff}}(p) - E_-^{\text{eff}}(p)}{2} \Big|_{p=0},$$

for the longitudinal shell slope.

The effective enriched coefficient field splits as

$$\hat{T}_{\text{eff}}^\alpha(G_*) = \hat{T}_{\text{bare}}^\alpha(G_*) + \hat{T}_{(II)}^\alpha(G_*) + \hat{T}_{(\text{sp})}^\alpha(G_*).$$

In the current refined locked isotropic/projector-stiffness route alone,

$$\hat{T}_{\text{bare}}^\alpha(G_*) = 0$$

at quadratic order, because the direct projector mixing source vanishes there.

Therefore the first genuinely nontrivial transverse shell matrices must come from the Class II mixed $J \cdot K$ route and/or its spectator renormalization.

For the Class II branch,

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q'),$$

and in the minimal first-shell-dominant channel,

$$\hat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*}(C_\alpha \cdot G_*).$$

Hence

$$W_{(II)}^\alpha \propto i A_{G_*}(C_\alpha \cdot G_*) s^\alpha.$$

So the full microscopic shell frontier is now compressed to one explicit finite Fourier-overlap problem:

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.$$

XV. EXPLICIT 12-MODE BCC FOURIER CONVOLUTIONS AT THE SELECTED SHELL

A. Goal

The remaining shell-level microscopic quantities are the first-shell Fourier coefficients

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*),$$

together with the longitudinal slope $v_{\parallel}$.

In this step we evaluate the actual BCC convolution sums for the 12-mode locked background and determine which classes of microscopic coefficient fields contribute at the selected shell vector G_* .

B. 12-mode BCC background and the shell combinatorial identity

Let

$$\phi_0(x) = \sum_{Q \in S_{\text{BCC}}} A_Q e^{iQ \cdot x}, \quad A_{-Q} = \overline{A_Q}, \quad (121)$$

with first reciprocal shell

$$S_{\text{BCC}} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

Fix $G_* \in S_{\text{BCC}}$.

Define

$$N(G_*) := \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\}. \quad (122)$$

For every $G_* \in S_{\text{BCC}}$,

$$\boxed{N(G_*) = 4.} \quad (123)$$

For example, if

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

then the four ordered pairs are

$$\frac{q_*}{\sqrt{2}}(1, 0, 1) + \frac{q_*}{\sqrt{2}}(0, 1, -1) = G_*, \quad (124)$$

$$\frac{q_*}{\sqrt{2}}(1, 0, -1) + \frac{q_*}{\sqrt{2}}(0, 1, 1) = G_*, \quad (125)$$

together with the two order-reversed pairs.

Hence the entire shell-level Fourier problem is controlled by a finite set of explicitly known ordered shell pairs.

C. Quadratic scalar coefficient: the actual Bragg-mass channel

Let the scalar periodic coefficient contain a quadratic first-shell term

$$U_{\text{quad}}(x) = u_2 \phi_0(x)^2. \quad (126)$$

Then

$$\widehat{U}_{\text{quad}}(G_*) = u_2 \sum_{Q_1 + Q_2 = G_*} A_{Q_1} A_{Q_2}. \quad (127)$$

In the equal-amplitude locked branch,

$$A_Q = A \quad (Q \in S_{\text{BCC}}),$$

this gives

$$\boxed{\widehat{U}_{\text{quad}}(G_*) = u_2 N(G_*) A^2 = 4u_2 A^2.} \quad (128)$$

Therefore the direct scalar Bragg mass is

$$\boxed{M_0^{\text{quad}} = \widehat{U}_{\text{quad}}(G_*) \mathbf{1}_2 = 4u_2 A^2 \mathbf{1}_2.} \quad (129)$$

So a naive quadratic scalar shell coefficient does *not* vanish in the 12-mode equal-amplitude BCC branch. Instead, it produces a nonzero constant pair-mixing obstruction.

D. Editorial consequence of the quadratic scalar sum

This means that the condition

$$\widehat{U}(G_*) = 0$$

cannot be treated as automatic merely from the statement

$$U_{\text{per}}(x) \propto \phi_0(x)^2$$

in the full 12-mode equal-amplitude BCC sum.

Hence the massless shell condition requires one of the following:

1. the actual scalar coefficient is *not* of the naive quadratic form $u_2\phi_0^2$;
2. or its G_* -coefficient cancels against other microscopic scalar contributions;
3. or a symmetry removes the constant intervalley scalar block altogether;
4. or the final shell reduction projects out the dangerous scalar pair-mixing channel.

Thus the constant-mass problem is now sharply localized rather than heuristic.

E. Linear gradient-enriched coefficients

Now consider the minimal enriched class

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha, \quad \alpha = 1, 2. \quad (130)$$

Since

$$\partial_j \phi_0(x) = \sum_{Q \in S_{\text{BCC}}} iQ_j A_Q e^{iQ \cdot x},$$

its G_* -Fourier coefficient is immediately

$$\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha. \quad (131)$$

Therefore, under the constant-envelope shell reduction,

$$W^\alpha = \widehat{\mathcal{T}}^\alpha(G_*) s^\alpha = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha. \quad (132)$$

So the transverse shell matrices are nonzero provided

$$A_{G_*} \neq 0, \quad C^\alpha \cdot G_* \neq 0. \quad (133)$$

This is exactly the minimal microscopic target criterion for the transverse shell Dirac channel.

F. Quadratic gradient-enriched coefficients

Now consider a quadratic derivative class, for example

$$\mathcal{T}_{\text{quad}}^\alpha(x) = \lambda_\alpha \phi_0(x) (C^\alpha_j \partial_j \phi_0(x)) \otimes s^\alpha. \quad (134)$$

Then

$$\widehat{\mathcal{T}}_{\text{quad}}^\alpha(G_*) = i\lambda_\alpha \sum_{Q_1 + Q_2 = G_*} (C^\alpha \cdot Q_2) A_{Q_1} A_{Q_2} s^\alpha. \quad (135)$$

In the equal-amplitude branch $A_Q = A$, the ordered-pair symmetry implies

$$\sum_{Q_1+Q_2=G_*} Q_1 = \sum_{Q_1+Q_2=G_*} Q_2,$$

and since

$$\sum_{Q_1+Q_2=G_*} (Q_1 + Q_2) = N(G_*)G_* = 4G_*,$$

we obtain

$$\sum_{Q_1+Q_2=G_*} Q_2 = 2G_*. \quad (136)$$

Hence

$$\widehat{\mathcal{T}}_{\text{quad}}^\alpha(G_*) = 2i\lambda_\alpha A^2 (C^\alpha \cdot G_*) s^\alpha. \quad (137)$$

Therefore, at least in the 12-mode equal-amplitude BCC branch, some quadratic gradient-enriched couplings do retain first-shell support. They are not minimal, but they are not automatically killed.

G. Class II mixed $J \cdot K$ route

Now turn to the actual Class II mediator branch. After integrating out the heavy mediator, the shell-relevant enriched derivative coefficient is generated by the mixed term

$$-\frac{\alpha_X \beta_X}{M_X^2} J_i^A[\Psi_0] K_{i,\alpha}^A[P, \psi].$$

Therefore

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q'). \quad (138)$$

For the periodic locked background,

$$\widehat{J}_i^A(Q) = iQ_i A_Q j^A. \quad (139)$$

Hence the Class II shell coefficient is

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} iQ_i A_Q j^A \widehat{K}_{i,\alpha}^A(Q'). \quad (140)$$

a. Zero-mode dominated K -branch. If the shell-relevant K -coefficient is dominated by its zero Fourier mode,

$$\widehat{K}_{i,\alpha}^A(Q') \approx \widehat{K}_{i,\alpha}^A(0) \delta_{Q',0},$$

then only $Q = G_*$ survives and

$$\widehat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \sum_A iG_{*i} A_{G_*} j^A \widehat{K}_{i,\alpha}^A(0). \quad (141)$$

This is of the canonical form

$$\widehat{T}_{(II)}^\alpha(G_*) = iA_{G_*} (C_\alpha \cdot G_*), \quad (142)$$

for an effective microscopic coefficient vector C_α .

b. First-shell K-branch. If $\widehat{K}_{i,\alpha}^A(Q')$ also has first-shell support, then the surviving terms are all ordered shell pairs with $Q + Q' = G_*$, and the coefficient acquires the same finite BCC combinatorial enhancement controlled by $N(G_*) = 4$.

Thus the mixed Class II route reproduces the same structural shell criterion:

$$\boxed{\widehat{T}_{(II)}^\alpha(G_*) \neq 0 \iff A_{G_*} \neq 0, \quad C_\alpha \cdot G_* \neq 0 \text{ modulo accidental cancellations.}} \quad (143)$$

H. What survives in the current working branch

The refined locked isotropic/projector-stiffness branch alone does not produce the needed transverse enriched source at quadratic order. So in the current working branch the first genuinely nontrivial transverse matrices come from:

$$\boxed{\widehat{T}_{\text{eff}}^\alpha(G_*) = \widehat{T}_{(II)}^\alpha(G_*) + \widehat{T}_{(\text{sp})}^\alpha(G_*), \quad \alpha = 1, 2.} \quad (144)$$

By contrast, the constant scalar Bragg channel already exists at the periodic-core level:

$$\boxed{M_0 = \widehat{U}_{\text{bare}}(G_*)\mathbf{1}_2 + \widehat{U}_{(II)}(G_*)\mathbf{1}_2 + \widehat{U}_{(\text{sp})}(G_*)\mathbf{1}_2.} \quad (145)$$

So the two tasks are sharply different:

1. generate nonzero $\widehat{T}_{\text{eff}}^1(G_*), \widehat{T}_{\text{eff}}^2(G_*)$;
2. force the scalar combination $\widehat{U}_{\text{eff}}(G_*)$ to vanish.

I. The actual bottleneck exposed by the convolution

At the end of the explicit BCC convolution, the shell-level full-PDE frontier is no longer vague.

The transverse side is structurally easy: the first-shell Fourier coefficient of a gradient-linear enriched coefficient field is automatically of the form

$$i(C^\alpha \cdot G_*)A_{G_*},$$

and the Class II branch naturally realizes this structure.

The real bottleneck is the constant Bragg mass: in the equal-amplitude 12-mode BCC branch, a naive quadratic scalar coefficient gives

$$\widehat{U}_{\text{quad}}(G_*) = 4u_2A^2 \neq 0.$$

Therefore the condition

$$M_0 = 0$$

must be imposed by an actual microscopic cancellation or symmetry mechanism, not by wishful first-shell rhetoric.

XVI. FINAL SUMMARY

For the 12-mode BCC background

$$\phi_0(x) = \sum_{Q \in S_{\text{BCC}}} A_Q e^{iQ \cdot x},$$

the selected shell vector $G_* \in S_{\text{BCC}}$ receives contributions from exactly

$$N(G_*) = 4$$

ordered first-shell pairs satisfying

$$Q_1 + Q_2 = G_*.$$

Therefore, for a quadratic scalar coefficient

$$U_{\text{quad}}(x) = u_2 \phi_0(x)^2,$$

one has

$$\hat{U}_{\text{quad}}(G_*) = u_2 \sum_{Q_1+Q_2=G_*} A_{Q_1} A_{Q_2}.$$

In the equal-amplitude branch $A_Q = A$,

$$\boxed{\hat{U}_{\text{quad}}(G_*) = 4u_2 A^2,}$$

hence

$$\boxed{M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0}$$

generically.

So a naive quadratic scalar shell coefficient produces a nonzero constant Bragg mass obstruction.

By contrast, for the minimal gradient-enriched class

$$\mathcal{T}^\alpha(x) = C^\alpha_j \partial_j \phi_0(x) \otimes s^\alpha,$$

the selected-shell Fourier coefficient is

$$\boxed{\widehat{\mathcal{T}}^\alpha(G_*) = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha.}$$

Hence

$$\boxed{W^\alpha = i(C^\alpha \cdot G_*) A_{G_*} s^\alpha,}$$

so the transverse Dirac matrices are nonzero whenever

$$A_{G_*} \neq 0, \quad C^\alpha \cdot G_* \neq 0.$$

For the actual Class II branch,

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q'),$$

with

$$\hat{J}_i^A(Q) = iQ_i A_Q j^A.$$

Thus the Class II shell coefficient has the same canonical structure

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) \sim iA_{G_*}(C_\alpha \cdot G_*),}$$

up to explicit microscopic overlap constants and possible finite BCC combinatorial factors.

Therefore the explicit 12-mode BCC convolution exposes the true shell-level bottleneck:

$$\boxed{\text{nonzero transverse Dirac coefficients are structurally natural,}}$$

but

$$\boxed{M_0 = 0 \text{ is not automatic and must come from an actual microscopic cancellation or symmetry.}}$$

XVII. ELIMINATION OF THE CONSTANT BRAGG MASS: THE ACTUAL FULL-PDE BOTTLENECK

A. Goal

At the present stage, the full periodic Bloch program has isolated the true microscopic obstruction:

$$\boxed{M_0 = \widehat{U}_{\text{eff}}(G_*) \neq 0}$$

at the selected shell pair. The transverse shell Dirac data W^1, W^2 are structurally available, but the constant scalar pair-mixing block generically gaps the crossing.

Thus the correct next task is no longer to discuss abstract witness conditions. It is to determine the actual microscopic mechanism enforcing

$$\boxed{M_0 = 0.}$$

B. What the explicit 12-mode computation already showed

In the locked equal-amplitude 12-mode BCC branch, the literal quadratic scalar periodic coefficient

$$U_{\text{per}}(x) = c \phi_0(x)^2$$

gives

$$\widehat{U}(G_*) = 4cA^2 \neq 0$$

generically. Therefore the projected constant pair-mixing block

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2$$

is nonzero in the plain scalar branch.

Hence:

$$\boxed{\text{the condition } M_0 = 0 \text{ is not automatic in the full 12-mode BCC reduction.}}$$

C. General decomposition of the constant pair-mixing block

Let the constant pair-off-diagonal block on the internal doublet be

$$M_0 = m_0 \mathbf{1}_2 + m_1 s_1 + m_2 s_2 + m_3 s_3. \quad (146)$$

Then the constant part of the enriched shell-pair Hamiltonian is

$$H_0^{\text{pair}} = \tau_1 \otimes \Re M_0 - \tau_2 \otimes \Im M_0. \quad (147)$$

The crucial distinction is:

- $m_0 \mathbf{1}_2$: scalar identity-channel Bragg mass;
- $m_1 s_1 + m_2 s_2$: traceless transverse internal pieces;
- $m_3 s_3$: genuine internal mass direction.

The scalar Bragg obstruction found in the plain periodic shell Hessian is precisely the identity-channel case

$$M_0^{\text{scalar}} = B_0^{\text{tot}} \mathbf{1}_2, \quad B_0^{\text{tot}} \neq 0. \quad (148)$$

D. Why Class II cannot directly cancel the scalar identity channel

Inside the explicit Class II branch, every induced constant pair-mixing correction carries an internal Pauli generator. Hence, if nonzero,

$$\delta M_0^{(\text{II})} \in \text{span}\{s_1, s_2, s_3\},$$

so in the shell-pair block it contributes only to

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

never to

$$\tau_1 \otimes \mathbf{1}_2.$$

Because the identity channel and the traceless Pauli channel are orthogonal under the internal trace, a Class II contribution cannot directly cancel the scalar Bragg piece

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Moreover, under the natural shell-Dirac odd/reflection parity, the gradient-sector constant Class II pair-mixing vanishes at principal order.

Therefore:

Class II renormalizes the derivative Weyl/Dirac witness sector, but does not remove the scalar Bragg-mass obstruction.
--

(149)

E. Exact decomposition of the shell mass problem

The actual full shell coefficient splits into

$$\hat{U}_{\text{eff}}(G_*) = \hat{U}_{\text{bare}}(G_*) + \hat{U}_{(\text{II})}(G_*) + \hat{U}_{(\text{sp})}(G_*). \quad (150)$$

But by the identity/traceless orthogonality just explained, the Class II contribution does not enter the scalar identity channel directly. Hence the actual scalar mass condition reduces to

$\hat{U}_{\text{scalar}}(G_*) = \hat{U}_{\text{bare}}(G_*) + \hat{U}_{(\text{sp})}(G_*) = 0.$

(151)

So the full-PDE massless condition is no longer vague: one must either prove the exact scalar cancellation (151), or else replace the scalar-shell route by a different microscopic operator class.

F. Three and only three honest routes

We now classify the only honest possibilities for achieving $M_0 = 0$.

a. Route A: exact scalar-channel cancellation The direct shell scalar block vanishes because

$\hat{U}_{\text{bare}}(G_*) + \hat{U}_{(\text{sp})}(G_*) = 0.$
--

(152)

This is the sharp cancellation route inside the literal projected Bloch framework. It requires an actual microscopic equality, not a heuristic statement.

b. Route B: symmetry removes constant pair-even mixing Suppose there is a shell reflection/exchange symmetry $\mathcal{R}$ such that

$$\mathcal{R}: \quad p \mapsto -p$$

in the pair-exchange channel, and the coefficient fields transform so that

$$B_0 \mapsto -B_0$$

while

$$p_\alpha W^\alpha \mapsto p_\alpha W^\alpha.$$

Then the constant pair-even block is forbidden, but the odd linear shell-mixing terms survive. This is the correct parity-separation mechanism: kill the constant shell-even block, keep the linear shell-odd block.

c. Route C: bypass the literal scalar Bragg channel Instead of starting from a projected scalar pair channel that inevitably produces

$$B_0 \tau_1 \otimes \mathbf{1}_2,$$

one derives a different microscopic projected operator whose principal low-energy block is already grading-based or block-off-diagonal from the outset. Then the dangerous scalar Bragg identity channel is absent rather than cancelled.

Thus the only honest alternatives are:

$$\boxed{\text{exact cancellation,} \quad \text{symmetry exclusion,} \quad \text{or operator-class bypass.}} \quad (153)$$

G. Residual valley symmetry and the doubled Dirac mass space

Once one has already produced an isolated doubled local Dirac block

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i),$$

the full constant Dirac-mass space is exactly

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}. \quad (154)$$

Hence mass protection is the finite algebraic condition

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (155)$$

An exact residual valley $U(1)_V$ or exact residual valley $\mathbb{Z}_2^V$ is a sufficient mechanism: under $U(1)_V$, the two mass generators rotate as a charged doublet, and under $\mathbb{Z}_2^V$ they are odd. So in either case no nonzero element of $\mathcal{M}_D$ is symmetry-allowed.

However, this is still a theorem-level reduction, not yet a microscopic derivation. The actual open task is whether the final corrected TECT/Class II realization indeed preserves such a residual symmetry.

H. What this means for the present full-PDE program

Combining the shell-mass obstruction with the mass-protection theorem gives the following exact logical separation:

1. the transverse Dirac witness problem is already localized to the explicit Class II/enriched shell coefficients;
2. the scalar Bragg obstruction is a separate identity-channel problem;
3. Class II does not solve that identity-channel problem directly;
4. if one obtains an actual doubled Dirac block, then constant Dirac masses are excluded by an exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$;
5. therefore the next full-PDE task is to identify the *microscopic* mechanism that removes the scalar identity-channel block while preserving the derivative shell symbol.

I. The true next finite calculation

So the actual microscopic Step F is:

$$\boxed{\text{compute } \hat{U}_{\text{bare}}(G_*) \text{ and } \hat{U}_{(\text{sp})}(G_*) \text{ explicitly, and test whether exact scalar cancellation occurs.}} \quad (156)$$

In parallel, one must identify the residual symmetry of the isolated 4×4 shell block:

$$\boxed{\text{determine whether the projected full microscopic block preserves exact } U(1)_V \text{ or } \mathbb{Z}_2^V.} \quad (157)$$

These two finite tasks are now the only genuine microscopic bottlenecks. Everything else has already been reduced.

XVIII. FINAL SUMMARY

The full periodic Bloch program has isolated the true microscopic obstruction:

$$M_0 = \widehat{U}_{\text{eff}}(G_*) \neq 0.$$

For the literal 12-mode equal-amplitude scalar branch,

$$\widehat{U}(G_*) = 4cA^2 \neq 0,$$

so the constant Bragg mass is generically nonzero. Thus the condition

$$M_0 = 0$$

is not automatic in the projected scalar shell route.

Inside the explicit Class II branch, the induced constant pair-mixing lies in the traceless internal Pauli sector:

$$\delta M_0^{(\text{II})} \in \text{span}\{s_1, s_2, s_3\},$$

so it contributes to

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

not to the scalar identity channel

$$\tau_1 \otimes \mathbf{1}_2.$$

Therefore:

Class II renormalizes the derivative witness sector, but does not directly cancel the scalar Bragg mass.

Hence the actual scalar mass condition reduces to

$$\widehat{U}_{\text{bare}}(G_*) + \widehat{U}_{(\text{sp})}(G_*) = 0.$$

There are only three honest routes for achieving a massless shell block:

(A) exact scalar-channel cancellation,

(B) symmetry exclusion of constant pair-even mixing,

(C) bypassing the literal scalar Bragg channel by a different projected operator class.

After an isolated doubled Dirac block has been obtained, the genuine constant mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \},$$

and mass protection is the finite condition

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Exact residual valley $U(1)_V$ or exact residual valley $\mathbb{Z}_2^V$ is a sufficient protection mechanism, but its actual microscopic realization remains to be verified.

Therefore the true remaining full-PDE bottlenecks are exactly:

compute $\widehat{U}_{\text{bare}}(G_*) + \widehat{U}_{(\text{sp})}(G_*)$ and test cancellation,

and

identify the actual residual symmetry of the projected 4×4 shell block.

XIX. ONE-STEP SPECTATOR SCHUR CORRECTION IN THE LOCKED 12-MODE BCC BRANCH

A. Goal

The true remaining shell-level bottleneck is the scalar identity-channel Bragg mass

$$M_0 = B_0^{\text{tot}} \mathbf{1}_2.$$

The projected Bragg-mass formula states that the shell crossing is massless if and only if

$$B_0 = 0.$$

The plain scalar branch gives a nonzero B_0 , while Class II renormalizes the derivative witness sector and does not directly cancel the scalar identity channel.

Hence the only concrete remaining cancellation mechanism inside the current full-PDE framework is the spectator Schur correction.

B. Core/spectator block and exact Schur formula

Let the one-step shell closure be

$$S_{\text{cl}}^{(1)} = S_{\text{core}} \cup S_{\text{sp}}, \quad S_{\text{core}} = \{0, -G_*\}.$$

In the enriched shell basis, the finite shell matrix has the block form

$$H_{\text{shell}}(p) = \begin{pmatrix} E_+(p) & B(p) & C_+(p) \\ B(p)^\dagger & E_-(p) & C_-(p) \\ C_+(p)^\dagger & C_-(p)^\dagger & H_{ss}(p) \end{pmatrix}.$$

The exact core off-diagonal block after spectator elimination is

$$B^{\text{eff}}(E, p) = B(p) - C_+(p)(H_{ss}(p) - E)^{-1}C_-(p)^\dagger.$$

Therefore the spectator correction to the constant Bragg block is

$$\boxed{B_0^{(\text{sp})} = -C_+(0) R_0 C_-(0)^\dagger, \quad R_0 := (H_{ss}(0) - E_*)^{-1}.} \quad (158)$$

C. Which spectators actually contribute?

A spectator harmonic $\eta \in S_{\text{sp}}$ contributes to $B_0^{(\text{sp})}$ only if it couples to both core states:

$$\langle +|H|\eta \rangle \neq 0, \quad \langle -|H|\eta \rangle \neq 0.$$

For a periodic coefficient with first-shell support, these conditions are exactly

$$\eta \in S_{\text{BCC}}, \quad \eta + G_* \in S_{\text{BCC}}.$$

Define the bi-coupled spectator set

$$S_{\text{bi}}(G_*) := \{\eta \in \Gamma^* : \eta \in S_{\text{BCC}}, \eta + G_* \in S_{\text{BCC}}\}. \quad (159)$$

Choose

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0).$$

Then the explicit solutions are

$$\boxed{S_{\text{bi}}(G_*) = \frac{q_*}{\sqrt{2}} \{(-1, 0, 1), (-1, 0, -1), (0, -1, 1), (0, -1, -1)\}.} \quad (160)$$

So

$$\boxed{|S_{\text{bi}}(G_*)| = 4.} \quad (161)$$

This is the same fourfold shell combinatorics already found in the ordered-pair identity

$$N(G_*) = \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\} = 4.$$

D. Constant-envelope scalar spectator formula

Now assume the spectator coupling comes from a scalar periodic coefficient

$$V(x) = \sum_{Q \in \Gamma^*} \hat{V}(Q) e^{iQ \cdot x},$$

acting trivially on the internal doublet. Under the constant-envelope shell ansatz, the core-spectator matrix elements are

$$\langle +, a | V | \eta, b \rangle = \delta_{ab} \hat{V}(\eta), \quad (162)$$

$$\langle -, a | V | \eta, b \rangle = \delta_{ab} \hat{V}(\eta + G_*). \quad (163)$$

Hence (158) becomes

$$\boxed{\hat{U}_{(\text{sp})}(G_*) = - \sum_{\eta, \zeta \in S_{\text{bi}}} \hat{V}(\eta) (R_0)_{\eta\zeta} \overline{\hat{V}(\zeta + G_*)}.} \quad (164)$$

If R_0 is diagonal in the chosen spectator basis,

$$(R_0)_{\eta\zeta} = \delta_{\eta\zeta} \Delta_\eta^{-1},$$

then

$$\boxed{\hat{U}_{(\text{sp})}(G_*) = - \sum_{\eta \in S_{\text{bi}}} \frac{\hat{V}(\eta) \overline{\hat{V}(\eta + G_*)}}{\Delta_\eta}.} \quad (165)$$

This is the first explicit finite formula for the spectator cancellation channel.

E. Sign and cancellation structure

Assume for simplicity:

1. $V(x)$ is real, so $\hat{V}(-Q) = \overline{\hat{V}(Q)}$;
2. the relevant spectator energies satisfy $\Delta_\eta > 0$;
3. all first-shell spectator couplings have the same amplitude

$$\hat{V}(Q) = v_1 A \quad (Q \in S_{\text{BCC}}).$$

Then (165) reduces to

$$\hat{U}_{(\text{sp})}(G_*) = -|v_1|^2 A^2 \sum_{\eta \in S_{\text{bi}}} \frac{1}{\Delta_\eta}. \quad (166)$$

If the four bi-coupled spectators are approximately degenerate,

$$\Delta_\eta \approx \Delta_{\text{sp}}, \quad \eta \in S_{\text{bi}},$$

then

$$\boxed{\hat{U}_{(\text{sp})}(G_*) \approx - \frac{4|v_1|^2}{\Delta_{\text{sp}}} A^2.} \quad (167)$$

So the spectator Schur correction has exactly the right structure to cancel the positive bare scalar Bragg mass: it is an identity-channel contribution, it is of order A^2 , and it carries the same BCC factor 4.

F. Combination with the actual bare scalar block

In the current locked 12-mode BCC branch, the actual scalar Hessian gives

$$B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2,$$

while the alignment term gives no scalar Bragg block on the locked sector, and Class II does not cancel the scalar identity channel directly.

Therefore the actual scalar mass condition becomes

$$B_0^{\text{tot}} = \left[4uA^2 - \sum_{\eta \in S_{\text{bi}}} \frac{\widehat{V}(\eta) \overline{\widehat{V}(\eta + G_*)}}{\Delta_\eta} \right] \mathbf{1}_2 = 0. \quad (168)$$

In the equal-coupling degenerate-spectator approximation this reduces to

$$4uA^2 - \frac{4|v_1|^2}{\Delta_{\text{sp}}} A^2 = 0, \quad (169)$$

equivalently

$$u = \frac{|v_1|^2}{\Delta_{\text{sp}}}. \quad (170)$$

G. Generalized cancellation law

From the exact 12-mode scalar calculation, the bare scalar block can be written more generally as

$$B_0^{\text{bare}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.$$

Hence the spectator-corrected exact cancellation condition is

$$2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) - \sum_{\eta, \zeta \in S_{\text{bi}}} \widehat{V}(\eta) (R_0)_{\eta\zeta} \overline{\widehat{V}(\zeta + G_*)} = 0. \quad (171)$$

In the diagonal equal-spectator approximation this becomes

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = \frac{1}{2} \sum_{\eta \in S_{\text{bi}}} \frac{\widehat{V}(\eta) \overline{\widehat{V}(\eta + G_*)}}{A^2 \Delta_\eta}. \quad (172)$$

H. What is actually proved at this step

This step does *not* yet prove that full unrestricted TECT satisfies $M_0 = 0$. But it does prove something much sharper than before:

1. the one-step spectator Schur correction hits the *same scalar identity channel* as the bare Bragg mass;
2. its contribution is explicitly finite and controlled by exactly four bi-coupled BCC spectators;
3. in the natural diagonal-spectator approximation it has the opposite sign and the same A^2 -scaling as the bare scalar block;
4. therefore exact cancellation is structurally possible without touching the transverse Dirac matrices W^1, W^2 .

So the full-PDE frontier is no longer “whether cancellation is conceivable”. It is the much narrower question:

what are the actual spectator couplings $\widehat{V}(\eta)$ and spectator gaps Δ_η in the full microscopic TECT linearization?

XX. FINAL SUMMARY

The constant Bragg mass is the projected shell off-diagonal identity-channel block

$$B_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2,$$

and the shell crossing is massless iff

$$B_0 = 0.$$

In the locked equal-amplitude 12-mode BCC branch, the currently known microscopic sectors imply:

$$B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2,$$

$$B_0^{(\text{align})} = 0,$$

and the canonical Class II branch does not directly cancel the scalar identity channel.

Therefore the only remaining available cancellation mechanism inside the present full-PDE program is the spectator Schur correction

$$B_0^{(\text{sp})} = -C_+(0) R_0 C_-(0)^\dagger.$$

For the selected shell vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

the one-step bi-coupled spectator set is

$$S_{\text{bi}}(G_*) = \frac{q_*}{\sqrt{2}} \{(-1, 0, 1), (-1, 0, -1), (0, -1, 1), (0, -1, -1)\},$$

so

$$|S_{\text{bi}}(G_*)| = 4.$$

Under the constant-envelope scalar spectator reduction,

$$\widehat{U}_{(\text{sp})}(G_*) = - \sum_{\eta \in S_{\text{bi}}} \frac{\widehat{V}(\eta) \overline{\widehat{V}(\eta + G_*)}}{\Delta_\eta}.$$

If the four spectator gaps are approximately equal and the first-shell spectator couplings have common amplitude

$$\widehat{V}(Q) = v_1 A,$$

then

$$\widehat{U}_{(\text{sp})}(G_*) \approx -\frac{4|v_1|^2}{\Delta_{\text{sp}}} A^2.$$

Hence the exact scalar cancellation law becomes

$$\boxed{4uA^2 - \frac{4|v_1|^2}{\Delta_{\text{sp}}} A^2 = 0,}$$

equivalently

$$\boxed{u = \frac{|v_1|^2}{\Delta_{\text{sp}}}.}$$

More generally, combining the exact scalar-core coefficient formula with the spectator Schur term gives

$$\boxed{2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) - \sum_{\eta, \zeta \in S_{\text{bi}}} \widehat{V}(\eta) (R_0)_{\eta\zeta} \overline{\widehat{V}(\zeta + G_*)} = 0.}$$

Therefore the spectator correction does hit the same dangerous scalar identity channel, with the correct sign and scaling to cancel the bare Bragg mass. So the full-PDE frontier is now fully localized to the explicit microscopic data

$$\widehat{V}(\eta), \quad \Delta_\eta.$$

XXI. ACTUAL SCALAR SPECTATOR VERTICES AND BARE SPECTATOR DETUNING IN THE ONE-STEP BCC CLOSURE

A. Goal

The scalar Bragg obstruction is

$$B_0^{\text{tot}} = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2.$$

The previous step showed that the one-step spectator Schur correction hits the same identity channel. The purpose of the present step is to compute, inside the actual locked 12-mode microscopic branch,

$$C_+(\eta), \quad C_-(\eta), \quad \Delta_\eta,$$

for the bi-coupled spectator modes $\eta \in S_{\text{bi}}(G_*)$.

B. Actual scalar periodic Hessian in the locked branch

Inside the actual locked scalar Hessian sector, the only periodic scalar contribution is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2, \tag{173}$$

and on the locked branch this gives

$$B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2. \tag{174}$$

The alignment term vanishes on the locked sector, and the canonical Class II sector does not directly cancel the scalar identity channel.

Hence the actual scalar spectator vertices must be extracted from (173).

C. First-shell Fourier coefficient of the actual scalar Hessian

Write the locked equal-amplitude 12-mode BCC background as

$$\phi_0(x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}, \quad A_Q = A, \quad Q \in S_{\text{BCC}}. \tag{175}$$

Then

$$|\phi_0(x)|^2 = A^2 \sum_{Q_1, Q_2 \in S_{\text{BCC}}} e^{i(Q_1 - Q_2) \cdot x}. \tag{176}$$

Equivalently, since the set is inversion-symmetric, the first-shell Fourier coefficient is still governed by the same ordered-pair counting:

$$\widehat{|\phi_0|^2}(Q) = 4A^2 \quad (Q \in S_{\text{BCC}}). \tag{177}$$

Therefore

$$\boxed{\widehat{U}_{\text{per}}^{\text{tot}}(Q) = 4uA^2 \quad (Q \in S_{\text{BCC}}).} \tag{178}$$

D. Bi-coupled spectator set

Fix

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0), \quad k_* = \frac{G_*}{2}.$$

The spectators contributing to the core off-diagonal Schur term are the harmonics

$$\eta \in S_{\text{bi}}(G_*) := \{\eta \in \Gamma^* : \eta \in S_{\text{BCC}}, \eta + G_* \in S_{\text{BCC}}\}.$$

Explicitly,

$$S_{\text{bi}}(G_*) = \frac{q_*}{\sqrt{2}} \{(-1, 0, 1), (-1, 0, -1), (0, -1, 1), (0, -1, -1)\}. \quad (179)$$

Hence

$$|S_{\text{bi}}(G_*)| = 4. \quad (180)$$

E. Actual core–spectator vertices

Use the constant-envelope shell basis

$$|+, a\rangle = e^{ik_* \cdot x} \xi_a, \quad |-, a\rangle = e^{i(k_* - G_*) \cdot x} \xi_a, \quad |\eta, a\rangle = e^{i(k_* + \eta) \cdot x} \xi_a,$$

with $a = 1, 2$ the internal doublet index.

Then the scalar periodic coefficient acts trivially on the internal space, so

$$\langle +, a | U_{\text{per}}^{\text{tot}} | \eta, b \rangle = \delta_{ab} \hat{U}_{\text{per}}^{\text{tot}}(\eta), \quad (181)$$

$$\langle -, a | U_{\text{per}}^{\text{tot}} | \eta, b \rangle = \delta_{ab} \hat{U}_{\text{per}}^{\text{tot}}(\eta + G_*). \quad (182)$$

Since $\eta \in S_{\text{bi}}(G_*)$ implies $\eta \in S_{\text{BCC}}$ and $\eta + G_* \in S_{\text{BCC}}$, (178) gives

$$C_+(\eta) = 4uA^2 \mathbf{1}_2, \quad C_-(\eta) = 4uA^2 \mathbf{1}_2, \quad \eta \in S_{\text{bi}}(G_*). \quad (183)$$

Thus every bi-coupled spectator carries the same scalar core–spectator vertex in the diagonal equal-amplitude locked branch.

F. Bare spectator detuning

We now compute the bare spectator momentum magnitude. For the core state,

$$k_* = \frac{q_*}{2\sqrt{2}}(1, 1, 0),$$

so

$$|k_*|^2 = \frac{q_*^2}{4}. \quad (184)$$

Take for example

$$\eta = \frac{q_*}{\sqrt{2}}(-1, 0, 1) \in S_{\text{bi}}(G_*).$$

Then

$$k_* + \eta = \frac{q_*}{\sqrt{2}} \left(-\frac{1}{2}, \frac{1}{2}, 1 \right),$$

hence

$$|k_* + \eta|^2 = \frac{q_*^2}{2} \left(\frac{1}{4} + \frac{1}{4} + 1 \right) = \frac{3q_*^2}{4}.$$

By symmetry, the same value holds for all $\eta \in S_{\text{bi}}(G_*)$:

$$\boxed{|k_* + \eta|^2 = \frac{3q_*^2}{4}, \quad \eta \in S_{\text{bi}}(G_*).} \quad (185)$$

For the bare Brazovskii kernel

$$h_0(q) = r + Z|q|^2 + Y|q|^4, \quad q_*^2 = -\frac{Z}{2Y},$$

the bare spectator detuning relative to the folded core is therefore

$$\begin{aligned} \Delta_{\text{sp}}^{(0)} &:= h_0\left(\sqrt{\frac{3}{4}} q_*\right) - h_0\left(\frac{q_*}{2}\right) \\ &= Z\left(\frac{3q_*^2}{4} - \frac{q_*^2}{4}\right) + Y\left(\frac{9q_*^4}{16} - \frac{q_*^4}{16}\right) \\ &= \frac{1}{2}Zq_*^2 + \frac{1}{2}Yq_*^4 \\ &= \frac{1}{2}\left(-\frac{Z^2}{2Y} + \frac{Z^2}{4Y}\right) = -\frac{Z^2}{8Y}. \end{aligned} \quad (186)$$

Hence the bare magnitude is

$$\boxed{|\Delta_{\text{sp}}^{(0)}| = \frac{Z^2}{8Y}.} \quad (187)$$

a. Honest interpretation. The sign in (186) refers only to the bare diagonal ordering in the simplified folded basis. The actual Schur denominator must use the exact finite shell eigenvalue difference

$$\Delta_\eta = E_\eta - E_*,$$

after the one-step shell matrix is fully assembled. So (187) should be interpreted as the natural bare detuning scale, not yet the final exact sign-resolved denominator.

G. Actual one-step spectator Schur term

The exact scalar spectator correction is

$$B_0^{(\text{sp})} = - \sum_{\eta, \zeta \in S_{\text{bi}}} C_+(\eta) (R_0)_{\eta\zeta} C_-(\zeta)^\dagger.$$

Using (183),

$$B_0^{(\text{sp})} = -16u^2 A^4 \sum_{\eta, \zeta \in S_{\text{bi}}} (R_0)_{\eta\zeta} \mathbf{1}_2. \quad (188)$$

If the spectator resolvent is diagonal and approximately degenerate,

$$(R_0)_{\eta\zeta} \approx \delta_{\eta\zeta} \Delta_{\text{sp}}^{-1},$$

then

$$\boxed{B_0^{(\text{sp})} \approx -\frac{64u^2 A^4}{\Delta_{\text{sp}}} \mathbf{1}_2.} \quad (189)$$

H. Actual scalar cancellation law

Combining the bare scalar block with the spectator correction gives

$$B_0^{\text{tot}} = \left(4uA^2 - \frac{64u^2A^4}{\Delta_{\text{sp}}} \right) \mathbf{1}_2 \quad (\text{diagonal degenerate-spectator approximation}). \quad (190)$$

Hence the exact scalar cancellation condition becomes

$$\boxed{4uA^2 - \frac{64u^2A^4}{\Delta_{\text{sp}}} = 0 \iff \Delta_{\text{sp}} = 16uA^2.} \quad (191)$$

Using the bare detuning magnitude (187) as the first approximation,

$$\Delta_{\text{sp}} \sim \frac{Z^2}{8Y},$$

this would require

$$\boxed{A^2 \sim \frac{Z^2}{128uY}.} \quad (192)$$

I. Critical scaling consequence

The cancellation law (191) reveals the true scaling bottleneck.

The bare scalar Bragg mass scales as

$$B_0^{(\text{bare})} \sim O(A^2),$$

whereas the spectator Schur correction scales as

$$B_0^{(\text{sp})} \sim O\left(\frac{A^4}{\Delta_{\text{sp}}}\right).$$

Therefore, if Δ_{sp} remains finite and nonzero as $A \rightarrow 0$, then

$$B_0^{(\text{sp})} = o(B_0^{(\text{bare})}),$$

and no cancellation can occur near the onset of condensation.

Thus exact cancellation requires one of the following:

1. the spectator detuning softens with the condensate amplitude,

$$\boxed{\Delta_{\text{sp}} = O(A^2),}$$

so that the two terms become comparable;

2. or a separate symmetry removes the scalar identity channel altogether;
3. or the projected operator class is changed so that the dangerous scalar Bragg channel is absent from the start.

J. What is actually achieved

This step does not yet prove that the final complete TECT operator is massless at the shell crossing. But it does accomplish the decisive localization:

1. the actual scalar spectator vertices are now explicit,

$$C_{\pm}(\eta) = 4uA^2 \mathbf{1}_2;$$

2. the actual one-step bi-coupled spectator set has exactly four elements;
3. the natural bare spectator detuning scale is explicit,

$$|\Delta_{\text{sp}}^{(0)}| = \frac{Z^2}{8Y};$$

4. the spectator cancellation law is explicit,

$$\Delta_{\text{sp}} = 16uA^2;$$

5. therefore the full-PDE bottleneck is now sharpened to a single yes/no question:

does the exact finite shell matrix produce $\Delta_{\text{sp}} = O(A^2)$, or is a symmetry needed?

XXII. FINAL SUMMARY

Inside the actual locked 12-mode BCC microscopic branch, the scalar periodic Hessian is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2,$$

and its first-shell Fourier coefficient is

$$\widehat{U}_{\text{per}}^{\text{tot}}(Q) = 4uA^2 \quad (Q \in S_{\text{BCC}}).$$

For the selected shell vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0), \quad k_* = \frac{G_*}{2},$$

the bi-coupled spectator set is

$$S_{\text{bi}}(G_*) = \frac{q_*}{\sqrt{2}} \{(-1, 0, 1), (-1, 0, -1), (0, -1, 1), (0, -1, -1)\},$$

so

$$|S_{\text{bi}}(G_*)| = 4.$$

Therefore the actual scalar core-spectator vertices are

$$C_+(\eta) = 4uA^2 \mathbf{1}_2, \quad C_-(\eta) = 4uA^2 \mathbf{1}_2, \quad \eta \in S_{\text{bi}}(G_*).$$

For every such spectator,

$$|k_* + \eta|^2 = \frac{3q_*^2}{4}, \quad |k_*|^2 = \frac{q_*^2}{4},$$

so the natural bare spectator detuning scale is

$$|\Delta_{\text{sp}}^{(0)}| = \frac{Z^2}{8Y}.$$

In the diagonal degenerate-spectator approximation, the spectator Schur correction is

$$B_0^{(\text{sp})} \approx -\frac{64u^2A^4}{\Delta_{\text{sp}}} \mathbf{1}_2.$$

Hence the total scalar Bragg block is

$$B_0^{\text{tot}} = \left(4uA^2 - \frac{64u^2A^4}{\Delta_{\text{sp}}} \right) \mathbf{1}_2,$$

and the exact scalar cancellation law becomes

$$\Delta_{\text{sp}} = 16uA^2.$$

Therefore the full-PDE bottleneck is now fully explicit:

$$\text{spectator cancellation is structurally possible but not generic.}$$

Indeed,

$$B_0^{(\text{bare})} \sim O(A^2), \quad B_0^{(\text{sp})} \sim O\left(\frac{A^4}{\Delta_{\text{sp}}}\right),$$

so cancellation requires either

$$\Delta_{\text{sp}} = O(A^2),$$

or an independent exact symmetry forbidding the scalar identity channel.

XXIII. EXACT ONE-STEP SHELL-MATRIX DIAGONALIZATION AND THE FAILURE OF GENERIC SPECTATOR CANCELLATION

A. Goal

The remaining microscopic issue is whether the one-step spectator shell can dynamically soften enough to cancel the scalar Bragg mass. Equivalently, we must determine the scaling of the exact spectator detuning

$$\Delta_{\text{sp}}(A) = E_{\text{sp}}(A) - E_{\text{core}}(A)$$

in the actual locked 12-mode branch.

The decisive question is:

$$\text{Does } \Delta_{\text{sp}}(A) \text{ become } O(A^2) \text{ automatically, or does it remain } O(1)?$$

B. Actual scalar sector entering the one-step shell matrix

Inside the currently known locked scalar branch, the periodic scalar Hessian is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2,$$

and the first-shell Bragg block is

$$B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2.$$

At the same time, the projector-stiffness sector yields

$$J = 0$$

at quadratic order, so it does not modify the scalar identity channel.

Therefore, for the present one-step scalar shell analysis, the only relevant coupling is the scalar periodic mixing induced by $u|\phi_0|^2$.

C. Core and spectator states

Fix

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0), \quad k_* = \frac{G_*}{2}.$$

The two core harmonics are

$$|+\rangle := |k_*\rangle, \quad |-\rangle := |k_* - G_*\rangle.$$

The bi-coupled one-step spectators are

$$S_{\text{bi}}(G_*) = \{\eta_1, \eta_2, \eta_3, \eta_4\},$$

with

$$|k_* + \eta_a\rangle, \quad a = 1, \dots, 4.$$

Because the scalar periodic coefficient acts trivially on the internal doublet, the internal factor is spectator here and the shell-mass problem reduces to a scalar harmonic matrix tensored with $\mathbf{1}_2$.

D. Bare diagonal energies

Let

$$h_0(q) = r + Z|q|^2 + Y|q|^4, \quad q_*^2 = -\frac{Z}{2Y}.$$

Then the core bare momentum magnitude is

$$|k_*|^2 = \frac{q_*^2}{4},$$

while every bi-coupled spectator satisfies

$$|k_* + \eta_a|^2 = \frac{3q_*^2}{4}.$$

Hence the bare core and spectator energies are

$$E_c^{(0)} := h_0\left(\frac{q_*}{2}\right), \quad E_s^{(0)} := h_0\left(\sqrt{\frac{3}{4}}q_*\right),$$

and the bare detuning is

$$\Delta_0 := E_s^{(0)} - E_c^{(0)}. \tag{193}$$

Direct substitution gives

$$\Delta_0 = -\frac{Z^2}{8Y}, \quad |\Delta_0| = \frac{Z^2}{8Y}. \tag{194}$$

Thus the natural bare one-step spectator separation is finite and nonzero.

E. One-step shell matrix

By cubic symmetry and equal amplitudes, every bi-coupled spectator sees the same scalar vertex

$$g := 4uA^2.$$

Likewise the direct core Bragg mixing is the same scalar coefficient

$$m := 4uA^2.$$

So the scalar one-step shell matrix is

$$H_{\text{1step}} = \begin{pmatrix} E_c & m & g & g & g & g \\ m & E_c & g & g & g & g \\ g & g & E_s & 0 & 0 & 0 \\ g & g & 0 & E_s & 0 & 0 \\ g & g & 0 & 0 & E_s & 0 \\ g & g & 0 & 0 & 0 & E_s \end{pmatrix} \otimes \mathbf{1}_2 + O(A^4), \quad (195)$$

where

$$E_c = E_c^{(0)} + O(A^2), \quad E_s = E_s^{(0)} + O(A^2).$$

F. Symmetry reduction of the spectator sector

Introduce the normalized symmetric spectator combination

$$|S\rangle := \frac{1}{2} \sum_{a=1}^4 |\eta_a\rangle.$$

The three orthogonal spectator combinations decouple completely from the core pair and remain at energy E_s .

In the core sector, introduce pair-even and pair-odd combinations

$$|c_e\rangle := \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |c_o\rangle := \frac{|+\rangle - |-\rangle}{\sqrt{2}}.$$

Then

$$\langle c_o | H_{\text{1step}} | S \rangle = 0,$$

while

$$\langle c_e | H_{\text{1step}} | S \rangle = 2\sqrt{2}g.$$

Hence the one-step shell matrix reduces exactly to

$$H_{\text{red}} = \left[(E_c - m) |c_o\rangle \langle c_o| \right] \oplus \begin{pmatrix} E_c + m & 2\sqrt{2}g \\ 2\sqrt{2}g & E_s \end{pmatrix} \oplus E_s \mathbf{1}_3 \quad (\text{all tensored with } \mathbf{1}_2). \quad (196)$$

G. Exact eigenvalues

The odd core state is exact:

$$\lambda_o = E_c - m. \quad (197)$$

The coupled (c_e, S) -block has eigenvalues

$$\lambda_{\pm} = \frac{E_c + m + E_s}{2} \pm \sqrt{\frac{(E_s - E_c - m)^2}{4} + 8g^2}. \quad (198)$$

The spectator-like branch is λ_+ , and the even-core-like branch is λ_- when $E_s > E_c + m$. Expanding for small A with

$$m = 4uA^2, \quad g = 4uA^2, \quad E_s - E_c = \Delta_0 + O(A^2),$$

gives

$$\lambda_- = E_c + m - \frac{8g^2}{\Delta_0} + O(A^6), \quad (199)$$

$$\lambda_+ = E_s + \frac{8g^2}{\Delta_0} + O(A^6). \quad (200)$$

Therefore the exact one-step spectator detuning is

$$\begin{aligned} \Delta_{\text{sp}}(A) &:= \lambda_+ - \lambda_- \\ &= \Delta_0 - m + \frac{16g^2}{\Delta_0} + O(A^6) \\ &= \Delta_0 + O(A^2). \end{aligned} \quad (201)$$

Since $\Delta_0 \neq 0$, we conclude:

$$\boxed{\Delta_{\text{sp}}(A) \text{ does not soften to } O(A^2) \text{ automatically.}} \quad (202)$$

H. Effective Bragg mass after exact one-step diagonalization

The two core-like levels are λ_o and λ_- . Hence the effective scalar Bragg splitting is

$$\begin{aligned} B_0^{\text{eff}} &= \frac{\lambda_- - \lambda_o}{2} \\ &= m - \frac{4g^2}{\Delta_0} + O(A^6) \\ &= 4uA^2 - \frac{64u^2A^4}{\Delta_0} + O(A^6). \end{aligned} \quad (203)$$

So the direct Bragg mass is $O(A^2)$, whereas the spectator correction is only $O(A^4)$.

Therefore:

$$\boxed{B_0^{\text{eff}} = 4uA^2 + O(A^4),} \quad (204)$$

and spectator mixing cannot cancel the leading scalar Bragg mass near the onset $A \rightarrow 0$.

I. Decisive conclusion

The exact one-step shell diagonalization proves the following.

[Generic failure of spectator cancellation in the current locked scalar branch] In the equal-amplitude locked 12-mode BCC scalar branch with periodic Hessian

$$U_{\text{per}}^{\text{tot}}(x) = u|\phi_0(x)|^2,$$

the one-step spectator shell does not generically remove the scalar Bragg mass. More precisely:

1. the bare one-step spectator detuning satisfies

$$\Delta_0 = \frac{Z^2}{8Y} \neq 0$$

up to sign convention;

2. the exact one-step detuning obeys

$$\Delta_{\text{sp}}(A) = \Delta_0 + O(A^2);$$

3. the effective scalar Bragg mass obeys

$$B_0^{\text{eff}} = 4uA^2 + O(A^4).$$

Hence the spectator Schur correction is parametrically too small to cancel the leading scalar Bragg obstruction near condensation onset.

This follows directly from the exact reduction (196) and the small- A expansions (199)–(203).

J. What remains possible

Since generic spectator cancellation fails in the current safe operator library, the honest remaining options are:

1. exact scalar-sector cancellation by modifying the scalar periodic Hessian so that

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0;$$

2. exact residual valley symmetry $U(1)_V$ or $\mathbb{Z}_2^V$, which forbids constant Dirac masses once the doubled Dirac block has been isolated;

3. a different microscopic operator class that bypasses the literal scalar Bragg identity channel from the outset.

Thus the spectator route is no longer the leading microscopic frontier. The true next frontier is the residual-symmetry route or an exact scalar coefficient cancellation mechanism.

XXIV. FINAL SUMMARY

The actual one-step shell matrix in the current locked scalar 12-mode BCC branch reduces exactly to:

$$H_{\text{red}} = (E_c - m) \oplus \begin{pmatrix} E_c + m & 2\sqrt{2}g \\ 2\sqrt{2}g & E_s \end{pmatrix} \oplus E_s \mathbf{1}_3, \quad m = g = 4uA^2,$$

tensored with the internal identity.

Its exact eigenvalues are

$$\lambda_o = E_c - m,$$

$$\lambda_{\pm} = \frac{E_c + m + E_s}{2} \pm \sqrt{\frac{(E_s - E_c - m)^2}{4} + 8g^2}.$$

With

$$\Delta_0 := E_s^{(0)} - E_c^{(0)} = \frac{Z^2}{8Y} \neq 0$$

up to sign convention, one obtains

$$\boxed{\Delta_{\text{sp}}(A) = \Delta_0 + O(A^2),}$$

so the spectator gap remains generically $O(1)$.

Consequently,

$$\boxed{B_0^{\text{eff}} = 4uA^2 - \frac{64u^2 A^4}{\Delta_0} + O(A^6) = 4uA^2 + O(A^4),}$$

which proves that the spectator Schur correction is too small to cancel the leading scalar Bragg mass near onset.

Therefore:

generic spectator cancellation fails in the current locked scalar safe operator library.

This is fully consistent with the already established facts that:

$$B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2$$

survives in the current scalar branch, and

$$J = 0$$

for the projector-stiffness sector at quadratic order, so that sector does not remove the scalar identity-channel obstruction.

Hence the honest remaining microscopic routes are:

(i) exact scalar-sector cancellation $2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0$,

(ii) exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$,

or

(iii) a different microscopic operator class bypassing the scalar Bragg channel.

XXV. ACTUAL RESIDUAL-VALLEY SYMMETRY AUDIT ON THE PROJECTED 4×4 SHELL BLOCK

A. Goal

The previous steps reduced the shell-level problem to an actual projected 4×4 core block. The remaining question is no longer whether a mass-protection theorem exists in principle. That part is already closed abstractly: mass protection is equivalent to the finite exclusion test

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

and an exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ is sufficient.

The actual task here is sharper:

identify the residual symmetry of the concrete projected 4×4 shell block that we actually constructed.

B. Starting point: the actual massless canonical shell block

After the previous shell construction, assume that:

1. the scalar Bragg obstruction has been removed,

$$M_0 = 0;$$

2. the longitudinal coefficient is nonzero,

$$v_{\parallel} \neq 0;$$

3. the transverse traceless matrices are nondegenerate and have been brought to the canonical orthogonal form.

Then the projected 4×4 shell-core Hamiltonian takes the canonical massless form

$$\boxed{H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2.} \quad (205)$$

Define the normalized gamma matrices

$$\Gamma_3 := \tau_3 \otimes \mathbf{1}_2, \quad \Gamma_1 := \tau_1 \otimes s_1, \quad \Gamma_2 := \tau_1 \otimes s_2. \quad (206)$$

They satisfy

$$\{\Gamma_i, \Gamma_j\} = 2\delta_{ij} \mathbf{1}_4, \quad i, j = 1, 2, 3,$$

after rescaling the momenta by $v_{\parallel}, \nu_1, \nu_2$.

So (205) is already an exact massless Dirac principal symbol on the projected shell space.

C. Commutant of the actual projected gamma triple

We now solve the commutant problem: find all Hermitian $Q \in \text{Herm}(4)$ satisfying

$$[Q, \Gamma_i] = 0 \quad (i = 1, 2, 3). \quad (207)$$

Write a general Hermitian basis element as

$$Q = \tau_a \otimes s_b, \quad a, b \in \{0, 1, 2, 3\}.$$

A direct calculation shows:

- $\mathbf{1}_4$ commutes with all Γ_i ;
- the unique nontrivial traceless Hermitian solution is

$$\boxed{Q_* := \tau_3 \otimes s_3.} \quad (208)$$

Indeed,

$$[Q_*, \Gamma_3] = [\tau_3 \otimes s_3, \tau_3 \otimes \mathbf{1}_2] = 0,$$

and

$$[Q_*, \Gamma_1] = [\tau_3 \otimes s_3, \tau_1 \otimes s_1] = 0, \quad [Q_*, \Gamma_2] = [\tau_3 \otimes s_3, \tau_1 \otimes s_2] = 0,$$

because the minus sign from the τ -sector is exactly cancelled by the minus sign from the s -sector.

Therefore the commutant algebra of the actual projected Dirac symbol is

$$\boxed{\text{Comm}(\Gamma_1, \Gamma_2, \Gamma_3) = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, Q_*\}.} \quad (209)$$

This is the first genuinely microscopic residual-symmetry result for the concrete projected block.

D. Exact residual $U(1)$ on the projected block

Because $Q_*^\dagger = Q_*$ and $Q_*^2 = \mathbf{1}_4$, define

$$\boxed{U_\theta := e^{i\theta Q_*/2}.} \quad (210)$$

Since $[Q_*, \Gamma_i] = 0$, one has

$$[Q_*, H_{\text{core}}^{\text{lin}}(p)] = 0. \quad (211)$$

Therefore the actual massless projected shell block possesses an exact continuous residual symmetry:

$$\boxed{U(1)_V^{(\text{actual})} : \quad \Psi \mapsto U_\theta \Psi, \quad Q_* = \tau_3 \otimes s_3.} \quad (212)$$

So in the concrete projected block, the theorem-level valley $U(1)$ mechanism is not merely abstract: it is realized explicitly by the operator Q_* .

E. Mass space of the actual projected block

We now solve the mass equations

$$\{M, \Gamma_i\} = 0 \quad (i = 1, 2, 3) \quad (213)$$

inside $\text{Herm}(4)$.

A direct basis computation gives exactly two independent Hermitian masses:

$$\boxed{M_1 := \tau_2 \otimes \mathbf{1}_2, \quad M_2 := \tau_1 \otimes s_3.} \quad (214)$$

Indeed,

$$\{M_1, \Gamma_3\} = 0, \quad \{M_1, \Gamma_1\} = 0, \quad \{M_1, \Gamma_2\} = 0,$$

and similarly

$$\{M_2, \Gamma_3\} = 0, \quad \{M_2, \Gamma_1\} = 0, \quad \{M_2, \Gamma_2\} = 0.$$

Hence the actual mass space of the projected block is

$$\boxed{\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}.} \quad (215)$$

This is of course unitarily equivalent to the standard theorem-level mass space

$$\text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\},$$

but now written in the actual canonical shell basis generated by the previous full-PDE construction.

F. Action of the actual residual symmetry on the actual mass space

We now compute the action of Q_* on the two masses. First,

$$Q_* M_1 = (\tau_3 \otimes s_3)(\tau_2 \otimes \mathbf{1}_2) = -i \tau_1 \otimes s_3 = -i M_2,$$

while

$$M_1 Q_* = +i M_2.$$

Hence

$$[Q_*, M_1] = -2i M_2. \quad (216)$$

Likewise,

$$Q_* M_2 = (\tau_3 \otimes s_3)(\tau_1 \otimes s_3) = +i \tau_2 \otimes \mathbf{1}_2 = +i M_1,$$

while

$$M_2 Q_* = -i M_1.$$

Hence

$$[Q_*, M_2] = 2i M_1. \quad (217)$$

Therefore the two mass matrices form a charged doublet under $U(1)_V^{(\text{actual})}$. Exponentiating, one gets

$$\boxed{U_\theta M_1 U_\theta^{-1} = \cos \theta M_1 + \sin \theta M_2,} \quad (218)$$

$$\boxed{U_\theta M_2 U_\theta^{-1} = -\sin \theta M_1 + \cos \theta M_2.} \quad (219)$$

Hence no nonzero element of $\mathcal{M}_{\text{actual}}$ is invariant under the full $U(1)$:

$$\boxed{A_{U(1)_V^{(\text{actual})}} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (220)$$

So the actual projected 4×4 shell block is mass-protected by an exact continuous residual symmetry.

G. The weaker residual $\mathbb{Z}_2$ symmetry is already enough

Even if one keeps only the discrete subgroup generated by Q_* itself,

$$S_V := Q_* = \tau_3 \otimes s_3, \quad S_V^2 = \mathbf{1}_4, \quad (221)$$

then

$$S_V M_1 S_V^{-1} = -M_1, \quad S_V M_2 S_V^{-1} = -M_2. \quad (222)$$

Therefore the weaker exact residual

$$\boxed{\mathbb{Z}_2^V : \quad \Psi \mapsto S_V \Psi} \quad (223)$$

already forbids every constant mass in $\mathcal{M}_{\text{actual}}$:

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (224)$$

So the actual projected shell block has both:

$$\text{exact } U(1)_V^{(\text{actual})} \implies \text{exact } \mathbb{Z}_2^V.$$

H. Basis-invariant form

The result above can be written without choosing s_1, s_2, s_3 explicitly.

Let the traceless transverse matrices be

$$\widetilde{W}^1 = \vec{w}_1 \cdot \vec{s}, \quad \widetilde{W}^2 = \vec{w}_2 \cdot \vec{s},$$

with $\vec{w}_1, \vec{w}_2 \in \mathbb{R}^3$ linearly independent. After an $SO(2)$ rotation in momentum space and an internal $SU(2)$ basis choice, they can be made orthogonal. Define

$$\hat{q} := \frac{\vec{w}_1 \times \vec{w}_2}{|\vec{w}_1 \times \vec{w}_2|}.$$

Then the actual residual generator is

$$\boxed{Q_* = \tau_3 \otimes (\hat{q} \cdot \vec{s}).} \quad (225)$$

The two actual mass generators can be chosen as

$$\boxed{M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes (\hat{q} \cdot \vec{s}).} \quad (226)$$

Everything then proceeds exactly as above.

Thus the residual $U(1)$ is not an artifact of the special canonical basis. It is an invariant algebraic property of the actual massless projected 4×4 shell symbol.

I. What this does and does not prove microscopically

This step proves the following actual statement:

Once the projected shell block is brought to the massless canonical Dirac form, it possesses an exact residual $U(1)_V$ and hence (227)

However, one must still separate this from the final full microscopic claim.

The theorem-level classification already says that exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ is sufficient for mass protection, and that opposite-valley matching alone is not enough. It also says that Class II is compatible with mass protection but does not prove it by itself. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

What still remains to be checked microscopically is:

1. whether the exact constant $O(p^0)$ corrections of the final corrected shell block all commute with the same Q_* ;
2. whether the remote-sector Schur self-energy $\Sigma(0,0)$ also commutes with Q_* ;
3. equivalently, whether the full residual allowed constant algebra satisfies

$$A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

If the exact microscopic theory preserves the same residual symmetry through remote elimination, then theorem-level radiative stability applies and no mass can be regenerated. :contentReference[oaicite:8]index=8

J. Sharp honest conclusion of Step I

Therefore the actual projected 4×4 shell block has passed the symmetry audit at the principal-symbol level:

$$\boxed{\text{the concrete massless projected shell block itself carries an exact residual valley } U(1)_V.} \quad (228)$$

So the remaining mass-protection frontier is no longer to guess a symmetry. The symmetry has already been identified explicitly:

$$Q_* = \tau_3 \otimes s_3 \quad (\text{or basis-invariantly } Q_* = \tau_3 \otimes (\hat{q} \cdot s)).$$

The remaining frontier is only whether the *full corrected microscopic block*, including all constant corrections and remote elimination, preserves that same Q_* .

XXVI. FINAL SUMMARY

The theorem-level mass-protection problem is known to reduce to the finite test

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

and exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ is a sufficient mechanism, while opposite-valley matching alone is not enough and Class II is only compatible with protection.

For the *actual* projected massless shell block obtained in the full-PDE program,

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2,$$

the residual-symmetry audit can be done explicitly.

The commutant of the three gamma matrices is

$$\boxed{\text{Comm}(\Gamma_1, \Gamma_2, \Gamma_3) = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, Q_*\}, \quad Q_* = \tau_3 \otimes s_3.}$$

Hence the block possesses an exact continuous residual symmetry

$$\boxed{U(1)_V^{(\text{actual})} : \quad U_{\theta} = e^{i\theta Q_*/2}.$$

The actual constant mass space of this projected block is

$$\boxed{\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3.}$$

Under the exact residual $U(1)$,

$$U_{\theta} M_1 U_{\theta}^{-1} = \cos \theta M_1 + \sin \theta M_2,$$

$$U_{\theta} M_2 U_{\theta}^{-1} = -\sin \theta M_1 + \cos \theta M_2.$$

Therefore

$$A_{U(1)_V^{(\text{actual})}} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Even the discrete subgroup

$$S_V = Q_* = \tau_3 \otimes s_3$$

already sends both masses to minus themselves, so

$$A_{\mathbb{Z}_2^V} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Thus the concrete projected 4×4 massless shell block is already mass-protected at the principal-symbol level.

What remains open is only the final microscopic persistence question: do all exact constant $O(p^0)$ corrections and the remote Schur self-energy $\Sigma(0,0)$ commute with the same Q_* ? If yes, then the projected symmetry audit is fully completed and mass protection is radiatively stable.

XXVII. EXACT CONSTANT-CORRECTION ALGEBRA AUDIT ON THE ACTUAL PROJECTED SHELL BLOCK

A. Goal

The principal-symbol audit of the previous step identified the actual residual generator

$$Q_* = \tau_3 \otimes s_3$$

for the projected massless 4×4 shell block. The remaining microscopic question is now purely algebraic:

$$\boxed{\text{Do the exact constant } O(p^0) \text{ corrections preserve } Q_*?}$$

If yes, then no actual constant Dirac mass can appear in the corrected block.

B. Actual principal block and actual mass space

The actual canonical projected shell symbol is

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2. \quad (229)$$

Its actual mass space is

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}} \{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3. \quad (230)$$

By direct computation,

$$[Q_*, M_1] = -2iM_2, \quad [Q_*, M_2] = 2iM_1. \quad (231)$$

Hence the two masses form a charged doublet under the exact $U(1)$ generated by Q_* , and therefore no nonzero element of $\mathcal{M}_{\text{actual}}$ can be Q_* -invariant.

C. The full constant correction to be audited

Write the exact corrected shell block near the crossing as

$$H_{\text{core}}^{\text{exact}}(p) = H_{\text{core}}^{\text{lin}}(p) + \delta H_{\text{const}} + O(|p|^2), \quad (232)$$

with

$$\delta H_{\text{const}} = \delta H_{\text{scalar}} + \delta H_{\text{Class II}} + \Sigma(0,0). \quad (233)$$

The constant-correction audit is therefore the source-by-source classification of these three terms relative to Q_* and $\mathcal{M}_{\text{actual}}$.

D. Commutant of the actual residual generator

A general Hermitian 4×4 operator may be expanded as

$$K = \sum_{a,b=0}^3 k_{ab} \tau_a \otimes s_b.$$

The condition

$$[K, Q_*] = 0, \quad Q_* = \tau_3 \otimes s_3,$$

is equivalent to requiring that the τ -sector and s -sector have the same commutation parity with τ_3 and s_3 , respectively.

Hence the full Q_* -commuting constant algebra is

$$\boxed{\mathcal{C}_{Q_*} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \tau_3 \otimes \mathbf{1}_2, \mathbf{1}_2 \otimes s_3, Q_*, \tau_1 \otimes s_1, \tau_1 \otimes s_2, \tau_2 \otimes s_1, \tau_2 \otimes s_2\}.} \quad (234)$$

Since

$$M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3$$

do *not* belong to $\mathcal{C}_{Q_*}$, we have

$$\boxed{\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (235)$$

Therefore any exact constant correction that preserves Q_* is automatically non-mass-generating.

E. Source I: scalar constant sector

The plain scalar periodic shell obstruction was the identity-channel Bragg block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2,$$

and this was the dangerous constant mass-like channel in the projected shell pair. The previous steps already showed:

1. in the actual locked scalar branch, the bare scalar Bragg block is generically nonzero;
2. spectator cancellation generically fails near onset;
3. therefore the scalar identity channel must be removed by an actual microscopic cancellation law or by exact symmetry.

Once that dangerous scalar channel has been removed, the remaining scalar constant sector is Q_* -invariant and non-mass-generating. Equivalently, after transporting the standard theorem-level valley- $U(1)$ classification to the actual canonical basis, the symmetry-preserving scalar constant algebra reduces to a two-dimensional form

$$\boxed{\delta H_{\text{scalar}}^{(\text{allowed})} = a_0 \mathbf{1}_4 + a_* Q_*}. \quad (236)$$

This is the actual-basis version of the theorem-level result that an exact residual valley symmetry restricts the allowed constant scalar block to the symmetry-neutral sector only.

So the scalar audit is now sharp:

$$\boxed{\text{either the scalar identity channel has already been eliminated, or the full shell block is not massless.}} \quad (237)$$

F. Source II: Class II constant sector

The canonical Class II correction renormalizes only the derivative shell tensor coefficients A, B, C , i.e. the first-order Weyl/Dirac witness sector, and does not by construction insert an independent constant shell-level mass operator. Moreover, every Class II pair-mixing correction carries an explicit internal Pauli generator, so its constant block, if nonzero, lies in the traceless internal sector rather than the scalar identity channel:

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\}.$$

Under the natural shell-Dirac odd/reflection parity, the principal constant gradient block vanishes altogether:

$$\delta B_0^{\text{Class II}} = 0.$$

In the actual Q_* -basis this means:

$$\boxed{\delta H_{\text{Class II}} \in \text{span}\{\tau_1 \otimes s_1, \tau_1 \otimes s_2, \tau_2 \otimes s_1, \tau_2 \otimes s_2\} \subset \mathcal{C}_{Q_*}}, \quad (238)$$

or, under the natural odd shell parity,

$$\boxed{\delta H_{\text{Class II}}^{\text{const}} = 0}. \quad (239)$$

Crucially, neither case intersects the actual mass space $\mathcal{M}_{\text{actual}}$. Indeed,

$$\text{span}\{\tau_1 \otimes s_1, \tau_1 \otimes s_2, \tau_2 \otimes s_1, \tau_2 \otimes s_2\} \cap \text{span}\{\tau_2 \otimes \mathbf{1}_2, \tau_1 \otimes s_3\} = \{0\}.$$

Therefore:

$$\boxed{\text{Class II is fully compatible with mass protection in the actual } Q_*\text{-basis and does not generate } M_1 \text{ or } M_2.} \quad (240)$$

G. Source III: remote self-energy $\Sigma(0, 0)$

The theorem-level result states that if the full microscopic theory preserves the residual valley symmetry and remote-sector elimination preserves that symmetry, then

$$[Q_V, \Sigma(0, 0)] = 0,$$

and in the little-group scalar sector

$$\Sigma(0, 0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2$$

in the standard doubled Dirac basis. Transporting this statement by the unitary equivalence between the standard doubled basis and the actual canonical basis gives:

$$\boxed{[Q_*, \Sigma(0, 0)] = 0, \quad \Sigma(0, 0) = \tilde{a}_0 \mathbf{1}_4 + \tilde{a}_* Q_*}, \quad (241)$$

in the actual projected shell basis, provided the same residual symmetry is exact microscopically.

Therefore the remote self-energy cannot generate either of the actual masses M_1, M_2 :

$$\boxed{\Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\} \quad \text{if the exact } Q_*\text{-symmetry survives remote elimination.}} \quad (242)$$

So the theorem-level radiative stability statement becomes an explicit actual-basis statement: if the exact microscopic theory preserves the same Q_* , then no radiative constant mass can be generated on the isolated projected shell block.

H. Combined constant-correction audit theorem

We can now state the actual constant-correction audit in the form needed for the full PDE program.
[Actual constant-correction audit] Assume:

1. the projected shell block has been reduced to the massless canonical form (229);
2. the dangerous scalar identity-channel Bragg block has been removed;
3. the exact corrected microscopic block preserves the actual residual generator

$$Q_* = \tau_3 \otimes s_3;$$

4. remote-sector elimination preserves the same exact symmetry.

Then every exact constant correction belongs to the Q_* -commuting algebra

$$\delta H_{\text{const}} \in \mathcal{C}_{Q_*},$$

and therefore

$$\boxed{\delta H_{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (243)$$

Hence no exact constant Dirac mass can appear in the corrected projected shell block.

By source decomposition,

$$\delta H_{\text{const}} = \delta H_{\text{scalar}} + \delta H_{\text{Class II}} + \Sigma(0, 0).$$

After removing the scalar identity channel, the remaining scalar constant sector is Q_* -invariant and lies in $\text{span}\{\mathbf{1}_4, Q_*\}$. The Class II constant sector is either zero (under natural shell odd parity) or belongs to the traceless Q_* -commuting non-mass sector. The remote self-energy is Q_* -invariant by exact symmetry preservation and therefore also lies in $\mathcal{C}_{Q_*}$. Since $\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}$, no exact constant mass survives.

I. What is now really left open

This finishes the algebra. The only remaining microscopic question is no longer the classification of constant operators. That is done.

The only remaining yes/no input is:

$$\boxed{\text{Does the actual full microscopic corrected block preserve the same exact } Q_* = \tau_3 \otimes s_3 \text{ symmetry?}} \quad (244)$$

If yes, the mass-protection part of the full local Dirac program is closed. If no, one must identify the precise symmetry-breaking constant term and check whether it lies in $\mathcal{M}_{\text{actual}}$.

XXVIII. FINAL SUMMARY

The full constant-correction problem has now been reduced to an actual symmetry audit around the explicit residual generator

$$\boxed{Q_* = \tau_3 \otimes s_3.}$$

For the actual projected massless shell block

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2,$$

the actual constant mass space is

$$\boxed{\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3.}$$

The Q_* -commuting constant algebra is

$$\mathcal{C}_{Q_*} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \tau_3 \otimes \mathbf{1}_2, \mathbf{1}_2 \otimes s_3, Q_*, \tau_1 \otimes s_1, \tau_1 \otimes s_2, \tau_2 \otimes s_1, \tau_2 \otimes s_2\}.$$

Hence

$$\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

The exact constant correction splits into

$$\delta H_{\text{const}} = \delta H_{\text{scalar}} + \delta H_{\text{Class II}} + \Sigma(0, 0).$$

The source-by-source audit gives:

$$\text{scalar identity-channel Bragg mixing must already have been removed;}$$

$$\text{Class II only renormalizes the derivative shell tensor and does not generate an actual mass;}$$

$$\text{if the same residual symmetry is exact microscopically, then } [Q_*, \Sigma(0, 0)] = 0 \text{ and the remote self-energy is non-mass-generating}$$

Therefore:

$$\text{if the exact corrected microscopic block preserves } Q_*, \text{ then no constant actual mass } M_1 \text{ or } M_2 \text{ can appear.}$$

So the mass-protection frontier is no longer algebraic. It is now a single microscopic symmetry-preservation question:

$$\text{Does the final corrected full microscopic shell block preserve the exact } Q_* = \tau_3 \otimes s_3 \text{ symmetry?}$$

XXIX. FULL MICROSCOPIC SYMMETRY-PRESERVATION AUDIT

A. Goal

At this stage the local shell Dirac block, the actual mass space, and the actual residual generator

$$Q_* = \tau_3 \otimes s_3$$

have already been identified on the projected 4×4 shell block.

Therefore the final full-PDE mass-protection question is no longer a structural one. It is the explicit source-by-source symmetry audit:

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0. \quad (245)$$

If all three hold, then no exact constant mass can appear in the corrected projected shell block.

B. What is already known at theorem level

For the doubled local Dirac block

$$H_D(p) = \sum_i v_i p_i (\tau_z \otimes \sigma_i),$$

the genuine constant Dirac mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}. \quad (246)$$

Mass protection is exactly the finite test

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}}, \quad (247)$$

and exact residual valley $U(1)_V$ or exact residual valley $\mathbb{Z}_2^V$ is sufficient. Opposite-valley matching alone is not sufficient, and Class II is compatible only, not by itself a proof. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

If the full microscopic theory preserves the residual valley symmetry and remote elimination preserves it, then the Schur-complement shell Hamiltonian inherits the same symmetry and

$$[Q_V, \Sigma(0, 0)] = 0.$$

Under little-group scalarity, this restricts the zero-momentum self-energy to

$$\Sigma(0, 0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2, \quad (248)$$

so no radiatively generated term proportional to $\tau_x \otimes \mathbf{1}_2$ or $\tau_y \otimes \mathbf{1}_2$ can appear. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

C. Transport to the actual canonical shell basis

In the actual projected shell basis of the full-PDE construction, the massless canonical shell block was written as

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2. \quad (249)$$

Its actual residual generator is

$$\boxed{Q_* = \tau_3 \otimes s_3}, \quad (250)$$

and its actual mass space is

$$\boxed{\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3.} \quad (251)$$

There exists a unitary basis change U_{can} relating the standard doubled-Dirac basis and the actual canonical shell basis such that

$$Q_* = U_{\text{can}} Q_V U_{\text{can}}^{-1}, \quad \mathcal{M}_{\text{actual}} = U_{\text{can}} \mathcal{M}_D U_{\text{can}}^{-1}.$$

Therefore every theorem-level symmetry statement in the standard basis transports directly to the actual basis.

In particular, if the exact microscopic theory preserves the standard residual valley symmetry, then in the actual basis

$$\boxed{[Q_*, \Sigma(0, 0)] = 0}, \quad (252)$$

and under transported little-group scalarity

$$\boxed{\Sigma(0, 0) = \tilde{a}_0 \mathbf{1}_4 + \tilde{a}_* Q_*}. \quad (253)$$

Hence

$$\Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

D. Source I: scalar constant sector

The current locked scalar branch gives the periodic scalar Hessian

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2,$$

and hence the scalar identity-channel Bragg obstruction

$$B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2.$$

The alignment term contributes nothing on the locked branch, and the canonical Class II correction does not directly cancel the scalar identity channel.

Therefore the scalar source audit is sharp: either the dangerous identity-channel block has already been removed by an actual scalar cancellation law, or the shell block is not massless to begin with. Once that scalar identity-channel piece is removed, the remaining symmetry-preserving scalar constant sector lies in the neutral Q_* -commuting subalgebra:

$$\delta H_{\text{scalar}}^{(\text{allowed})} = a_0 \mathbf{1}_4 + a_* Q_*. \quad (254)$$

Hence the scalar sector is harmless *after* the identity-channel obstruction is eliminated.

E. Source II: Class II constant sector

The canonical minimal Class II representative

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J e_{1j} \hat{n}_i + c_I e_{2j} \hat{n}_i$$

renormalizes only the derivative shell tensor coefficients A, B, C , equivalently the first-order Weyl/Dirac witnesses. It does not by construction insert an independent constant 4×4 mass operator.

More sharply, every projected constant pair-mixing block induced by Class II, if nonzero, lies in the traceless internal Pauli channel

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\},$$

so in the shell-pair block it is of the form

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

not $\tau_1 \otimes \mathbf{1}_2$. Under the natural shell-Dirac odd/reflection parity, the gradient-sector constant pair-mixing vanishes at principal order:

$$\delta B_0^{\text{Class II}} = 0.$$

Hence Class II renormalizes the derivative witness sector but does not remove the scalar Bragg obstruction and does not directly generate an actual mass.

In the actual Q_* -basis this means:

$$\boxed{\delta H_{\text{Class II}}^{\text{const}} = 0 \quad (\text{natural odd shell parity}),} \quad (255)$$

or more weakly

$$\boxed{\delta H_{\text{Class II}}^{\text{const}} \in \mathcal{C}_{Q_*} \setminus \mathcal{M}_{\text{actual}}.} \quad (256)$$

So the Class II source passes the mass audit.

F. Source III: remote self-energy

The only genuinely open microscopic part is remote-sector inheritance. At theorem level, if the full microscopic theory admits a symmetry action

$$(U_\theta \oplus U_{R,\theta}) H_{\text{full}}(p) (U_\theta \oplus U_{R,\theta})^{-1} = H_{\text{full}}(p),$$

then the Schur-complement effective shell Hamiltonian inherits it, and

$$[Q_V, \Sigma(0, 0)] = 0.$$

Transported to the actual canonical basis, this becomes

$$\boxed{[Q_*, \Sigma(0, 0)] = 0.} \quad (257)$$

Under the transported little-group scalarity,

$$\Sigma(0, 0) = \tilde{a}_0 \mathbf{1}_4 + \tilde{a}_* Q_*,$$

so no term in $\mathcal{M}_{\text{actual}}$ can be generated. :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

Therefore the remote self-energy source passes the mass audit *if and only if* the exact same residual symmetry is actually preserved by the full microscopic theory and by remote elimination.

G. Combined actual audit theorem

We can now formulate the exact logical status in the actual basis.

[Actual full-microscopic mass-protection audit] Assume:

1. the projected shell block has been reduced to the actual canonical massless form (249);
2. the dangerous scalar identity-channel Bragg block has been removed;
3. the full microscopic corrected block preserves the actual residual generator

$$Q_* = \tau_3 \otimes s_3;$$

4. remote elimination preserves the same exact symmetry.

Then

$$\delta H_{\text{const}} = \delta H_{\text{scalar}} + \delta H_{\text{Class II}} + \Sigma(0, 0) \in \mathcal{C}_{Q_*}, \quad (258)$$

and

$$\boxed{\delta H_{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (259)$$

Hence no exact constant Dirac mass can appear in the corrected projected shell block.

After the scalar identity-channel obstruction is removed, the remaining scalar constant block is Q_* -neutral. The canonical Class II correction is either zero at constant order or lies in the traceless non-mass sector and does not intersect $\mathcal{M}_{\text{actual}}$. If the same residual symmetry is exact in the full microscopic theory, then the remote self-energy commutes with Q_* and is restricted to $\tilde{a}_0 \mathbf{1}_4 + \tilde{a}_* Q_*$. Thus every exact constant correction lies in $\mathcal{C}_{Q_*}$, while $\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}$.

H. Sharp honest conclusion

This finishes the algebra completely.

The only remaining full-PDE question is no longer the classification of masses or residual generators. Those are already explicit.

The only remaining microscopic yes/no input is:

$$\boxed{\text{Does the actual final corrected full microscopic shell block preserve the exact } Q_* = \tau_3 \otimes s_3 \text{ symmetry?}} \quad (260)$$

If yes, the mass-protection part of the local Dirac program is closed. If no, one must identify the explicit symmetry-breaking constant term and check whether it projects onto M_1 or M_2 .

XXX. FINAL SUMMARY

The full constant-correction problem is now completely reduced to a concrete symmetry-preservation audit around the actual residual generator

$$Q_* = \tau_3 \otimes s_3.$$

For the actual projected massless shell block

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2,$$

the actual constant mass space is

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{\tau_2 \otimes \mathbf{1}_2, \tau_1 \otimes s_3\}.$$

The theorem-level classification already establishes that:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}$$

is the exact finite mass-protection criterion, and exact residual valley $U(1)_V$ or exact $\mathbb{Z}_2^V$ is sufficient. Opposite-valley matching alone is not sufficient, and Class II is only compatible. :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

Transported to the actual canonical basis, this becomes:

if the corrected microscopic shell block preserves Q_* , then no constant actual mass can appear.

Source-by-source:

(i) the scalar identity-channel Bragg block must already have been removed;

(ii) Class II renormalizes only the derivative witness sector and does not generate an independent constant mass;

(iii) if the same residual symmetry is exact microscopically, then $[Q_*, \Sigma(0,0)] = 0$ and remote elimination is non-mass-generating

Therefore the algebra is finished. The only remaining full-PDE microscopic open question is:

Does the actual final corrected full microscopic block preserve the exact $Q_* = \tau_3 \otimes s_3$ symmetry?

If yes, mass protection is closed. If no, the symmetry-breaking constant term must be identified explicitly.

XXXI. ACTUAL SOURCE-LEVEL Q_* -INVARIANCE TEST AND FINAL COMPRESSION OF THE FULL-PDE FRONTIER

A. Goal

The actual projected massless shell block has already been reduced to the canonical 4×4 form

$$H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2, \quad (261)$$

with actual residual generator

$$Q_* = \tau_3 \otimes s_3. \quad (262)$$

The actual constant-mass space is

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}} \{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3. \quad (263)$$

The theorem-level finite criterion is already known:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

and exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ is sufficient for full constant mass protection. The actual remaining task is therefore purely source-by-source:

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0. \quad (264)$$

B. Source I: scalar core sector

Inside the currently known locked scalar branch, the periodic scalar Hessian is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2,$$

and the projected shell analysis gives a generic scalar identity-channel Bragg block of the form

$$\delta H_{\text{scalar}}^{\text{danger}} = B_0^{\text{scalar}} \tau_1 \otimes \mathbf{1}_2, \quad B_0^{\text{scalar}} \neq 0 \text{ generically.} \quad (265)$$

This is exactly the dangerous constant mass-like obstruction in the projected shell pair.

Therefore the scalar audit is sharp:

$$\boxed{\text{either } B_0^{\text{scalar}} \text{ has already been removed by an actual scalar cancellation law or exact symmetry,}} \quad (266)$$

or else the projected shell block is not massless to begin with.

Once the dangerous identity-channel block is removed, the remaining Q_* -preserving scalar constant sector is necessarily neutral:

$$\boxed{\delta H_{\text{scalar}}^{(\text{allowed})} = a_0 \mathbf{1}_4 + a_* Q_*}. \quad (267)$$

Hence after elimination of the identity-channel obstruction, the scalar source does not intersect $\mathcal{M}_{\text{actual}}$.

C. Source II: Class II source

For the canonical minimal Class II representative, the corrected shell block remains of the form

$$H_D(p) = \tau_z \otimes (A_{\text{eff}} p_{\parallel} \sigma_3 + B_{\text{eff}} p_1 \sigma_1 + C_{\text{eff}} p_2 \sigma_2), \quad (268)$$

that is, Class II only renormalizes the derivative shell tensor coefficients A, B, C , equivalently the first-order Weyl/Dirac witnesses. It does not by construction insert an independent constant shell-level mass operator.

Transporting (268) to the actual canonical basis (261), this means that Class II only renormalizes

$$v_{\parallel}, \quad \nu_1, \quad \nu_2,$$

and does not generate M_1 or M_2 .

So the actual constant Class II audit is:

$$\boxed{\delta H_{\text{Class II}}^{\text{const}} = 0 \quad (\text{canonical minimal Class II branch}).} \quad (269)$$

In particular,

$$[Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0$$

trivially, and

$$\delta H_{\text{Class II}}^{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Thus the Class II source fully passes the actual mass audit.

D. Source III: remote-sector source

There are in fact two nested statements here.

a. (A) Stronger source-level statement from the zero-mixing theorem. The actual Step-3 theorem states:

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.} \quad (270)$$

And the practical microscopic version is the sectorwise remote-label mismatch criterion: if every remote sector differs from

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2$$

in at least one shell-relevant label, then $V(0) = 0$ exactly.

Therefore, once the finite remote-label audit succeeds,

$$\boxed{\Sigma(0, 0) = V(0)^\dagger (-H_R(0))^{-1} V(0) = 0.} \quad (271)$$

This is strictly stronger than merely commuting with Q_* .

b. (B) Weaker theorem-level symmetry statement. Even without using $V(0) = 0$, the theorem-level radiative-stability result says that if the full microscopic theory preserves the residual valley symmetry and remote elimination preserves it, then

$$[Q_V, \Sigma(0, 0)] = 0$$

in the standard doubled basis, and under little-group scalarity

$$\Sigma(0, 0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2.$$

Transporting this statement to the actual canonical basis gives

$$\boxed{[Q_*, \Sigma(0, 0)] = 0, \quad \Sigma(0, 0) = \tilde{a}_0 \mathbf{1}_4 + \tilde{a}_* Q_*} \quad (272)$$

So the remote audit passes in either of the following ways:

1. strongest actual route:

$$V(0) = 0 \implies \Sigma(0, 0) = 0;$$

2. weaker symmetry route:

$$[Q_*, \Sigma(0, 0)] = 0 \implies \Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Hence the remote source is no longer the bottleneck once the Step-3 label audit is executed.

E. Combined actual source-level audit

We can now summarize the actual source-by-source result.

[Actual source-level mass audit] Assume:

1. the projected shell block has been reduced to the actual canonical massless form (261);
2. the dangerous scalar identity-channel block $B_0^{\text{scalar}} \tau_1 \otimes \mathbf{1}_2$ has been removed;
3. the canonical minimal Class II branch is used;
4. the remote-label mismatch audit succeeds, so that $V(0) = 0$.

Then:

$$\delta H_{\text{scalar}}^{\text{const}} = a_0 \mathbf{1}_4 + a_* Q_*, \quad \delta H_{\text{Class II}}^{\text{const}} = 0, \quad \Sigma(0, 0) = 0. \quad (273)$$

Therefore

$$\boxed{\delta H_{\text{const}} \cap \mathcal{M}_{\text{actual}} = \{0\}}. \quad (274)$$

Hence no exact constant Dirac mass can appear in the corrected projected shell block.

After removing the scalar identity-channel obstruction, the remaining scalar constant block is Q_* -neutral and lies in $\text{span}\{\mathbf{1}_4, Q_*\}$. The canonical minimal Class II sector changes only the first-order shell coefficients and contributes no independent constant shell-level mass operator. If the remote-label audit succeeds, then $V(0) = 0$, hence $\Sigma(0, 0) = 0$ exactly. None of these three sectors intersects $\mathcal{M}_{\text{actual}}$.

F. What is now genuinely left open

This compresses the full-PDE frontier further.

The source-level Class II audit is passed. The remote source audit is also passed once the Step-3 label mismatch theorem is actually verified for the chosen microscopic realization. Therefore the only genuinely dangerous constant source is the scalar identity-channel Bragg block.

Hence the full microscopic frontier is no longer:

“all constant corrections”,

but rather the single microscopic question:

$$\boxed{\text{How is the scalar identity-channel Bragg block removed in the final complete TECT realization?}} \quad (275)$$

There are only the already-isolated honest possibilities:

1. exact scalar coefficient cancellation in the microscopic scalar Hessian;
2. exact symmetry forbidding the scalar identity-channel block;
3. a different microscopic operator class bypassing the scalar Bragg channel from the outset.

XXXII. FINAL SUMMARY

The actual source-level Q_* -audit is now essentially complete.

The projected massless shell block has the actual residual generator

$$\boxed{Q_* = \tau_3 \otimes s_3},$$

and the actual constant mass space

$$\boxed{\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{\tau_2 \otimes \mathbf{1}_2, \tau_1 \otimes s_3\}}.$$

At theorem level, mass protection is already reduced to the finite test

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}},$$

with exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ as sufficient mechanisms, and radiative stability follows if the same symmetry survives remote elimination.

At the actual source level:

$$\boxed{\text{(i) after removing the dangerous scalar identity-channel Bragg block, the remaining scalar constant sector is } a_0 \mathbf{1}_4 + a_* Q_*};$$

(ii) canonical minimal Class II only renormalizes derivative shell coefficients and contributes no independent constant shell-level

(iii) if the Step-3 remote-label mismatch audit succeeds, then $V(0) = 0$ exactly, hence $\Sigma(0, 0) = 0$.

So the strongest honest actual-basis conclusion is:

Class II source audit: passed, remote source audit: passed once $V(0) = 0$,

and therefore

the only genuinely dangerous remaining constant source is the scalar identity-channel Bragg block.

Thus the full-PDE frontier is now compressed to one microscopic problem:

remove the scalar identity-channel Bragg block in the final complete TECT realization.

XXXIII. SCALAR IDENTITY-CHANNEL ELIMINATION IN THE ACTUAL MICROSCOPIC SCALAR SECTOR

A. Goal

After the source-level Q_* -audit, the only genuinely dangerous remaining constant source is the scalar identity-channel Bragg block

$$B_0^{\text{scalar}} \tau_1 \otimes \mathbf{1}_2.$$

The purpose of this step is to determine whether the currently known microscopic scalar sector of TECT can eliminate this obstruction *within the actual safe operator library*.

B. Exact scalar Bragg-mass formula

Take the most general shell-relevant scalar periodic contribution built from the three natural locked first-shell scalar operators:

$$U_{\text{tot}}(x) = c_0 \phi_0(x)^2 + c_1 |\nabla \phi_0(x)|^2 + c_2 \phi_0(x)(-\Delta)\phi_0(x). \quad (276)$$

For the locked 12-mode BCC branch, the first-shell Fourier coefficients are

$$\widehat{\phi_0^2}(G_*) = 4A^2, \quad (277)$$

$$\widehat{|\nabla \phi_0|^2}(G_*) = 2q_*^2 A^2, \quad (278)$$

$$\widehat{\phi_0(-\Delta)\phi_0}(G_*) = 4q_*^2 A^2. \quad (279)$$

Hence

$$\begin{aligned} \widehat{U}_{\text{tot}}(G_*) &= c_0(4A^2) + c_1(2q_*^2 A^2) + c_2(4q_*^2 A^2) \\ &= 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2). \end{aligned} \quad (280)$$

Therefore the projected scalar Bragg block is

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2. \quad (281)$$

The exact scalar cancellation condition is thus

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0. \quad (282)$$

C. Actual coefficients contributed by the currently known safe sectors

The actual uploaded microscopic analysis isolates the contributions of the presently known safe sectors.

a. Sector 1: actual scalar Hessian. Inside the actual Q -sector scalar Hessian, the only periodic scalar contribution is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2. \quad (283)$$

Therefore

$$\boxed{c_0^{(\text{scalar})} = u, \quad c_1^{(\text{scalar})} = 0, \quad c_2^{(\text{scalar})} = 0.} \quad (284)$$

So

$$\boxed{B_0^{(\text{scalar})} = 4uA^2 \mathbf{1}_2.} \quad (285)$$

b. Sector 2: alignment term. On the locked branch

$$\Psi = z\phi, \quad P = zz^\dagger = \text{const},$$

one has

$$\partial_i P = 0,$$

hence the refined alignment term vanishes identically on the locked sector:

$$\boxed{c_0^{(\text{align})} = 0, \quad c_1^{(\text{align})} = 0, \quad c_2^{(\text{align})} = 0.} \quad (286)$$

So

$$\boxed{B_0^{(\text{align})} = 0.} \quad (287)$$

c. Sector 3: canonical Class II. The exact Class II witness formulas modify only the derivative witness coefficients:

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

So Class II acts in the derivative witness sector $(W_{\parallel}, W_I, W_J)$, not in the scalar identity-channel Bragg block. Equivalently,

$$\boxed{c_0^{(\text{Class II})} = 0, \quad c_1^{(\text{Class II})} = 0, \quad c_2^{(\text{Class II})} = 0} \quad (288)$$

as a direct scalar cancellation channel.

D. Exact no-go theorem for the current safe operator library

Summing the known safe microscopic sectors gives

$$\boxed{(c_0, c_1, c_2)_{\text{known}} = (u, 0, 0).} \quad (289)$$

Substituting into (281),

$$\boxed{B_0^{\text{known}} = 4uA^2 \mathbf{1}_2.} \quad (290)$$

Hence, for $u \neq 0$,

$$\boxed{B_0^{\text{known}} \neq 0.} \quad (291)$$

Therefore:

[No-go inside the current safe operator library] Within the currently known safe microscopic sectors of the locked 12-mode BCC branch, namely:

1. the actual scalar Hessian,
2. the refined alignment term,
3. the canonical minimal Class II mediator,

the scalar identity-channel Bragg block cannot be eliminated. Equivalently,

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 2u \neq 0$$

for $u \neq 0$, so the exact scalar cancellation law cannot be satisfied.

Immediate from (282) and (289).

E. Why canonical Class II cannot help

The reason is structural, not numerical. Every Class II contribution carries an explicit internal Pauli generator in the projected shell block:

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\},$$

so the constant pair block is of the form

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

never

$$\tau_1 \otimes \mathbf{1}_2.$$

Hence the scalar identity channel and the traceless Pauli channel are orthogonal, and Class II cannot directly cancel

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

F. Exact classification of the remaining routes

The reduced operator analysis already isolates the only two logically distinct possibilities.

a. Route 1: direct identity-channel cancellation. To cancel the scalar Bragg mass directly, one must introduce an additional microscopic sector whose projected constant pair block has a nonzero identity component:

$$\delta M_0^{\text{id}} = \delta m_0 \mathbf{1}_2 + \delta \widetilde{M}_0, \quad \delta m_0 \neq 0, \quad (292)$$

with

$$\boxed{\delta m_0 = -B_0^{\text{tot}}}. \quad (293)$$

Any UV completion whose projected constant pair block is purely traceless cannot realize this route.

b. Route 2: graded 2 + 2 bypass. Instead of cancelling the scalar Bragg block inside the plain scalar shell-pair reduction, one passes to a J -graded 2 + 2 low-energy sector in which the relevant principal operator is:

1. J -odd,
2. first-order in momentum.

Then the reduced infrared operator is block-off-diagonal and first-order, so the dangerous constant scalar Bragg block is absent from the relevant low-energy block from the outset.

Therefore the exact classification is:

$$\boxed{\text{identity-carrying mediator route} \quad \text{or} \quad \text{graded 2 + 2 bypass route.}} \quad (294)$$

G. Sharp honest conclusion

At this point the local full-PDE program becomes completely explicit.

1. The derivative witness sector has already been isolated and Class II provides a concrete route for generating it.
2. The actual constant-mass algebra and residual generator have already been identified on the projected 4×4 shell block.
3. The remote-sector source becomes harmless once the zero-mixing label audit gives $V(0) = 0$.
4. The only unresolved microscopic source is the scalar identity-channel Bragg block.
5. Inside the current safe operator library, this scalar block cannot be removed.

So the full-PDE frontier is no longer broad. It is precisely the choice between:

adding an identity-carrying microscopic sector,

or

abandoning the plain scalar shell-pair channel in favor of the graded $2 + 2$ bypass route.

XXXIV. FINAL SUMMARY

The scalar identity-channel Bragg block is controlled by the exact shell-level formula

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.$$

Hence the exact scalar cancellation condition is

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.$$

For the currently known safe microscopic sectors, the uploaded analysis gives:

$$(c_0, c_1, c_2)_{\text{scalar}} = (u, 0, 0),$$

$$(c_0, c_1, c_2)_{\text{align}} = (0, 0, 0),$$

$$(c_0, c_1, c_2)_{\text{Class II}} = (0, 0, 0)$$

as a direct scalar-cancellation channel.

Therefore

$$(c_0, c_1, c_2)_{\text{known}} = (u, 0, 0),$$

and so

$$B_0^{\text{known}} = 4uA^2 \mathbf{1}_2 \neq 0 \quad (u \neq 0).$$

Thus there is a genuine no-go result:

inside the current safe microscopic operator library, the scalar identity-channel Bragg block cannot be removed.

Moreover, canonical Class II cannot help because its projected constant pair block is purely traceless:

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\},$$

so it cannot cancel the identity channel

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Therefore the remaining microscopic routes are exactly two:

(i) direct identity-channel cancellation by an identity-carrying mediator or sector,

or

(ii) a graded $2 + 2$ bypass in which the dangerous scalar pair block is absent from the relevant low-energy operator.

So the full-PDE frontier is no longer diffuse. It is the exact decision:

identity-carrying completion vs. graded $2 + 2$ bypass.

XXXV. ROUTE B MAINLINE: GRADED $2 + 2$ BYPASS AS THE ACTUAL FULL-PDE PROGRAM

A. Route B replaces scalar-channel cancellation

The previous scalar-shell analysis established a genuine no-go inside the current safe operator library: the scalar identity-channel Bragg block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2$$

cannot be eliminated there. Therefore the correct mainline is no longer to cancel B_0^{tot} inside the plain scalar shell-pair channel.

Instead, the correct mainline is the graded $2 + 2$ bypass: the relevant low-energy sector is not the plain scalar Bragg pair, but an isolated four-dimensional graded infrared sector

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2,$$

whose leading reduced operator is first-order and odd under the grading.

B. Abstract graded structure

Let J be a grading involution on the low-energy sector:

$$J^\dagger = J, \quad J^2 = \mathbf{1}, \quad J|_{K_{\pm}} = \pm \mathbf{1}. \quad (295)$$

Let the projected reduced operator be

$$D_{\text{eff}}(p) = \Pi_K L(p) \Pi_K = D_1(p) + O(|p|^2), \quad (296)$$

and assume the leading term is:

1. first-order in momentum,
2. J -odd:

$$J D_1(p) J^{-1} = -D_1(p). \quad (297)$$

Then in the J -graded basis one has

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_i M_i.$$

(298)

This is the rigorous reduced bypass structure already isolated in the manuscripts.

C. Why the scalar Bragg mass is absent

The dangerous obstruction in the plain scalar shell-pair route was a momentum-independent constant block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

But in the graded bypass route, the relevant principal infrared operator is by definition first-order:

$$A_1(p) = p_i M_i.$$

Hence

$$\boxed{A_1(0) = 0, \quad D_1(0) = 0.} \quad (299)$$

Therefore the scalar Bragg block is not cancelled inside D_1 ; it is absent because the relevant principal operator starts at first order in momentum.

This is the precise meaning of the bypass mechanism.

D. General mixed symbol and finite nondegeneracy matrix

Write the off-diagonal mixed symbol in Pauli form:

$$\boxed{A_1(p) = \ell_0(p) \mathbf{1}_2 + \ell_a(p) \sigma_a, \quad \ell_a(p) = C_a^i p_i.} \quad (300)$$

The identity part $\ell_0(p) \mathbf{1}_2$ is not a mass obstruction because it is still linear in p . The actual load-bearing data are the traceless coefficient vectors C_a^i .

Define the finite nondegeneracy matrix

$$\boxed{G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j.} \quad (301)$$

Then

$$\boxed{\det A_1(p) = -p_i G_{ij} p_j, \quad \det D_1(p) = (p_i G_{ij} p_j)^2.} \quad (302)$$

Thus the graded bypass route reduces the Dirac question to the finite test

$$\boxed{G_{ij} > 0.}$$

E. Actual local CP^2 projector-defect operator class

The manuscripts already identify the actual local TECT projector-defect geometry as belonging to this graded operator class.

For a local defect background

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1,$$

the physical tangent fluctuation lives in the rank-two complex bundle

$$\boxed{Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp.} \quad (303)$$

Thus the defect fluctuation problem is already non-scalar.

For a reflection-symmetric representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

the leading worldvolume transport kernel is built linearly from the transverse gradient, hence is odd:

$$\mathcal{V}^\mu(-\xi) = -\mathcal{V}^\mu(\xi).$$

Therefore the actual local projector Hessian matches the abstract graded operator class: static rank-two bundle operator + even/odd low channels + odd first-order transport kernel.

F. Natural local symmetry-compatible branch: S2

Inside the actual local projector-defect transport class, the natural local branch is S2:

$$\boxed{A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).} \quad (304)$$

Equivalently, in terms of projected transport matrices,

$$M^{\mu} = C_0^{\mu} \mathbf{1}_2 + C_a^{\mu} \sigma_a.$$

For the actual local projector-defect transport ansatz one has

$$C_0^i = 0,$$

and

$$(C_1^1, C_2^1, C_3^1) = (I_{\theta} \Lambda_E, I_{\theta} \Lambda_O, 0),$$

$$(C_1^2, C_2^2, C_3^2) = (-I_{\theta} \Lambda_O, I_{\theta} \Lambda_E, 0),$$

$$(C_1^{\parallel}, C_2^{\parallel}, C_3^{\parallel}) = (0, 0, I_{\theta} \Lambda_{\parallel}).$$

Therefore

$$\boxed{G_{ij} = I_{\theta}^2 \begin{pmatrix} \Lambda_E^2 + \Lambda_O^2 & 0 & 0 \\ 0 & \Lambda_E^2 + \Lambda_O^2 & 0 \\ 0 & 0 & \Lambda_{\parallel}^2 \end{pmatrix}.} \quad (305)$$

Hence the graded bypass nondegeneracy condition is exactly

$$\boxed{G_{ij} > 0 \iff \Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (306)$$

Equivalently,

$$\boxed{G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (307)$$

So the graded bypass route is not an independent mystery: inside the actual local projector-defect operator class, it reduces exactly to the same three transport coefficients that already control the witness problem.

G. Concrete minimal transport ansatz

A concrete minimal microscopic transport action compatible with the actual local projector-defect geometry is

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^{\dagger} \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} + g_E \theta'(\xi) E^{\mu}_a \Sigma_a + g_O \theta'(\xi) J^{\mu}_a \Sigma_a \right] \partial_{y^{\mu}} \eta. \quad (308)$$

Its projected coefficients are

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.} \quad (309)$$

Therefore the full local S2 nondegeneracy condition becomes exactly

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (310)$$

So the graded bypass route has now been compressed to a finite coefficient-extraction problem.

H. What is proved and what remains open

What is now genuinely closed:

1. the plain scalar Bragg route is not the correct mainline;
2. the graded $2 + 2$ reduced Dirac theorem is rigorous;
3. the actual local CP^2 projector-defect geometry belongs to this operator class;
4. the natural local branch is S2 ;
5. the dangerous scalar Bragg block is absent from the relevant graded principal operator because

$$D_1(0) = 0;$$

6. the full principal Dirac nondegeneracy problem is reduced to the finite test

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

What remains open:

1. derive the actual transport kernel $V^\mu(\xi)$ from the *full* microscopic TECT operator rather than postulate it as a minimal local ansatz;
2. prove that the actual reduced leading perturbation is indeed grading-odd in the full microscopic defect/valley reduction;
3. compute the actual microscopic values of

$$g_c, \quad g_E, \quad g_O;$$

4. verify directly that complete TECT indeed gives

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

I. Sharp honest conclusion

Thus the graded bypass route is now the correct non-repetitive mainline.
The decisive unresolved microscopic problem is no longer

“how do we cancel B_0 ?”

but rather

Does the actual defect/projector reduced operator admit the required $2 + 2$ grading and produce nonzero g_c, g_E, g_O ?

(311)

If yes, then the principal infrared operator is Dirac automatically.

XXXVI. FINAL SUMMARY

The scalar identity-channel Bragg block cannot be removed inside the current safe scalar/Class II operator library.
Therefore the correct mainline is no longer scalar-channel cancellation, but the graded $2 + 2$ bypass route.

In the graded route, the relevant infrared sector is

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2,$$

with a grading involution J such that the leading projected operator is first-order and J -odd:

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_i M_i.$$

Hence

$$D_1(0) = 0,$$

so the dangerous scalar Bragg block is absent from the relevant principal operator rather than cancelled inside it. The full nondegeneracy problem reduces to the finite matrix

$$G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j,$$

with

$$\det A_1(p) = -p_i G_{ij} p_j.$$

In the actual local projector-defect transport class, the natural branch is S2,

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

and

$$G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Equivalently,

$$G_{ij} > 0 \iff g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

So the graded bypass route is not a separate mystery; it reduces exactly to the same three transport coefficients that already control the witness problem.

The strongest honest current conclusion is:

$$\text{Dirac emergence is rigorously proved at the reduced graded-spectral level,}$$

$$\text{the actual local CP}^2 \text{ projector-defect geometry belongs to this operator class,}$$

but

$$\text{the full microscopic TECT closure is now reduced to extracting } g_c, g_E, g_O \text{ from the actual transport kernel } V^\mu(\xi).$$

XXXVII. ACTUAL GRADED 2 + 2 COEFFICIENT EXTRACTION

A. Goal

In the graded bypass route, the remaining microscopic task is no longer to cancel the scalar Bragg block. The actual remaining task is to extract the transport coefficients of the projected first-order mixed symbol from the defect/projector transport kernel.

The full extraction chain is:

$$P_B(\xi) \implies V^\mu(\xi) \implies u_a^\pm(\xi) \implies (M^\mu)_{ab} \implies \Lambda_{\parallel}, \Lambda_E, \Lambda_O \implies g_c, g_E, g_O. \quad (312)$$

B. Actual local defect background and the four-mode infrared sector

Let

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1, \quad (313)$$

be a static defect background depending only on the transverse coordinate ξ . Projector fluctuations satisfy

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0,$$

hence

$$\boxed{\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2.} \quad (314)$$

For a reflection-symmetric defect background, the transverse defect operator admits even/odd low modes

$$f_+(-\xi) = f_+(\xi), \quad f_-(-\xi) = -f_-(\xi).$$

Choosing a local orthonormal frame $e_a(\xi)$ of Q_B , the low modes are

$$\boxed{u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2,} \quad (315)$$

and therefore

$$\boxed{\mathcal{K} = \text{span}\{u_a^\pm\}, \quad \dim_{\mathbb{C}} \mathcal{K} = 4.} \quad (316)$$

C. Projected mixed symbol

The actual projected overlap matrices are

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi).} \quad (317)$$

These are the coefficient matrices of the first-order graded symbol.

Expand each M^μ in the Pauli basis:

$$\boxed{M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i.} \quad (318)$$

Then the reduced off-diagonal symbol is

$$A_1(p) = p_\mu M^\mu, \quad (319)$$

and the graded nondegeneracy matrix is

$$\boxed{G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j.} \quad (320)$$

D. Actual local projector-defect transport class

The actual local projector-defect transport class is odd and Q_B -endomorphism valued. Its minimal S2-compatible form is

$$\boxed{V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3 + \dots} \quad (321)$$

where:

1. n^μ is the worldvolume singlet direction,
2. $E^\mu{}_a$ is the transverse worldvolume doublet,
3. $J^\mu{}_a = \epsilon_{ab} E^\mu{}_b$,

4. Σ_a is the canonical Q_B -fiber Pauli basis,
5. $\Lambda_{\parallel}^{(0)}$ is the possible constant-leading longitudinal branch,
6. λ_c is the primitive commutator-square longitudinal coefficient.

The honest notes explicitly distinguish these two longitudinal branches: in the broad actual-local class, a constant-leading longitudinal $\Lambda_{\parallel}^{(0)}$ is symmetry-allowed, whereas in the exact concrete Class II branch the longitudinal term starts only in the primitive $I_{\theta'^3}$ channel. :contentReference[oaicite:5]index=5

E. Projection integrals

Define the two universal defect overlap integrals

$$I_{\theta} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi), \quad (322)$$

$$I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (323)$$

Then projecting (321) onto the graded low modes gives

$$M^{\mu} = I_{\theta} \left[\Lambda_{\parallel}^{(0)} n^{\mu} \sigma_3 + \Lambda_E E^{\mu}{}_a \sigma_a + \Lambda_O J^{\mu}{}_a \sigma_a \right] + \frac{\lambda_c}{2} I_{\theta'^3} n^{\mu} \sigma_3 + \dots \quad (324)$$

where the omitted terms are higher-order or symmetry-subleading.

Thus the reduced S2 symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2), \quad (325)$$

with

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (326)$$

F. Exact concrete Class II branch

In the exact concrete Class II local formulas, the transverse sector starts in the I_{θ} channel while the longitudinal sector starts only in the primitive $I_{\theta'^3}$ channel. Equivalently,

$$\Lambda_{\parallel}^{(0)} = 0 \quad (\text{exact concrete Class II branch}), \quad (327)$$

so

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (328)$$

The notes also record the explicit local overlap values for the concrete representative:

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (329)$$

Hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (330)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (331)$$

G. Coefficient vectors and nondegeneracy matrix

From (324), the coefficient vectors are

$$C_0^i = 0, \quad (332)$$

$$(C_1^1, C_2^1, C_3^1) = (I_\theta \Lambda_E, I_\theta \Lambda_O, 0), \quad (333)$$

$$(C_1^2, C_2^2, C_3^2) = (-I_\theta \Lambda_O, I_\theta \Lambda_E, 0), \quad (334)$$

$$(C_1^\parallel, C_2^\parallel, C_3^\parallel) = (0, 0, \lambda_\parallel). \quad (335)$$

Therefore

$$G_{ij} = \begin{pmatrix} I_\theta^2(\Lambda_E^2 + \Lambda_O^2) & 0 & 0 \\ 0 & I_\theta^2(\Lambda_E^2 + \Lambda_O^2) & 0 \\ 0 & 0 & \lambda_\parallel^2 \end{pmatrix}. \quad (336)$$

Hence the graded bypass nondegeneracy condition is exactly

$$G_{ij} > 0 \iff \lambda_\parallel \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (337)$$

Equivalently, in the exact concrete Class II branch,

$$G_{ij} > 0 \iff \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (338)$$

H. Dictionary to (g_c, g_E, g_O)

The local Class II shell/defect dictionary identifies

$$g_c \leftrightarrow \lambda_c, \quad g_E \leftrightarrow \Lambda_E, \quad g_O \leftrightarrow \Lambda_O,$$

up to the already-isolated overall normalization fixed by the final shell/defect overlap completion. Thus

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0) \quad (339)$$

is exactly the same finite microscopic test as (338).

Moreover, inside the explicit seed space of the concrete Class II reduction,

$$g_c \propto c_\parallel, \quad g_E \propto c_I, \quad g_O \propto c_J,$$

so explicit witness seeds exist already at the reduced level. What remains open is whether the *true complete TECT action* realizes that nontrivial sector in the physically relevant defect background.

I. What is actually closed and what remains open

What is now closed:

1. the graded $2 + 2$ bypass route is the correct mainline;
2. the actual local CP^2 projector-defect geometry provides the right 4-mode graded sector;
3. the projected coefficient extraction formulas are explicit:

$$\lambda_\parallel, \alpha, \beta$$

are given by (326);

4. in the exact concrete Class II branch, the longitudinal/transverse split is explicit:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O;$$

5. the entire Dirac nondegeneracy problem is reduced to the finite test

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

What remains honestly open:

1. derive the actual full microscopic transport kernel $V^{\mu}(\xi)$ from complete TECT, rather than postulate the minimal local ansatz;
2. decide which longitudinal branch the complete microscopic theory realizes:

$$\Lambda_{\parallel}^{(0)} \neq 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with primitive } I_{\theta^3} \text{ only};$$

3. compute the actual microscopic values of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O;$$

4. verify directly that complete TECT indeed yields

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

J. Sharp honest conclusion

Therefore the graded bypass frontier is no longer conceptual. It is exactly the coefficient-extraction problem:

$$\boxed{\text{derive } V^{\mu}(\xi) \text{ from full TECT and extract } \lambda_c, \Lambda_E, \Lambda_O.} \quad (340)$$

Once this is done, the principal Dirac operator follows immediately.

XXXVIII. FINAL SUMMARY

In the graded $2+2$ bypass route, the remaining microscopic extraction chain is

$$\boxed{P_B(\xi) \rightarrow V^{\mu}(\xi) \rightarrow u_a^{\pm}(\xi) \rightarrow (M^{\mu})_{ab} \rightarrow \Lambda_{\parallel}, \Lambda_E, \Lambda_O \rightarrow g_c, g_E, g_O.}$$

The actual local defect/projector geometry gives a natural four-mode infrared sector:

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp}, \quad u_a^{\pm}(\xi) = f_{\pm}(\xi)e_a(\xi), \quad \dim_{\mathbb{C}} \mathcal{K} = 4.$$

The projected mixed symbol is

$$\boxed{(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle.}$$

For the actual local S2-compatible transport class,

$$V^{\mu}(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^{\mu} \Sigma_3 + \Lambda_E E^{\mu}{}_a \Sigma_a + \Lambda_O J^{\mu}{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \Sigma_3 + \cdots,$$

the projected coefficients are

$$\boxed{\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.}$$

In the exact concrete Class II branch,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

Using the explicit overlap coefficients,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

one gets

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.$$

Hence the full graded nondegeneracy problem reduces to the finite test

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Equivalently,

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

So the strongest honest current conclusion is:

the graded bypass route is fully reduced to extracting $\lambda_c, \Lambda_E, \Lambda_O$ from the actual full TECT transport kernel $V^{\mu}(\xi)$.

XXXIX. ACTUAL TRANSPORT-KERNEL DERIVATION PROTOCOL FROM THE FULL MICROSCOPIC HESSIAN

A. Goal

The graded $2 + 2$ bypass route has reduced the full microscopic frontier to a single explicit calculation: derive the actual odd first-order transport kernel

$$L_1 = -iV^{\mu}(\xi)\partial_{y^{\mu}}$$

from the complete TECT Hessian around the shell-/defect-adapted background, and then project it to obtain the coefficients

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

This section fixes the operator protocol for that extraction.

B. Background split

The correct minimal microscopic background is shell-/defect-adapted rather than constant:

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

or, if the gauge sector is kept explicitly,

$$\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)).$$

This is exactly the background choice already isolated in the notes for obtaining a genuinely nontrivial background-dependent Hessian and nonzero cubic vertices. :contentReference[oaicite:3]index=3

Write the fluctuation split as

$$\Phi = \bar{\Phi}_{\min} + \varphi. \quad (341)$$

Then the microscopic quadratic operator is

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \left. \frac{\delta^2 \Gamma}{\delta \varphi \delta \varphi} \right|_{\bar{\Phi}_{\min}}. \quad (342)$$

C. Projector parametrization of the defect fluctuations

For the local projector-defect sector, the background projector is

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1, \quad (343)$$

and fluctuations are parameterized by

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0. \quad (344)$$

Thus

$$\boxed{\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2.} \quad (345)$$

This is the actual rank-two bundle structure of the local reduced problem.

D. Step 1: identify the static transverse operator

After substituting (344) into the full Hessian and separating transverse from worldvolume derivatives

$$y^\mu = (t, x_1, x_2), \quad \xi = \text{transverse coordinate},$$

the quadratic operator takes the schematic form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi) \partial_{y^\mu} + W^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots, \quad (346)$$

where the static transverse operator is

$$\boxed{H_B = -\partial_\xi^2 + U_B(\xi)} \quad (347)$$

acting on the rank-two bundle Q_B . This is exactly the operator whose low even/odd modes define the graded $2+2$ infrared sector.

E. Step 2: isolate the odd first-order transport kernel

The actual first-order transport piece is the coefficient of $-i\partial_{y^\mu}$ in the background-dependent Hessian:

$$\boxed{L_1 = -iV^\mu(\xi)\partial_{y^\mu}.} \quad (348)$$

The structural notes already fix its microscopic origin:

$$\boxed{V^\mu(\xi) \sim (\partial_\xi P_B)T^\mu,} \quad (349)$$

for some microscopic bundle endomorphism-valued tensors T^μ . Because

$$P_B(-\xi) = P_B(\xi), \quad \partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi),$$

one gets

$$\boxed{V^\mu(-\xi) = -V^\mu(\xi).} \quad (350)$$

Thus the microscopic TECT linearization naturally supplies the odd transport kernel required by the graded reduced theorem. :contentReference[oaicite:6]index=6

a. Operational extraction rule. Given the explicit Hessian $\Gamma^{(2)}[\bar{\Phi}_{\min}]$, one extracts $V^\mu(\xi)$ by:

1. expanding the Hessian to first order in worldvolume derivatives;
2. collecting all terms linear in ∂_{y^μ} ;
3. moving all ξ -dependent coefficient matrices to the left of ∂_{y^μ} ;
4. projecting the result onto the tangent bundle Q_B .

The resulting bundle endomorphism coefficient is precisely $V^\mu(\xi)$.

F. Step 3: construct the isolated graded 2 + 2 sector

Solve the static transverse problem

$$H_B u = \varepsilon u$$

on $Q_B(\xi)$. For a reflection-symmetric defect background, H_B decomposes into even and odd transverse parity channels, so the natural low modes are

$$\boxed{u_a^+(\xi), \quad u_a^-(\xi), \quad a = 1, 2,} \quad (351)$$

with

$$u_a^+(-\xi) = +u_a^+(\xi), \quad u_a^-(-\xi) = -u_a^-(\xi).$$

This yields the isolated graded sector

$$\boxed{\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim_{\mathbb{C}} \mathcal{K}_{\pm} = 2.} \quad (352)$$

The notes already identify this as the actual next calculation gap rather than a conceptual gap. :contentReference[oaicite:7]index=7

G. Step 4: project the transport kernel

The mixed first-order overlap matrices are

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle = \int d\xi (u_a^+)^\dagger V^\mu(\xi) u_b^-. \quad (353)$$

Because

$$u_a^+ \text{ is even, } \quad u_b^- \text{ is odd, } \quad V^\mu(\xi) \text{ is odd,}$$

the integrand is even, so the overlap is generically nonzero. This is the direct microscopic origin of the mixed first-order symbol. :contentReference[oaicite:8]index=8

Expand

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i. \quad (354)$$

Then the reduced mixed symbol is

$$A_1(p) = p_\mu M^\mu, \quad (355)$$

and its load-bearing nondegeneracy data are the traceless coefficient vectors

$$b^\mu = (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3.$$

The already-established criterion is

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3 \iff A_1(p) \text{ is nondegenerate for generic } p \neq 0. \quad (356)$$

Equivalently,

$$G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j, \quad G_{ij} > 0. \quad (357)$$

The identity part $C_0^\mu \mathbf{1}_2$ is not load-bearing once the traceless symbol is nondegenerate.

H. Step 5: convert to the actual S2 coefficients

The local CP^2 projector-defect geometry naturally supports the S2 branch:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (358)$$

The actual local overlap formulas are already known:

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (359)$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (360)$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (361)$$

These are exactly the coefficient-extraction formulas we need.

a. Meaning. So the microscopic task is not to guess $\lambda_{\parallel}, \alpha, \beta$. It is to derive the coefficient functions

$$h_{\parallel}(\theta), \quad h_{11}(\theta), \quad h_{12}(\theta), \quad h_{21}(\theta), \quad h_{22}(\theta)$$

from the full background-dependent Hessian.

I. Step 6: exact concrete Class II/local branch

Inside the explicit local Class II-compatible branch, one may write

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)).$$

The exact local reduction then yields

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O, \quad (362)$$

with

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

Thus

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3,} \quad (363)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.} \quad (364)$$

Hence the full S2 nondegeneracy condition becomes

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (365)$$

This is already isolated as the exact remaining coefficient problem in the explicit local operator class.

J. The actual Hessian-extraction protocol

We can now summarize the operator protocol that must be executed on the complete TECT PDE.

a. Protocol O1. Background insertion. Insert the shell-/defect-adapted background

$$\bar{\Phi}_{\min} = (\bar{A}_B, \bar{n}_B, \bar{\chi}_B)$$

into the full microscopic action or equations of motion.

b. Protocol O2. Tangent parametrization. Rewrite projector fluctuations as

$$\delta P = \eta z_B^{\dagger} + z_B \eta^{\dagger}, \quad \eta \in Q_B.$$

c. Protocol O3. Hessian construction. Compute the full background-dependent Hessian

$$\Gamma^{(2)}[\bar{\Phi}_{\min}]$$

as an operator on $\eta(\xi, y)$.

d. Protocol O4. First-order derivative split. Separate the operator into

$$H_B(\xi) - iV^{\mu}(\xi) \partial_{y^{\mu}} + W^{\mu\nu}(\xi) \partial_{y^{\mu}} \partial_{y^{\nu}} + \cdots$$

and extract the odd first-order coefficient $V^{\mu}(\xi)$.

e. Protocol O5. Low-mode problem. Solve the static transverse bundle problem

$$H_B u = \varepsilon u$$

to obtain the graded low modes $u_a^\pm$.

f. Protocol O6. Projection. Compute

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

g. Protocol O7. Pauli decomposition. Write

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i.$$

h. Protocol O8. S2 reduction. Match the extracted matrices to the S2 form and read off

$$\lambda_\parallel, \quad \alpha, \quad \beta$$

or equivalently

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

i. Protocol O9. Final finite test. Check

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \tag{366}$$

K. What is actually open

At this point the strongest honest statement is exact and narrow.
What is already closed:

1. the graded $2 + 2$ reduced Dirac theorem;
2. the structural matching of the actual local CP^2 projector-defect Hessian to this operator class;
3. the explicit overlap formulas for the local S2 coefficients;
4. the finite nondegeneracy criterion in terms of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

What remains genuinely open:

1. the final full microscopic TECT condensate/projector PDE in closed background-dependent Hessian form suitable for direct spectral analysis;
2. the explicit derivation of the coefficient functions

$$h_\parallel, \quad h_{11}, \quad h_{12}, \quad h_{21}, \quad h_{22}$$

or equivalently the actual transport kernel $V^\mu(\xi)$;

3. the direct microscopic verification that

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0)$$

in complete TECT.

These are exactly the open items isolated in the notes.

L. Sharp honest conclusion

So the full-PDE frontier has now been compressed as far as it can be compressed without inventing new microscopic dynamics:

derive the actual background-dependent Hessian of complete TECT, extract $V^\mu(\xi)$, and evaluate the three coefficients $\lambda_c, \Lambda_E,$
(367)

Once that is done, the graded bypass route either closes immediately or fails explicitly.

XL. FINAL SUMMARY

The graded 2 + 2 bypass route is now reduced to an explicit operator-extraction protocol.
The actual local defect/projector geometry supplies the rank-two bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp,$$

and the isolated graded infrared sector

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-, \quad \dim_{\mathbb{C}} \mathcal{K}_\pm = 2,$$

with low modes $u_a^\pm(\xi)$.

The first-order transport kernel is extracted from the background-dependent Hessian as

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) \sim (\partial_\xi P_B)T^\mu, \quad V^\mu(-\xi) = -V^\mu(\xi).$$

Projecting onto the graded low modes gives

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle, \quad M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i.$$

For the actual local S2 branch, the reduced symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

with overlap formulas

$$\lambda_\parallel = \int d\xi f_+ f_- \theta'(\xi) h_\parallel(\theta(\xi)),$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))),$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))).$$

In the exact concrete local Class II-compatible branch,

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,$$

so the finite graded nondegeneracy test is simply

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Therefore the strongest honest current status is:

the graded bypass route is fully reduced to deriving $V^\mu(\xi)$ from the complete TECT Hessian and extracting $\lambda_c, \Lambda_E, \Lambda_O$.

This is no longer a conceptual gap. It is an explicit background-Hessian calculation gap.

XLI. BACKGROUND-DEPENDENT FULL HESSIAN TEMPLATE FOR THE GRADED 2 + 2 ROUTE

A. Goal

The graded bypass route has already reduced the full microscopic frontier to a single operator-extraction problem: derive the actual odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}$$

from the complete TECT Hessian around the shell-/defect-adapted background.

The purpose of this section is to fix the full Hessian template and the source-by-source protocol for reading off $V^\mu(\xi)$.

B. Minimal shell-/defect-adapted background

The correct minimal background is not a uniform constant configuration, but the shell-/defect-adapted one

$$\boxed{\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),} \quad (368)$$

or, if the gauge sector is kept explicitly,

$$\boxed{\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)).} \quad (369)$$

This is the minimal admissible choice because it:

1. preserves the shell/defect geometry,
2. keeps the reflection selection structure,
3. and produces nontrivial background-dependent vertices

$$\Gamma^{(2)}[\bar{\Phi}_{\min}], \quad \Gamma^{(3)}[\bar{\Phi}_{\min}], \quad \Gamma^{(4)}[\bar{\Phi}_{\min}].$$

C. Background split and Hessian

Expand

$$\Phi = \bar{\Phi}_{\min} + \varphi, \quad \varphi = (a, \nu, \zeta), \quad (370)$$

so that

$$A = a, \quad n = \bar{n}_B + \nu, \quad \chi = \bar{\chi}_B + \zeta.$$

The full quadratic fluctuation operator is

$$\boxed{\Gamma^{(2)}[\bar{\Phi}_{\min}] = \left. \frac{\delta^2 \Gamma}{\delta \varphi \delta \varphi} \right|_{\bar{\Phi}_{\min}}.} \quad (371)$$

D. Projector-defect parametrization

For the local defect/projector sector, write

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1, \quad (372)$$

and parametrize the physical tangent fluctuations by

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0. \quad (373)$$

Hence

$$\boxed{\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2.} \quad (374)$$

This converts the defect fluctuation problem into a rank-two bundle problem on $Q_B(\xi)$.

E. Coordinate split

Let

$$x = (y^\mu, \xi), \quad y^\mu = (t, x_1, x_2)$$

be worldvolume coordinates y^μ plus one transverse coordinate ξ . The full Hessian can then be expanded in worldvolume derivatives:

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots \quad (375)$$

This is the full Hessian template that the complete TECT operator must be reduced to.

F. Interpretation of the blocks

The three relevant pieces in (375) are:

a. (i) *Static transverse channel operator*

$$H_B(\xi) = -\partial_\xi^2 + \mathcal{U}_B(\xi) + \mathcal{W}_B(\xi) + \dots, \quad (376)$$

where $\mathcal{U}_B, \mathcal{W}_B$ are Q_B -endomorphism-valued background potentials induced by the complete TECT Hessian.

This operator determines the even/odd low modes

$$u_a^\pm(\xi) = f_\pm(\xi)e_a(\xi), \quad a = 1, 2.$$

b. (ii) *Odd first-order transport kernel*

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}. \quad (377)$$

This is the load-bearing piece for the graded $2 + 2$ route.

c. (iii) *Higher-order worldvolume kinetic block*

$$L_2 = W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu}. \quad (378)$$

This influences dispersion corrections and spectral normalization, but not the principal first-order graded symbol.

G. Source-by-source decomposition of the Hessian

The complete microscopic Hessian should be organized as

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \Gamma_{\text{grad}}^{(2)} + \Gamma_{\text{pot}}^{(2)} + \Gamma_{\text{ani}}^{(2)} + \Gamma_{\text{gf/proj}}^{(2)} + \Gamma_{\text{med}}^{(2)} + \dots \quad (379)$$

The source-to-block mapping is then:

a. *Gradient/background kinetic sector* Contributes to all of

$$H_B, \quad V^\mu, \quad W^{\mu\nu},$$

through background-covariant derivatives and mixed (ξ, y) -derivative terms.

b. *Potential / static background sector* Contributes only to

$$H_B,$$

through ξ -dependent channel mass/potential matrices.

c. *Derivative anisotropy / transport sector* This is the primary microscopic source for

$$V^\mu(\xi),$$

because it couples background gradients to worldvolume derivatives.

d. Gauge-fixing / projector-constraint sector Contributes to

$$H_B, \quad V^\mu, \quad W^{\mu\nu}$$

through bundle connection terms, frame-covariant derivative corrections, and projection constraints.

e. Mediator sector If the complete microscopic TECT realization contains the effective Class II carrier sector in the defect background, then its reduction can contribute to $V^\mu(\xi)$ through induced traceless transport tensors, but not to the dangerous scalar identity-channel of the plain shell-pair route.

H. Microscopic origin of the odd transport kernel

The actual structural theorem already fixes the leading microscopic form:

$$\boxed{V^\mu(\xi) \sim (\partial_\xi P_B(\xi)) T^\mu \quad \text{or equivalently} \quad (\partial_\xi z_B(\xi)) \tilde{T}^\mu \sim \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).} \quad (380)$$

Since $P_B(\xi)$ is reflection-even,

$$P_B(-\xi) = P_B(\xi),$$

one has

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi), \quad \theta'(-\xi) = -\theta'(\xi),$$

and therefore

$$\boxed{V^\mu(-\xi) = -V^\mu(\xi).} \quad (381)$$

This oddness is the entire reason why same-parity overlaps vanish while mixed even/odd overlaps survive.

I. Low-mode problem

Once $H_B(\xi)$ is extracted, one solves

$$H_B u = \varepsilon u \quad (382)$$

on the rank-two bundle $Q_B(\xi)$. For a reflection-symmetric defect background, the transverse spectrum decomposes into even and odd parity channels, giving the graded low modes

$$\boxed{u_a^+(\xi), \quad u_a^-(\xi), \quad a = 1, 2,} \quad (383)$$

with

$$u_a^+(-\xi) = +u_a^+(\xi), \quad u_a^-(-\xi) = -u_a^-(\xi).$$

J. Projection formula for the mixed symbol

The projected mixed symbol is

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle = \int d\xi (u_a^+)^\dagger(\xi) V^\mu(\xi) u_b^-(\xi).} \quad (384)$$

Because u_a^+ is even, u_b^- is odd, and V^μ is odd, the integrand is even, so the mixed overlap is generically nonzero.

By contrast,

$$\langle u_a^\pm, V^\mu u_b^\pm \rangle = 0$$

by parity, so the leading reduced operator is automatically grading-odd:

$$\boxed{D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu.} \quad (385)$$

In particular,

$$\boxed{D_1(0) = 0.} \quad (386)$$

K. Conversion to S2 coefficients

Expand

$$M^\mu = C_0{}^\mu \mathbf{1}_2 + C_i{}^\mu \sigma_i. \quad (387)$$

The natural local branch is the S2 form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (388)$$

Equivalently,

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are read off from the corresponding components of M^μ .

The actual overlap formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (389)$$

$$\alpha = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (390)$$

$$\beta = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (391)$$

So the real microscopic job is to derive the coefficient functions

$$h_{\parallel}, \quad h_{11}, \quad h_{12}, \quad h_{21}, \quad h_{22}$$

from the complete Hessian.

L. Equivalent transport-coefficient form

In the explicit local transport language one may write

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3 + \dots \quad (392)$$

Projecting gives

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O, \quad (393)$$

with

$$I_{\theta} = \int d\xi f_+ f_- \theta'(\xi), \quad I_{\theta'^3} = \int d\xi f_+ f_- \theta'(\xi)^3.$$

In the exact concrete local Class II-compatible branch one has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (394)$$

M. Final finite test

Therefore the full graded nondegeneracy problem is reduced to the finite microscopic extraction of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O,$$

or equivalently

$$\lambda_{\parallel}, \quad \alpha, \quad \beta.$$

The final test is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (395)$$

N. Protocol summary

The actual full-Hessian extraction protocol is:

1. Insert the shell-/defect-adapted background $\bar{\Phi}_{\min}$ into the complete TECT action.
2. Expand to second order and compute the background-dependent Hessian $\Gamma^{(2)}[\bar{\Phi}_{\min}]$.
3. Rewrite projector fluctuations in tangent-bundle form $\delta P = \eta z_B^\dagger + z_B \eta^\dagger, z_B^\dagger \eta = 0$.
4. Split the Hessian into the form

$$H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots$$

5. Solve the static bundle eigenproblem for H_B and identify the even/odd low modes $u_a^\pm$.
6. Compute the projected overlaps

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

7. Decompose M^μ in the Pauli basis and match to the S2 form.
8. Read off $\lambda_{\parallel}, \alpha, \beta$ or equivalently $\lambda_c, \Lambda_E, \Lambda_O$.
9. Check the finite nondegeneracy criterion (395).

XLII. FINAL SUMMARY

The full microscopic graded-bypass frontier is now reduced to a single Hessian-extraction protocol. The correct shell-/defect-adapted background is

$$\boxed{\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),}$$

or

$$\boxed{\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)).}$$

Projector fluctuations are written as

$$\boxed{\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0, \quad \eta(\xi, y) \in Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp.}$$

The complete background-dependent Hessian must be reduced to

$$\boxed{\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots}$$

The load-bearing piece is the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) \sim (\partial_\xi P_B)T^\mu, \quad V^\mu(-\xi) = -V^\mu(\xi).$$

Solving the static transverse bundle problem for H_B gives the graded low modes $u_a^\pm$, and the mixed symbol is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

Same-parity overlaps vanish, so the leading reduced operator is automatically grading-odd:

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu, \quad D_1(0) = 0.$$

Matching to the natural S2 branch gives

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Equivalently, in the local transport-coefficient language,

$$\lambda_\parallel = I_\theta \Lambda_\parallel^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

In the exact concrete local Class II-compatible branch,

$$\Lambda_\parallel^{(0)} = 0, \quad \lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

Therefore the full graded nondegeneracy problem is reduced to the finite microscopic test

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

So the strongest honest current statement is:

the only remaining full-PDE gap is to derive $V^\mu(\xi)$ from the complete background-dependent Hessian and evaluate $\lambda_c, \Lambda_E, \Lambda_O$.

XLIII. SOURCE-BY-SOURCE HESSIAN BLOCK ENUMERATION FOR THE FULL TECT GRADED ROUTE

A. Goal

The graded $2 + 2$ bypass route is already reduced to the extraction of the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

What remains is not a conceptual theorem, but a source-by-source decomposition of the complete background-dependent Hessian:

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] \longrightarrow H_B(\xi), \quad V^\mu(\xi), \quad W^{\mu\nu}(\xi).$$

The purpose of this section is to fix that decomposition in a way that can be used directly for the next explicit differentiation step.

B. Universal Hessian template

For the shell-/defect-adapted background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi))$$

or

$$\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

the full quadratic operator on tangent fluctuations $\eta(\xi, y) \in Q_B(\xi)$ is organized as

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots. \quad (396)$$

This is the actual operator template already structurally matched to the local projector Hessian. The remaining gap is only the explicit calculation of the coefficient blocks. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

C. Projector-defect geometry block

The fluctuation bundle is

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp, \quad \delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0. \quad (397)$$

Thus every Hessian source must first be projected to the tangent bundle Q_B . This bundle reduction is not optional; it is the actual microscopic low-energy field space. :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

D. Source decomposition of the full Hessian

We write the complete microscopic Hessian schematically as

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \Gamma_{\text{grad}}^{(2)} + \Gamma_{\text{pot}}^{(2)} + \Gamma_{\text{ani}}^{(2)} + \Gamma_{\text{gf/proj}}^{(2)} + \Gamma_{\text{med}}^{(2)} + \dots, \quad (398)$$

where each source class has a distinct block signature in (396).

E. Block A: gradient/background kinetic sector

This is the sector originating from the background-covariant derivative structure of the complete action. After inserting the ξ -dependent defect background, the gradient terms split into:

1. pure ξ -derivative pieces:

$$\partial_\xi^\dagger \partial_\xi \Rightarrow H_B(\xi);$$

2. mixed (ξ, y^μ) -derivative pieces:

$$(\partial_\xi \bar{\Phi}_B) \partial_{y^\mu} \Rightarrow V^\mu(\xi);$$

3. pure worldvolume second-derivative pieces:

$$\partial_{y^\mu} \partial_{y^\nu} \Rightarrow W^{\mu\nu}(\xi).$$

Hence:

$$\Gamma_{\text{grad}}^{(2)} \rightsquigarrow H_B \oplus V^\mu \oplus W^{\mu\nu}. \quad (399)$$

This is the unavoidable source of the odd transport kernel, because the leading local candidates are linear in

$$\partial_\xi P_B(\xi), \quad \partial_\xi z_B(\xi), \quad \theta'(\xi),$$

all of which are reflection-odd. :contentReference[oaicite:12]index=12

F. Block B: static potential / defect pinning sector

Any source with no worldvolume derivatives contributes only to the transverse static channel operator. Examples include:

1. projector pinning / anisotropy potentials $U(P)$,
2. defect-background-dependent channel masses,
3. local static scalar self-interactions evaluated on $\bar{\Phi}_{\min}$.

Therefore:

$$\boxed{\Gamma_{\text{pot}}^{(2)} \rightsquigarrow H_B(\xi) \quad \text{only.}} \quad (400)$$

This is exactly why the open problem list includes explicit derivation of the background-dependent potential matrix $\mathcal{U}_B$: it belongs to H_B , not to V^μ . :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

G. Block C: anisotropic / transport sector

This is the primary source for the load-bearing first-order transport kernel. Its generic structure is

$$(\text{background tensor}) \cdot \partial_{y^\mu}$$

projected to Q_B .

At the local effective level this is the sector encoded by

$$V^\mu(\xi) \sim \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

or more explicitly by the S2-compatible transport ansatz

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3 + \cdots.$$

Hence:

$$\boxed{\Gamma_{\text{ani}}^{(2)} \rightsquigarrow V^\mu(\xi) \quad (\text{primary load-bearing source}).} \quad (401)$$

This is the source class whose projection determines

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O,$$

and therefore the actual graded nondegeneracy test. :contentReference[oaicite:15]index=15

H. Block D: gauge-fixing / projector-constraint / bundle-connection sector

Because fluctuations live in the moving rank-two bundle $Q_B(\xi)$, the Hessian must be expressed in a local frame $e_a(\xi)$. Differentiating this moving frame generates connection-like terms. Thus the projector-constraint sector contributes:

1. bundle-covariant ξ -channel terms $\rightarrow H_B$,
2. mixed bundle-connection transport terms $\rightarrow V^\mu$,
3. possible worldvolume connection-squared pieces $\rightarrow W^{\mu\nu}$.

Therefore:

$$\boxed{\Gamma_{\text{gf/proj}}^{(2)} \rightsquigarrow H_B \oplus V^\mu \oplus W^{\mu\nu}.} \quad (402)$$

This sector is structurally unavoidable once the tangent bundle Q_B and the moving frame $e_a(\xi)$ are used. :contentReference[oaicite:16]index=16

I. Block E: mediator sector

If the full microscopic realization contains the effective Class II carrier sector in the defect background, then its Hessian contribution is not primarily a static scalar mass source in the graded route. Rather, the canonical minimal Class II representative renormalizes the derivative shell tensor coefficients while leaving the constant shell-level mass sector absent. :contentReference[oaicite:17]index=17 :contentReference[oaicite:18]index=18

Therefore, in the graded bypass route:

$$\boxed{\Gamma_{\text{med}}^{(2)} \rightsquigarrow V^\mu(\xi) \quad (\text{traceless transport sector}),} \quad (403)$$

and not the dangerous scalar identity-channel Bragg block of the discarded plain shell-pair route.

J. Parity selection rule

For a reflection-symmetric defect background,

$$P_B(-\xi) = P_B(\xi), \quad \partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi),$$

hence the leading transport kernel is odd:

$$\boxed{V^\mu(-\xi) = -V^\mu(\xi).} \quad (404)$$

Let the low transverse modes be

$$u_a^+(-\xi) = +u_a^+(\xi), \quad u_a^-(-\xi) = -u_a^-(\xi).$$

Then

$$\langle u_a^\pm, V^\mu u_b^\pm \rangle = 0, \quad \langle u_a^+, V^\mu u_b^- \rangle \neq 0$$

generically.

Thus the leading reduced symbol is automatically grading-odd:

$$\boxed{D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu, \quad D_1(0) = 0.} \quad (405)$$

This is the concrete reason the graded bypass route avoids the scalar Bragg obstruction. :contentReference[oaicite:19]index=19 :contentReference[oaicite:20]index=20

K. Projection block and final coefficient extraction

Once $H_B(\xi)$ is constructed, solve

$$H_B u = \varepsilon u$$

to obtain the low even/odd modes $u_a^\pm$. Then compute

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.} \quad (406)$$

Expanding

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i,$$

the natural S2 reduction is

$$\boxed{A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).} \quad (407)$$

Thus the only load-bearing microscopic outputs are

$$\lambda_{\parallel}, \quad \alpha, \quad \beta,$$

or equivalently

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

In the exact concrete local Class II-compatible branch,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

So the full graded nondegeneracy test is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (408)$$

L. Source map summary

The full source map is therefore:

$$\boxed{\Gamma_{\text{grad}}^{(2)} \rightarrow H_B \oplus V^{\mu} \oplus W^{\mu\nu},} \quad (409)$$

$$\boxed{\Gamma_{\text{pot}}^{(2)} \rightarrow H_B,} \quad (410)$$

$$\boxed{\Gamma_{\text{ani}}^{(2)} \rightarrow V^{\mu},} \quad (411)$$

$$\boxed{\Gamma_{\text{gt/proj}}^{(2)} \rightarrow H_B \oplus V^{\mu} \oplus W^{\mu\nu},} \quad (412)$$

$$\boxed{\Gamma_{\text{med}}^{(2)} \rightarrow V^{\mu} \quad (\text{in the graded bypass route}).} \quad (413)$$

M. Sharp honest conclusion

This finishes the source-level organization.

What remains open is no longer “which mechanism might work.” It is the explicit differentiation of the complete TECT action to obtain the coefficient blocks in (396), especially the odd transport kernel:

$$\boxed{\text{compute } V^{\mu}(\xi) \text{ from the full background-dependent Hessian and evaluate } (M^{\mu})_{ab}.} \quad (414)$$

Once that is done, the graded bypass route either closes immediately or fails explicitly.

XLIV. FINAL SUMMARY

The complete TECT Hessian around the shell-/defect-adapted background must be organized as

$$\boxed{\Gamma^{(2)}[\bar{\Phi}_{\text{min}}] = H_B(\xi) - iV^{\mu}(\xi)\partial_{y^{\mu}} + W^{\mu\nu}(\xi)\partial_{y^{\mu}}\partial_{y^{\nu}} + \cdots .}$$

Here:

$$H_B(\xi)$$

is the static transverse channel operator on the rank-two bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp,$$

$$V^\mu(\xi)$$

is the odd first-order transport kernel built from the ξ -variation of the defect background, and

$$W^{\mu\nu}(\xi)$$

contains higher-order worldvolume kinetic corrections.

The source-by-source Hessian map is:

$$\Gamma_{\text{grad}}^{(2)} \rightarrow H_B \oplus V^\mu \oplus W^{\mu\nu},$$

$$\Gamma_{\text{pot}}^{(2)} \rightarrow H_B,$$

$$\Gamma_{\text{ani}}^{(2)} \rightarrow V^\mu,$$

$$\Gamma_{\text{gf/proj}}^{(2)} \rightarrow H_B \oplus V^\mu \oplus W^{\mu\nu},$$

$$\Gamma_{\text{med}}^{(2)} \rightarrow V^\mu \quad (\text{graded bypass route}).$$

For a reflection-symmetric defect background, the leading transport kernel is odd:

$$V^\mu(-\xi) = -V^\mu(\xi).$$

Hence same-parity overlaps vanish and mixed even/odd overlaps survive:

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

This automatically gives a grading-odd principal symbol

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad D_1(0) = 0.$$

Matching to the natural S2 branch yields the final finite microscopic test

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Therefore the graded bypass frontier is now completely localized:

$$\text{differentiate the complete TECT action, extract } V^\mu(\xi), \text{ project it to } (M^\mu)_{ab}, \text{ and evaluate } \lambda_c, \Lambda_E, \Lambda_O.$$

XIV. ACTUAL TERM-BY-TERM DIFFERENTIATION RULES FOR THE FULL TECT HESSIAN

A. Goal

The graded $2 + 2$ bypass route has already reduced the local Dirac problem to the extraction of the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}$$

from the background-dependent Hessian of complete TECT.

At this point, the correct next step is not another structural theorem. It is to fix the term-by-term differentiation rules: given a microscopic source term in the complete TECT action, which part of its second variation contributes to

$$H_B(\xi), \quad V^\mu(\xi), \quad W^{\mu\nu}(\xi)?$$

B. Background and tangent fluctuation

We work with the shell-/defect-adapted background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \quad \text{or} \quad \bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \quad (415)$$

and project the defect fluctuations to the physical tangent bundle

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0, \quad \eta(\xi, y) \in Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp. \quad (416)$$

This is the actual low-energy field space of the defect/projector problem. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

C. Normal-ordered Hessian decomposition

After inserting (415) and (416) into the full microscopic action and taking the second variation, every quadratic term can be written, after integration by parts in ξ , in the normal-ordered form

$$\delta^2 \Gamma = \int d\xi d^3y \eta^\dagger \left[\mathcal{A}(\xi) + \mathcal{B}(\xi)\partial_\xi + \mathcal{C}(\xi)\partial_\xi^2 + \mathcal{D}^\mu(\xi)\partial_{y^\mu} + \mathcal{E}^\mu(\xi)\partial_\xi\partial_{y^\mu} + \mathcal{F}^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \cdots \right] \eta. \quad (417)$$

The Hermitianized block decomposition is then

$$\boxed{\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \cdots} \quad (418)$$

D. Universal reading rule

The extraction rule is purely algebraic:

a. (R1) Static transverse block All terms with no worldvolume derivative belong to

$$\boxed{H_B(\xi)}. \quad (419)$$

b. (R2) First-order transport block All terms with exactly one worldvolume derivative belong to the operator

$$\boxed{L_1 = -iV^\mu(\xi)\partial_{y^\mu}}. \quad (420)$$

Only the Hermitianized coefficient of $-i\partial_{y^\mu}$ matters.

c. (R3) Higher kinetic block All terms with exactly two worldvolume derivatives belong to

$$\boxed{W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu}.} \quad (421)$$

Thus the only load-bearing microscopic term for the graded bypass route is the coefficient of exactly one worldvolume derivative.

E. Term-by-term rule for the kinetic / gradient sector

Take a generic two-derivative kinetic term

$$\Gamma_{\text{grad}} = \int d^4x \mathcal{K}_{AB}(\Phi) \partial_M \Phi^A \partial^M \Phi^B. \quad (422)$$

Expanding $\Phi = \bar{\Phi}_{\text{min}} + \varphi$, the quadratic fluctuation operator contains schematically:

1. $(\partial_\xi \varphi)^2$ and background-dependent potential pieces

$$\Rightarrow H_B;$$

2. $(\partial_{y^\mu} \varphi)^2$

$$\Rightarrow W^{\mu\nu};$$

3. mixed background-gradient pieces

$$(\partial_\xi \bar{\Phi}_{\text{min}}) \varphi \partial_{y^\mu} \varphi \Rightarrow V^\mu.$$

Hence:

$$\boxed{\delta^2 \Gamma_{\text{grad}} \rightsquigarrow H_B^{(\text{grad})} \oplus V_{\text{grad}}^\mu \oplus W_{\text{grad}}^{\mu\nu}.} \quad (423)$$

The actual local projector notes already indicate that the odd transport kernel is built from background gradients such as

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

which come precisely from this mixed background-gradient channel. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

F. Term-by-term rule for the static potential sector

Take a local potential term

$$\Gamma_{\text{pot}} = \int d^4x V(\Phi). \quad (424)$$

Its second variation contains no worldvolume derivatives at all. Therefore:

$$\boxed{\delta^2 \Gamma_{\text{pot}} \rightsquigarrow H_B^{(\text{pot})} \quad \text{only}.} \quad (425)$$

This is exactly why the open problem list separates the explicit derivation of the potential matrix $\mathcal{U}_B$ from the transport kernel $\mathcal{V}^\mu$: the former belongs to H_B , not to V^μ . :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

G. Term-by-term rule for the anisotropic / transport sector

Take a microscopic anisotropic transport term whose local reduced form is schematically

$$\Gamma_{\text{ani}} = \int d^4x \mathcal{T}^\mu(\Phi, \partial_\xi \Phi) \varphi^\dagger(-i\partial_{y^\mu})\varphi. \quad (426)$$

After background insertion,

$$\mathcal{T}^\mu(\bar{\Phi}_{\text{min}}, \partial_\xi \bar{\Phi}_{\text{min}})$$

is a ξ -dependent bundle endomorphism, so its second variation contributes directly to the odd first-order transport kernel:

$$\boxed{\delta^2\Gamma_{\text{ani}} \rightsquigarrow V_{\text{ani}}^\mu(\xi) \quad (\text{primary load-bearing source}).} \quad (427)$$

This is the microscopic source class encoded by the actual-local transport ansatz

$$V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)),$$

whose leading invariant coefficients $\Lambda_{\parallel}^{(0)}, \Lambda_E^{(0)}, \Lambda_O^{(0)}$ were already isolated as the true microscopic coefficient problem. :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

H. Term-by-term rule for the projector-constraint / moving-frame sector

Because the physical fluctuations live in the moving bundle $Q_B(\xi)$, one must express them in a local orthonormal frame $e_a(\xi)$:

$$\eta(\xi, y) = \eta^a(\xi, y)e_a(\xi).$$

Differentiating this frame produces bundle-connection terms:

$$\partial_\xi \eta = (\partial_\xi \eta^a)e_a + \eta^a \partial_\xi e_a.$$

Hence the projector-constraint sector contributes:

1. static bundle-connection terms $\rightarrow H_B$,
2. mixed connection/transport terms $\rightarrow V^\mu$,
3. worldvolume kinetic corrections $\rightarrow W^{\mu\nu}$.

So:

$$\boxed{\delta^2\Gamma_{\text{gf/proj}} \rightsquigarrow H_B^{(\text{proj})} \oplus V_{\text{proj}}^\mu \oplus W_{\text{proj}}^{\mu\nu}.} \quad (428)$$

This is structurally unavoidable once the tangent-bundle parametrization is used. :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

I. Term-by-term rule for the mediator sector

If the full microscopic realization contains the effective Class II carrier sector, then after integrating out the heavy mediator the induced quadratic operator has the schematic form

$$\delta^2\Gamma_{\text{med}} \sim -\frac{1}{M_X^2} \mathcal{C}[\bar{\Phi}_{\text{min}}] \mathcal{C}[\bar{\Phi}_{\text{min}}]^\dagger. \quad (429)$$

In the graded bypass route, this sector contributes to the traceless transport kernel rather than the discarded scalar Bragg channel:

$$\boxed{\delta^2 \Gamma_{\text{med}} \rightsquigarrow V_{\text{med}}^\mu(\xi),} \quad (430)$$

with the exact local compatible form

$$V_{\text{med}}^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3 + \dots \quad (431)$$

In the exact concrete local Class II-compatible branch one has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

This is precisely the coefficient dictionary already isolated in the notes. :contentReference[oaicite:14]index=14 :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

J. Oddness filter

For a reflection-symmetric background,

$$P_B(-\xi) = P_B(\xi), \quad \partial_{\xi} P_B(-\xi) = -\partial_{\xi} P_B(\xi).$$

Therefore any candidate term contributing to V^μ must be odd in ξ . This provides a strict filter:

$$\boxed{\text{Only terms linear in } \partial_{\xi} P_B, \partial_{\xi} z_B, \theta'(\xi) \text{ (or odd descendants thereof) can contribute to } V^\mu.} \quad (432)$$

As a consequence,

$$\langle u_a^{\pm}, V^\mu u_b^{\pm} \rangle = 0, \quad \langle u_a^+, V^\mu u_b^- \rangle \neq 0$$

generically, so the leading reduced operator is automatically grading-odd:

$$\boxed{D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad D_1(0) = 0.} \quad (433)$$

This is the concrete microscopic reason the graded route bypasses the scalar Bragg obstruction. :contentReference[oaicite:17]index=17 :contentReference[oaicite:18]index=18

K. What is actually to be differentiated next

After the source map above, the next explicit calculation is completely determined.

One must take each microscopic source term in the complete TECT action and compute only the coefficient of exactly one worldvolume derivative in the tangent-bundle Hessian:

$$\boxed{[\delta^2 \Gamma_{\text{grad}}]_{1y}, \quad [\delta^2 \Gamma_{\text{ani}}]_{1y}, \quad [\delta^2 \Gamma_{\text{gf/proj}}]_{1y}, \quad [\delta^2 \Gamma_{\text{med}}]_{1y}.} \quad (434)$$

Their sum is

$$\boxed{V^\mu(\xi) = V_{\text{grad}}^\mu(\xi) + V_{\text{ani}}^\mu(\xi) + V_{\text{proj}}^\mu(\xi) + V_{\text{med}}^\mu(\xi).} \quad (435)$$

The potential sector does not enter V^μ .

L. Final finite extraction stage

Once $V^\mu(\xi)$ is known, the remaining steps are already fixed:

1. solve $H_B u = \varepsilon u$ on $Q_B(\xi)$ to get $u_a^\pm$;
2. compute

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle;$$

3. decompose

$$M^\mu = C_0{}^\mu \mathbf{1}_2 + C_i{}^\mu \sigma_i;$$

4. match to the S2 form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2);$$

5. test

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

M. Sharp honest conclusion

So the project has now been compressed to an actual differentiation checklist.
The open problem is not:

“which route should we use?”

That is settled.

The open problem is:

differentiate the complete TECT action term-by-term, extract the odd first-order tangent-bundle kernel $V^\mu(\xi)$, and evaluate it	(436)
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This is now the only remaining microscopic gap in the graded bypass mainline.

XLVI. FINAL SUMMARY

The full TECT Hessian around the shell-/defect-adapted background is organized as

$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots$
--

The term-by-term differentiation rule is:

0 worldvolume derivatives $\implies H_B(\xi),$
--

1 worldvolume derivative $\implies V^\mu(\xi),$

2 worldvolume derivatives $\implies W^{\mu\nu}(\xi).$

Accordingly, the full Hessian source map is:

$\delta^2 \Gamma_{\text{grad}} \rightsquigarrow H_B^{(\text{grad})} \oplus V_{\text{grad}}^\mu \oplus W_{\text{grad}}^{\mu\nu}$

$$\delta^2 \Gamma_{\text{pot}} \rightsquigarrow H_B^{(\text{pot})} \quad \text{only,}$$

$$\delta^2 \Gamma_{\text{ani}} \rightsquigarrow V_{\text{ani}}^\mu \quad (\text{primary load-bearing source}),$$

$$\delta^2 \Gamma_{\text{gf/proj}} \rightsquigarrow H_B^{(\text{proj})} \oplus V_{\text{proj}}^\mu \oplus W_{\text{proj}}^{\mu\nu},$$

$$\delta^2 \Gamma_{\text{med}} \rightsquigarrow V_{\text{med}}^\mu \quad (\text{graded bypass route}).$$

The reflection-odd filter is

$$V^\mu(-\xi) = -V^\mu(\xi),$$

so only terms linear in

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi)$$

or odd descendants thereof can contribute to the load-bearing first-order transport kernel.

Hence the actual next calculation is fully fixed:

$$V^\mu(\xi) = V_{\text{grad}}^\mu(\xi) + V_{\text{ani}}^\mu(\xi) + V_{\text{proj}}^\mu(\xi) + V_{\text{med}}^\mu(\xi),$$

with the potential sector absent from V^μ .

Therefore the only remaining microscopic gap in the graded-bypass mainline is:

$$\text{differentiate the complete TECT action term-by-term, extract } V^\mu(\xi), \text{ and project it to obtain } \lambda_c, \Lambda_E, \Lambda_O.$$

XLVII. SECOND VARIATION OF THE GRADIENT/BACKGROUND KINETIC SECTOR

A. Goal

We now differentiate the gradient/background kinetic sector explicitly and determine whether the pure isotropic kinetic core alone can generate the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

The answer will be:

$$\text{the bare isotropic quadratic gradient core generates } H_B \text{ and } W^{\mu\nu}, \text{ but not a genuine } V^\mu \text{ at raw local Hessian level.}$$

B. Minimal isotropic projector gradient sector

Take the minimal local projector gradient action

$$\Gamma_{\text{grad}}[P] = \frac{\rho}{4} \int d\xi d^3y (\partial_M P \partial^M P), \quad M = (\xi, \mu), \quad (437)$$

with a static defect background

$$P_B = P_B(\xi), \quad P_B(-\xi) = P_B(\xi).$$

Physical tangent fluctuations are parameterized by

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0, \quad \eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp. \quad (438)$$

Choose a local orthonormal frame $e_a(\xi)$ of Q_B and write

$$\eta(\xi, y) = \eta^a(\xi, y) e_a(\xi).$$

Then the bundle connection is

$$\mathcal{A}_\xi = -i e^\dagger \partial_\xi e.$$

For the explicit defect frame already identified in the local projector analysis,

$$\boxed{\mathcal{A}_\xi = 0.} \quad (439)$$

C. Quadratic variation of the isotropic core

Expanding (437) to quadratic order around P_B , the tangent-bundle fluctuation operator takes the standard bundle-Schrödinger form

$$\delta^2 \Gamma_{\text{grad}} = \int d\xi d^3 y \, \eta^\dagger \left[-\rho D_\xi^2 + \mathcal{U}_B^{(\rho)}(\xi) - \rho \partial_{y^\mu} \partial_{y^\mu} \right] \eta, \quad (440)$$

where

$$D_\xi = \partial_\xi + i \mathcal{A}_\xi$$

and $\mathcal{U}_B^{(\rho)}(\xi)$ is the background-dependent Hermitian bundle potential induced by the ξ -dependent defect geometry.

Using (439), one gets in the chosen frame

$$\boxed{\delta^2 \Gamma_{\text{grad}} = \int d\xi d^3 y \, \eta^\dagger \left[-\rho \partial_\xi^2 + \mathcal{U}_B^{(\rho)}(\xi) - \rho \partial_{y^\mu} \partial_{y^\mu} \right] \eta.} \quad (441)$$

D. Block identification

Comparing (441) with the universal Hessian decomposition

$$\Gamma^{(2)}[\bar{\Phi}_{\text{min}}] = H_B(\xi) - i V^\mu(\xi) \partial_{y^\mu} + W^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots,$$

we read off:

$$\boxed{H_B^{(\text{grad})} = -\rho \partial_\xi^2 + \mathcal{U}_B^{(\rho)}(\xi),} \quad (442)$$

$$\boxed{V_{\text{grad}}^\mu(\xi) = 0 \quad \text{for the bare isotropic quadratic core in the explicit } \mathcal{A}_\xi = 0 \text{ frame,}} \quad (443)$$

$$\boxed{W_{\text{grad}}^{\mu\nu}(\xi) = \rho \delta^{\mu\nu} \mathbf{1}_2.} \quad (444)$$

Thus the pure isotropic two-derivative core supplies the static transverse operator and the second-order worldvolume kinetic block, but no genuine odd first-order transport kernel.

E. Why this does not contradict the local matching theorem

The local projector-defect matching theorem says that, for a reflection-symmetric defect background, the leading transport kernel is odd and is built from background gradients such as

$$\partial_\xi P_B(\xi), \quad \partial_\xi z_B(\xi), \quad \theta'(\xi),$$

so same-parity overlaps vanish and mixed even/odd overlaps survive. This is a structural operator-class statement, not yet the explicit derivation of the final microscopic coefficient functions. The notes explicitly mark the derivation of the concrete transport kernel $\mathcal{V}^\mu$ and its coefficient functions as still open. :contentReference[oaicite:4]index=4

So (443) means only:

the raw isotropic quadratic core by itself is not enough to produce the load-bearing first-order mixed symbol.

It does *not* say that the final defect/projector operator class lacks such a symbol.

F. Comparison with the shell-isotropic no-go

This result is fully consistent with the previously established shell-level no-go: for the naive free isotropic shell dispersion, and even in the locked one-mode slow-envelope expansion, the first-order transport coefficient vanishes identically:

$$\left[ZD_{G_*}^2 - Y(D_{G_*}^2)^2 \right]_{\text{1st order}} = 0.$$

Hence the actual coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

cannot arise from the pure free isotropic shell dispersion alone. They must come from more structured transport sectors. :contentReference[oaicite:5]index=5

Equation (443) is the local projector analogue of that same obstruction.

G. What kind of terms can generate a nonzero V^μ ?

To generate a genuine first-order transport kernel, one needs a term in the quadratic Hessian that is:

1. linear in one worldvolume derivative ∂_{y^μ} ,
2. odd under $\xi \mapsto -\xi$,
3. bundle-endomorphism valued on Q_B .

Thus the minimal candidate sources are terms linear in

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

multiplied by an internal/worldvolume tensor structure. This is precisely the structural content of the local transport ansatz

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Therefore the genuinely load-bearing source classes are:

$$\boxed{V_{\text{ani}}^\mu(\xi), \quad V_{\text{proj}}^\mu(\xi), \quad V_{\text{med}}^\mu(\xi),} \tag{445}$$

rather than the bare isotropic quadratic core.

H. Explicitly compatible structured sectors

The notes already isolate the concrete structured sources that can realize the needed coefficients:

1. the local projector-defect transport matrix

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

whose microscopic coefficient functions remain open in full TECT;

2. the primitive longitudinal commutator-square channel

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu,$$

which gives in the explicit representative

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3;$$

:contentReference[oaicite:7]index=7

3. the explicit transverse odd-gradient channels

$$V_{(\perp, E)}^\mu(\xi) = \theta'(\xi) E_a^\mu \sigma_a, \quad V_{(\perp, O)}^\mu(\xi) = \theta'(\xi) J_a^\mu \sigma_a;$$

:contentReference[oaicite:8]index=8

4. admissible mixed (i, a) -carrier sectors such as Class II, whose local seed ansatz yields

$$g_c \propto c_\parallel, \quad g_E \propto c_I, \quad g_O \propto c_J$$

generically inside the Class II truncation. :contentReference[oaicite:9]index=9

I. Practical conclusion for the Hessian program

The result of the present differentiation step is therefore:

$$\boxed{\delta^2 \Gamma_{\text{grad}} = H_B^{(\text{grad})} + W_{\text{grad}}^{\mu\nu} \partial_{y^\mu} \partial_{y^\nu}, \quad V_{\text{grad}}^\mu = 0 \text{ for the bare isotropic quadratic core.}} \quad (446)$$

Hence the actual next source to differentiate is not the isotropic core again, but the first structured transport-producing sector:

$$\boxed{\delta^2 \Gamma_{\text{ani}} \quad \text{and/or} \quad \delta^2 \Gamma_{\text{med}}.} \quad (447)$$

Only these can determine the genuine microscopic coefficient functions

$$h_\parallel, \quad h_{11}, \quad h_{12}, \quad h_{21}, \quad h_{22}$$

or equivalently

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

XLVIII. FINAL SUMMARY

For the minimal isotropic projector gradient sector

$$\Gamma_{\text{grad}}[P] = \frac{\rho}{4} \int d\xi d^3y (\partial_M P \partial^M P),$$

expanded around the shell-/defect-adapted background and projected to the tangent bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp,$$

the quadratic fluctuation operator in the explicit defect frame with

$$\mathcal{A}_\xi = 0$$

takes the form

$$\delta^2 \Gamma_{\text{grad}} = \int d\xi d^3y \eta^\dagger \left[-\rho \partial_\xi^2 + \mathcal{U}_B^{(\rho)}(\xi) - \rho \partial_{y^\mu} \partial_{y^\mu} \right] \eta.$$

Therefore its Hessian blocks are

$$H_B^{(\text{grad})} = -\rho \partial_\xi^2 + \mathcal{U}_B^{(\rho)}(\xi),$$

$$V_{\text{grad}}^\mu(\xi) = 0 \quad \text{for the bare isotropic quadratic core,}$$

$$W_{\text{grad}}^{\mu\nu}(\xi) = \rho \delta^{\mu\nu} \mathbf{1}_2.$$

This is fully consistent with the already-established shell-level result that the naive free isotropic shell dispersion and locked one-mode isotropic kinetic truncation produce no nontrivial first-order witness data. :contentReference[oaicite:10]index=10

It is also consistent with the honest local matching theorem: the theorem proves that the *operator class* naturally admits an odd transport kernel built from defect gradients, but the exact microscopic coefficient functions of that kernel remain open. :contentReference[oaicite:12]index=12

Hence the practical conclusion is:

$$\text{the bare isotropic quadratic gradient core is not the load-bearing source of } V^\mu(\xi).$$

The genuine microscopic transport sources must instead come from structured gradient-sensitive sectors:

$$V_{\text{ani}}^\mu(\xi), \quad V_{\text{proj}}^\mu(\xi), \quad V_{\text{med}}^\mu(\xi).$$

So the Hessian program has now advanced to a sharper form:

$$\text{differentiate the first structured transport-producing sector next, not the isotropic core again.}$$

XLIX. SECOND VARIATION OF THE ANISOTROPIC / TRANSPORT SECTOR

A. Goal

The previous step established that the bare isotropic quadratic gradient core gives

$$H_B^{(\text{grad})} \neq 0, \quad W_{\text{grad}}^{\mu\nu} \neq 0, \quad V_{\text{grad}}^\mu = 0$$

at raw local Hessian level.

Therefore the first genuinely nonzero microscopic source of the graded first-order transport kernel must come from a structured transport-producing sector. The purpose of the present step is to differentiate that sector explicitly and identify its contribution to

$$V^\mu(\xi).$$

B. Important distinction: static anisotropy potential versus transport anisotropy

A static anisotropy potential such as

$$V(P) = \frac{\lambda}{2} ([H, P]^2), \quad H = \text{diag}(1, 0, -1), \quad (448)$$

does *not* itself produce the odd first-order worldvolume transport block. Its second variation contributes only to the static bundle Schrödinger operator H_B , yielding the explicit channel-diagonal Hessian

$$\mathcal{H}_B^{\text{stat}} = -\rho \partial_\xi^2 + \begin{pmatrix} \lambda \cos^2 2\theta(\xi) - 2\rho \theta'(\xi)^2 & 0 \\ 0 & \lambda - \rho \theta'(\xi)^2 \end{pmatrix},$$

while the leading odd transport kernel is a separate background-gradient-built structure.

Hence the source we must differentiate now is not the static anisotropy potential (448), but the actual first-order transport sector.

C. Strongest honest local transport source

The strongest honest local transport form already isolated in the notes is

$$\boxed{V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).} \quad (449)$$

Equivalently, at the action level one writes the transport source as

$$\boxed{\Gamma_{\text{trans}} = -i \int d\xi d^3y \, \eta^\dagger(\xi, y) V^\mu(\xi) \partial_{y^\mu} \eta(\xi, y).} \quad (450)$$

This is already a quadratic fluctuation term on the tangent bundle $Q_B(\xi)$, so in the Hessian language it is itself a second-variation contribution.

Thus the second variation is immediate:

$$\boxed{\delta^2 \Gamma_{\text{trans}} = -i \int d\xi d^3y \, \eta^\dagger(\xi, y) V^\mu(\xi) \partial_{y^\mu} \eta(\xi, y).} \quad (451)$$

Comparing with the master Hessian decomposition

$$\Gamma^{(2)}[\bar{\Phi}_{\text{min}}] = H_B(\xi) - iV^\mu(\xi) \partial_{y^\mu} + W^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots,$$

we read off immediately:

$$\boxed{H_B^{(\text{ani/trans})} = 0, \quad V_{\text{ani/trans}}^\mu(\xi) = V^\mu(\xi), \quad W_{\text{ani/trans}}^{\mu\nu} = 0} \quad (452)$$

at principal order.

So this source is purely load-bearing for the graded first-order symbol.

D. Reflection parity

For the explicit defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

one has

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad \partial_\xi P_B(\xi) = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Therefore $\theta'(-\xi) = -\theta'(\xi)$, so the kernel (449) is automatically odd:

$$\boxed{V^\mu(-\xi) = -V^\mu(\xi).} \quad (453)$$

This is exactly the parity property required by the graded even/odd overlap theorem.

E. S2-compatible decomposition

The broad actual-local transport class admits the shell-compatible singlet $\oplus$ doublet decomposition

$$\mathcal{M}^\mu(\theta) = \Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a + O(\theta^2). \quad (454)$$

Hence

$$V_{\text{broad}}^\mu(\xi) = \theta'(\xi) \left[\Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a \right] + O(\theta' \theta^2). \quad (455)$$

Projecting this kernel to the even/odd low modes gives the leading witnesses

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad (456)$$

$$\alpha = \Lambda_E^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad \beta = \Lambda_O^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad (457)$$

with

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0.$$

So in the broad actual-local class, a constant-leading longitudinal branch is symmetry-allowed. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

F. Exact concrete local Class II-compatible branch

The exact concrete local Class II-compatible branch is sharper. The transverse sector already appears at order $\theta'(\xi)$, but the longitudinal sector is not of I_{θ} -type. Instead, it arises from the primitive commutator-square channel

$$V_{(\parallel)}^\mu(\xi) = \lambda_c \theta'(\xi) \Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q n^\mu. \quad (458)$$

Using

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

this becomes

$$V_{(\parallel),\text{tr}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (459)$$

Thus the longitudinal witness is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad (460)$$

with

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0.$$

The transverse channels are

$$V_{(\perp,E)}^\mu(\xi) = \theta'(\xi) \Lambda_E E^\mu{}_a \sigma_a, \quad V_{(\perp,O)}^\mu(\xi) = \theta'(\xi) \Lambda_O J^\mu{}_a \sigma_a, \quad (461)$$

giving

$$\alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (462)$$

So in the exact concrete local Class II-compatible branch:

$$V_{\text{ani/trans}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \dots. \quad (463)$$

This is the first genuinely nonzero microscopic source of the full S2 witness triplet. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10

G. Projection and reduced symbol

Let $u_a^\pm(\xi) = f_\pm(\xi)e_a(\xi)$ be the graded low modes of the static bundle operator. Then

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle. \quad (464)$$

Since V^μ is odd, same-parity overlaps vanish, while the mixed overlaps survive. Therefore the reduced first-order symbol is automatically grading-odd:

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (465)$$

Thus the anisotropic/transport sector is precisely the first source that can produce a genuinely nonzero V^μ and hence nontrivial $(\lambda_\parallel, \alpha, \beta)$.

H. Final finite test

The full graded nondegeneracy problem is therefore reduced to

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0), \quad (466)$$

or equivalently

$$\lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (467)$$

I. What is actually proved and what remains open

What is proved at this step:

1. the first genuinely nonzero microscopic source of the odd first-order transport kernel is the anisotropic / transport sector, not the bare isotropic gradient core;
2. its second variation contributes directly to $V^\mu(\xi)$ and not primarily to H_B or $W^{\mu\nu}$;
3. the broad actual-local operator class allows a constant-leading longitudinal branch;
4. the exact concrete local Class II-compatible branch realizes the transverse channels at I_θ -order and the longitudinal channel in the primitive I_{θ^3} -branch.

What remains honestly open:

1. the actual complete microscopic TECT transport kernel has not yet been derived from the final full action;
2. therefore the actual coefficient point

$$\Lambda_\parallel^{(0)}, \Lambda_E^{(0)}, \Lambda_O^{(0)}$$

or equivalently

$$\lambda_c, \Lambda_E, \Lambda_O$$

is not yet known;

3. consequently, the full microscopic verification of

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0)$$

remains the final coefficient-identification problem.

L. FINAL SUMMARY

The bare isotropic quadratic gradient core is not the load-bearing source of the odd first-order transport kernel. Its second variation gives

$$H_B^{(\text{grad})} \neq 0, \quad W_{\text{grad}}^{\mu\nu} \neq 0, \quad V_{\text{grad}}^\mu = 0$$

at raw local Hessian level. This is fully consistent with the previously established shell-level isotropic first-order no-go.

The first genuinely nonzero microscopic source of $V^\mu(\xi)$ is instead the anisotropic / transport sector, whose quadratic Hessian contribution is already of the form

$$\delta^2 \Gamma_{\text{ani/trans}} = -i \int \eta^\dagger V^\mu(\xi) \partial_{y^\mu} \eta.$$

So at principal order,

$$H_B^{(\text{ani/trans})} = 0, \quad V_{\text{ani/trans}}^\mu(\xi) = V^\mu(\xi), \quad W_{\text{ani/trans}}^{\mu\nu} = 0.$$

In the broad actual-local transport class,

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

with shell-compatible leading invariant coefficients

$$\Lambda_{\parallel}^{(0)}, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)},$$

so a constant-leading longitudinal branch is symmetry-allowed.

In the exact concrete local Class II-compatible branch, however, the transverse sector starts at I_θ -order while the longitudinal sector starts only in the primitive commutator-square channel:

$$V_{\text{ani/trans}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \dots$$

Hence

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

Therefore the anisotropic/transport sector is the first explicit microscopic source that can produce the full nontrivial S2 witness triplet, and the full graded nondegeneracy test remains

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

So the honest remaining microscopic gap is now completely localized to the coefficient-identification problem: derive the actual complete TECT transport kernel and determine whether it lands at a nondegenerate coefficient point.

LI. SECOND VARIATION OF THE CLASS II MEDIATOR SECTOR

A. Goal

The graded $2+2$ bypass route has isolated the genuinely load-bearing microscopic source classes for the first-order transport kernel. Among them, Class II is the strongest currently explicit microscopic class because it packages all three invariant coordinates at once:

$$g_c, \quad g_E, \quad g_O.$$

The purpose of this step is to compute the quadratic Class II reduction far enough to identify the actual mediator contribution to

$$-i V_{\text{med}}^\mu(\xi) \partial_{y^\mu}.$$

B. UV Class II action and heavy-field elimination

The minimal Class II mediated UV Lagrangian is

$$\mathcal{L}_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi]. \quad (468)$$

Here X_i^a carries exactly the needed (i, a) -carrier structure, and the minimal choices are

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \text{h.c.}, \quad K_i^a[P, \psi] = (T_a \partial_i P) \mathcal{O}[\psi], \quad (469)$$

or after shell/tangent reduction

$$K_i^a \sim \partial_i \delta P^a \psi.$$

All of this is already explicit in the notes. :contentReference[oaicite:5]index=5

Eliminating the heavy mediator gives

$$X_i^a = -\frac{\alpha}{M_X^2} J_i^a[\Psi] - \frac{\beta}{M_X^2} K_i^a[P, \psi], \quad (470)$$

hence the exact effective action contains

$$S_{\text{eff}}^{(II)} = S_0 - \frac{\alpha^2}{2M_X^2} \int J_i^a J_i^a - \frac{\alpha\beta}{M_X^2} \int J_i^a K_i^a - \frac{\beta^2}{2M_X^2} \int K_i^a K_i^a. \quad (471)$$

The load-bearing mixed term is

$$S_{\text{mix}}^{(II)} = -\frac{\alpha\beta}{M_X^2} \int J_i^a[\Psi] K_i^a[P, \psi]. \quad (472)$$

This is exactly the mixed source already identified as the relevant one.

C. Background expansion

Expand around the shell-/defect-adapted background

$$\bar{\Phi}_{\text{min}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

or, in the reduced locked notation,

$$\Psi = (\phi_0 + \sigma) z_0 + \psi, \quad P = P_B + \delta P, \quad \delta P \sim z_B \xi^\dagger + \xi z_B^\dagger.$$

Then decompose

$$J_i^a = \bar{J}_i^a(\xi) + J_i^{a(1)}[\eta] + J_i^{a(2)}[\eta] + \dots, \quad (473)$$

$$K_i^a = \bar{K}_i^a(\xi) + K_i^{a(1)}[\eta] + K_i^{a(2)}[\eta] + \dots, \quad (474)$$

where $\eta \in Q_B(\xi)$ is the tangent fluctuation field.

For the defect background, the background-gradient content enters through

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

so any first-order transport contribution extracted from $J^{(1)}$ or $K^{(1)}$ is automatically odd in ξ . This parity fact is already closed.

D. Normal form of the first variations

The only pieces relevant to the graded principal symbol are those linear in η and containing exactly one worldvolume derivative. So we write the first variations in normal form:

$$J_i^{a(1)}[\eta] = \mathcal{J}_i^{a(0)}(\xi) \eta + \mathcal{J}_i^{a\mu}(\xi) (-i\partial_{y^\mu})\eta + \cdots, \quad (475)$$

$$K_i^{a(1)}[\eta] = \mathcal{K}_i^{a(0)}(\xi) \eta + \mathcal{K}_i^{a\mu}(\xi) (-i\partial_{y^\mu})\eta + \cdots. \quad (476)$$

Here the superscript (0) means no worldvolume derivative, while the μ -part carries exactly one $-i\partial_{y^\mu}$.

E. Quadratic mediator Hessian

Substituting (473) and (474) into (471), the quadratic fluctuation operator from the mediator sector is schematically

$$\delta^2\Gamma_{\text{med}} = -\frac{\alpha^2}{2M_X^2} \delta^2(JJ) - \frac{\alpha\beta}{M_X^2} \delta^2(JK) - \frac{\beta^2}{2M_X^2} \delta^2(KK). \quad (477)$$

At the level of the graded principal symbol, the only load-bearing contributions are those with exactly one worldvolume derivative. These arise from the first variations (475)–(476).

Keeping only the $1y$ -terms, one obtains

$$\delta^2\Gamma_{\text{med}}\Big|_{1y} = -i \int d\xi d^3y \eta^\dagger \left[V_{JJ}^\mu(\xi) + V_{JK}^\mu(\xi) + V_{KK}^\mu(\xi) \right] \partial_{y^\mu} \eta, \quad (478)$$

with

$$V_{JJ}^\mu(\xi) = -\frac{\alpha^2}{M_X^2} [\bar{J}_i^a(\xi) \mathcal{J}_i^{a\mu}(\xi)]_{\text{tan}}, \quad (479)$$

$$V_{JK}^\mu(\xi) = -\frac{\alpha\beta}{M_X^2} [\bar{J}_i^a(\xi) \mathcal{K}_i^{a\mu}(\xi) + \bar{K}_i^a(\xi) \mathcal{J}_i^{a\mu}(\xi)]_{\text{tan}}, \quad (480)$$

$$V_{KK}^\mu(\xi) = -\frac{\beta^2}{M_X^2} [\bar{K}_i^a(\xi) \mathcal{K}_i^{a\mu}(\xi)]_{\text{tan}}. \quad (481)$$

The subscript “tan” means projection to the tangent bundle endomorphism sector on $Q_B(\xi)$.

Hence

$$\boxed{V_{\text{med}}^\mu(\xi) = V_{JJ}^\mu(\xi) + V_{JK}^\mu(\xi) + V_{KK}^\mu(\xi).} \quad (482)$$

F. Why the mixed $J \cdot K$ channel is the load-bearing one

The honest class-by-class coefficient audit already showed:

1. pure scalar/projector/amplitude-only classes are excluded;
2. a naive even singlet such as $\theta'^2\sigma_3$ is excluded by parity;
3. the transverse doublet is explicitly allowed;
4. the primitive odd longitudinal singlet is the minimal surviving longitudinal mechanism;
5. Class II is the strongest explicit class because it packages all three directions (a, b, c) at once.

Translated into (482), this means:

- V_{JJ}^μ and V_{KK}^μ may renormalize static or non-load-bearing sectors, but by themselves they do not provide the sharpest explicit unified three-coordinate realization. - The mixed channel V_{JK}^μ is the cleanest admissible source carrying both:

background/shell gradient structure and internal tangent projector structure.

So the actual load-bearing microscopic prototype is

$$V_{\text{med, lb}}^\mu(\xi) = -\frac{\alpha\beta}{M_X^2} [\bar{J}_i^a(\xi) \mathcal{K}_i^{a\mu}(\xi) + \bar{K}_i^a(\xi) \mathcal{J}_i^{a\mu}(\xi)]_{\text{tan}}. \quad (483)$$

G. Exact concrete quadratic reduction template

Inside the exact concrete Class II quadratic reduction already extracted in the notes, one has

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.} \quad (484)$$

Integrating out X_i^a yields the induced enriched tensor

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (485)$$

For the shell-adapted seed family

$$m^a = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

this gives

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J. \quad (486)$$

Thus Class II is the unique currently explicit microscopic class realizing all three directions at once.

In the local defect/projector transport language, this reduction matches the explicit S2 kernel

$$V_{\text{med}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) [\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a] + \dots, \quad (487)$$

and therefore

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (488)$$

This exact local coefficient dictionary is already closed. :contentReference[oaicite:10]index=10

H. What is actually proved at this step

What is proved:

1. the mediator sector contributes to the Hessian at quadratic order through the heavy-field Schur-complement structure (471);
2. its load-bearing first-order transport contribution is the $1y$ -part of $\delta^2(JK)$, namely (483);
3. inside the exact concrete Class II quadratic template, this source reduces to the explicit three-coordinate transport kernel (487);

4. hence the graded nondegeneracy problem is reduced to

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

What is still honestly open:

1. the unreduced final complete TECT expressions for $J_i^a[\Psi]$ and $K_i^a[P, \psi]$ in the physically relevant defect background;
2. whether the true complete TECT action actually realizes a nontrivial Class II sector there;
3. the final absolute normalization of (g_c, g_E, g_O) , equivalently $(\lambda_c, \Lambda_E, \Lambda_O)$, in the complete dynamics.

This is exactly the current honest status recorded in the notes.

I. Sharp honest conclusion

Therefore the mediator differentiation step is now as closed as it can be without inventing extra microscopic data. The result is:

$$\boxed{\delta^2 \Gamma_{\text{med}} \text{ contributes a genuine odd first-order transport kernel,}} \quad (489)$$

and inside the concrete Class II reduction template this kernel already carries the full S2 triplet

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

So the remaining full-PDE frontier is not the existence of a transport-producing mediator mechanism. That is already explicit.
It is only:

$$\boxed{\text{whether the final complete TECT action realizes that nontrivial Class II sector in the relevant defect background, and with wh}} \quad (490)$$

LII. FINAL SUMMARY

The Class II mediator sector starts from the explicit UV action

$$\boxed{\mathcal{L}_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi].}$$

Eliminating the heavy field gives

$$\boxed{S_{\text{eff}}^{(II)} = S_0 - \frac{\alpha^2}{2M_X^2} \int J_i^a J_i^a - \frac{\alpha\beta}{M_X^2} \int J_i^a K_i^a - \frac{\beta^2}{2M_X^2} \int K_i^a K_i^a.}$$

The load-bearing source is the mixed term

$$\boxed{S_{\text{mix}}^{(II)} = -\frac{\alpha\beta}{M_X^2} \int J_i^a K_i^a.}$$

After expanding around the shell-/defect-adapted background and keeping only the quadratic terms with exactly one worldvolume derivative, the mediator contribution to the graded transport kernel is

$$\boxed{V_{\text{med}}^\mu(\xi) = V_{JJ}^\mu(\xi) + V_{JK}^\mu(\xi) + V_{KK}^\mu(\xi),}$$

with the actual load-bearing microscopic prototype

$$V_{\text{med, lb}}^\mu(\xi) = -\frac{\alpha\beta}{M_X^2} [\bar{J}_i^a(\xi) \mathcal{K}_i^{a\mu}(\xi) + \bar{K}_i^a(\xi) \mathcal{J}_i^{a\mu}(\xi)]_{\text{tan}}.$$

Inside the exact concrete Class II quadratic reduction already closed in the notes,

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.},$$

integrating out X_i^a yields

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.$$

For the shell-adapted seed family this gives

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J,$$

so Class II is the only currently explicit microscopic class that packages all three invariant coordinates at once.

In the local defect/projector transport language this becomes

$$V_{\text{med}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \cdots,$$

hence

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

Therefore the full graded nondegeneracy test is again

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

The strongest honest conclusion is:

$$\delta^2 \Gamma_{\text{med}} \text{ provides a genuinely explicit odd first-order transport source,}$$

but

it is still open whether the final complete TECT action realizes this nontrivial Class II sector in the physically relevant defect b

LIII. ACTUAL SOURCE SYNTHESIS FOR THE GRADED TRANSPORT KERNEL

A. Goal

The graded 2 + 2 bypass route has already reduced the full microscopic frontier to the extraction of the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi) \partial_{y^\mu}.$$

The remaining task is now to combine all genuinely nontrivial microscopic sources into a single transport kernel and separate:

1. the broad actual-local transport class,
2. the exact concrete Class II-compatible branch.

B. Master source synthesis

From the previous steps, the bare isotropic quadratic core contributes only to

$$H_B(\xi), \quad W^{\mu\nu}(\xi),$$

and not to a genuine odd first-order transport kernel. Hence the actual transport source is the sum of the structured sectors:

$$V^\mu(\xi) = V_{\text{trans}}^\mu(\xi) + V_{\text{proj}}^\mu(\xi) + V_{\text{med}}^\mu(\xi). \quad (491)$$

Here:

- V_{trans}^μ denotes the direct odd transport/anisotropic sector;
- V_{proj}^μ denotes bundle-connection/projector-frame induced transport;
- V_{med}^μ denotes the effective Class II mediator contribution.

All three are Q_B -endomorphism valued and reflection-odd in ξ .

C. Broad actual-local transport class

The broad actual-local transport class already isolated in the notes is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (492)$$

with the leading shell-compatible invariant expansion

$$\mathcal{M}^\mu(\theta) = \Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a + O(\theta^2). \quad (493)$$

Thus the synthesized broad transport kernel is

$$V_{\text{broad}}^\mu(\xi) = \theta'(\xi) \left[\Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a \right] + O(\theta' \theta^2). \quad (494)$$

Projecting onto the graded low modes gives

$$M^\mu = I_\theta \left[\Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a \right] + O(\theta_0^3 \kappa), \quad (495)$$

hence

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa), \quad \alpha = \Lambda_E^{(0)} I_\theta + O(\theta_0^3 \kappa), \quad \beta = \Lambda_O^{(0)} I_\theta + O(\theta_0^3 \kappa). \quad (496)$$

Since $I_\theta \neq 0$, the broad-class nondegeneracy test is

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0). \quad (497)$$

This shows that in the broad actual-local operator class a constant-leading longitudinal branch is allowed. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

D. Exact concrete Class II-compatible branch

The exact concrete Class II-compatible branch is sharper. Its key feature is that the transverse channels start at the I_θ -level, while the longitudinal witness does not come from the constant-leading I_θ -branch. Instead it starts in the primitive commutator-square channel:

$$\boxed{V_{(\parallel)}^\mu(\xi) = \lambda_c \theta'(\xi) S_\xi n^\mu, \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.} \quad (498)$$

For the explicit representative,

$$\boxed{S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3,} \quad (499)$$

so

$$\boxed{V_{(\parallel),\text{tr}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3.} \quad (500)$$

The transverse channels are

$$\boxed{V_{(\perp,E)}^\mu(\xi) = \theta'(\xi) \Lambda_E E^\mu{}_a \sigma_a, \quad V_{(\perp,O)}^\mu(\xi) = \theta'(\xi) \Lambda_O J^\mu{}_a \sigma_a.} \quad (501)$$

Therefore the exact concrete Class II-compatible transport kernel is

$$\boxed{V_{\text{Class II}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \dots} \quad (502)$$

Projecting to the low modes gives

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (503)$$

Hence the exact concrete Class II nondegeneracy test is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (504)$$

equivalently

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (505)$$

This is the exact finite witness test already recorded in the notes. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

E. Source decomposition of the coefficients

The actual source synthesis can therefore be written coefficientwise as

$$\Lambda_\parallel^{\text{act}} = \Lambda_{\parallel,\text{trans}} + \Lambda_{\parallel,\text{proj}} + \Lambda_{\parallel,\text{med}}, \quad (506)$$

$$\Lambda_E^{\text{act}} = \Lambda_{E,\text{trans}} + \Lambda_{E,\text{proj}} + \Lambda_{E,\text{med}}, \quad (507)$$

$$\Lambda_O^{\text{act}} = \Lambda_{O,\text{trans}} + \Lambda_{O,\text{proj}} + \Lambda_{O,\text{med}}. \quad (508)$$

In the broad actual-local class, the actual kernel may already have a nonzero

$$\Lambda_\parallel^{(0)}$$

at leading I_θ -order. In the exact concrete Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0$$

and the first nonzero longitudinal contribution is the primitive λ_c -branch:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Thus the full longitudinal decision reduces to the two-branch audit:

$$\boxed{\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \neq 0.} \quad (509)$$

This is exactly the branch-sensitive leading-order audit already isolated in the notes. :contentReference[oaicite:9]index=9
:contentReference[oaicite:10]index=10

F. What is already closed

At this point the following are closed:

1. the graded $2 + 2$ reduced theorem;
2. the structural matching of the local CP^2 projector-defect Hessian to the graded operator class;
3. the existence of the natural S2 branch;
4. the explicit overlap formulas for

$$\lambda_{\parallel}, \quad \alpha, \quad \beta;$$

5. the exact concrete Class II coefficient dictionary

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O;$$

6. the finite witness test

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

All of this is already explicitly recorded. :contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12

G. What remains honestly open

What is still open is much narrower:

1. derive the actual full microscopic TECT transport kernel instead of choosing the minimal local ansatz;
2. determine whether the actual coefficient point lies in the broad constant-leading branch or in the primitive longitudinal branch;
3. compute the actual microscopic values of

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O;$$

4. verify directly that the extracted point satisfies the nondegeneracy condition.

Equivalently, the open problem is the actual microscopic coefficient extraction from the full TECT dynamics, not the reduced spectral logic. :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14 :contentReference[oaicite:15]index=15

H. Sharp honest conclusion

Therefore the full source synthesis has now compressed the remaining frontier to a single branch-sensitive coefficient audit:

$$\boxed{\text{Determine whether complete TECT produces } \Lambda_{\parallel}^{(0)}, \Lambda_E^{(0)}, \Lambda_O^{(0)} \text{ at leading } I_\theta \text{ order,}} \quad (510)$$

or instead

$$\boxed{\Lambda_{\parallel}^{(0)} = 0 \text{ and the first nonzero longitudinal coefficient is the primitive } \lambda_c\text{-branch.}} \quad (511)$$

Once this actual coefficient point is extracted, the graded-bypass Dirac audit is finished.

LIV. FINAL SUMMARY

The actual transport kernel of the graded $2 + 2$ route is now synthesized as

$$\boxed{V^\mu(\xi) = V_{\text{trans}}^\mu(\xi) + V_{\text{proj}}^\mu(\xi) + V_{\text{med}}^\mu(\xi).}$$

The broad actual-local operator class has the leading invariant form

$$\boxed{V_{\text{broad}}^\mu(\xi) = \theta'(\xi) \left[\Lambda_0^{(0)} n^\mu \mathbf{1}_2 + \Lambda_{\parallel}^{(0)} n^\mu \sigma_3 + \Lambda_E^{(0)} E^\mu{}_a \sigma_a + \Lambda_O^{(0)} J^\mu{}_a \sigma_a \right] + O(\theta' \theta^2).}$$

Hence the broad-class leading witness test is

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0).}$$

The exact concrete Class II-compatible branch is sharper:

$$\boxed{V_{\text{Class II}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \cdots,}$$

so

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.}$$

Therefore the exact concrete nondegeneracy test is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),}$$

equivalently

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).}$$

Thus the remaining full-PDE frontier is not structural. It is the branch-sensitive coefficient audit:

$$\boxed{\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with first nonzero longitudinal term in the primitive } \lambda_c\text{-branch.}}$$

So the strongest honest current statement is:

$$\boxed{\text{the graded-bypass Dirac problem has been reduced to extracting the actual microscopic coefficient point } (\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).}$$

IV. BRANCH-SENSITIVE COEFFICIENT AUDIT FOR THE GRADED 2 + 2 ROUTE

A. Goal

At this stage the graded-bypass reduction is already complete at the spectral level. The remaining microscopic ambiguity is no longer whether a longitudinal witness can exist in principle. That part is already established.

The actual unresolved question is only:

$$\boxed{\text{Which longitudinal branch does the complete microscopic TECT transport extraction realize?}} \quad (512)$$

B. What is already closed

The following are already explicitly closed in the uploaded notes:

1. The graded 2 + 2 reduced theorem is rigorous.
2. The actual local CP^2 projector-defect geometry belongs to this operator class.
3. The natural local branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (513)$$

4. In the actual local projector-defect transport class, the finite nondegeneracy test is exactly

$$\boxed{\Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (514)$$

5. In the exact concrete Class II branch,

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},} \quad (515)$$

so the reduced S2 nondegeneracy condition is

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (516)$$

These points are already explicit in the notes. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

C. Broad actual-local branch

In the broad actual-local operator class the transport kernel has the shell-compatible form

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad \mathcal{M}^{\mu}(\theta) = \Lambda_0^{\mu} \mathbf{1}_2 + \Lambda_i^{\mu} \sigma_i + O(\theta^2). \quad (517)$$

The branch-sensitive theorem states that none of the leading invariant coefficients

$$\Lambda_{\parallel}^{(0)}, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)}$$

is symmetry-forbidden in this broad class. In particular, a constant-leading longitudinal branch is allowed. :contentReference[oaicite:7]index=7

Projecting to the graded low modes gives

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa), \quad \alpha = I_{\theta} \Lambda_E^{(0)} + O(\theta_0^3 \kappa), \quad \beta = I_{\theta} \Lambda_O^{(0)} + O(\theta_0^3 \kappa), \quad (518)$$

with $I_{\theta} \neq 0$.

Therefore the broad constant-leading branch is:

$$\boxed{\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0.} \quad (519)$$

Its finite decision test is

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0).} \quad (520)$$

D. Exact concrete Class II primitive branch

In the exact currently constructed concrete Class II witness formulas, the transverse sector begins at I_θ -order, but the longitudinal sector does *not* begin in the constant-leading I_θ -branch. Instead it appears only through the primitive commutator-square channel:

$$V_{\parallel, \text{tr}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (521)$$

Therefore

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (522)$$

So the primitive longitudinal branch is:

$$\textbf{Branch B: } \Lambda_\parallel^{(0)} = 0 \quad \text{and} \quad \lambda_c \neq 0. \quad (523)$$

Its finite decision test is

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (524)$$

This exact branch-sensitive distinction is already explicitly isolated in the notes. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

E. Unified longitudinal decision protocol

The notes already give the exact practical execution order. We can rewrite it as the following deterministic decision tree.

- a. (W1) *Extract the shell-compatible local transport matrix.* Compute

$$\mathcal{M}^\mu(\theta)$$

from the actual complete TECT Hessian extraction.

- b. (W2) *Test the constant-leading longitudinal coefficient.* Compute

$$\Lambda_\parallel^{(0)} = \frac{1}{2} [\sigma_3 n_\mu \mathcal{M}^\mu(0)]. \quad (525)$$

- c. (W3) *If $\Lambda_\parallel^{(0)} \neq 0$, stop.* Then the actual theory lies on Branch A, and

$$\lambda_\parallel = I_\theta \Lambda_\parallel^{(0)} + O(\theta_0^3 \kappa).$$

- d. (W4) *If $\Lambda_\parallel^{(0)} = 0$, compute the primitive channel.* Define

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}, \quad (526)$$

and project the actual longitudinal transport onto

$$\theta'(\xi) S_\xi n^\mu$$

to extract λ_c .

- e. (W5) *If $\lambda_c \neq 0$, stop.* Then the actual theory lies on Branch B, and

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.$$

f. (W6) If both vanish, the current branch library is exhausted. That is, if

$$\Lambda_{\parallel}^{(0)} = 0 \quad \text{and} \quad \lambda_c = 0,$$

then the currently established longitudinal seed library contains no surviving branch, and one must inspect higher-order even longitudinal channels explicitly.

This exact protocol is already stated in the notes. :contentReference[oaicite:10]index=10

F. Why the longitudinal sector is the true bottleneck

The notes are explicit that once the transverse channel survives microscopically, the real bottleneck is the longitudinal one:

$$\theta'(\xi)^3 \sigma_3 \implies \lambda_{\parallel} \neq 0.$$

Equivalently, once $(\alpha, \beta) \neq (0, 0)$ is under control, the only remaining witness problem is the survival of one admissible longitudinal seed direction inside the actual microscopic transport space V_{mic} . :contentReference[oaicite:11]index=11

Thus the full witness problem is no longer symmetric among the three coefficients. The actual audit order is:

$$\boxed{\text{transverse already largely reduced} \implies \text{longitudinal branch selection is the remaining bottleneck.}} \quad (527)$$

G. Exact final finite tests

The branch-sensitive final tests are therefore:

a. Branch A test

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0).} \quad (528)$$

b. Branch B test

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (529)$$

If one wants a single combined formulation, it is:

$$\boxed{[\Lambda_{\parallel}^{(0)} \neq 0 \text{ or } (\Lambda_{\parallel}^{(0)} = 0 \text{ and } \lambda_c \neq 0)] \quad \text{and} \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (530)$$

H. Strongest honest conclusion

We can now state the exact honest branch-sensitive status:

$$\boxed{\text{the longitudinal witness problem is reduced to a two-branch finite decision protocol.}} \quad (531)$$

What remains open is not the reduced spectral logic, but the actual microscopic coefficient extraction:

$$\boxed{\text{the complete microscopic TECT transport extraction has not yet determined whether } \Lambda_{\parallel}^{(0)} \neq 0 \text{ or instead } \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \neq 0} \quad (532)$$

This is already explicitly isolated in the notes. :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

LVI. FINAL SUMMARY

The graded-bypass Dirac problem is now reduced to a branch-sensitive coefficient audit. What is already closed:

1. The reduced graded $2 + 2$ theorem is rigorous.
2. The actual local projector-defect geometry supports the natural S2 branch

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

3. In the broad actual-local operator class, the nondegeneracy test is

$$\Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

4. In the exact concrete Class II branch, the witness coefficients are

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

so the nondegeneracy test becomes

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

The actual unresolved question is only which longitudinal branch the complete microscopic TECT transport extraction realizes:

Branch A: $\Lambda_{\parallel}^{(0)} \neq 0 \implies \lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa),$

or

Branch B: $\Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0 \implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}.$

Thus the exact final finite tests are:

Branch A: $\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0),$

or

Branch B: $\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$

Therefore the strongest honest current conclusion is:

the longitudinal witness problem is fully reduced to a two-branch finite decision protocol,

but

the complete microscopic TECT transport extraction has not yet determined which branch the actual coefficient point lies on.

LVII. ACTUAL LONGITUDINAL BRANCH TEST

A. Goal

The graded $2 + 2$ route has already reduced the full local Dirac problem to the S2 witness condition

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

The transverse sector is already in explicit I_{θ} -form, so the true remaining bottleneck is the longitudinal sector.

The actual microscopic task is therefore not another structural reduction. It is the finite branch selection problem:

$$\boxed{\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \stackrel{?}{\neq} 0.}$$

B. Input data from the local transport extraction

Assume the full microscopic extraction has produced the shell-compatible local transport kernel

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad (533)$$

with invariant decomposition

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}_a \sigma_a + h_O(\theta) J^{\mu}_a \sigma_a. \quad (534)$$

Then the actual witness formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (535)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_O(\theta(\xi)). \quad (536)$$

This is already explicitly fixed in the notes. :contentReference[oaicite:4]index=4

C. Step 0: transverse sector is already reduced

The transverse sector is already completely in I_{θ} -form:

$$\alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O, \quad I_{\theta} \neq 0.$$

Hence the only remaining transverse condition is

$$\boxed{(\Lambda_E, \Lambda_O) \neq (0, 0)}. \quad (537)$$

So the actual remaining ambiguity is exclusively longitudinal.

D. Branch A: constant-leading longitudinal test

In the broad actual-local operator class, the leading invariant longitudinal coefficient

$$\Lambda_{\parallel}^{(0)}$$

is not symmetry-forbidden. If it is nonzero, then

$$\boxed{\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa)}. \quad (538)$$

Since $I_\theta \neq 0$, it follows immediately that

$$\Lambda_{\parallel}^{(0)} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (539)$$

Thus the Branch A decision criterion is

$$\textbf{Branch A test: } \Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (540)$$

This branch is allowed in the broad actual-local class, but has not yet been proven to be the one realized by the complete microscopic TECT extraction.

E. Branch B: primitive longitudinal test

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

one must pass to the primitive longitudinal channel already explicitly isolated in the notes.

Define

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (541)$$

Then the primitive longitudinal seed is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu. \quad (542)$$

Factor the actual primitive longitudinal component as

$$V_{\parallel, \text{act}}^\mu(\xi) = \lambda_c \theta'(\xi) S_\xi n^\mu + V_{\parallel, \text{rem}}^\mu(\xi), \quad (543)$$

where $V_{\parallel, \text{rem}}^\mu$ is orthogonal to the primitive target covector in the reduced witness pairing. This is the exact protocol already written in the notes.

In the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad (544)$$

hence

$$V_{\parallel, \text{tr}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (545)$$

The corresponding primitive overlap is

$$J_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) S_\xi = \frac{1}{2} I_{\theta'^3}, \quad (546)$$

with

$$I_{\theta'^3} \neq 0 \quad (\theta_0 \kappa \neq 0). \quad (547)$$

Therefore

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (548)$$

So the Branch B decision criterion is

$$\textbf{Branch B test: } \Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (549)$$

Since $I_{\theta'^3} \neq 0$, this implies immediately

$$\lambda_c \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (550)$$

All of this is already explicitly established for the concrete primitive/Class II-compatible branch.

F. Unified decision tree

The two branches combine into a single deterministic audit protocol:

a. (X1) *Extract the invariant transport matrix.* Compute $\mathcal{M}^\mu(\theta)$ from the actual microscopic TECT linearization and reduce it to the invariant profile functions

$$h_{\parallel}(\theta), \quad h_E(\theta), \quad h_O(\theta).$$

b. (X2) *Test the leading longitudinal invariant.* Define

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2}[\sigma_3 n_\mu \mathcal{M}^\mu(0)]. \quad (551)$$

c. (X3) *If $\Lambda_{\parallel}^{(0)} \neq 0$, stop.* Then the actual coefficient point lies on Branch A and

$$\lambda_{\parallel} \neq 0$$

immediately.

d. (X4) *If $\Lambda_{\parallel}^{(0)} = 0$, test the primitive channel.* Compute

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}$$

and extract the primitive coefficient λ_c from the projection of the actual longitudinal transport onto

$$\theta'(\xi) S_\xi n^\mu.$$

e. (X5) *If $\lambda_c \neq 0$, stop.* Then the actual coefficient point lies on Branch B and

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0.$$

f. (X6) *If both vanish, the current branch library is exhausted.* That is,

$$\Lambda_{\parallel}^{(0)} = 0 \quad \text{and} \quad \lambda_c = 0$$

means the presently established longitudinal seed library contains no surviving witness and one must inspect higher-order even longitudinal channels explicitly. This is already the honest closure statement in the notes.

G. Why this is the true bottleneck

The transverse sector already sits in the I_θ -branch and is reduced to a finite nonzero test on (Λ_E, Λ_O) . The longitudinal sector is different because a naive even singlet source proportional to $\theta'^2 \sigma_3$ fails by parity, so the primitive commutator-square channel is the first admissible odd singlet.

Therefore the remaining actual microscopic problem is no longer the full 3-parameter witness problem symmetrically. It is only the longitudinal branch selection:

$$\boxed{\text{Does the complete microscopic TECT transport extraction realize Branch A or Branch B?}} \quad (552)$$

H. Strongest honest conclusion

The longitudinal witness chain is now closed *conditional on branch choice*:

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0 \implies \lambda_{\parallel} \neq 0,} \quad (553)$$

and

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (554)$$

What is still open is only the actual microscopic determination of which branch the extracted coefficient point belongs to.

LVIII. FINAL SUMMARY

The graded-bypass Dirac problem is now reduced to a branch-sensitive longitudinal audit.
The transverse sector is already closed at the finite-test level:

$$(\Lambda_E, \Lambda_O) \neq (0, 0).$$

The only remaining ambiguity is the longitudinal branch.

a. Branch A. In the broad actual-local transport class, the constant-leading longitudinal coefficient is allowed:

$$\Lambda_{\parallel}^{(0)} \neq 0 \implies \lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa) \neq 0.$$

b. Branch B. If

$$\Lambda_{\parallel}^{(0)} = 0,$$

the primitive commutator-square channel must be tested:

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}, \quad V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) S_{\xi} n^{\mu}.$$

In the explicit defect representative,

$$S_{\xi} = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad I_{\theta'^3} \neq 0,$$

so

$$\lambda_c \neq 0 \implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0.$$

Therefore the exact unified decision tree is:

$$(1) \text{ test } \Lambda_{\parallel}^{(0)}; \quad (2) \text{ if it vanishes, test } \lambda_c.$$

So the strongest honest current conclusion is:

$$\text{the longitudinal witness problem is fully reduced to a two-branch finite decision protocol,}$$

but

$$\text{the complete microscopic TECT extraction has not yet determined whether the actual coefficient point lies on Branch A or Branch B.}$$

LIX. ACTUAL COEFFICIENT-POINT EXTRACTION: LONGITUDINAL BRANCH TEST

A. Goal

At this stage the graded $2 + 2$ route is already reduced to the S2 witness test

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

The transverse sector is already in explicit overlap form, so the only remaining ambiguity is the longitudinal branch.

Thus the actual microscopic problem is:

$$\text{Does the extracted coefficient point satisfy } \Lambda_{\parallel}^{(0)} \neq 0 \text{ (Branch A), or instead } \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \neq 0 \text{ (Branch B)?}$$

B. Input from the full Hessian extraction

Assume the background-dependent Hessian extraction has already produced the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Write the invariant decomposition

$$\mathcal{M}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a. \quad (556)$$

Then the projected witness coefficients are

$$\lambda_\parallel = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_\parallel(\theta(\xi)), \quad (557)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_O(\theta(\xi)). \quad (558)$$

C. Transverse sector is already reduced

The transverse sector is already explicit:

$$\boxed{\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad I_\theta \neq 0.} \quad (559)$$

Hence the only remaining transverse condition is simply

$$\boxed{(\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (560)$$

So the true bottleneck is longitudinal only.

D. Branch A test: constant-leading longitudinal coefficient

Define the constant-leading longitudinal invariant by

$$\boxed{\Lambda_\parallel^{(0)} := \frac{1}{2} [\sigma_3 n_\mu \mathcal{M}^\mu(0)].} \quad (561)$$

If this is nonzero, then the extracted coefficient point lies on the broad constant-leading branch:

$$\boxed{\textbf{Branch A: } \Lambda_\parallel^{(0)} \neq 0.} \quad (562)$$

In that case,

$$\boxed{\lambda_\parallel = I_\theta \Lambda_\parallel^{(0)} + O(\theta_0^3 \kappa),} \quad (563)$$

and since $I_\theta \neq 0$,

$$\boxed{\Lambda_\parallel^{(0)} \neq 0 \implies \lambda_\parallel \neq 0.} \quad (564)$$

Therefore the finite Branch A pass condition is

$$\boxed{\Lambda_\parallel^{(0)} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (565)$$

E. Branch B test: primitive longitudinal channel

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

the broad constant-leading branch is absent and one must test the primitive longitudinal seed.

Introduce

$$\boxed{\mathcal{T}_{\xi} := i[P_B, \partial_{\xi} P_B], \quad S_{\xi} := (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}.} \quad (566)$$

Then the primitive longitudinal channel is

$$\boxed{V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) S_{\xi} n^{\mu}.} \quad (567)$$

For the explicit defect representative already used in the local reduction,

$$\boxed{S_{\xi} = \frac{1}{2} \theta'(\xi)^2 \sigma_3,} \quad (568)$$

hence

$$\boxed{V_{(\parallel), \text{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.} \quad (569)$$

The projected primitive overlap is

$$J_{\parallel} = \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) S_{\xi} = \frac{1}{2} I_{\theta'^3}, \quad (570)$$

with

$$\boxed{I_{\theta'^3} \neq 0.} \quad (571)$$

Therefore the Branch B formula is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (572)$$

So

$$\boxed{\lambda_c \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (573)$$

Thus the finite Branch B pass condition is

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (574)$$

F. Operational decision tree

The actual longitudinal branch test is now completely deterministic.

- a. Step X1.* Extract the invariant transport matrix $\mathcal{M}^{\mu}(\theta)$ from the full background-dependent Hessian.
- b. Step X2.* Compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} [\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)].$$

- c. Step X3.* If $\Lambda_{\parallel}^{(0)} \neq 0$, stop: the theory lies on Branch A.

d. *Step X4.* If $\Lambda_{\parallel}^{(0)} = 0$, compute the primitive coefficient λ_c by projecting the actual longitudinal transport onto

$$\theta'(\xi) S_{\xi} n^{\mu}.$$

e. *Step X5.* If $\lambda_c \neq 0$, stop: the theory lies on Branch B.

f. *Step X6.* If both vanish, then the presently established longitudinal seed library is exhausted, and one must inspect higher-order even longitudinal channels explicitly.

G. What is already explicit and what is still open

The current honest status is therefore:

1. In the broad actual-local operator class, Branch A is symmetry-allowed.
2. In the exact currently constructed concrete Class II branch, the explicit realized branch is Branch B:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3 = \frac{g_c}{2} I_{\theta^3}.$$

3. What remains open is not the reduced logic, but the actual microscopic coefficient extraction: which branch the final complete TECT coefficient point belongs to has not yet been determined.

H. Sharp honest conclusion

Thus the longitudinal witness chain is now closed *conditional on branch choice*:

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0 \implies \lambda_{\parallel} \neq 0,} \quad (575)$$

and

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (576)$$

The only genuinely open microscopic question is:

$$\boxed{\text{Does the actual complete TECT coefficient point lie on Branch A or on Branch B?}} \quad (577)$$

LX. FINAL SUMMARY

The graded-bypass Dirac problem is now reduced to a branch-sensitive longitudinal coefficient audit. The transverse sector is already closed at the finite-test level:

$$\boxed{(\Lambda_E, \Lambda_O) \neq (0, 0).}$$

The only remaining ambiguity is the longitudinal branch.

a. *Branch A.* In the broad actual-local operator class,

$$\boxed{\Lambda_{\parallel}^{(0)} = \frac{1}{2} [\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)]}$$

is symmetry-allowed, and

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0 \implies \lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa) \neq 0.}$$

b. *Branch B.* If

$$\Lambda_{\parallel}^{(0)} = 0,$$

one must test the primitive longitudinal seed

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}, \quad V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) S_{\xi} n^{\mu}.$$

In the explicit defect representative,

$$S_{\xi} = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad I_{\theta'^3} \neq 0,$$

so

$$\lambda_c \neq 0 \implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0.$$

Therefore the exact decision tree is:

$$\text{first test } \Lambda_{\parallel}^{(0)}; \quad \text{if it vanishes, test } \lambda_c.$$

The strongest honest current status is:

$$\text{Branch A is allowed in the broad actual-local class,}$$

but

$$\text{the explicitly realized currently constructed concrete Class II branch is Branch B.}$$

So the only genuinely open microscopic question is:

$$\text{which branch the actual complete TECT coefficient point belongs to.}$$

LXI. OPERATOR CHECKLIST FOR READING $\mathcal{M}^{\mu}(0)$ AND S_{ξ} FROM THE FULL HESSIAN

A. Goal

The reduced graded $2 + 2$ logic is already closed. The only remaining microscopic task is to determine which longitudinal branch the actual complete TECT coefficient point lies on:

$$\Lambda_{\parallel}^{(0)} \neq 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \neq 0.$$

So the only thing still needed is an executable extraction protocol from the full background-dependent Hessian.

B. Starting operator

Begin from the shell-/defect-adapted background-dependent Hessian

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^{\mu}(\xi)\partial_{y^{\mu}} + W^{\mu\nu}(\xi)\partial_{y^{\mu}}\partial_{y^{\nu}} + \cdots \quad (578)$$

acting on tangent fluctuations

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C}z_B(\xi))^{\perp}.$$

The actual transport extraction assumes the shell-compatible local form

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad (579)$$

with invariant decomposition

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}{}_a \sigma_a + h_O(\theta) J^{\mu}{}_a \sigma_a. \quad (580)$$

This is exactly the local transport input already isolated in the notes.

C. Step Y1: isolate the transport kernel from the Hessian

Compute the full quadratic operator and collect *only* the terms with exactly one worldvolume derivative:

$$\left[\Gamma^{(2)}[\bar{\Phi}_{\min}] \right]_{1y}.$$

Move all ξ -dependent coefficient matrices to the left of ∂_{y^μ} , project to $Q_B(\xi)$, and write

$$\boxed{\left[\Gamma^{(2)}[\bar{\Phi}_{\min}] \right]_{1y} = -iV^\mu(\xi)\partial_{y^\mu}.} \quad (581)$$

This is the only block from which $\Lambda_{\parallel}^{(0)}$ and λ_c are to be read.

D. Step Y2: divide out the universal odd factor

Because the defect background is reflection-even and the transport kernel is reflection-odd, the leading local transport kernel factors through $\theta'(\xi)$:

$$\boxed{\mathcal{M}^\mu(\theta(\xi)) = \frac{V^\mu(\xi)}{\theta'(\xi)}} \quad (582)$$

on the support where $\theta'(\xi) \neq 0$, followed by smooth continuation to $\theta = 0$.

This is the precise operator meaning of the transport reduction

$$V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).$$

E. Step Y3: read the constant-leading longitudinal coefficient

Evaluate the invariant transport matrix at $\theta = 0$:

$$\mathcal{M}^\mu(0).$$

Then project onto the shell-compatible longitudinal singlet direction:

$$\boxed{\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].} \quad (583)$$

This is the unique constant-leading longitudinal invariant coefficient already defined in the notes.

a. Branch A criterion. If

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0,} \quad (584)$$

then the actual coefficient point lies on Branch A and

$$\boxed{\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa).} \quad (585)$$

Since $I_\theta \neq 0$, this immediately gives $\lambda_{\parallel} \neq 0$.

F. Step Y4: if Branch A fails, construct the primitive longitudinal projector

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

define the commutator channel

$$\boxed{\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].} \quad (586)$$

Then form the quadratic traceless Q_B -response

$$\boxed{S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.} \quad (587)$$

This is exactly the primitive longitudinal seed channel already isolated in the notes.

For the explicit defect representative,

$$\boxed{S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3.} \quad (588)$$

G. Step Y5: project the actual transport kernel onto the primitive branch

Take the actual longitudinal part of the transport kernel

$$V_{\parallel, \text{act}}^\mu(\xi) := \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(\theta(\xi))] \theta'(\xi) n^\mu \sigma_3 + \cdots,$$

and factor it against the primitive covector

$$\theta'(\xi) S_\xi n^\mu.$$

Operationally, define λ_c as the coefficient of the projection of $V_{\parallel, \text{act}}^\mu$ onto this primitive channel:

$$\boxed{V_{\parallel, \text{act}}^\mu(\xi) = \lambda_c \theta'(\xi) S_\xi n^\mu + V_{\parallel, \text{rem}}^\mu(\xi),} \quad (589)$$

where $V_{\parallel, \text{rem}}^\mu$ is orthogonal to the primitive target in the reduced witness pairing.

Using the explicit seed

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3,$$

this becomes

$$\boxed{V_{\parallel, \text{act}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + V_{\parallel, \text{rem}}^\mu(\xi).} \quad (590)$$

H. Step Y6: evaluate the primitive overlap

The primitive longitudinal overlap is

$$J_\parallel := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) S_\xi. \quad (591)$$

In the explicit seed representative,

$$\boxed{J_\parallel = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.} \quad (592)$$

Therefore

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (593)$$

a. *Branch B criterion.* If

$$\boxed{\Lambda_{\parallel}^{(0)} = 0 \quad \text{and} \quad \lambda_c \neq 0,} \quad (594)$$

then the actual coefficient point lies on Branch B and $\lambda_{\parallel} \neq 0$ follows immediately.

I. Step Y7: the complete decision tree

The full longitudinal branch test is now a deterministic checklist:

- a. Y7.1 Extract $V^{\mu}(\xi)$ from the one-world-volume-derivative block of the full Hessian.
- b. Y7.2 Divide by $\theta'(\xi)$ to obtain $\mathcal{M}^{\mu}(\theta)$.
- c. Y7.3 Compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)].$$

- d. Y7.4 If $\Lambda_{\parallel}^{(0)} \neq 0$, conclude:

$$\boxed{\text{Branch A.}}$$

- e. Y7.5 If $\Lambda_{\parallel}^{(0)} = 0$, compute

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}$$

and project the actual longitudinal transport onto

$$\theta'(\xi) S_{\xi} n^{\mu}$$

to extract λ_c .

- f. Y7.6 If $\lambda_c \neq 0$, conclude:

$$\boxed{\text{Branch B.}}$$

g. Y7.7 If both coefficients vanish, the presently established longitudinal seed library is exhausted, and one must inspect higher-order even longitudinal channels explicitly.

J. What is already explicit and what remains open

The current honest status is:

1. Branch A is allowed in the broad actual-local class.
2. Branch B is explicitly realized in the currently constructed concrete Class II witness formulas:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3 = \frac{g_c}{2} I_{\theta'^3}.$$

3. The actual complete TECT coefficient point has not yet been extracted, so the final branch choice remains open.

K. Sharp honest conclusion

Thus the remaining microscopic problem is no longer structural. It is exactly the executable coefficient-point audit:

$$\boxed{\text{extract } V^{\mu}(\xi) \text{ from the full Hessian, compute } \Lambda_{\parallel}^{(0)}, \text{ and if it vanishes, compute } \lambda_c.} \quad (595)$$

Once that is done, the longitudinal branch is fixed.

LXII. FINAL SUMMARY

The graded-bypass problem is now reduced to a fully executable longitudinal branch test. Starting from the full background-dependent Hessian, extract the odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).$$

Then the longitudinal branch is determined in the following order.

a. Branch A test. Compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].$$

If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_\theta + O(\theta_0^3 \kappa) \neq 0,$$

so the coefficient point lies on Branch A.

b. Branch B test. If

$$\Lambda_{\parallel}^{(0)} = 0,$$

define

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

In the explicit defect seed,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad I_{\theta'^3} \neq 0.$$

Project the actual longitudinal transport onto

$$\theta'(\xi) S_\xi n^\mu$$

to extract λ_c . If

$$\lambda_c \neq 0,$$

then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0,$$

so the coefficient point lies on Branch B.

Therefore the exact decision tree is:

$$\boxed{\text{first compute } \Lambda_{\parallel}^{(0)}; \quad \text{if it vanishes, compute } \lambda_c.}$$

The strongest honest current status is:

$$\boxed{\text{Branch A is allowed in the broad actual-local operator class,}}$$

$$\boxed{\text{the explicitly realized currently constructed concrete Class II branch is Branch B,}}$$

but

$$\boxed{\text{the actual complete TECT coefficient point has not yet been extracted, so the final branch choice remains open.}}$$

LXIII. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (ROUTE B MAINLINE)

LXIV. CURRENT GLOBAL STATUS

The plain scalar Bragg-pair route is no longer the correct mainline. Inside the currently known safe microscopic operator library, the scalar identity-channel Bragg block remains nonzero:

$$B_0^{\text{scalar}} = 4uA^2 \mathbf{1}_2 \neq 0,$$

while the alignment term vanishes on the locked branch and canonical Class II only renormalizes the derivative witness sector.

Therefore the correct current mainline is the graded $2 + 2$ bypass route.

LXV. WHAT IS ALREADY CLOSED

A. Minimal locked isotropic PDE no-go

For the minimal locked isotropic complex-triplet Brazovskii PDE:

1. the linearized quartic term does not directly generate an enriched first-order Pauli-sector shell tensor;
2. the isotropic kinetic first-order shell term cancels exactly at the Brazovskii shell;
3. the locked zeroth-order cross-shell quartic projection vanishes.

Hence the minimal locked isotropic PDE alone does not close the full witness problem.

B. Scalar-channel no-go inside the current safe operator library

The shell-relevant scalar constant sector can be parameterized by

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.$$

The exact scalar cancellation condition is therefore

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.$$

For the currently known safe microscopic sectors:

$$(c_0, c_1, c_2)_{\text{scalar}} = (u, 0, 0), \quad (c_0, c_1, c_2)_{\text{align}} = (0, 0, 0), \quad (c_0, c_1, c_2)_{\text{Class II}} = (0, 0, 0)$$

as a direct scalar-cancellation channel.

Therefore

$$(c_0, c_1, c_2)_{\text{known}} = (u, 0, 0),$$

and so

$$B_0^{\text{known}} = 4uA^2 \mathbf{1}_2 \neq 0 \quad (u \neq 0).$$

So the scalar identity-channel Bragg block cannot be removed inside the current safe operator library.

C. Class II does not cancel the identity channel

Every Class II constant pair-mixing contribution carries an explicit traceless internal Pauli generator:

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\},$$

hence it is orthogonal to the scalar identity channel

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Therefore Class II renormalizes derivative witnesses but does not remove the scalar Bragg obstruction.

LXVI. WHY ROUTE B IS THE CORRECT MAINLINE

The correct mainline is the graded $2 + 2$ reduced route:

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2,$$

with grading-odd first-order projected symbol

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu.$$

Because the relevant principal operator is first-order and grading-odd,

$$D_1(0) = 0,$$

so the dangerous scalar Bragg block is absent from the relevant principal infrared operator rather than cancelled inside it.

LXVII. ACTUAL LOCAL $\mathbb{CP}^2$ PROJECTOR-DEFECT REALIZATION

For a local projector defect background

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1,$$

the physical tangent fluctuation lives in the rank-two complex bundle

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp.$$

Using a reflection-symmetric explicit representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

one has

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad \partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Hence every leading local transport kernel linear in the defect gradient factors through

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

The low transverse bundle problem gives even/odd modes

$$u_a^+(\xi), \quad u_a^-(\xi), \quad a = 1, 2,$$

and the mixed symbol is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

By parity,

$$\langle u_a^\pm, V^\mu u_b^\pm \rangle = 0, \quad \langle u_a^+, V^\mu u_b^- \rangle \neq 0$$

generically, so the leading reduced operator is automatically grading-odd.

LXVIII. NATURAL S2 BRANCH

The natural local symmetry-compatible branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

The transverse coefficients are already reduced to

$$\alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O, \quad I_{\theta} \neq 0.$$

So the transverse finite test is simply

$$(\Lambda_E, \Lambda_O) \neq (0, 0).$$

LXIX. BROAD ACTUAL-LOCAL BRANCH

In the broad actual-local operator class, the transport kernel has invariant decomposition

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}_a \sigma_a + h_O(\theta) J^{\mu}_a \sigma_a.$$

Assuming

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + \Lambda_{\parallel}^{(2)} \theta^2 + O(\theta^4),$$

the longitudinal overlap is

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + \Lambda_{\parallel}^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa).$$

Therefore, if

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then

$$\lambda_{\parallel} \neq 0$$

already on the constant-leading I_{θ} -branch.

This is **Branch A**.

LXX. EXACT CONCRETE CLASS II PRIMITIVE BRANCH

In the exact concrete currently constructed Class II-compatible branch, the transverse sector starts at I_{θ} -order, but the longitudinal sector does not come from the constant-leading I_{θ} -branch.

Instead, define

$$\mathcal{T}_{\xi} := i[P_B, \partial_{\xi} P_B], \quad S_{\xi} := (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}.$$

For the explicit defect representative,

$$S_{\xi} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Hence the primitive longitudinal seed is

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) S_{\xi} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.$$

Its reduced witness is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0.$$

The transverse coefficients remain

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

Equivalently, in the explicit concrete Class II witness notation,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_\theta, \quad \beta = g_O I_\theta.$$

Therefore the exact concrete Class II finite test is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

This is **Branch B**.

LXXI. LONGITUDINAL TWO-BRANCH DECISION THEOREM

The actual unresolved question is only which branch the complete microscopic TECT transport extraction realizes:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0,$$

or

$$\text{Branch B: } \Lambda_{\parallel}^{(0)} = 0 \text{ with } \lambda_c \neq 0.$$

The executable decision protocol is:

1. Extract the odd transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}$$

from the full background-dependent Hessian.

2. Divide by $\theta'(\xi)$ to obtain $\mathcal{M}^\mu(\theta)$.

3. Compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2}[\sigma_3 n_\mu \mathcal{M}^\mu(0)].$$

4. If $\Lambda_{\parallel}^{(0)} \neq 0$, stop: Branch A holds.

5. If $\Lambda_{\parallel}^{(0)} = 0$, compute the primitive coefficient λ_c by projecting the actual longitudinal transport onto

$$\theta'(\xi) S_\xi n^\mu.$$

6. If $\lambda_c \neq 0$, stop: Branch B holds.

7. If both vanish, the presently established longitudinal seed library is exhausted and higher-order even longitudinal channels must be inspected explicitly.

LXXII. MASS PROTECTION STATUS

Once an isolated doubled local Dirac block exists, the constant Dirac mass space is

$$\mathcal{M} = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Mass protection is equivalent to

$$\mathcal{A}_{\text{res}} \cap \mathcal{M} = \{0\}.$$

Exact residual valley $U(1)_V$ is sufficient:

$$Q_V = \tau_z \otimes \mathbf{1}_2, \quad U_\theta = e^{i\theta Q_V/2},$$

because the two mass generators rotate as a charged doublet under $U(1)_V$.

The weaker remnant

$$\mathbb{Z}_2^V : \quad S_V = \tau_z \otimes \mathbf{1}_2$$

is already enough, because both mass generators are odd under S_V .

If the same residual symmetry survives remote-sector elimination, then no radiatively generated constant mass can appear.

What remains open microscopically is whether the actual full TECT/Class II realization indeed preserves exact residual $U(1)_V$ or at least $\mathbb{Z}_2^V$.

LXXIII. WHAT IS FULLY CLOSED

1. plain scalar Bragg route is insufficient;
2. current safe scalar/Class II library cannot remove the scalar identity-channel Bragg block;
3. graded $2+2$ reduced Dirac theorem is rigorous;
4. actual local CP^2 projector-defect geometry belongs to this operator class;
5. the natural local branch is S2;
6. transverse witness sector is already reduced to a finite nonzero test;
7. exact concrete Class II branch explicitly realizes the primitive longitudinal branch;
8. theorem-level mass protection by exact residual valley symmetry is closed.

LXXIV. WHAT IS STILL OPEN

1. derive the final complete microscopic TECT projector/condensate PDE in a form suitable for direct spectral analysis;
2. derive the actual transport kernel $V^\mu(\xi)$ from the complete background-dependent Hessian, rather than postulate the minimal local ansatz;
3. extract the actual coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O);$$

4. determine whether the actual complete TECT coefficient point lies on Branch A or Branch B;
5. verify the final finite test:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0),$$

or

$$\text{Branch B: } \lambda_c \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0);$$

6. verify microscopically whether the actual full realization preserves exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$.

LXXV. STRONGEST HONEST CONCLUSION

The strongest justified current statement is:

Dirac emergence is rigorously proved as a reduced graded spectral theorem,

the actual local CP^2 projector-defect geometry is naturally compatible with the S2 branch,

and the full local witness problem is reduced to the finite coefficient-point audit $(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O)$.

The actual remaining microscopic frontier is no longer conceptual. It is the extraction of the actual coefficient point from the complete background-dependent TECT Hessian.

LXXVI. EXACT NEXT TASK

The exact next task is:

derive $V^\mu(\xi)$ from the full Hessian, compute $\Lambda_{\parallel}^{(0)}$, and if it vanishes, compute λ_c .

Equivalently, the next decision tree is:

$$(1) \Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0, \quad (2) \text{ if no, } \lambda_c \stackrel{?}{\neq} 0.$$

Once this is done, the longitudinal branch is fixed.

LXXVII. FINAL SUMMARY

The plain scalar Bragg-pair route is no longer the correct mainline. Inside the current safe microscopic operator library,

$$B_0^{\text{scalar}} = 4uA^2 \mathbf{1}_2 \neq 0,$$

and canonical Class II cannot cancel the scalar identity channel.

Therefore the correct mainline is the graded $2 + 2$ bypass route.

The reduced local Dirac problem is already closed at the spectral level: the actual local CP^2 projector-defect geometry yields a rank-two bundle problem on

$$Q_B = (\mathbb{C}z_B)^\perp,$$

with a grading-odd first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

The natural local branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

The transverse witness sector is already reduced to

$$(\Lambda_E, \Lambda_O) \neq (0, 0).$$

The only remaining ambiguity is longitudinal and is fully reduced to the two-branch decision:

Branch A: $\Lambda_{\parallel}^{(0)} \neq 0$,

or

Branch B: $\Lambda_{\parallel}^{(0)} = 0$ with $\lambda_c \neq 0$.

In the broad actual-local class, Branch A is allowed. In the exact concrete currently constructed Class II branch, the explicitly realized branch is Branch B:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Therefore the full microscopic frontier is now exactly:

extract the actual coefficient point $(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O)$ from the complete background-dependent TECT Hessian.

LXXVIII. EXACT FULL-HESSIAN EXTRACTION FORMULA FOR THE ROUTE B LONGITUDINAL AUDIT

We now stop discussing the reduced theorem abstractly and derive the exact operator-level extraction formulas needed for the actual microscopic branch decision.

The goal is to reduce the full microscopic TECT linearization to the finite coefficient audit

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

LXXIX. MICROSCOPIC STARTING POINT AND HONEST SCOPE

Let the microscopic TECT field be the complex triplet

$$\Psi(x, t) \in \mathbb{C}^3,$$

with minimal local core

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi,$$

possibly supplemented by alignment and/or Class II enriched terms.

We do *not* assume that the minimal isotropic core alone is sufficient. Indeed, that route is already known to be insufficient for generating the full shell witness sector. The present task is different: given the *final complete microscopic operator*, derive the exact formulas that extract the actual local transport matrix and decide whether the theory lands on Branch A or Branch B.

LXXX. DEFECT BACKGROUND AND BUNDLE-VALUED FLUCTUATION SECTOR

Work in the frozen-amplitude projector regime

$$\Psi_B(\xi) = \rho_0 z_B(\xi), \quad P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1,$$

where the defect depends only on the normal coordinate ξ , while the worldvolume coordinates are denoted by

$$y^\mu, \quad \mu \in \{\parallel, 1, 2\}.$$

The physical tangent fluctuation lives in the rank-two complex bundle

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp, \quad \Pi_Q(\xi) = \mathbf{1}_3 - P_B(\xi).$$

Choose a local orthonormal frame

$$e(\xi) = (e_1(\xi), e_2(\xi))$$

for $Q_B(\xi)$, so that every tangent fluctuation can be written as

$$\eta(\xi, y) = e(\xi) \chi(\xi, y), \quad \chi(\xi, y) \in \mathbb{C}^2.$$

LXXXI. FULL HESSIAN AND EXACT FIRST-ORDER TRANSPORT EXTRACTION

Let

$$\mathfrak{E}[\Psi] = 0$$

be the full microscopic TECT equation of motion. Linearizing around Ψ_B gives the Hessian operator

$$\mathbb{H}_B := \left. \frac{\delta \mathfrak{E}}{\delta \Psi^\dagger} \right|_{\Psi_B}.$$

Because the background is translationally invariant along the worldvolume coordinates y^μ , the linearized operator admits the exact tangential derivative expansion

$$\mathbb{H}_B = \mathbb{H}_0(\xi, \partial_\xi) + \sum_\mu \mathbb{H}_1^\mu(\xi, \partial_\xi) (-i\partial_{y^\mu}) + O(\partial_y^2). \quad (596)$$

The exact first-order coefficient operators are therefore

$$\boxed{\mathbb{H}_1^\mu = \left. \frac{\partial \mathbb{H}_B}{\partial (-i\partial_{y^\mu})} \right|_{\partial_y=0}}. \quad (597)$$

Equation (597) is the first key point: to obtain the local transport kernel, one does *not* guess it. One differentiates the full Hessian with respect to the tangential derivative operator and then sets all worldvolume derivatives to zero.

LXXXII. REDUCTION TO THE SHELL-COMPATIBLE LOCAL TRANSPORT KERNEL

After restriction to the tangent bundle and rewriting in the local frame $e(\xi)$, define

$$\widetilde{\mathbb{H}}_1^\mu := e^\dagger \Pi_Q \mathbb{H}_1^\mu \Pi_Q e. \quad (598)$$

The shell-compatible local transport coefficient is the multiplication part of $\widetilde{\mathbb{H}}_1^\mu$ after all ∂_ξ -derivatives are pushed, by integration by parts, onto the low-mode profiles. Equivalently, the actual reduced mixed symbol is obtained directly from the even/odd low modes.

Let

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2,$$

be the even/odd low transverse modes. Then the reduced first-order matrices are

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, \mathbb{H}_1^\mu u_b^- \rangle_\xi}, \quad (599)$$

and the reduced principal symbol is

$$A_1(p) = p_\mu M^\mu. \quad (600)$$

Thus the microscopic audit is finite once the operators $\mathbb{H}_1^\mu$ are known.

LXXXIII. REFLECTION SYMMETRY AND GRADING-ODDNESS

Assume a reflection-symmetric defect:

$$P_B(-\xi) = P_B(\xi).$$

Then

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi).$$

Hence every leading transport contribution that is linear in the defect gradient is odd under $\xi \mapsto -\xi$:

$$\mathbb{H}_1^\mu(-\xi) = -\mathbb{H}_1^\mu(\xi). \quad (601)$$

Therefore the same-parity matrix elements vanish:

$$\langle u_a^\pm, \mathbb{H}_1^\mu u_b^\pm \rangle_\xi = 0,$$

while the opposite-parity matrix elements

$$\langle u_a^+, \mathbb{H}_1^\mu u_b^- \rangle_\xi$$

are generically allowed. This is exactly the microscopic origin of the grading-odd $2 + 2$ reduced Dirac structure.

LXXXIV. EXACT FACTORIZATION THROUGH THE DEFECT GRADIENT

For the explicit local representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi),$$

one has

$$\partial_\xi z_B = \theta'(\xi) e_1(\xi), \quad \partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Therefore every leading local transport term linear in the defect gradient must factor through $\theta'(\xi)$, and the restricted transport kernel takes the form

$$\boxed{V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi))}, \quad (602)$$

where

$$\mathcal{M}^\mu(\theta) \in \text{Herm}(2).$$

Equivalently, on the open set where $\theta'(\xi) \neq 0$,

$$\boxed{\mathcal{M}^\mu(\theta(\xi)) = \theta'(\xi)^{-1} V^\mu(\xi)}. \quad (603)$$

This is the second key point: the actual coefficient-point audit is the audit of the Hermitian 2×2 matrix-valued function $\mathcal{M}^\mu(\theta)$.

LXXXV. PAULI DECOMPOSITION AND EXACT INVARIANT COEFFICIENT FORMULAS

Because $\mathcal{M}^\mu(\theta)$ is Hermitian,

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i. \quad (604)$$

Now impose the S_2 -compatible invariant decomposition:

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a, \quad a = 1, 2. \quad (605)$$

Assuming the basis tensors are orthonormal in the standard way, the scalar profile functions are extracted exactly by traces:

$$\boxed{h_0(\theta) = \frac{1}{2} \text{Tr}[n_\mu \mathcal{M}^\mu(\theta)]}, \quad (606)$$

$$\boxed{h_\parallel(\theta) = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(\theta)]}, \quad (607)$$

$$\boxed{h_E(\theta) = \frac{1}{2} E_{\mu a} \text{Tr}[\sigma_a \mathcal{M}^\mu(\theta)]}, \quad (608)$$

$$\boxed{h_O(\theta) = \frac{1}{2} J_{\mu a} \text{Tr}[\sigma_a \mathcal{M}^\mu(\theta)]}. \quad (609)$$

LXXXVI. EXACT BRANCH A COEFFICIENT

Expand the longitudinal invariant near $\theta = 0$:

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + \Lambda_{\parallel}^{(2)}\theta^2 + O(\theta^4). \quad (610)$$

Then the exact constant-leading longitudinal coefficient is

$$\Lambda_{\parallel}^{(0)} = h_{\parallel}(0) = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)]. \quad (611)$$

This is the exact Branch A test. No further reduction is needed.
Likewise, the transverse constant-leading coefficients are

$$\Lambda_E = h_E(0), \quad \Lambda_O = h_O(0). \quad (612)$$

LXXXVII. REDUCED WITNESS FORMULAS

Define the even-odd weight

$$w(\xi) := f_+(\xi)f_-(\xi)\theta'(\xi).$$

Then the reduced witness coefficients are

$$\lambda_{\parallel} = \int d\xi w(\xi) h_{\parallel}(\theta(\xi)), \quad (613)$$

$$\alpha = \int d\xi w(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi w(\xi) h_O(\theta(\xi)). \quad (614)$$

On Branch A,

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + \Lambda_{\parallel}^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa), \quad (615)$$

where

$$I_{\theta} := \int d\xi f_+ f_- \theta', \quad I_{\theta^2 \theta'} := \int d\xi f_+ f_- \theta^2 \theta'.$$

Therefore

$$\Lambda_{\parallel}^{(0)} \neq 0$$

immediately implies

$$\lambda_{\parallel} \neq 0$$

provided $I_{\theta} \neq 0$.

LXXXVIII. EXACT BRANCH B COEFFICIENT

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

we pass to the primitive longitudinal channel.

Define

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (616)$$

For the explicit representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (617)$$

The primitive target kernel is

$$K_{\text{prim}}^\mu(\xi) := \theta'(\xi) S_\xi n^\mu. \quad (618)$$

Now define the weighted pairing

$$\langle A, B \rangle_w := \int d\xi f_+(\xi) f_-(\xi) \text{Tr}[A(\xi)^\dagger B(\xi)]. \quad (619)$$

Then the exact primitive coefficient extracted from the actual transport kernel is

$$\lambda_c = \frac{\langle K_{\text{prim}}^\mu, V_{\parallel, \text{act}, \mu} \rangle_w}{\langle K_{\text{prim}}^\mu, K_{\text{prim}, \mu} \rangle_w}. \quad (620)$$

Equivalently,

$$V_{\parallel, \text{act}}^\mu(\xi) = \lambda_c \theta'(\xi) S_\xi n^\mu + V_{\parallel, \text{rem}}^\mu(\xi), \quad \langle K_{\text{prim}}^\mu, V_{\parallel, \text{rem}, \mu} \rangle_w = 0. \quad (621)$$

Using (617), the projected primitive piece becomes

$$V_{\parallel, \text{proj}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (622)$$

Hence the reduced longitudinal witness on Branch B is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (623)$$

Thus the exact Branch B decision criterion is simply

$$\lambda_c \neq 0.$$

LXXXIX. UNIFIED THEOREM-LEVEL EXTRACTION STATEMENT

[Exact full-Hessian branch-decision formula] Assume the full microscopic TECT Hessian around the projector defect admits the shell-compatible local first-order reduction

$$\mathbb{H}_B = \mathbb{H}_0 - iV^\mu(\xi) \partial_{y^\mu} + O(\partial_y^2), \quad V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Then:

1. the exact Branch A coefficient is

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_\mu \mathcal{M}^\mu(0)];$$

2. if $\Lambda_{\parallel}^{(0)} \neq 0$, the longitudinal witness survives on the I_θ -branch;
3. if $\Lambda_{\parallel}^{(0)} = 0$, the exact Branch B coefficient is given by the weighted projection formula (620);

4. if $\lambda_c \neq 0$, then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0;$$

5. if both $\Lambda_{\parallel}^{(0)}$ and λ_c vanish, then the currently established branch library is exhausted and higher even longitudinal channels must be audited explicitly.

Items (1)–(2) follow immediately from the invariant decomposition (605) and the expansion (610). Items (3)–(4) follow from the orthogonal projection of the actual longitudinal transport onto the primitive target kernel K_{prim}^{μ} , together with (617). Item (5) is just the statement that no nonzero longitudinal seed remains inside the currently established two-branch library.

XC. WHAT THIS DOES AND DOES NOT CLOSE

This calculation closes the *extraction formalism* completely.
What is now fully fixed:

1. how to obtain $\mathbb{H}_1^{\mu}$ from the full Hessian;
2. how to pass from $\mathbb{H}_1^{\mu}$ to the local transport matrix $\mathcal{M}^{\mu}(\theta)$;
3. how to compute the Branch A scalar $\Lambda_{\parallel}^{(0)}$;
4. how to compute the Branch B scalar λ_c ;
5. how to reconstruct the reduced witnesses $\lambda_{\parallel}, \alpha, \beta$.

What is still not yet numerically or symbolically completed is only this:

insert the final complete microscopic TECT operator and perform the term-by-term Hessian audit.

So the frontier is no longer conceptual. It is a finite, explicit coefficient extraction problem.

XCI. FINAL SUMMARY: EXACT ROUTE B EXTRACTION FORMULA

Let

$$\mathbb{H}_B = \mathbb{H}_0 + \mathbb{H}_1^{\mu}(-i\partial_{y^{\mu}}) + O(\partial_y^2)$$

be the full microscopic Hessian around the projector defect background.

The exact first-order transport operator is

$$\mathbb{H}_1^{\mu} = \left. \frac{\partial \mathbb{H}_B}{\partial(-i\partial_{y^{\mu}})} \right|_{\partial_y=0}.$$

After restriction to the rank-two tangent bundle and local low-mode reduction, the transport kernel has the form

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)), \quad \mathcal{M}^{\mu}(\theta) \in \text{Herm}(2).$$

Its invariant decomposition is

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}{}_a \sigma_a + h_O(\theta) J^{\mu}{}_a \sigma_a.$$

Therefore the exact Branch A coefficient is

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2} \text{Tr}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}(0)].$$

If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

the longitudinal witness survives immediately on the I_{θ} -branch.

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

define

$$\mathcal{T}_{\xi} = i[P_B, \partial_{\xi} P_B], \quad S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}, \quad K_{\text{prim}}^{\mu} = \theta' S_{\xi} n^{\mu}.$$

Then the exact primitive coefficient is

$$\lambda_c = \frac{\langle K_{\text{prim}}^{\mu}, V_{\parallel, \text{act}, \mu} \rangle_w}{\langle K_{\text{prim}}^{\mu}, K_{\text{prim}, \mu} \rangle_w}.$$

For the explicit defect representative,

$$S_{\xi} = \frac{1}{2} \theta'^2 \sigma_3,$$

hence

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Thus the exact two-branch decision protocol is:

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0, \quad \text{if not, } \lambda_c \stackrel{?}{\neq} 0.$$

What remains open is no longer the logical structure of the audit, but only the explicit term-by-term insertion of the final complete microscopic TECT operator into this extraction formula.

XCII. TERM-BY-TERM OPERATOR AUDIT FOR THE ACTUAL ROUTE B MAINLINE

We now freeze the no-loop rule and perform the actual source decomposition of the strongest currently explicit microscopic branch.

The honest starting point is not the unknown final complete TECT action in full generality, but the strongest explicit branch presently constructed:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (624)$$

The shell-/defect-adapted quadratic operator is organized as

$$\Gamma^{(2)}[\bar{\Phi}_{\text{min}}] = H_B(\xi) - iV^{\mu}(\xi)\partial_{y^{\mu}} + W^{\mu\nu}(\xi)\partial_{y^{\mu}}\partial_{y^{\nu}} + \cdots, \quad (625)$$

with

$$V^{\mu}(\xi) = V_{\text{grad}}^{\mu}(\xi) + V_{\text{ani}}^{\mu}(\xi) + V_{\text{proj}}^{\mu}(\xi) + V_{\text{med}}^{\mu}(\xi). \quad (626)$$

The potential sector contributes only to H_B and does not enter V^{μ} .

XCIII. SECTOR-BY-SECTOR STATUS

A. Minimal isotropic core

For the isotropic Brazovskii bare kernel,

$$h_0(q) = r_0 + Z|q|^2 + Y|q|^4, \quad (627)$$

one has

$$\partial_{q_i} h_0(q) = 2(Z + 2Y|q|^2)q_i. \quad (628)$$

At the Brazovskii shell,

$$|q| = q_*, \quad q_*^2 = -\frac{Z}{2Y}, \quad (629)$$

hence

$$\partial_{q_i} h_0(q) \Big|_{|q|=q_*} = 0. \quad (630)$$

Therefore the minimal isotropic first-order shell transport vanishes on-shell.

So the minimal isotropic core does not by itself generate a nonzero constant-leading longitudinal coefficient $\Lambda_{\parallel}^{(0)}$.

B. Alignment sector

On the locked branch,

$$\Psi = z\phi, \quad P = zz^\dagger = \text{const}, \quad (631)$$

so

$$\partial_i P = 0. \quad (632)$$

Hence the refined alignment term vanishes identically on the locked sector and contributes neither to the scalar Bragg cancellation nor to the odd first-order transport kernel:

$$V_{\text{align}}^\mu(\xi) = 0. \quad (633)$$

C. Projector / gauge-fixing sector

Because the physical tangent fluctuations live in the moving bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp,$$

we must write

$$\eta(\xi, y) = \eta^a(\xi, y)e_a(\xi).$$

Then

$$\partial_\xi \eta = (\partial_\xi \eta^a)e_a + \eta^a \partial_\xi e_a, \quad (634)$$

so bundle-connection terms are unavoidable.

Therefore the projector/gauge-fixing sector contributes

$$\delta^2 \Gamma_{\text{gf/proj}} \rightsquigarrow H_B^{(\text{proj})} \oplus V_{\text{proj}}^\mu(\xi) \oplus W_{\text{proj}}^{\mu\nu}(\xi). \quad (635)$$

This is the main currently available candidate for a genuine Branch A source, namely for a nonzero constant-leading coefficient in the invariant component

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + \Lambda_{\parallel}^{(2)}\theta^2 + O(\theta^4).$$

D. Mediator / Class II sector

Inside each Bloch fiber, the heavy mediator is eliminated by the Schur complement:

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k) H_{XX}(k)^{-1} \mathcal{C}(k)^\dagger, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k). \quad (636)$$

In the heavy-mass expansion,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}). \quad (637)$$

After shell projection,

$$H_{\text{shell}}(p) = H_{\text{shell}}(0) + p_i K_i + O(|p|^2), \quad (638)$$

with

$$K_i = K_i^{(\Psi\Psi)} + K_i^{(\text{II})} + O(M_X^{-4}), \quad (639)$$

where

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}. \quad (640)$$

Thus the mediator sector is the strongest explicit currently calculable source of the nontrivial first-order shell tensor.

XCIV. WHAT IS ALREADY DECIDED FOR THE LONGITUDINAL BRANCH

The total constant-leading longitudinal coefficient decomposes as

$$\Lambda_{\parallel}^{(0)} = \Lambda_{\parallel, \Psi\Psi}^{(0)} + \Lambda_{\parallel, \text{ani}}^{(0)} + \Lambda_{\parallel, \text{proj}}^{(0)} + \Lambda_{\parallel, \text{med}}^{(0)}. \quad (641)$$

At the current explicit level, the following are already fixed:

$$\Lambda_{\parallel, \text{min}}^{(0)} = 0, \quad \Lambda_{\parallel, \text{align}}^{(0)} = 0. \quad (642)$$

Moreover, in the exact concrete Class II-compatible local branch,

$$\Lambda_{\parallel, \text{med}}^{(0)} = 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (643)$$

Therefore, within the strongest explicit currently constructed branch, the concrete mediator realization is Branch B, not Branch A.

XCV. IMMEDIATE CONSEQUENCE

Hence the actual remaining candidate sources for

$$\Lambda_{\parallel}^{(0)} \neq 0$$

are reduced to:

1. the projector/gauge-fixing transport sector;
2. any still-unspecified background-dependent anisotropic pieces inside L_{Ψ}^{ref} .

By contrast, if one stays inside the exact currently constructed concrete Class II branch, the longitudinal audit should proceed directly through

$$\lambda_c \stackrel{?}{\neq} 0 \quad (644)$$

rather than through

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0. \quad (645)$$

XCVI. ACTUAL NEXT CALCULATION

Thus the next explicit computation is not an abstract theorem. It is the operator derivative audit:

$$[\delta^2 \Gamma_{\text{grad}}]_{1y}, \quad [\delta^2 \Gamma_{\text{ani}}]_{1y}, \quad [\delta^2 \Gamma_{\text{gf/proj}}]_{1y}, \quad [\delta^2 \Gamma_{\text{med}}]_{1y}, \quad (646)$$

followed by

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle. \quad (647)$$

If one works with the explicit Bloch branch, the practical order is:

$$H_{\Psi\Psi}(k), \mathcal{J}(k), \mathcal{K}(k) \implies H_{\text{eff}}(k) \implies H_{\text{shell}}(p) \implies K_i \implies P_* K_i P_* \implies (\lambda_{\parallel}, \alpha, \beta). \quad (648)$$

This is now the unique correct mainline.

XCVII. FINAL SUMMARY

The current operator-level status is:

$$\Gamma^{(2)}[\bar{\Phi}_{\text{min}}] = H_B(\xi) - iV^\mu(\xi)\partial_{y^\mu} + W^{\mu\nu}(\xi)\partial_{y^\mu}\partial_{y^\nu} + \dots, \quad V^\mu = V_{\text{grad}}^\mu + V_{\text{ani}}^\mu + V_{\text{proj}}^\mu + V_{\text{med}}^\mu. \quad (649)$$

The potential sector does not contribute to V^μ .

At the currently explicit level:

$$V_{\text{min}}^\mu \text{ does not generate on-shell isotropic first-order shell transport,} \quad (650)$$

$$V_{\text{align}}^\mu = 0 \quad \text{on the locked branch,} \quad (651)$$

while

$$V_{\text{proj}}^\mu \text{ and } V_{\text{med}}^\mu \text{ are the only nontrivial source sectors for the odd transport kernel.} \quad (652)$$

For the strongest explicit periodic microscopic branch,

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi], \quad (653)$$

the Bloch-fiber elimination of the heavy mediator gives

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k)H_{XX}(k)^{-1}\mathcal{C}(k)^\dagger, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad (654)$$

and the shell-linear coefficient is

$$K_i = K_i^{(\Psi\Psi)} - \frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*} + O(M_X^{-4}). \quad (655)$$

The total constant-leading longitudinal coefficient satisfies

$$\Lambda_{\parallel}^{(0)} = \Lambda_{\parallel, \Psi\Psi}^{(0)} + \Lambda_{\parallel, \text{ani}}^{(0)} + \Lambda_{\parallel, \text{proj}}^{(0)} + \Lambda_{\parallel, \text{med}}^{(0)}. \quad (656)$$

What is already fixed is:

$$\Lambda_{\parallel, \text{min}}^{(0)} = 0, \quad \Lambda_{\parallel, \text{align}}^{(0)} = 0. \quad (657)$$

Moreover, in the exact currently constructed concrete Class II-compatible branch,

$$\Lambda_{\parallel, \text{med}}^{(0)} = 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (658)$$

Therefore the explicit currently constructed mediator branch is Branch B.

Hence the only remaining candidate sources for a genuine Branch A realization are:

$V_{\text{proj}}^\mu \text{ and/or still-unspecified background-dependent anisotropic pieces in } L_{\Psi}^{\text{ref}}.$

(659)

So the actual next computation is:

$H_{\Psi\Psi}(k), \mathcal{J}(k), \mathcal{K}(k) \implies H_{\text{eff}}(k) \implies H_{\text{shell}}(p) \implies K_i \implies P_* K_i P_* \implies (\lambda_{\parallel}, \alpha, \beta).$

(660)

XCVIII. FINITE HARMONIC-MATRIX TEMPLATE FOR THE FULL PERIODIC ROUTE B AUDIT

XCIX. GOAL

The remaining microscopic task is no longer a conceptual theorem. It is the explicit construction of the Bloch-harmonic matrix of the refined locked TECT branch with the Class II mediator, followed by shell projection and first-derivative extraction.

The correct no-loop sequence is

$$\text{full periodic microscopic operator} \rightarrow \text{Bloch fiber} \rightarrow \text{shell truncation} \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

We now write the exact finite-dimensional template needed for that computation.

C. FOLDED BRAGG CONVENTION AND BCC BACKGROUND

We work in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|. \quad (661)$$

Choose the first BCC reciprocal shell

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |\mathcal{S}_{12}| = 12, \quad (662)$$

and the locked periodic background

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger. \quad (663)$$

Let

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs}, \quad \mathbb{C}^3 = \mathbb{C}z_0 \oplus Q_{z_0}, \quad (664)$$

be the internal basis adapted to the locked decomposition.

CI. BLOCH-HARMONIC EXPANSION OF THE FLUCTUATIONS

For the triplet fluctuation, write

$$\delta \Psi_k(x) = e^{ik \cdot x} \sum_{G \in \mathcal{G}} \sum_{r=0}^2 c_{G,r} u_r e^{iG \cdot x}, \quad (665)$$

where $\mathcal{G}$ is the reciprocal lattice generated by the periodic BCC background.

For the Class II mediator,

$$\delta X_i^a(x) = e^{ik \cdot x} \sum_{G \in \mathcal{G}} x_{G,i,a} e^{iG \cdot x}. \quad (666)$$

Thus the Bloch fiber acts on the coefficient vector

$$\mathbf{v}(k) = \begin{pmatrix} c_{G,r} \\ x_{G,i,a} \end{pmatrix}_{G \in \mathcal{G}}, \quad (667)$$

and has the block form

$$\boxed{\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}}. \quad (668)$$

CII. TRIPLET BLOCK $H_{\Psi\Psi}(k)$

A. Universal structure

The refined locked triplet block always splits as

$$H_{\Psi\Psi}(k) = H_{\Psi\Psi}^{\text{bare}}(k) + H_{\Psi\Psi}^{\text{per}}(k), \quad (669)$$

where the bare translationally invariant part is diagonal in reciprocal harmonics and the periodic background produces convolution terms.

In the adapted basis (G, r) , write

$$(H_{\Psi\Psi}(k))_{(G,r),(G',s)} = \delta_{G,G'} h_{rs}^{(0)}(k+G) + U_{G-G'}^{rs} + A_{G,G'}^{rs}(k). \quad (670)$$

Here:

1. $h_{rs}^{(0)}(q)$ is the bare triplet symbol coming from the refined locked kinetic part;
2. $U_{G-G'}^{rs}$ is the periodic Hessian generated by the background Ψ_0 ;
3. $A_{G,G'}^{rs}(k)$ collects explicitly k -dependent refined derivative/anisotropic terms.

B. Minimal isotropic benchmark

For the minimal isotropic triplet Brazovskii core,

$$h_{rs}^{(0)}(q) = h_0(q) \delta_{rs}, \quad h_0(q) = r_* + K(|q|^2 - q_*^2)^2, \quad (671)$$

so

$$\partial_{q_i} h_0(q) = 4K(|q|^2 - q_*^2) q_i. \quad (672)$$

Hence on the shell $|q| = q_*$,

$$\boxed{\partial_{q_i} h_0(q)|_{|q|=q_*} = 0.} \quad (673)$$

This is the operator-level reason why the bare isotropic scalar part is not the load-bearing source for the first-order shell witness.

CIII. MEDIATOR BLOCK $H_{XX}(k)$

If the Class II mediator is purely heavy and algebraic at leading order, then

$$(H_{XX}(k))_{(G,i,a),(G',j,b)} = M_X^2 \delta_{G,G'} \delta_{ij} \delta^{ab} + D_{ij}^{ab}(k+G) \delta_{G,G'} + V_{ij;G-G'}^{ab}, \quad (674)$$

where the last two terms are optional refinements.

At the strict minimal heavy-mediator level,

$$\boxed{H_{XX}(k) = M_X^2 \mathbf{1}_X.} \quad (675)$$

If gradient or periodic corrections of the mediator are present, they can be included in D_{ij}^{ab} and $V_{ij;G-G'}^{ab}$, but they do not change the Schur-complement logic below.

CIV. MIXED BLOCK $H_{\Psi X}(k)$

The Class II UV action is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (676)$$

Therefore the mixed fiber block is

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad (677)$$

with

$$\mathcal{C}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k). \quad (678)$$

The harmonic matrix elements are obtained by unit-cell projection:

$$\mathcal{J}_{(G,r),(G',i,a)}(k) = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-i(k+G)\cdot x} u_r^\dagger \left(\frac{\delta J_i^a}{\delta \Psi} \right)_{\Psi_0} e^{i(k+G')\cdot x}, \quad (679)$$

and similarly

$$\mathcal{K}_{(G,r),(G',i,a)}(k) = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-i(k+G)\cdot x} u_r^\dagger \left(\frac{\delta K_i^a}{\delta \Psi} \right)_{\Psi_0, P_0} e^{i(k+G')\cdot x}. \quad (680)$$

These matrices are finite after truncation to a finite harmonic set.

CV. MEDIATOR ELIMINATION AND EFFECTIVE TRIPLET BLOCH OPERATOR

Eliminating the heavy mediator by the Schur complement gives

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (681)$$

With the shorthand $\mathcal{C}(k) = H_{\Psi X}(k)$, this is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k) H_{XX}(k)^{-1} \mathcal{C}(k)^\dagger. \quad (682)$$

At strict leading order in the heavy-mass expansion,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}). \quad (683)$$

CVI. NEAR-BRAGG SHELL TRUNCATION

At the folded Bragg point $k_* = G_*/2$, the two near-degenerate harmonics are

$$|+, a\rangle \equiv |G=0, a\rangle, \quad |-, a\rangle \equiv |G=-G_*, a\rangle, \quad a=1, 2, \quad (684)$$

where $a=1, 2$ spans the internal tangent doublet.

Thus the shell space is

$$\mathcal{V}_* = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\}, \quad \dim_{\mathbb{C}} \mathcal{V}_* = 4. \quad (685)$$

Let Π_{S_*} denote the harmonic projector onto this four-dimensional near-Bragg shell sector. Then the shell-truncated operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}. \quad (686)$$

CVII. DEFINITION OF THE FIRST-DERIVATIVE MATRICES K_i

Expand $H_{\text{shell}}(p)$ around $p = 0$:

$$H_{\text{shell}}(p) = H_{\text{shell}}(0) + p_i K_i + O(|p|^2). \quad (687)$$

Therefore

$$K_i = \left. \frac{\partial H_{\text{shell}}(p)}{\partial p_i} \right|_{p=0} = \Pi_{S_*} [\partial_{k_i} H_{\text{eff}}(k)]_{k=k_*} \Pi_{S_*}. \quad (688)$$

Using (683),

$$K_i = K_i^{(\Psi\Psi)} + K_i^{(\text{II})} + O(M_X^{-4}), \quad (689)$$

with

$$K_i^{(\Psi\Psi)} = \Pi_{S_*} [\partial_{k_i} H_{\Psi\Psi}(k)]_{k=k_*} \Pi_{S_*}, \quad (690)$$

and

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} [(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger)]_{k=k_*} \Pi_{S_*}. \quad (691)$$

This is the exact finite harmonic matrix formula required by the full Route B audit.

CVIII. WHAT VANISHES IMMEDIATELY AND WHAT CAN ACTUALLY SURVIVE

A. Bare isotropic scalar diagonal part

If

$$H_{\Psi\Psi}^{\text{bare}}(k) = \delta_{G,G'} h_0(k+G) \mathbf{1}_{\text{int}},$$

then by (673),

$$\Pi_{S_*} [\partial_{k_i} H_{\Psi\Psi}^{\text{bare}}(k)]_{k=k_*} \Pi_{S_*} = 0. \quad (692)$$

B. Purely scalar periodic potential part

If $U_{G-G'}^{rs}$ is k -independent, then

$$\partial_{k_i} U_{G-G'}^{rs} = 0, \quad (693)$$

hence such a purely scalar background-convolution piece contributes to $H_{\text{shell}}(0)$, but not to K_i .

Therefore the genuinely load-bearing sources for K_i must come from:

1. explicitly k -dependent refined internal/anisotropic pieces $A_{G,G'}^{rs}(k)$ inside $H_{\Psi\Psi}(k)$,
2. the mediator correction $K_i^{(\text{II})}$.

CIX. FINAL COEFFICIENT EXTRACTION

Let P_* be the projector onto the isolated 2×2 internal shell block after the shell-space identification. Then the actual microscopic coefficients are extracted by

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{V}_*} [\sigma_3 P_* (\hat{n}_i K_i) P_*], \quad (694)$$

$$\alpha = \frac{1}{2} \text{Tr}_{\mathcal{V}_*} [\sigma_1 P_* (e_{1i} K_i) P_* + \sigma_2 P_* (e_{2i} K_i) P_*], \quad (695)$$

$$\beta = \frac{1}{2} \text{Tr}_{\mathcal{V}_*} [\sigma_2 P_* (e_{1i} K_i) P_* - \sigma_1 P_* (e_{2i} K_i) P_*]. \quad (696)$$

Thus the full operator-construction program is now reduced to a finite matrix computation.

CX. EXACT LOGICAL CONSEQUENCE

The true remaining microscopic question is no longer “what kind of mechanism might work?” It is only this finite calculation:

$$H_{\Psi\Psi}(k), \quad \mathcal{J}(k), \quad \mathcal{K}(k) \implies H_{\text{eff}}(k) \implies H_{\text{shell}}(p) \implies K_i \implies (\lambda_{\parallel}, \alpha, \beta). \quad (697)$$

In particular, the present template shows that:

1. the bare isotropic scalar piece is not the load-bearing source of K_i ;
2. a purely k -independent scalar periodic potential is not the load-bearing source of K_i ;
3. the actual microscopic load-bearing sectors are the refined k -dependent internal triplet block and the Class II Schur-complement term.

So the problem is now genuinely finite and explicit.

CXI. FINAL SUMMARY

We have fixed the finite harmonic-matrix template required for the full Route B microscopic audit. The folded Bragg convention is

$$k_* = \frac{G_*}{2}, \quad k = k_* + p.$$

The locked 12-mode BCC background is

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger.$$

The Bloch fiber of the refined locked + Class II microscopic branch has the block form

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix},$$

with

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad H_{XX}(k) \simeq M_X^2 \mathbf{1}_X$$

at leading heavy-mediator order.

After mediator elimination,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k)H_{XX}(k)^{-1}\mathcal{C}(k)^\dagger, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

The shell space is the four-dimensional near-Bragg sector

$$\mathcal{V}_* = \text{span}\{|+, a\rangle, |-, a\rangle\}, \quad a = 1, 2,$$

with

$$|+, a\rangle \equiv |G = 0, a\rangle, \quad |-, a\rangle \equiv |G = -G_*, a\rangle.$$

The shell-truncated operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*},$$

and its first derivatives are

$$K_i = \Pi_{S_*} [\partial_{k_i} H_{\text{eff}}(k)]_{k=k_*} \Pi_{S_*}.$$

At leading order,

$$K_i = K_i^{(\Psi\Psi)} + K_i^{(\text{II})} + O(M_X^{-4}),$$

where

$$K_i^{(\Psi\Psi)} = \Pi_{S_*} [\partial_{k_i} H_{\Psi\Psi}(k)]_{k=k_*} \Pi_{S_*},$$

and

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} [(\partial_{k_i} \mathcal{C})\mathcal{C}^\dagger + \mathcal{C}(\partial_{k_i} \mathcal{C}^\dagger)]_{k=k_*} \Pi_{S_*}.$$

The key structural consequence is:

the bare isotropic scalar part does not provide the load-bearing K_i contribution,

a purely k -independent scalar periodic potential does not provide the load-bearing K_i contribution,

the actual microscopic load-bearing sources are the refined k -dependent internal block and the Class II Schur-complement term

Finally,

$$\lambda_{\parallel}, \alpha, \beta$$

are extracted from $P_* K_i P_*$ by the standard shell-projector trace formulas.

Therefore the remaining microscopic frontier is now reduced to a finite explicit matrix calculation:

$$H_{\Psi\Psi}(k), \mathcal{J}(k), \mathcal{K}(k) \implies H_{\text{eff}}(k) \implies H_{\text{shell}}(p) \implies K_i \implies (\lambda_{\parallel}, \alpha, \beta).$$

CXII. TERM-BY-TERM SURVIVAL TEST FOR $K_i^{(\Psi\Psi)}$

CXIII. GOAL

We now isolate, inside the triplet Bloch block

$$H_{\Psi\Psi}(k),$$

which terms can actually contribute to

$$K_i^{(\Psi\Psi)} = \Pi_{S_*} [\partial_{k_i} H_{\Psi\Psi}(k)]_{k=k_*} \Pi_{S_*}.$$

This is the exact next microscopic question, because the correct no-loop Route B program is already fixed as

periodic refined locked microscopic operator $\rightarrow$ Bloch fiber $\rightarrow$ shell truncation $\rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta)$.

CXIV. CANONICAL DECOMPOSITION OF THE TRIPLET BLOCH BLOCK

Write the triplet Bloch block as

$$H_{\Psi\Psi}(k) = H_{\text{bare}}(k) + H_{\text{per}}^{(0)} + H_{\text{per}}^{(\text{int})}(k), \quad (698)$$

where:

1. $H_{\text{bare}}(k)$ is the translation-invariant bare symbol;
2. $H_{\text{per}}^{(0)}$ is the k -independent periodic Hessian induced by the background;
3. $H_{\text{per}}^{(\text{int})}(k)$ is the genuinely refined k -dependent internal/anisotropic part.

Accordingly,

$$K_i^{(\Psi\Psi)} = K_i^{\text{bare}} + K_i^{\text{per},0} + K_i^{\text{per,int}}, \quad (699)$$

with

$$K_i^{\text{bare}} = \Pi_{S_*} [\partial_{k_i} H_{\text{bare}}(k)]_{k_*} \Pi_{S_*},$$

$$K_i^{\text{per},0} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(0)} \right]_{k_*} \Pi_{S_*},$$

$$K_i^{\text{per,int}} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(\text{int})}(k) \right]_{k_*} \Pi_{S_*}.$$

CXV. VANISHING OF THE BARE ISOTROPIC CONTRIBUTION

For the minimal isotropic Brazovskii symbol,

$$h_0(q) = r_* + K(|q|^2 - q_*^2)^2, \quad (700)$$

one has

$$\partial_{q_i} h_0(q) = 4K(|q|^2 - q_*^2)q_i. \quad (701)$$

At the selected shell,

$$|q| = q_*, \quad (702)$$

hence

$$\boxed{\partial_{q_i} h_0(q)|_{|q|=q_*} = 0.} \quad (703)$$

Equivalently, in the slow-envelope operator language, the first-order shell transport coefficient of

$$Z\nabla^2 - Y\nabla^4$$

is proportional to

$$2i(Z + 2Yq_*^2)G \cdot \nabla,$$

which vanishes exactly at

$$q_*^2 = -\frac{Z}{2Y}.$$

Therefore

$$\boxed{K_i^{\text{bare}} = 0.} \quad (704)$$

This is not approximate. It is an exact on-shell cancellation.

CXVI. VANISHING OF THE PURELY SCALAR PERIODIC CONVOLUTION CONTRIBUTION

Now consider the background-induced periodic Hessian contribution

$$H_{\text{per}}^{(0)},$$

coming from scalar periodic terms such as the quartic Hessian background potential. In the reciprocal-harmonic basis it has the generic form

$$(H_{\text{per}}^{(0)})_{(G,r),(G',s)} = U_{G-G'}^{rs}. \quad (705)$$

If the periodic background term is scalar in the internal doublet and contains no explicit Bloch momentum dependence, then

$$\partial_{k_i} U_{G-G'}^{rs} = 0. \quad (706)$$

Hence

$$\boxed{K_i^{\text{per},0} = 0.} \quad (707)$$

So a purely k -independent scalar periodic convolution can affect

$$H_{\text{shell}}(0),$$

for example the constant Bragg block, but it cannot by itself generate the first-order shell derivative matrix K_i .

CXVII. WHAT CAN SURVIVE: REFINED INTERNAL / ANISOTROPIC PIECES

The only triplet-sector source that can survive inside $K_i^{(\Psi\Psi)}$ is therefore the genuinely k -dependent refined block

$$H_{\text{per}}^{(\text{int})}(k).$$

Write its harmonic matrix elements as

$$(H_{\text{per}}^{(\text{int})}(k))_{(G,r),(G',s)} = A_{G,G'}^{rs}(k). \quad (708)$$

Then

$$\boxed{K_i^{\text{per,int}} = \Pi_{S_*} [\partial_{k_i} A_{G,G'}^{rs}(k)]_{k=k_*} \Pi_{S_*}.} \quad (709)$$

Therefore the full triplet contribution reduces to

$$\boxed{K_i^{(\Psi\Psi)} = \Pi_{S_*} [\partial_{k_i} H_{\text{per}}^{(\text{int})}(k)]_{k=k_*} \Pi_{S_*}.} \quad (710)$$

CXVIII. NECESSARY CONDITION FOR BRANCH A FROM THE TRIPLET SECTOR

Recall that Branch A requires a nonzero constant-leading longitudinal coefficient

$$\Lambda_{\parallel}^{(0)} \neq 0.$$

Inside the triplet sector, this can only happen if

$$H_{\text{per}}^{(\text{int})}(k)$$

contains a shell-compatible k -linear term whose shell projection produces a σ_3 -type longitudinal block.

In other words, a necessary condition for Branch A coming from the triplet block is:

$$\boxed{\Pi_{S_*} [\partial_{k_{\parallel}} H_{\text{per}}^{(\text{int})}(k)]_{k=k_*} \Pi_{S_*} \not\propto 0} \quad (711)$$

and, more specifically, its projected traceless internal part must have a nonzero σ_3 component.

If this fails, then the triplet block does not realize Branch A, and one must pass either to:

1. the projector/gauge-fixing transport sector, or
2. the Class II primitive longitudinal branch.

CXIX. TERM-BY-TERM AUDIT CRITERION

The practical finite audit is therefore:

$$H_{\Psi\Psi}(k) = \underbrace{H_{\text{bare}}(k)}_{\text{dead in } K_i} + \underbrace{H_{\text{per}}^{(0)}}_{\text{dead in } K_i} + \underbrace{H_{\text{per}}^{(\text{int})}(k)}_{\text{only triplet-sector live source}}. \quad (712)$$

Hence the triplet-sector Route B computation is reduced to the finite question:

What are the explicit k -linear harmonic matrix elements of $H_{\text{per}}^{(\text{int})}(k)$ at $k = k_*$?

(713)

This is now the exact next subtask.

CXX. CONSEQUENCE FOR THE FULL OPERATOR AUDIT

Combining the triplet result with the mediator Schur-complement formula, the full first-derivative shell matrix satisfies

$$K_i = \underbrace{K_i^{\text{per,int}}}_{\text{triplet refined source}} + \underbrace{K_i^{(\text{II})}}_{\text{Class II Schur source}} + O(M_X^{-4}). \quad (714)$$

Thus the load-bearing microscopic sources are now sharply localized:

(I) refined k -dependent internal/anisotropic triplet block,

(715)

(II) Class II mediator Schur complement.

(716)

No other term in the currently explicit minimal/refined decomposition can produce the required first-order shell witness by itself.

CXXI. HONEST CONCLUSION

At this stage, the strongest honest statement is:

$K_i^{\text{bare}} = 0, \quad K_i^{\text{per},0} = 0,$

(717)

so

$K_i^{(\Psi\Psi)} = K_i^{\text{per,int}}.$

(718)

Therefore the remaining triplet-sector microscopic frontier is no longer conceptual. It is the explicit extraction of the k -linear refined internal harmonic block

$$H_{\text{per}}^{(\text{int})}(k)$$

at the folded Bragg point k_* .

CXXII. FINAL SUMMARY

The triplet Bloch block decomposes as

$$H_{\Psi\Psi}(k) = H_{\text{bare}}(k) + H_{\text{per}}^{(0)} + H_{\text{per}}^{(\text{int})}(k).$$

Accordingly,

$$K_i^{(\Psi\Psi)} = \Pi_{S_*} [\partial_{k_i} H_{\Psi\Psi}(k)]_{k_*} \Pi_{S_*} = K_i^{\text{bare}} + K_i^{\text{per},0} + K_i^{\text{per,int}}.$$

For the minimal isotropic Brazovskii symbol,

$$h_0(q) = r_* + K(|q|^2 - q_*^2)^2,$$

one has

$$\partial_{q_i} h_0(q) \Big|_{|q|=q_*} = 0,$$

equivalently the slow-envelope first-order coefficient

$$2i(Z + 2Yq_*^2)G \cdot \nabla$$

vanishes exactly at

$$q_*^2 = -\frac{Z}{2Y}.$$

Hence

$$\boxed{K_i^{\text{bare}} = 0.}$$

If the periodic background Hessian is k -independent,

$$(H_{\text{per}}^{(0)})_{(G,r),(G',s)} = U_{G-G'}^{rs}, \quad \partial_{k_i} U_{G-G'}^{rs} = 0,$$

then

$$\boxed{K_i^{\text{per},0} = 0.}$$

Therefore the only triplet-sector contribution that can survive in $K_i^{(\Psi\Psi)}$ is the refined k -dependent internal/anisotropic block:

$$\boxed{K_i^{(\Psi\Psi)} = K_i^{\text{per,int}} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(\text{int})}(k) \right]_{k_*} \Pi_{S_*}.$$

Thus the full first-derivative shell matrix is reduced to

$$\boxed{K_i = K_i^{\text{per,int}} + K_i^{(\text{II})} + O(M_X^{-4}),}$$

where the only load-bearing microscopic sources are:

$$\boxed{\text{(I) refined } k\text{-dependent internal/anisotropic triplet block,}}$$

$$\boxed{\text{(II) Class II mediator Schur complement.}}$$

Hence the next exact microscopic subtask is to write the explicit harmonic matrix elements of

$$H_{\text{per}}^{(\text{int})}(k)$$

and extract their k -linear shell projection at $k = k_*$.

CXXIII. MINIMAL k -LINEAR HARMONIC TENSOR FORM OF THE REFINED INTERNAL BLOCK

CXXIV. GOAL

We now determine the minimal shell-compatible k -linear structure of the refined internal/anisotropic Bloch block

$$H_{\text{per}}^{(\text{int})}(k)$$

that can generate the three reduced S2 witness coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta.$$

The previous audit already established that

$$K_i^{(\Psi\Psi)} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(\text{int})}(k) \right]_{k=k_*} \Pi_{S_*},$$

because both the bare isotropic part and the k -independent scalar periodic convolution are dead in the first shell derivative.

So the remaining triplet-sector question is purely finite: what is the minimal k -linear tensor form of

$$H_{\text{per}}^{(\text{int})}(k)$$

at the folded Bragg point k_* ?

CXXV. SHELL-ADAPTED DECOMPOSITION

Fix the adapted frame

$$k = k_* + p, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2, \quad \hat{n} = \frac{G_*}{|G_*|}.$$

After projection to the isolated shell doublet block H_* , every Hermitian 2×2 first-order symbol has a unique Pauli decomposition

$$P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = \mathcal{E}_0 + p_i \left(u_i \mathbf{1}_2 + v_i^a \sigma_a \right) + O(|p|^2), \quad (719)$$

with real coefficients u_i, v_i^a .

The scalar term $u_i \mathbf{1}_2$ is irrelevant for the local Weyl data. Therefore the actual reduced transport content is carried by

$$p_i v_i^a \sigma_a. \quad (720)$$

CXXVI. S2 SYMMETRY REDUCTION

The worldvolume momentum representation splits as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\},$$

while the traceless internal fiber splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

Thus the most general symmetry-compatible intertwiner is exactly the natural S2 form:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (721)$$

Equivalently, the shell-derivative matrices must satisfy

$$P_*(\hat{n}_i K_i^{\text{per, int}}) P_* = \lambda_{\parallel} \sigma_3, \quad (722)$$

$$P_*(e_{1i} K_i^{\text{per, int}}) P_* = \alpha \sigma_1 - \beta \sigma_2, \quad (723)$$

$$P_*(e_{2i} K_i^{\text{per, int}}) P_* = \beta \sigma_1 + \alpha \sigma_2. \quad (724)$$

CXXVII. MINIMAL k -LINEAR HARMONIC TENSOR ANSATZ

The unique minimal shell-compatible ansatz for the refined internal block is therefore

$$\boxed{H_{\text{per}}^{(\text{int})}(k) = H_0 + (k - k_*)_i \mathcal{N}_i^a s^a + O(|k - k_*|^2)}, \quad (725)$$

where s^a denotes the internal Pauli basis before shell projection and $\mathcal{N}_i^a$ is a finite harmonic operator on the shell carrier space.

After shell projection, define

$$\boxed{\tilde{\mathcal{N}}_i^a := P_* \mathcal{N}_i^a P_*}. \quad (726)$$

Then

$$P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = P_* H_0 P_* + p_i \tilde{\mathcal{N}}_i^a \sigma_a + O(|p|^2). \quad (727)$$

Thus

$$\boxed{K_i^{\text{per}, \text{int}} = \tilde{\mathcal{N}}_i^a \sigma_a}. \quad (728)$$

CXXVIII. MINIMAL SHELL TENSOR BASIS

To realize the S2 target exactly, $\tilde{\mathcal{N}}_i^a$ must lie in the three-dimensional shell-adapted tensor span

$$\boxed{\tilde{\mathcal{N}}_i^a = \lambda_{\parallel} \hat{n}_i \delta^{a3} + \alpha (e_{1i} \delta^{a1} + e_{2i} \delta^{a2}) + \beta (-e_{1i} \delta^{a2} + e_{2i} \delta^{a1})}. \quad (729)$$

Equivalently,

$$\tilde{\mathcal{N}}_{\parallel}^a = \lambda_{\parallel} \delta^{a3}, \quad \tilde{\mathcal{N}}_1^a = \alpha \delta^{a1} - \beta \delta^{a2}, \quad \tilde{\mathcal{N}}_2^a = \beta \delta^{a1} + \alpha \delta^{a2}. \quad (730)$$

Substituting (729) into (728) gives

$$p_i K_i^{\text{per}, \text{int}} = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2), \quad (731)$$

which is exactly the required reduced symbol.

CXXIX. MICROSCOPIC INTERPRETATION OF THE THREE CHANNELS

The ansatz (729) separates the three physically distinct microscopic channels.

A. Longitudinal singlet channel

The coefficient $\lambda_{\parallel}$ multiplies

$$\hat{n}_i \sigma_3.$$

This is the unique shell singlet-to-singlet intertwiner and therefore the only possible first-order longitudinal source inside the S2 branch.

In the local projector-defect geometry, the canonical microscopic realization of this channel is the commutator-square seed

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad (732)$$

which leads to the primitive transport seed

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3. \quad (733)$$

Therefore a triplet-sector Branch A realization requires that the projected k -linear refined block itself already produces a nonzero coefficient of $\hat{n}_i \sigma_3$. If not, the theory must fall back to the primitive Branch B channel.

B. Transverse even doublet channel

The coefficient α multiplies

$$e_{1i}\sigma_1 + e_{2i}\sigma_2.$$

This is the even transverse intertwiner and is the minimal source of the term

$$p_1\sigma_1 + p_2\sigma_2.$$

C. Transverse odd doublet channel

The coefficient β multiplies

$$-e_{1i}\sigma_2 + e_{2i}\sigma_1.$$

This is the odd transverse intertwiner and is the minimal source of the term

$$-p_2\sigma_1 + p_1\sigma_2.$$

CXXX. DIRECT PROJECTOR FORMULAS

Because the projected first-order symbol is already in the Pauli basis, the coefficients are extracted by

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{H_*} \left(\sigma_3 P_* (\hat{n}_i K_i^{\text{per,int}}) P_* \right), \quad (734)$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}_{H_*} \left(\sigma_a P_* (e_{ai} K_i^{\text{per,int}}) P_* \right), \quad (735)$$

$$\beta = \frac{1}{4} \left[\text{Tr}_{H_*} \left(\sigma_2 P_* (e_{1i} K_i^{\text{per,int}}) P_* \right) - \text{Tr}_{H_*} \left(\sigma_1 P_* (e_{2i} K_i^{\text{per,int}}) P_* \right) \right]. \quad (736)$$

CXXXI. NECESSARY AND SUFFICIENT TENSOR CRITERION

Inside the triplet refined block, the S2 witnesses survive iff the projected k -linear tensor

$$\tilde{\mathcal{N}}_i^a$$

has nonzero overlap with the three shell target tensors

$$\hat{n}_i \delta^{a3}, \quad e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, \quad -e_{1i} \delta^{a2} + e_{2i} \delta^{a1}.$$

Equivalently:

$$\lambda_{\parallel} \neq 0 \iff \tilde{\mathcal{N}}_i^a \not\propto \hat{n}_i \delta^{a3}, \quad (737)$$

$$(\alpha, \beta) \neq (0, 0) \iff \tilde{\mathcal{N}}_i^a \not\propto \text{span} \{ e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, -e_{1i} \delta^{a2} + e_{2i} \delta^{a1} \}. \quad (738)$$

Thus the triplet-sector audit is reduced to one finite tensor-overlap question:

$$\boxed{\text{Does the actual } k\text{-linear refined internal block contain the shell tensor basis } \{ \hat{n}_i \delta^{a3}, e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, -e_{1i} \delta^{a2} + e_{2i} \delta^{a1} \} ?} \quad (739)$$

CXXXII. HONEST CONCLUSION

At this stage the triplet refined block is no longer mysterious.

Its minimal shell-compatible k -linear form is exactly the three-parameter tensor ansatz (729), and nothing else is needed at first order.

Therefore the only remaining microscopic question is not whether some abstract S2 symbol might exist, but whether the actual harmonic matrix elements of the final refined TECT linearization produce nonzero coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

in that explicit shell tensor basis.

CXXXIII. FINAL SUMMARY

The remaining triplet-sector contribution to the first shell derivative is

$$K_i^{(\Psi\Psi)} = K_i^{\text{per,int}} = \Pi_{S_*} \left[\partial_{k_i} H_{\text{per}}^{(\text{int})}(k) \right]_{k=k_*} \Pi_{S_*}.$$

After projection to the isolated shell doublet block H_* , every Hermitian k -linear term has the form

$$P_* H_{\text{per}}^{(\text{int})}(k_* + p) P_* = \mathcal{E}_0 + p_i (u_i \mathbf{1}_2 + v_i^a \sigma_a) + O(|p|^2).$$

The scalar piece $u_i \mathbf{1}_2$ is irrelevant for the local Weyl data.

By S2 symmetry, the unique minimal shell-compatible traceless k -linear form is

$$p_i K_i^{\text{per,int}} = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2).$$

Equivalently, the projected shell tensor must be

$$\tilde{\mathcal{N}}_i^a = \lambda_{\parallel} \hat{n}_i \delta^{a3} + \alpha (e_{1i} \delta^{a1} + e_{2i} \delta^{a2}) + \beta (-e_{1i} \delta^{a2} + e_{2i} \delta^{a1}).$$

Thus the three first-order microscopic channels are:

$$\hat{n}_i \sigma_3 \quad \leftrightarrow \quad \text{longitudinal singlet,}$$

$$e_{1i} \sigma_1 + e_{2i} \sigma_2 \quad \leftrightarrow \quad \text{transverse even doublet,}$$

$$-e_{1i} \sigma_2 + e_{2i} \sigma_1 \quad \leftrightarrow \quad \text{transverse odd doublet.}$$

The coefficients are extracted by the projector traces

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{H_*} (\sigma_3 P_* (\hat{n}_i K_i^{\text{per,int}}) P_*),$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}_{H_*} (\sigma_a P_* (e_{ai} K_i^{\text{per,int}}) P_*),$$

$$\beta = \frac{1}{4} \left[\text{Tr}_{H_*} (\sigma_2 P_* (e_{1i} K_i^{\text{per,int}}) P_*) - \text{Tr}_{H_*} (\sigma_1 P_* (e_{2i} K_i^{\text{per,int}}) P_*) \right].$$

Hence the remaining triplet-sector microscopic audit is finite and explicit:

compute the actual k -linear harmonic coefficients of $H_{\text{per}}^{(\text{int})}(k)$ at $k = k_*$ and test their overlap with the shell tensor basis $\{\hat{n}_i \delta^{a3}$

CXXXIV. WHAT IS BEING DONE RIGHT NOW

The current task is not yet the complete derivation of the final full PDE itself. What is being done is the exact carrier audit of the future full PDE:

$$\Gamma^{(2)}[\bar{\Phi}] = H_B - iV^\mu \partial_{y^\mu} + W^{\mu\nu} \partial_{y^\mu} \partial_{y^\nu} + \dots$$

with the goal of identifying which microscopic sectors actually contribute to

$$K_i, \quad \Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \alpha, \quad \beta.$$

At the currently explicit level, the audit has established that

$$K_i^{\text{bare}} = 0, \quad K_i^{\text{per},0} = 0,$$

so that the load-bearing sources are reduced to

$$K_i = K_i^{\text{per,int}} + K_i^{(\text{II})}.$$

Therefore the present stage is:

not yet the full final PDE derivation,

but the exact decomposition of which terms in the future full PDE can actually carry the Dirac witness.

CXXXV. WORKING FULL PDE FIXING AND ACTUAL FORWARD REDUCTION

We now stop waiting for a more complete future action and fix the strongest explicit microscopic branch as the working full PDE:

$$L_{\text{work}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi].$$

(740)

This is not claimed to be the final complete TECT action in full generality. But it is the strongest explicit branch in which the full-PDE derivation can be pushed concretely.

CXXXVI. IMMEDIATE QUADRATIC-ORDER SIMPLIFICATION

Around the locked constant-projector background

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger = \text{const},$$

the current refined projector sector is block diagonal at quadratic order:

$$\delta^2 F_{\text{ref}} = \delta^2 F_\sigma + \delta^2 F_\xi, \quad J = 0. \tag{741}$$

Therefore, at this order,

$$\delta N_{ij}^a = 0 \tag{742}$$

from the direct projector channel.

Hence the projector sector is not a load-bearing source for the shell witness at the present order.

CXXXVII. ACTUAL WORKING BLOCH LINEARIZATION

The genuine working full-PDE linearization is therefore the Bloch fiber Hessian on the fluctuation pair

$$(\delta\Psi, \delta X),$$

namely

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}. \quad (743)$$

Using the folded Bragg convention

$$k_* = G_*/2, \quad k = k_* + p,$$

the shell-projected effective operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad (744)$$

where

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (745)$$

CXXXVIII. ACTUAL CARRIER REDUCTION

At the present order, the surviving microscopic witness carriers are reduced to exactly two sources:

$$\text{(I) refined internal } k\text{-dependence inside } H_{\Psi\Psi}(k), \quad (746)$$

$$\text{(II) the Class II Schur-complement term } H_{\Psi X} H_{XX}^{-1} H_{X\Psi}. \quad (747)$$

No separate projector-induced tree-level witness survives at this order.

CXXXIX. ACTUAL NEXT FINITE COMPUTATION

The first shell derivatives are

$$K_i = \partial_{p_i} H_{\text{shell}}(0), \quad (748)$$

and the reduced Weyl/Dirac coefficients are read from

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{H_*} (\sigma_\mu P_* K_i P_*). \quad (749)$$

Equivalently,

$$\lambda_{\parallel} = M_{3,\parallel}, \quad \alpha, \beta \text{ are the coefficients of } (p_1\sigma_1 + p_2\sigma_2), \quad (-p_2\sigma_1 + p_1\sigma_2). \quad (750)$$

Therefore the operator-construction phase is now genuinely fixed as the finite computation

$$H_{\Psi\Psi}(k), \quad H_{\Psi X}(k), \quad H_{XX}(k) \rightarrow H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow K_i \rightarrow (\lambda_{\parallel}, \alpha, \beta). \quad (751)$$

CXL. EXACT CHANNEL DECOMPOSITION OF THE CLASS II SCHUR-COMPLEMENT CONTRIBUTION

CXLI. GOAL

We now stop discussing the Class II sector abstractly and derive the exact shell-level channel decomposition of the Schur-complement contribution

$$K_i^{(\text{II})}.$$

The purpose is to determine which of the three reduced S2 witness channels

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are supplied automatically by the mediator sector, and in which leading-order branch they appear.

CXLI. CLASS II UV COMPLETION AND EFFECTIVE QUADRATIC OPERATOR

The explicit Class II UV-completed branch is

$$S_{\text{UV}}^{(\text{II})}[\Psi, P, \psi, X] = S_0[\Psi] + \int dt d^3x \left[\frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] \right]. \quad (752)$$

Linearizing in Bloch momentum k , the fiber Hessian has the block form

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}, \quad H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k). \quad (753)$$

After integrating out the heavy mediator X , the effective triplet block is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (754)$$

At leading heavy-mass order,

$$H_{XX}(k) = M_X^2 \mathbf{1}_X + O(k, \text{periodic}), \quad (755)$$

hence

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}), \quad \mathcal{C}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k). \quad (756)$$

CXLI. FIRST SHELL DERIVATIVE

At the folded Bragg point $k = k_* + p$, the shell-projected operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad (757)$$

so the Class II contribution to the first shell derivative is

$$\boxed{K_i^{(\text{II})} = -\Pi_{S_*} \left[\partial_{k_i} (H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k)) \right]_{k=k_*} \Pi_{S_*}.} \quad (758)$$

At strict leading order in M_X^{-2} ,

$$\boxed{K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}.} \quad (759)$$

This is the exact microscopic carrier formula.

CXLIV. SHELL-ADAPTED TENSOR REDUCTION

The shell witness theorem already fixes the only relevant microscopic carrier structure: a first-order enriched tensor must carry simultaneously

one spatial index i and one internal reduced-shell index a .

Therefore, after shell projection, $K_i^{(\text{II})}$ can only contribute through an effective (i, a) -tensor

$$N_{ij}^{a(\text{II})} \quad (760)$$

entering the shell operator in the standard form

$$\delta L_{\text{enr}}^{(\text{II})} = N_{ij}^{a(\text{II})} (\partial_i \phi_0) (-i \partial_j) \otimes s^a. \quad (761)$$

By the already-closed axial/S2 classification, the minimal admissible family is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (762)$$

Thus the projected Class II correction must reduce to

$$\boxed{N_{ij}^{a(\text{II})} = c_{\parallel} B_{\parallel ij}^a + c_I B_{I ij}^a + c_J B_{J ij}^a}, \quad (763)$$

where the three shell seed tensors are exactly

$$B_{\parallel ij}^a = \hat{n}_i \hat{n}_j \delta^{a3}, \quad (764)$$

$$B_{I ij}^a = \hat{n}_i e_{1j} \delta^{a1} + \hat{n}_i e_{2j} \delta^{a2}, \quad (765)$$

$$B_{J ij}^a = \hat{n}_i e_{2j} \delta^{a1} - \hat{n}_i e_{1j} \delta^{a2}. \quad (766)$$

CXLV. WHY ONLY THESE THREE CHANNELS SURVIVE

The reason is simple and exact:

1. $K_i^{(\text{II})}$ is first order in external shell momentum;
2. it carries one spatial derivative index through ∂_{k_i} ;
3. it carries one internal adjoint index through the mediator coupling X_i^a ;
4. after shell projection to the natural S2 branch, the only symmetry-compatible traceless channels are the longitudinal singlet and the two transverse doublet directions.

Hence no fourth independent first-order Class II witness channel exists at this order.

CXLVI. EXTRACTION OF THE REDUCED WITNESS COEFFICIENTS

The shell-matching formulas give

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \quad (767)$$

$$\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right), \quad (768)$$

$$\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \right). \quad (769)$$

Substituting (763) gives

$$\boxed{\lambda_{\parallel}^{(\text{II})} \propto c_{\parallel}, \quad \alpha^{(\text{II})} \propto c_I, \quad \beta^{(\text{II})} \propto c_J.} \quad (770)$$

Using the local defect/projector dictionary, one identifies

$$\boxed{g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.} \quad (771)$$

Therefore the shell-level Class II correction is completely equivalent to the coefficient triple

$$(g_c, g_E, g_O).$$

CXLVII. BRANCH STRUCTURE OF THE THREE CHANNELS

Now the crucial point.

The explicit local Class II witness formulas are not symmetric between longitudinal and transverse sectors. They are exactly

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.} \quad (772)$$

Hence:

1. the transverse channels g_E, g_O appear already at the ordinary odd-gradient I_{θ} -order;
2. the longitudinal channel g_c does *not* appear at I_{θ} -order; it appears only through the primitive commutator-square channel at I_{θ^3} -order.

Equivalently, the concrete Class II realization has the seed transport basis

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3, \quad (773)$$

$$V_{(\perp, E)}^{\mu}(\xi) = \theta'(\xi) E^{\mu}_a \sigma_a, \quad (774)$$

$$V_{(\perp, O)}^{\mu}(\xi) = \theta'(\xi) J^{\mu}_a \sigma_a. \quad (775)$$

Therefore the Class II sector automatically supplies:

$$\boxed{\text{transverse } S2 \text{ witness channels through } B_I, B_J,} \quad (776)$$

and

$$\boxed{\text{the longitudinal witness only through the primitive seed } B_{\parallel}.} \quad (777)$$

CXLVIII. EXACT MEANING OF THE SCHUR-COMPLEMENT COMPUTATION

The Schur-complement derivative (759) does *not* directly prove that

$$c_{\parallel}, c_I, c_J$$

are numerically nonzero for the full complete TECT action. That would require the explicit harmonic matrices $\mathcal{J}(k)$ and $\mathcal{K}(k)$.

But it *does* prove the following exact forward reduction:

$$\boxed{K_i^{(\Pi)} \implies N_{ij}^{a(\Pi)} \implies (c_{\parallel}, c_I, c_J) \iff (g_c, g_E, g_O).} \quad (778)$$

So the remaining microscopic Class II problem is finite and explicit: compute the projected coefficients

$$c_{\parallel}, \quad c_I, \quad c_J$$

from the actual Bloch harmonic matrices $H_{\Psi X}(k)$ and $H_{XX}(k)$.

CXLIX. NONDEGENERACY CRITERION

Because

$$I_{\theta} \neq 0, \quad I_{\theta'3} \neq 0,$$

the exact concrete Class II nondegeneracy criterion is

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (779)$$

Equivalently,

$$\boxed{c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).} \quad (780)$$

Thus the mediator sector closes the witness problem *inside the explicit seed space*: the transverse sector is already solved, while the longitudinal sector is solved on the primitive branch.

CL. SHARP HONEST CONCLUSION

The Class II Schur-complement contribution has now been reduced as far as it can be reduced without inserting the explicit harmonic matrix elements of $\mathcal{J}(k)$ and $\mathcal{K}(k)$.

What is already closed:

1. the exact shell-level tensor channels of $K_i^{(\Pi)}$ are only $B_{\parallel}, B_I, B_J$;
2. their coefficients are exactly the reduced microscopic couplings g_c, g_E, g_O ;
3. g_E, g_O supply the transverse I_{θ} -branch;
4. g_c supplies the longitudinal primitive $I_{\theta'3}$ -branch.

What remains open is only the finite coefficient extraction:

$$\boxed{\text{compute } (g_c, g_E, g_O) \text{ from the actual Bloch harmonic matrices.}} \quad (781)$$

CLI. FINAL SUMMARY

The Class II UV-completed working branch is

$$S_{\text{UV}}^{(II)} = S_0[\Psi] + \int \left[\frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] \right].$$

After integrating out the heavy mediator X , the effective Bloch operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k),$$

and the mediator contribution to the first shell derivative is

$$K_i^{(\text{II})} = -\Pi_{S_*} \left[\partial_{k_i} (H_{\Psi X} H_{XX}^{-1} H_{X\Psi}) \right]_{k=k_*} \Pi_{S_*}.$$

At leading heavy-mass order,

$$K_i^{(\text{II})} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}, \quad \mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

After shell reduction, the induced first-order Class II tensor has only the three S2-compatible seed channels:

$$N_{ij}^{a(\text{II})} = c_{\parallel} B_{\parallel ij}^a + c_I B_{I ij}^a + c_J B_{J ij}^a,$$

with

$$B_{\parallel ij}^a = \hat{n}_i \hat{n}_j \delta^{a3}, \quad B_{I ij}^a = \hat{n}_i e_{1j} \delta^{a1} + \hat{n}_i e_{2j} \delta^{a2}, \quad B_{J ij}^a = \hat{n}_i e_{2j} \delta^{a1} - \hat{n}_i e_{1j} \delta^{a2}.$$

Their coefficients are exactly the reduced Class II couplings:

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

The exact local Class II witness formulas are

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

Therefore:

the transverse S2 witness channels are supplied by g_E, g_O at I_{θ} -order,

the longitudinal Class II witness is supplied by g_c only on the primitive I_{θ^3} -branch.

Hence the exact concrete Class II nondegeneracy condition is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0),$$

equivalently

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).$$

Thus the remaining microscopic frontier is no longer structural. It is the finite coefficient extraction problem:

compute (g_c, g_E, g_O) from the actual harmonic matrices $H_{\Psi X}(k), H_{XX}(k)$.

CLII. EXPLICIT RECIPROCAL-HARMONIC MATRIX OF $J_i^a[\Psi]$ ON THE LOCKED 12-MODE BCC BACKGROUND

CLIII. SCOPE

At the current document level, the Class II UV action is explicitly

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (782)$$

The current notes give the exact microscopic definition

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi), \quad (783)$$

while the K -sector is only available in shell-relevant template form

$$K_i^a[P, \psi] \sim (T^a \partial_i P) \mathcal{O}[\psi], \quad K_i^a[P, \psi] \sim \partial_i \delta P^a \psi. \quad (784)$$

Therefore the next honest forward step is:

compute $\mathcal{J}(k)$ explicitly, keep $\mathcal{K}(k)$ as a reduced template.

CLIV. LOCKED PERIODIC BACKGROUND AND BLOCH FLUCTUATION BASIS

Take the locked 12-mode BCC background

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad P_0 = z_0 z_0^\dagger. \quad (785)$$

Choose an internal orthonormal basis

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs}. \quad (786)$$

Write the Bloch fluctuation as

$$\delta \Psi_k(x) = e^{ik \cdot x} \sum_{G \in \mathcal{G}} \sum_{r=0}^2 c_{G,r} u_r e^{iG \cdot x}. \quad (787)$$

For the heavy mediator,

$$\delta X_i^a(x) = e^{ik \cdot x} \sum_{G \in \mathcal{G}} x_{G,i,a} e^{iG \cdot x}. \quad (788)$$

CLV. LINEARIZATION OF J_i^a

Expand $\Psi = \Psi_0 + \delta \Psi$. Since J_i^a is bilinear in Ψ , its linearization around the background is

$$\delta J_i^a[\delta \Psi] = (\partial_i \delta \Psi^\dagger) T^a \Psi_0 + (\partial_i \Psi_0^\dagger) T^a \delta \Psi + \delta \Psi^\dagger T^a (\partial_i \Psi_0) + \Psi_0^\dagger T^a (\partial_i \delta \Psi). \quad (789)$$

Now use

$$\partial_i \Psi_0(x) = z_0 \partial_i \phi_0(x) = iA z_0 \sum_{Q \in \mathcal{S}_{12}} Q_i e^{iQ \cdot x}, \quad (790)$$

and

$$\partial_i \delta \Psi_k(x) = e^{ik \cdot x} \sum_{G,r} i(k+G)_i c_{G,r} u_r e^{iG \cdot x}. \quad (791)$$

Substituting (785)–(791) into (789), every term becomes a finite sum of harmonics of the form

$$e^{i(k+G+Q) \cdot x} \quad \text{or} \quad e^{i(k+G-Q) \cdot x}.$$

Hence the linearized current is a convolution operator in reciprocal space.

CLVI. EXACT HARMONIC MATRIX ELEMENTS OF $\mathcal{J}(k)$

Define the Bloch-space matrix $\mathcal{J}(k)$ by

$$\delta J_i^a(x) = e^{ik \cdot x} \sum_{G',i,a} (\mathcal{J}(k) c)_{G',i,a} e^{iG' \cdot x}. \quad (792)$$

Then the coefficient of $e^{i(k+G') \cdot x}$ is obtained by matching

$$G' = G + Q \quad \text{or} \quad G' = G - Q.$$

Introduce the internal constants

$$t_{rs}^a := u_r^\dagger T^a u_s. \quad (793)$$

Because T^a is Hermitian,

$$t_{sr}^a = \overline{t_{rs}^a}.$$

A direct substitution gives the harmonic matrix elements

$$\left(\mathcal{J}_i^a(k) \right)_{G',:,G,r} = iA \sum_{Q \in \mathcal{S}_{12}} \left[\delta_{G',G+Q} (k+G)_i t_{r0}^a + \delta_{G',G-Q} Q_i t_{0r}^a + \delta_{G',G-Q} Q_i t_{r0}^a + \delta_{G',G+Q} (k+G)_i t_{0r}^a \right]. \quad (794)$$

Grouping the $G' = G + Q$ and $G' = G - Q$ channels yields the cleaner form

$$\left(\mathcal{J}_i^a(k) \right)_{G',:,G,r} = iA \sum_{Q \in \mathcal{S}_{12}} \left[\delta_{G',G+Q} (k+G)_i (t_{r0}^a + t_{0r}^a) + \delta_{G',G-Q} Q_i (t_{r0}^a + t_{0r}^a) \right]. \quad (795)$$

Thus $\mathcal{J}(k)$ is explicitly known once the locked internal direction z_0 and the tangent basis (u_1, u_2) are fixed.

CLVII. SELECTION RULES ON THE LOCKED BRANCH

The tangent-shell sector relevant for the reduced $2+2$ problem uses only $r = 1, 2$. Therefore the physically relevant matrix elements are controlled by

$$t_{\alpha 0}^a = u_\alpha^\dagger T^a z_0, \quad \alpha = 1, 2. \quad (796)$$

Hence the first exact microscopic criterion is:

$$\boxed{\mathcal{J}(k) \text{ couples the shell-tangent doublet to the mediator iff } u_\alpha^\dagger T^a z_0 \neq 0 \text{ for some } (\alpha, a).} \quad (797)$$

This is now a concrete finite internal-algebra test, not a qualitative guess.

CLVIII. DERIVATIVE WITH RESPECT TO BLOCH MOMENTUM

Since only the $(k+G)_i$ -terms carry explicit k -dependence, one immediately obtains

$$\partial_{k_j} (\mathcal{J}_i^a(k))_{G',:,G,r} = iA \sum_{Q \in \mathcal{S}_{12}} \delta_{G',G+Q} \delta_{ij} (t_{r0}^a + t_{0r}^a). \quad (798)$$

This formula is especially important because the Class II Schur contribution to the first shell derivative is

$$K_i^{(\text{II})} \sim -\frac{1}{M_X^2} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right], \quad \mathcal{C} = \alpha_X \mathcal{J} + \beta_X \mathcal{K}.$$

So the J -part of the derivative is already fully explicit.

CLIX. NEAR-BRAGG SHELL RESTRICTION

At the folded Bragg point $k_* = G_*/2$, the shell basis is

$$|+, \alpha\rangle \equiv |G = 0, \alpha\rangle, \quad |-, \alpha\rangle \equiv |G = -G_*, \alpha\rangle, \quad \alpha = 1, 2. \quad (799)$$

The matrix element

$$\langle +, \alpha | \mathcal{J}_i^a(k_*) | -, \beta \rangle$$

is nonzero only if the shell difference is supplied by one background harmonic:

$$0 = (-G_*) + Q \iff Q = G_*. \quad (800)$$

Therefore only those $Q \in \mathcal{S}_{12}$ satisfying the Bragg matching condition contribute directly to the shell off-diagonal block.

Hence the direct shell-coupling part of $\mathcal{J}(k_*)$ is

$$\langle +, \alpha | \mathcal{J}_i^a(k_*) | -, \beta \rangle = iA (k_* - G_*)_i (t_{\beta 0}^a + t_{0\beta}^a) \delta_{Q, G_*}, \quad (801)$$

up to the conventional shell normalization and possible finite truncation of equivalent reciprocal representatives.

Since

$$k_* - G_* = -\frac{G_*}{2},$$

this is generically nonzero whenever the internal overlap

$$t_{\beta 0}^a + t_{0\beta}^a$$

is nonzero.

CLX. WHAT IS CLOSED AND WHAT REMAINS OPEN

The J -sector is now explicit:

1. the exact harmonic matrix $\mathcal{J}_i^a(k)$ is given by (795);
2. its Bloch derivative $\partial_{k_j} \mathcal{J}_i^a(k)$ is given by (798);
3. the shell selection rule is the single-harmonic Bragg condition $Q = G_*$.

By contrast, the K -sector is not yet unreduced explicitly in the current notes. All that is honestly available is the shell-relevant template

$$K_i^a[P, \psi] \sim (T^a \partial_i P) \mathcal{O}[\psi], \quad K_i^a \sim \partial_i \delta P^a \psi,$$

together with the already-reduced tensor summary and the induced coefficient dictionary

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

Therefore the current frontier is sharply split:

$$\boxed{\mathcal{J}(k) \text{ is explicit,} \quad \mathcal{K}(k) \text{ remains a reduced template.}} \quad (802)$$

CLXI. IMMEDIATE NEXT USE INSIDE THE SCHUR COMPLEMENT

The exact next finite computation is now

$$\boxed{K_i^{(\Pi, J)} = -\frac{\alpha_X^2}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{J}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_i} \mathcal{J}^\dagger) \right]_{k=k_*} \Pi_{S_*},} \quad (803)$$

together with the mixed JK -piece

$$\boxed{K_i^{(\Pi, JK)} = -\frac{\alpha_X \beta_X}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{J}) \mathcal{K}^\dagger + (\partial_{k_i} \mathcal{K}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_i} \mathcal{K}^\dagger) + \mathcal{K} (\partial_{k_i} \mathcal{J}^\dagger) \right]_{k=k_*} \Pi_{S_*},} \quad (804)$$

where only the $\mathcal{K}$ -insertions remain unreduced.

CLXII. FINAL SUMMARY

The Class II current is explicitly defined by

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi),$$

while the K -sector is currently available only through the reduced shell-relevant template

$$K_i^a[P, \psi] \sim (T^a \partial_i P) \mathcal{O}[\psi], \quad K_i^a[P, \psi] \sim \partial_i \delta P^a \psi.$$

On the locked 12-mode BCC background

$$\Psi_0 = z_0 \phi_0, \quad \phi_0 = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x},$$

with Bloch fluctuation

$$\delta \Psi_k(x) = e^{ik \cdot x} \sum_{G,r} c_{G,r} u_r e^{iG \cdot x},$$

the linearized current δJ_i^a is a reciprocal-space convolution operator.

Defining

$$t_{rs}^a := u_r^\dagger T^a u_s,$$

its exact harmonic matrix elements are

$$\left(\mathcal{J}_i^a(k) \right)_{G', ; G, r} = iA \sum_{Q \in \mathcal{S}_{12}} \left[\delta_{G', G+Q} (k+G)_i (t_{r0}^a + t_{0r}^a) + \delta_{G', G-Q} Q_i (t_{r0}^a + t_{0r}^a) \right].$$

Its Bloch derivative is therefore explicit:

$$\partial_{k_j} \left(\mathcal{J}_i^a(k) \right)_{G', ; G, r} = iA \sum_{Q \in \mathcal{S}_{12}} \delta_{G', G+Q} \delta_{ij} (t_{r0}^a + t_{0r}^a).$$

At the folded Bragg point $k_* = G_*/2$, direct shell coupling occurs only through the single-harmonic Bragg condition

$$Q = G_*.$$

Thus the J -sector is already explicit enough to be inserted into the Class II Schur-complement derivative.

By contrast, $\mathcal{K}(k)$ is not yet unreduced explicitly in the current notes. Hence the honest current frontier is

$$\boxed{\mathcal{J}(k) \text{ is explicit, while } \mathcal{K}(k) \text{ remains a reduced template.}}$$

Therefore the exact next finite computation is the explicit JJ -part

$$K_i^{(\Pi, J)} = -\frac{\alpha_X^2}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{J}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_i} \mathcal{J}^\dagger) \right]_{k=k_*} \Pi_{S_*},$$

followed by the mixed JK -part, where only the $\mathcal{K}$ -insertions remain open.

CLXIII. EXPLICIT SHELL EVALUATION OF THE JJ PART OF THE CLASS II SCHUR COMPLEMENT

CLXIV. GOAL

We now compute the part of the Class II shell derivative that is already fully explicit from the current notes, namely the JJ -piece:

$$K_i^{(\Pi, J)} = -\frac{\alpha_X^2}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{J}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_i} \mathcal{J}^\dagger) \right]_{k=k_*} \Pi_{S_*}. \quad (805)$$

The purpose is not to determine the final numerical values of all coefficients, but to identify which reduced S2 channels are already generated by the explicit J -sector.

CLXV. EXPLICIT HARMONIC MATRIX OF $\mathcal{J}(k)$

On the locked 12-mode BCC background

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x},$$

the exact reciprocal-harmonic matrix elements of $\mathcal{J}_i^a(k)$ are

$$(\mathcal{J}_i^a(k))_{G', ; G, r} = iA \sum_{Q \in \mathcal{S}_{12}} [\delta_{G', G+Q} (k+G)_i u_r^a + \delta_{G', G-Q} Q_i u_r^a], \quad (806)$$

where

$$u_r^a := t_{r0}^a + t_{0r}^a, \quad t_{r0}^a = u_r^\dagger T^a z_0. \quad (807)$$

The explicit Bloch derivative is

$$\partial_{k_j} (\mathcal{J}_i^a(k))_{G', ; G, r} = iA \sum_{Q \in \mathcal{S}_{12}} \delta_{G', G+Q} \delta_{ij} u_r^a. \quad (808)$$

Thus the derivative part is diagonal in the spatial indices:

$$\partial_{k_j} \mathcal{J}_i^a \propto \delta_{ij}.$$

CLXVI. SHELL BASIS AND BRAGG MATCHING

The folded shell basis is

$$|+, \alpha\rangle \equiv |G=0, \alpha\rangle, \quad |-, \alpha\rangle \equiv |G=-G_*, \alpha\rangle, \quad \alpha = 1, 2. \quad (809)$$

Direct shell coupling occurs only when one background harmonic matches the shell separation:

$$0 = (-G_*) + Q \iff Q = G_*. \quad (810)$$

Therefore both $\mathcal{J}(k_*)$ and $\partial_{k_i} \mathcal{J}(k_*)$ reduce on the shell to rank-one Bragg-transfer operators between the $\pm$ valleys.

CLXVII. STRUCTURE OF THE SHELL MATRIX PRODUCT

Because $\partial_{k_i} \mathcal{J}$ carries a factor δ_{ij} , the product

$$(\partial_{k_i} \mathcal{J}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_i} \mathcal{J}^\dagger)$$

inherits the same external spatial direction as the residual $\mathcal{J}$ -matrix element.

At the shell level, every matrix element is proportional to one factor of the Bragg vector:

$$(K_i^{(\Pi, J)})_{\alpha\beta} \propto \hat{n}_i \sum_a u_\alpha^a \overline{u_\beta^a}, \quad (811)$$

up to the common scalar normalization

$$\frac{\alpha_X^2 A^2 |G_*|}{M_X^2}.$$

Hence the JJ -piece is *axial*: it carries only the shell normal direction $\hat{n}_i$ and no intrinsic e_1, e_2 structure.

CLXVIII. INTERNAL 2×2 MATRIX INDUCED BY THE J -SECTOR

Define the Hermitian tangent-doublet matrix

$$M_{\alpha\beta} := \sum_a u_\alpha^a \overline{u_\beta^a}. \quad (812)$$

Then the shell derivative takes the form

$$\boxed{K_i^{(\Pi, J)} = -\Gamma_J \hat{n}_i M, \quad \Gamma_J > 0 \text{ up to the overall conventional sign normalization.}} \quad (813)$$

Now decompose M in the Pauli basis:

$$M = m_0 \mathbf{1}_2 + m_1 \sigma_1 + m_2 \sigma_2 + m_3 \sigma_3, \quad (814)$$

with

$$m_\mu = \frac{1}{2}(\sigma_\mu M), \quad \sigma_0 := \mathbf{1}_2.$$

Therefore

$$p_i K_i^{(\Pi, J)} = -\Gamma_J p_{\parallel} (m_0 \mathbf{1}_2 + m_1 \sigma_1 + m_2 \sigma_2 + m_3 \sigma_3). \quad (815)$$

CLXIX. WHAT SURVIVES AFTER S2 REDUCTION

The reduced S2 target symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (816)$$

Comparing (815) with (816), we obtain:

1. the JJ -piece carries only $p_{\parallel}$, so it cannot by itself generate the transverse S2 channels proportional to p_1, p_2 ;
2. the scalar $m_0 \mathbf{1}_2$ part is irrelevant to the reduced Weyl witness;
3. the only potentially load-bearing piece is the traceless longitudinal part

$$-\Gamma_J m_3 p_{\parallel} \sigma_3.$$

Thus the JJ -sector alone contributes at most to the longitudinal shell singlet channel:

$$\boxed{\lambda_{\parallel}^{(\Pi, J)} \propto m_3, \quad \alpha^{(\Pi, J)} = 0, \quad \beta^{(\Pi, J)} = 0.} \quad (817)$$

CLXX. WHY THIS DOES NOT CLOSE THE FINAL LONGITUDINAL PROBLEM

At first sight, (817) seems to generate a longitudinal $p_{\parallel} \sigma_3$ term already at the ordinary I_θ -type shell order. However, the exact local Class II transport analysis in the current notes shows that the concrete longitudinal coefficient is instead

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad (818)$$

whereas the transverse coefficients are

$$\alpha = g_E I_\theta, \quad \beta = g_O I_\theta. \quad (819)$$

Therefore the explicit local theory states that the true longitudinal witness does *not* come from the ordinary I_θ -branch of the toy J -type extraction. It comes from the distinct primitive commutator-square channel.

So the correct interpretation of (817) is:

$$\boxed{\text{the pure } JJ \text{ shell term can only probe a longitudinal-type direction,}} \quad (820)$$

but

$$\boxed{\text{it does not by itself establish the true concrete Class II longitudinal coefficient } g_c.} \quad (821)$$

CLXXI. IMMEDIATE CONSEQUENCE

Because the JJ -piece is purely axial, it cannot generate the explicit transverse seed directions

$$B_I, \quad B_J.$$

Hence the transverse Class II coefficients g_E, g_O cannot come from the pure JJ piece alone. They must instead come from one of the following:

1. the JK mixed term,
2. the KK term,
3. the already-established reduced local transport completion.

Equivalently,

$$\boxed{K_i^{(\Pi, J)} \text{ is longitudinal-only at the shell level.}} \quad (822)$$

CLXXII. SHARP HONEST RESULT

The pure explicit J -sector allows us to go one real step forward:

$$\boxed{\text{the } JJ\text{-part of the Class II Schur complement is now explicitly reduced to an axial shell singlet matrix.}} \quad (823)$$

But it also shows a hard limitation:

$$\boxed{\text{the transverse witness channels } (g_E, g_O) \text{ are not generated by } JJ \text{ alone.}} \quad (824)$$

Hence the next actual microscopic computation must be

$$\boxed{K_i^{(\Pi, JK)} \quad \text{and/or} \quad K_i^{(\Pi, KK)},} \quad (825)$$

because only there can the explicit projector-derived tensor structure enter and produce the B_I, B_J directions.

CLXXIII. FINAL SUMMARY

For the explicit working branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi],$$

the pure JJ -part of the Class II Schur complement is

$$K_i^{(\Pi, J)} = -\frac{\alpha_X^2}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{J}) \mathcal{J}^\dagger + \mathcal{J} (\partial_{k_i} \mathcal{J}^\dagger) \right]_{k=k_*} \Pi_{S_*}.$$

Using the explicit harmonic matrix of $\mathcal{J}(k)$, one finds that:

$$\partial_{k_j} \mathcal{J}_i^a \propto \delta_{ij},$$

and the shell coupling is fixed by the single-harmonic Bragg condition

$$Q = G_*.$$

Therefore the shell-reduced JJ -piece has the universal form

$$K_i^{(\Pi, J)} = -\Gamma_J \hat{n}_i M,$$

where M is a Hermitian 2×2 matrix on the tangent doublet space.

After Pauli decomposition

$$M = m_0 \mathbf{1}_2 + m_1 \sigma_1 + m_2 \sigma_2 + m_3 \sigma_3,$$

the shell symbol becomes

$$p_i K_i^{(\Pi, J)} = -\Gamma_{JJ} p_{\parallel} (m_0 \mathbf{1}_2 + m_1 \sigma_1 + m_2 \sigma_2 + m_3 \sigma_3).$$

Hence the pure JJ -sector is *axial only*: it carries only the shell-normal direction $p_{\parallel}$ and no intrinsic transverse p_1, p_2 structure.

Therefore:

$$\boxed{\lambda_{\parallel}^{(\Pi, J)} \propto m_3, \quad \alpha^{(\Pi, J)} = 0, \quad \beta^{(\Pi, J)} = 0.}$$

So the explicit JJ -piece can probe only a longitudinal-type shell singlet direction. It cannot by itself generate the transverse Class II witness channels g_E, g_O .

This is fully consistent with the exact local Class II transport formulas:

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

which show that the true concrete longitudinal witness belongs to the primitive branch, while the transverse witness belongs to the ordinary odd-gradient branch.

Hence the next genuinely load-bearing microscopic computation is not JJ anymore, but

$$\boxed{K_i^{(\Pi, JK)} \quad \text{and/or} \quad K_i^{(\Pi, KK)},}$$

because only those terms can inject explicit projector-derived tensor structure and generate the transverse seed directions B_I, B_J .

CLXXIV. WORKING REDUCED PRINCIPAL OPERATOR OF THE EXPLICIT TECT + CLASS II BRANCH

We now stop the carrier-audit phase and write the actual working reduced first-order operator for the strongest explicit microscopic branch.

The explicit working microscopic branch is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (826)$$

After eliminating the heavy mediator X_i^a , the shell-relevant enriched tensor induced by the mixed JK -sector is, at quadratic template level,

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (827)$$

For the shell-adapted seed family

$$m^a = m \delta^{a3}, \quad (828)$$

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (829)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}, \quad (830)$$

this gives

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J, \quad \Gamma := \frac{\lambda g_X m}{M_X^2}. \quad (831)$$

Using the already-closed shell/defect dictionary,

$$g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J, \quad (832)$$

where $\nu_{\parallel}, \nu_{\perp}$ are finite nonzero overlap normalizations.

Therefore the local reduced first-order symbol of the explicit working branch is

$$A_1^{\text{work}}(p) = \lambda_{\parallel}^{\text{work}} p_{\parallel} \sigma_3 + \alpha^{\text{work}} (p_1 \sigma_1 + p_2 \sigma_2) + \beta^{\text{work}} (-p_2 \sigma_1 + p_1 \sigma_2), \quad (833)$$

with

$$\lambda_{\parallel}^{\text{work}} = \lambda_{\parallel}^{(\Psi)} + \frac{1}{2} g_c I_{\theta^3}, \quad (834)$$

$$\alpha^{\text{work}} = \alpha^{(\Psi)} + g_E I_{\theta}, \quad \beta^{\text{work}} = \beta^{(\Psi)} + g_O I_{\theta}. \quad (835)$$

Equivalently,

$$\lambda_{\parallel}^{\text{work}} = \lambda_{\parallel}^{(\Psi)} + \frac{\nu_{\parallel} \Gamma c_{\parallel}}{2} I_{\theta^3}, \quad (836)$$

$$\alpha^{\text{work}} = \alpha^{(\Psi)} + \nu_{\perp} \Gamma c_I I_{\theta}, \quad \beta^{\text{work}} = \beta^{(\Psi)} + \nu_{\perp} \Gamma c_J I_{\theta}. \quad (837)$$

Hence the explicit Class II truncation by itself already gives a nondegenerate local Weyl symbol whenever

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0), \quad (838)$$

provided the overlap constants remain finite and nonzero.

In the pure Class II working truncation, setting the unresolved triplet-refined contribution to zero,

$$\lambda_{\parallel}^{(\Psi)} = \alpha^{(\Psi)} = \beta^{(\Psi)} = 0, \quad (839)$$

one obtains the fully explicit reduced principal symbol

$$A_1^{\text{Class II}}(p) = \frac{\nu_{\parallel} \Gamma c_{\parallel}}{2} I_{\theta^3} p_{\parallel} \sigma_3 + \nu_{\perp} \Gamma I_{\theta} \left[c_I (p_1 \sigma_1 + p_2 \sigma_2) + c_J (-p_2 \sigma_1 + p_1 \sigma_2) \right]. \quad (840)$$

This is now the actual working reduced PDE data. The next stage is no longer to search for a carrier, but to impose the remaining finite Dirac conditions:

$$V(0) = 0, \quad \Delta > 0, \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (841)$$

CLXXV. DIRAC STEP III AS AN ACTUAL WORKING ZERO-MIXING THEOREM

CLXXVI. GOAL

We now stop discussing the zero-mixing step abstractly and write the actual working form needed for the explicit TECT + Class II truncation.

The shell-relevant local Dirac carrier is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2, \quad (842)$$

where:

- $\mathbb{C}_{\text{valley}}^2$ is the selected opposite shell pair $\pm G_*$,
- $\mathbb{C}_{\text{int}}^2$ is the enriched transverse/internal doublet.

At $p = 0$, the full projected block has the form

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^{\dagger} & H_R(0) \end{pmatrix}, \quad V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D). \quad (843)$$

Therefore

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0. \quad (844)$$

CLXXVII. MINIMAL DANGEROUS TRUNCATION

For the working explicit branch, the remote sector should be organized as

$$\boxed{\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}.} \quad (845)$$

These sectors mean:

1. $\mathcal{H}_{\text{vol}}$: volumetric / longitudinal sector,
2. $\mathcal{H}_{\text{high}}$: higher-shell sector not belonging to the selected pair $\pm G_*$,
3. $\mathcal{H}_{\text{non-enr}}$: same shell but lacking the enriched internal doublet,
4. $\mathcal{H}_{\text{wrong-int}}$: nonmatching internal representation,
5. $\mathcal{H}_{\text{extra}}$: any remaining sector not covered by the previous canonical exclusions.

CLXXVIII. IMMEDIATE EXCLUSIONS

The shell-relevant mismatch labels are:

$$\boxed{\pm G_*, \quad \text{transverse/non-volumetric character,} \quad \text{presence of enriched internal doublet,} \quad \text{grading/parity/chirality,} \quad \text{extra conservation}} \quad (846)$$

Hence the first four canonical remote sectors are excluded immediately:

- a.* $\mathcal{H}_{\text{vol}}$. This sector fails the transverse/non-volumetric label. Therefore

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{vol}}, \mathcal{H}_D) = 0. \quad (847)$$

- b.* $\mathcal{H}_{\text{high}}$. This sector fails the shell-pair label $\pm G_*$. Therefore

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{high}}, \mathcal{H}_D) = 0. \quad (848)$$

- c.* $\mathcal{H}_{\text{non-enr}}$. This sector lacks the enriched internal doublet $\mathbb{C}_{\text{int}}^2$. Therefore

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{non-enr}}, \mathcal{H}_D) = 0. \quad (849)$$

- d.* $\mathcal{H}_{\text{wrong-int}}$. This sector has the wrong internal representation. Therefore

$$\text{Hom}_{H_*}(\mathcal{H}_{\text{wrong-int}}, \mathcal{H}_D) = 0. \quad (850)$$

Combining these,

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}, \mathcal{H}_D) = 0.} \quad (851)$$

CLXXIX. COMPRESSION OF THE ACTUAL FRONTIER

Therefore the entire zero-mixing problem compresses to the remaining sector

$$\mathcal{H}_{\text{extra}}.$$

Indeed,

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \text{Hom}_{H_*}(\mathcal{H}_{\text{extra}}, \mathcal{H}_D). \quad (852)$$

So the actual Step III frontier is no longer broad. It is only the following finite question:

$$\boxed{\text{Does there exist any extra remote sector } K \subset \mathcal{H}_{\text{extra}} \text{ that matches } \mathcal{H}_D \text{ in all shell-relevant labels?}} \quad (853)$$

If the answer is no, then

$$\boxed{V(0) = 0.} \quad (854)$$

CLXXX. OPERATIONAL FORM FOR THE WORKING TECT PROGRAM

Thus the zero-mixing step for the working explicit TECT + Class II branch is now:

1. construct the shell-relevant carrier

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2;$$

2. decompose the remote sector into

$$\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}};$$

3. eliminate the first four sectors by the canonical label mismatches;
4. inspect only $\mathcal{H}_{\text{extra}}$.

Therefore the actual working deliverable for Step III is not a vague symmetry discussion. It is the finite claim

$$\boxed{\mathcal{H}_{\text{extra}} = \emptyset \quad \text{or} \quad \forall K \subset \mathcal{H}_{\text{extra}}, K \text{ mismatches } \mathcal{H}_D \text{ in at least one label.}} \quad (855)$$

CLXXXI. EXACT CONSEQUENCE FOR THE EFFECTIVE DIRAC BLOCK

If (855) holds, then

$$V(0) = 0.$$

Combined with the remote gap condition $\Delta > 0$, the Schur self-energy satisfies

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger = O\left(\frac{|p|^2}{\Delta}\right), \quad (856)$$

and therefore

$$\boxed{H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right)}, \quad (857)$$

with no $O(p^0)$ constant mixing from the remote sector.

This is the actual forward-completion meaning of Step III.

CLXXXII. FINAL SUMMARY

The zero-constant-mixing step is now compressed to a concrete working theorem.
The shell-relevant Dirac carrier is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

and at $p = 0$,

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Hence

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.$$

For the working explicit TECT + Class II truncation, decompose

$$\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}.$$

The shell-relevant mismatch labels are:

$\pm G_*$, transverse/non-volumetric character, presence of enriched internal doublet, grading/parity/chirality, extra conserv

The first four sectors are excluded immediately:

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{vol}}, \mathcal{H}_D) = 0,$$

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{high}}, \mathcal{H}_D) = 0,$$

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{non-enr}}, \mathcal{H}_D) = 0,$$

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{wrong-int}}, \mathcal{H}_D) = 0.$$

Therefore the entire Step III frontier reduces to a single finite question:

$$\boxed{\text{does any } K \subset \mathcal{H}_{\mathrm{extra}} \text{ match } \mathcal{H}_D \text{ in all shell-relevant labels?}}$$

If no such extra dangerous sector exists, then

$$\boxed{V(0) = 0.}$$

Then, together with a positive remote gap $\Delta > 0$,

$$\Sigma(E, p) = O(|p|^2/\Delta),$$

and the effective shell Hamiltonian becomes

$$\boxed{H_{\mathrm{eff}}(E, p) = H_D(p) + O(|p|^2/\Delta),}$$

with no constant remote-sector mixing.

CLXXXIII. WORKING CLOSURE OF STEP III

At the level of the explicit working TECT + Class II truncation, we now impose

$$\boxed{\mathcal{H}_R = \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\mathrm{non-enr}} \oplus \mathcal{H}_{\mathrm{wrong-int}}, \quad \mathcal{H}_{\mathrm{extra}} = \emptyset.} \quad (858)$$

Since each of the four displayed remote sectors mismatches the Dirac carrier

$$\mathcal{H}_D \cong \mathbb{C}_{\mathrm{valley}}^2 \otimes \mathbb{C}_{\mathrm{int}}^2$$

in at least one shell-relevant label, it follows that

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0. \quad (859)$$

Therefore

$$\boxed{V(0) = 0.} \quad (860)$$

Thus, for the working truncation, Step III is now closed by definition, and the actual next task is Step IV:

$$\boxed{\Delta := \mathrm{dist}(\mathrm{spec} H_R(0), E_D(0)) > 0.} \quad (861)$$

CLXXXIV. STEP IV IN WORKING FORM: EXPLICIT FINITE REMOTE-GAP INEQUALITY

CLXXXV. GOAL

We now stop repeating the abstract spectral-isolation theorem and write the actual working form of Step IV for the explicit TECT + Class II truncation.

The remote-gap problem is not solved by the shell determinant

$$\det M = A(B^2 + C^2) \neq 0.$$

That determinant controls only the nondegeneracy of the local shell Dirac block. The remote gap

$$\Delta_\rho > 0$$

is an independent spectral property of the remote block $H_R(p)$.

What is already closed at theorem level is that the full Step IV reduces to a finite matrix inequality on the remote block.

CLXXXVI. REMOTE-BLOCK DECOMPOSITION

For the working truncation, write

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad \mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}. \quad (862)$$

Decompose the remote block as

$$H_R(p) = D_R(p) + R_R(p), \quad (863)$$

where $D_R(p)$ is the diagonal benchmark part and $R_R(p)$ is the residual Hermitian mixing.

Define the four sector-center energies

$$\mu_{\text{vol}}(p), \quad \mu_{\text{high}}(p), \quad \mu_{\text{non-enr}}(p), \quad \mu_{\text{wrong-int}}(p), \quad (864)$$

and set

$$\delta_0(p) = \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\}, \quad (865)$$

$$\eta_R(p) = \|R_R(p)\|. \quad (866)$$

On the neighborhood

$$N_\rho = \{|p| \leq \rho\},$$

define

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \delta_0(p), \quad \eta_{R,\rho} := \sup_{|p| \leq \rho} \eta_R(p). \quad (867)$$

CLXXXVII. WORKING REMOTE-GAP THEOREM

The exact theorem-level sufficient condition is

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \implies \text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset \quad (|p| \leq \rho),} \quad (868)$$

with

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.} \quad (869)$$

Equivalently, a Gershgorin sufficient bound is

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_a \left(|(H_R)_{aa}(p)| - \sum_{b \neq a} |(H_R)_{ab}(p)| \right) > 0. \quad (870)$$

Thus Step IV is no longer conceptual. It is a finite lower-bound problem on $H_R(p)$.

CLXXXVIII. IMMEDIATE EXPLICIT INPUT ALREADY AVAILABLE

Among the four sector centers, the current uploaded notes already provide an explicit formula for the wrong-internal shell separation.

The longitudinal–transverse splitting is

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right]. \quad (871)$$

Hence the wrong-internal benchmark gap is

$$m_{\text{int}}^{(0)} = |E_L - E_*| = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|. \quad (872)$$

Therefore a sufficient condition for the wrong-internal sector to stay separated is

$$\left(\lambda + \mu \right) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \neq 0. \quad (873)$$

A stronger sufficient condition is

$$\left(\lambda + \mu \right) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} > 0, \quad (874)$$

which implies $E_L > E_*$ and hence $m_{\text{int}}^{(0)} > 0$.

Under a bounded perturbation of the wrong-internal block,

$$\|Q \delta H Q\| \leq \varepsilon_{\text{int}}, \quad (875)$$

one has the stable lower bound

$$m_{\text{int}} \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}. \quad (876)$$

CLXXXIX. WORKING CLOSURE STRATEGY

We now impose the following working Step IV strategy.

First, define the diagonal benchmark gap by

$$\Delta_{\text{bench},\rho} := \inf_{|p| \leq \rho} \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, m_{\text{int}}(p)\}. \quad (877)$$

Second, define the total remote mixing budget

$$\eta_{\text{rem},\rho} := \sup_{|p| \leq \rho} \|R_R(p)\|. \quad (878)$$

Then the actual working deliverable of Step IV is the single inequality

$$\Delta_{\text{bench},\rho} > \eta_{\text{rem},\rho}. \quad (879)$$

If (879) holds, then

$$\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho} > 0. \quad (880)$$

CXC. CONSEQUENCES FOR THE DIRAC BLOCK

Once Step III and Step IV both hold, namely

$$V(0) = 0, \quad \Delta_{\text{full},\rho} > 0, \quad \|V(p)\| \leq c|p|,$$

the Schur-complement resolvent bound gives

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta_{\text{full},\rho}} \quad (|E| < \Delta_{\text{full},\rho}/2), \quad (881)$$

and hence

$$\|\Sigma(E, p)\| \leq \frac{2c^2}{\Delta_{\text{full},\rho}} |p|^2. \quad (882)$$

Therefore

$$\boxed{H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta_{\text{full},\rho}}\right)}. \quad (883)$$

CXCI. SHARP HONEST STATUS

What is closed now is the exact working form of Step IV: the remote-gap problem has been reduced to the finite inequality

$$\Delta_{\text{bench},\rho} > \eta_{\text{rem},\rho}.$$

What is not yet fully evaluated in the uploaded material is the full set of explicit formulas for

$$\mu_{\text{vol}}(p), \quad \mu_{\text{high}}(p), \quad \mu_{\text{non-enr}}(p),$$

and the final upper bound for

$$\eta_{\text{rem},\rho}.$$

So the remaining Step IV frontier is now exactly this:

$$\boxed{\text{compute the three remaining benchmark sector centers and one remote mixing norm bound.}} \quad (884)$$

CXCII. FINAL SUMMARY

The remote-gap condition

$$\Delta_\rho > 0$$

is an independent spectral input. It is not implied by the shell determinant

$$\det M = A(B^2 + C^2) \neq 0,$$

which controls only the local Dirac-block nondegeneracy.

For the working truncation,

$$\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

and

$$H_R(p) = D_R(p) + R_R(p).$$

Define the benchmark sector-center energies

$$\mu_{\text{vol}}(p), \quad \mu_{\text{high}}(p), \quad \mu_{\text{non-enr}}(p), \quad \mu_{\text{wrong-int}}(p),$$

together with

$$\delta_0(p) = \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\},$$

$$\eta_R(p) = \|R_R(p)\|.$$

On $|p| \leq \rho$,

$$\delta_{0,\rho} = \inf_{|p| \leq \rho} \delta_0(p), \quad \eta_{R,\rho} = \sup_{|p| \leq \rho} \eta_R(p).$$

Then the exact working sufficient criterion is

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

Equivalently one may use the Gershgorin bound

$$\Delta_{G,\rho} > 0.$$

The currently uploaded notes already provide one explicit benchmark piece: the wrong-internal gap

$$m_{\text{int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|.$$

Hence a sufficient condition for wrong-internal separation is

$$(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \neq 0,$$

and a stronger sufficient condition is positivity of the same quantity.

Therefore the remaining Step IV frontier is finite and explicit:

$$\boxed{\text{compute } \mu_{\text{vol}}(p), \mu_{\text{high}}(p), \mu_{\text{non-enr}}(p), \text{ and an upper bound for } \eta_{\text{rem},\rho}.$$

Once

$$V(0) = 0, \quad \Delta_{\text{full},\rho} > 0, \quad \|V(p)\| \leq c|p|$$

all hold, the Schur complement gives

$$H_{\text{eff}}(E, p) = H_D(p) + O(|p|^2 / \Delta_{\text{full},\rho}),$$

so the remote sector only contributes subleading curvature corrections.

CXCIII. WORKING BENCHMARK FORMULAS FOR THE REMAINING STEP-IV SECTOR CENTERS

CXCIV. GOAL

We now stop repeating the abstract remote-gap theorem and write the actual working benchmark formulas for the three remaining remote-sector centers:

$$\mu_{\text{vol}}(p), \quad \mu_{\text{high}}(p), \quad \mu_{\text{non-enr}}(p).$$

The remote-gap criterion is already fixed:

$$\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

So the only forward task is to give explicit lower bounds for the sector centers.

CXCV. VOLUMETRIC SECTOR BENCHMARK

The volumetric sector is controlled by the heavy scalar locking mode σ , which couples only to the longitudinal displacement and decouples from the transverse sector at quadratic order. The coupled longitudinal–volumetric block is

$$L_{L,\sigma}^{(2)} = \frac{1}{2}\rho|\dot{u}_L|^2 - \frac{1}{2}(\lambda + 2\mu)k^2|u_L|^2 + \frac{1}{2}|\dot{\sigma}|^2 - \frac{1}{2}(M_\sigma^2 + c_\sigma^2 k^2)|\sigma|^2 + g\sigma(-\mathbf{k})ik u_L(\mathbf{k}).$$

Therefore the natural remote benchmark energy of the volumetric sector is set by the heavy volumetric mode:

$$\boxed{\mu_{\text{vol}}(p) \simeq \sqrt{M_\sigma^2 + c_\sigma^2 |p|^2}.} \quad (885)$$

On $|p| \leq \rho$, a uniform lower bound is

$$\boxed{\Delta_{\text{vol},\rho}^{(0)} \geq \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}.} \quad (886)$$

In a weaker model-independent form one may simply use

$$\Delta_{\text{vol},\rho}^{(0)} \geq c_{\text{vol}} M_\sigma > 0 \quad (887)$$

for some finite $c_{\text{vol}} > 0$.

CXCVI. HIGHER-SHELL BENCHMARK

The higher-shell sector consists of harmonics not belonging to the selected pair $\pm G_*$. Hence it is separated radially from the selected shell by the Brazovskii penalty

$$r_* + K(|q|^2 - q_*^2)^2.$$

Accordingly, the working higher-shell sector center is

$$\boxed{\mu_{\text{high}}(p) \simeq r_* + K(|k_* + p + G|^2 - q_*^2)^2, \quad G \notin \{0, -G_*\}.} \quad (888)$$

A sufficient benchmark lower bound is

$$\Delta_{\text{high},\rho}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*|, \quad (889)$$

where δ_{shell} is the minimal radial separation from the selected shell.

In the current explicit benchmark notes, this is specialized to

$$\boxed{\Delta_{\text{high},\rho}^{(0)} \geq 4K - |r_*|.} \quad (890)$$

CXCVII. NON-ENRICHED SAME-SHELL BENCHMARK

The non-enriched sector lies on the same shell, but it does not carry the enriched internal doublet that defines the Dirac block. Therefore its separation is not controlled by radial shell distance, but by the internal block eigenvalue splitting inside the same-shell truncation.

Thus the correct working definition is:

$$\boxed{\mu_{\text{non-enr}}(p) = \text{lowest eigenvalue of the same-shell non-enriched internal block.}} \quad (891)$$

At the finite truncation level, the notes already reduce this to the benchmark lower bound

$$\boxed{\Delta_{\text{non-enr},\rho}^{(0)} \geq \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.} \quad (892)$$

Therefore this is the only genuinely non-automatic same-shell piece that still requires an actual finite matrix eigenvalue computation.

CXCVIII. WRONG-INTERNAL BENCHMARK

For completeness, the already-explicit wrong-internal gap is

$$m_{\text{int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|. \quad (893)$$

CXCIX. FINAL BENCHMARK SUMMARY

Combining the four sectorwise lower bounds, the final benchmark gap is

$$\Delta_{\text{bench},\rho}^{\text{fin}} := \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}. \quad (894)$$

Therefore the actual Step-IV closure condition for the working truncation is the single finite inequality

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}. \quad (895)$$

If (895) holds, then

$$\Delta_\rho \geq \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0. \quad (896)$$

CC. WHAT IS NOW GENUINELY LEFT

After this reduction, the remaining work is no longer structural.

Three of the four sector benchmarks are already fixed in usable working form:

$$\Delta_{\text{vol},\rho}^{(0)}, \quad \Delta_{\text{high},\rho}^{(0)}, \quad \Delta_{\text{wrong-int},\rho}^{(0)}.$$

The only genuinely model-specific unresolved benchmark is

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)},$$

together with the residual remote mixing norm

$$\eta_{R,\rho}.$$

Thus the final Step-IV frontier has been compressed to:

$$\boxed{\text{one finite same-shell non-enriched eigenvalue computation} \quad + \quad \text{one remote-mixing norm bound.}} \quad (897)$$

CCI. FINAL SUMMARY

The Step-IV remote-gap criterion is already fixed:

$$\delta_{0,\rho} > \eta_{R,\rho} \implies \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

The remaining forward task was to write explicit working benchmark formulas for

$$\mu_{\text{vol}}(p), \quad \mu_{\text{high}}(p), \quad \mu_{\text{non-enr}}(p).$$

These are now fixed as follows.

For the volumetric sector,

$$\mu_{\text{vol}}(p) \simeq \sqrt{M_\sigma^2 + c_\sigma^2 |p|^2},$$

with uniform lower bound

$$\Delta_{\text{vol},\rho}^{(0)} \geq \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}.$$

For the higher-shell sector,

$$\mu_{\text{high}}(p) \simeq r_* + K(|k_* + p + G|^2 - q_*^2)^2, \quad G \notin \{0, -G_*\},$$

with explicit benchmark

$$\Delta_{\text{high},\rho}^{(0)} \geq 4K - |r_*|.$$

For the non-enriched same-shell sector,

$$\mu_{\text{non-enr}}(p) = \text{lowest eigenvalue of the same-shell non-enriched internal block},$$

with finite-truncation lower bound

$$\Delta_{\text{non-enr},\rho}^{(0)} \geq \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

The wrong-internal sector is already explicit:

$$m_{\text{int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|.$$

Hence the final working benchmark gap is

$$\Delta_{\text{bench},\rho}^{\text{fin}} := \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

Therefore the actual Step-IV closure condition is the single finite inequality

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.$$

After this reduction, the only genuinely unresolved Step-IV data are:

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \quad \text{and} \quad \eta_{R,\rho}.$$

So the final frontier is no longer broad: it is one same-shell non-enriched finite matrix eigenvalue computation plus one remote-mixing norm bound.

CCII. WORKING FINITE MATRIX FOR THE SAME-SHELL NON-ENRICHED SECTOR

CCIII. GOAL

The only genuinely non-automatic Step-IV slot is the same-shell non-enriched sector. Its correct benchmark is not a universal symbolic formula, but an actual finite truncated eigenvalue computation.

The exact definition is already fixed:

$$H_{\text{non-enr}}^{(N)}(p) := P_{\text{non-enr}}^{(N)} H_{\text{shell}}^{(N)}(p) P_{\text{non-enr}}^{(N)}, \quad (898)$$

$$\Delta_{\text{non-enr},\rho}^{(N)} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_{\text{non-enr}}^{(N)}(p))). \quad (899)$$

We now write the actual working matrix form that must be computed.

CCIV. FINITE SHELL-TRUNCATED SPACE

Let the shell-truncated Bloch space be

$$\mathcal{H}_{\text{shell}}^{(N)} = \mathcal{H}_D^{(N)} \oplus \mathcal{H}_{\text{non-enr}}^{(N)} \oplus \mathcal{H}_{\text{rest}}^{(N)}. \quad (900)$$

Here:

- $\mathcal{H}_D^{(N)}$ is the enriched shell doublet sector producing the local Dirac block;
- $\mathcal{H}_{\text{non-enr}}^{(N)}$ is the complementary same-shell sector without the enriched internal doublet;
- $\mathcal{H}_{\text{rest}}^{(N)}$ contains all other truncated states.

Choose an orthonormal basis of same-shell harmonics

$$\mathcal{B}_{\text{shell}}^{(N)} = \{|G_a, \chi_\alpha\rangle\}, \quad a = 1, \dots, N_s, \quad \alpha = 1, \dots, n_{\text{int}}, \quad (901)$$

with G_a restricted to the selected shell orbit and χ_α spanning the internal sector.

The enriched Dirac carrier occupies a distinguished 4-dimensional subspace

$$\mathcal{H}_D^{(N)} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

so the same-shell non-enriched complement is the orthogonal remainder inside the same shell:

$$\mathcal{H}_{\text{non-enr}}^{(N)} = \left(\bigoplus_{a=1}^{N_s} \text{span}\{|G_a, \chi_\alpha\rangle\} \right) \ominus \mathcal{H}_D^{(N)}. \quad (902)$$

CCV. MINIMAL WORKING BASIS

For the minimal dangerous truncation, the smallest useful basis is:

$$\mathcal{B}_{\text{non-enr}}^{(N)} = \{|G_+, \eta_A\rangle, |G_-, \eta_A\rangle\}_{A=1}^m, \quad (903)$$

where:

- $G_\pm = \pm G_*$ are the selected opposite shell modes;
- η_A span the internal same-shell complement orthogonal to the enriched doublet.

Thus the minimal same-shell non-enriched block has dimension

$$2m \times 2m. \quad (904)$$

CCVI. BLOCK FORM OF THE MATRIX

In this basis, the projected same-shell non-enriched block takes the form

$$H_{\text{non-enr}}^{(N)}(p) = \begin{pmatrix} D_+(p) & M(p) \\ M(p)^\dagger & D_-(p) \end{pmatrix}, \quad (905)$$

where each entry is an $m \times m$ matrix.

The diagonal valley blocks are

$$(D_\pm(p))_{AB} = \langle G_\pm, \eta_A | H_{\text{shell}}^{(N)}(p) | G_\pm, \eta_B \rangle, \quad (906)$$

and the opposite-valley same-shell mixing block is

$$M_{AB}(p) = \langle G_+, \eta_A | H_{\text{shell}}^{(N)}(p) | G_-, \eta_B \rangle. \quad (907)$$

CCVII. LEADING EXPANSION NEAR THE CROSSING

Near $p = 0$, the working expansion is

$$D_{\pm}(p) = D_{\pm}^{(0)} + p_i D_{\pm,i}^{(1)} + O(|p|^2), \quad (908)$$

$$M(p) = M^{(0)} + p_i M_i^{(1)} + O(|p|^2). \quad (909)$$

Hence

$$H_{\text{non-enr}}^{(N)}(p) = H_{\text{non-enr}}^{(N)}(0) + p_i K_i^{\text{non-enr}} + O(|p|^2), \quad (910)$$

with

$$H_{\text{non-enr}}^{(N)}(0) = \begin{pmatrix} D_+^{(0)} & M^{(0)} \\ (M^{(0)})^\dagger & D_-^{(0)} \end{pmatrix}. \quad (911)$$

CCVIII. EXACT BENCHMARK TO COMPUTE

The exact finite same-shell benchmark is the smallest singular spectral distance of this block from zero:

$$\Delta_{\text{non-enr},\rho}^{(N)} = \inf_{|p| \leq \rho} \sigma_{\min} \left(H_{\text{non-enr}}^{(N)}(p) \right), \quad (912)$$

where $\sigma_{\min}$ is the smallest absolute eigenvalue since the block is Hermitian.

At $p = 0$, a simpler sufficient condition is

$$\text{dist}(0, \text{spec}(H_{\text{non-enr}}^{(N)}(0))) > 0. \quad (913)$$

If, in addition,

$$\sup_{|p| \leq \rho} \|p_i K_i^{\text{non-enr}} + O(|p|^2)\| < \text{dist}(0, \text{spec}(H_{\text{non-enr}}^{(N)}(0))), \quad (914)$$

then

$$\Delta_{\text{non-enr},\rho}^{(N)} > 0.$$

CCIX. MINIMAL PRACTICAL ROUTE

Therefore the actual practical computation is:

1. construct $H_{\text{shell}}^{(N)}(p)$;
2. identify the enriched 4-dimensional Dirac carrier $\mathcal{H}_D^{(N)}$;
3. take the orthogonal same-shell complement $\mathcal{H}_{\text{non-enr}}^{(N)}$;
4. compute the finite matrix $H_{\text{non-enr}}^{(N)}(0)$;
5. check whether its spectrum avoids zero;
6. then bound the p -dependent perturbation on $|p| \leq \rho$.

So the non-enriched same-shell slot is reduced to one explicit finite matrix plus one perturbative radius estimate.

CCX. INSERTION INTO THE FINAL STEP-IV BENCHMARK

The final working benchmark remains

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}. \quad (915)$$

Hence the complete Step-IV closure condition is still the single inequality

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}}. \quad (916)$$

The difference is that the same-shell non-enriched slot is now no longer abstract: it is a concrete $2m \times 2m$ Hermitian matrix eigenvalue problem.

CCXI. FINAL SUMMARY

The same-shell non-enriched sector is the only non-automatic Step-IV benchmark slot. Its exact finite definition is

$$H_{\text{non-enr}}^{(N)}(p) = P_{\text{non-enr}}^{(N)} H_{\text{shell}}^{(N)}(p) P_{\text{non-enr}}^{(N)},$$

$$\Delta_{\text{non-enr},\rho}^{(N)} = \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_{\text{non-enr}}^{(N)}(p))).$$

In the minimal working basis

$$\mathcal{B}_{\text{non-enr}}^{(N)} = \{|G_+, \eta_A\rangle, |G_-, \eta_A\rangle\}_{A=1}^m,$$

the block has the explicit form

$$H_{\text{non-enr}}^{(N)}(p) = \begin{pmatrix} D_+(p) & M(p) \\ M(p)^\dagger & D_-(p) \end{pmatrix},$$

with $m \times m$ valley-diagonal blocks $D_\pm(p)$ and opposite-valley same-shell block $M(p)$.

Near $p = 0$,

$$D_\pm(p) = D_\pm^{(0)} + p_i D_{\pm,i}^{(1)} + O(|p|^2), \quad M(p) = M^{(0)} + p_i M_i^{(1)} + O(|p|^2),$$

so the benchmark reduces to the smallest absolute eigenvalue of a concrete finite Hermitian matrix.

A sufficient practical route is:

$$\text{dist}(0, \text{spec}(H_{\text{non-enr}}^{(N)}(0))) > 0$$

plus a small- p perturbation bound on $|p| \leq \rho$.

Therefore the final Step-IV benchmark remains

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\},$$

and Step IV is closed if

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.$$

So the same-shell non-enriched slot is now reduced to one concrete $2m \times 2m$ Hermitian matrix eigenvalue problem.

CCXII. MINIMAL $m = 1$ CLOSURE OF THE SAME-SHELL NON-ENRICHED SLOT

CCXIII. GOAL

We now take the minimal same-shell non-enriched truncation $m = 1$ and close the corresponding benchmark by an explicit 2×2 Hermitian matrix criterion.

This is the first truly concrete version of the remaining Step-IV slot.

CCXIV. MINIMAL BASIS AND BLOCK

Let the minimal same-shell non-enriched basis be

$$\mathcal{B}_{\text{non-enr}}^{(1)} = \{|G_+, \eta\rangle, |G_-, \eta\rangle\},$$

where η is one internal state orthogonal to the enriched shell doublet.

Then the projected non-enriched block is

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}, \quad (917)$$

with

$$d_{\pm}(p) = \langle G_{\pm}, \eta | H_{\text{shell}}^{(1)}(p) | G_{\pm}, \eta \rangle, \quad m(p) = \langle G_+, \eta | H_{\text{shell}}^{(1)}(p) | G_-, \eta \rangle. \quad (918)$$

Since the block is Hermitian, its eigenvalues are real and given exactly by

$$\lambda_{\pm}(p) = \frac{d_+(p) + d_-(p)}{2} \pm \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}. \quad (919)$$

Therefore the same-shell non-enriched benchmark is

$$\Delta_{\text{non-enr}, \rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|\lambda_+(p)|, |\lambda_-(p)|\}. \quad (920)$$

CCXV. EXACT ZERO-AVOIDANCE CONDITION

The block (917) has a zero eigenvalue iff

$$\det H_{\text{non-enr}}^{(1)}(p) = d_+(p)d_-(p) - |m(p)|^2 = 0. \quad (921)$$

Hence a pointwise necessary and sufficient condition for spectral separation from zero is

$$d_+(p)d_-(p) - |m(p)|^2 \neq 0 \quad \text{and} \quad \lambda_{\pm}(p) \neq 0. \quad (922)$$

For uniform positivity on $|p| \leq \rho$, one may of course work directly with (920), but a simpler closed sufficient criterion is preferable.

CCXVI. CLOSED SUFFICIENT CRITERION I: SAME-SIGN DIAGONAL SECTOR

Assume first that the two diagonal entries have the same sign on $|p| \leq \rho$, e.g.

$$d_+(p) \geq 0, \quad d_-(p) \geq 0.$$

Then the smaller eigenvalue is

$$\lambda_-(p) = \frac{d_+(p) + d_-(p)}{2} - \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}.$$

Using the elementary inequality

$$\lambda_-(p) \geq \min\{d_+(p), d_-(p)\} - |m(p)|,$$

we obtain the uniform lower bound

$$\Delta_{\text{non-enr}, \rho}^{(1)} \geq \inf_{|p| \leq \rho} \min\{d_+(p), d_-(p)\} - \sup_{|p| \leq \rho} |m(p)|. \quad (923)$$

Therefore a closed sufficient condition is

$$\inf_{|p| \leq \rho} \min\{d_+(p), d_-(p)\} > \sup_{|p| \leq \rho} |m(p)|. \quad (924)$$

Under (924), one has

$$\Delta_{\text{non-enr}, \rho}^{(1)} > 0. \quad (925)$$

CCXVII. CLOSED SUFFICIENT CRITERION II: GERSHGORIN FORM

A more basis-robust sufficient condition follows immediately from Gershgorin. For the 2×2 Hermitian matrix (917), each eigenvalue lies in the union of the discs

$$\mathbb{D}(d_+(p), |m(p)|), \quad \mathbb{D}(d_-(p), |m(p)|).$$

Hence zero is excluded if

$$|d_+(p)| > |m(p)|, \quad |d_-(p)| > |m(p)|. \quad (926)$$

Uniformly on $|p| \leq \rho$, this becomes

$$\inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\} > \sup_{|p| \leq \rho} |m(p)|. \quad (927)$$

Then

$$\Delta_{\text{non-enr}, \rho}^{(1)} \geq \inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\} - \sup_{|p| \leq \rho} |m(p)| > 0. \quad (928)$$

This is the cleanest practical sufficient criterion.

CCXVIII. CONNECTION WITH THE COERCIVE DECOMPOSITION THEOREM

The current notes already state the general coercive route: if

$$H_{\text{non-enr}}^{(N)}(p) = D_{\text{non-enr}}^{(N)}(p) + R_{\text{non-enr}}^{(N)}(p)$$

with

$$\delta_{\text{non-enr}, \rho}^{(N)} > \eta_{\text{non-enr}, \rho}^{(N)},$$

then

$$\Delta_{\text{non-enr}, \rho}^{(N)} > 0.$$

For $m = 1$, this theorem becomes completely explicit by taking

$$D_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & 0 \\ 0 & d_-(p) \end{pmatrix}, \quad R_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} 0 & m(p) \\ \frac{0}{m(p)} & 0 \end{pmatrix}. \quad (929)$$

Then

$$\delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\}, \quad \eta_{\text{non-enr},\rho}^{(1)} = \sup_{|p| \leq \rho} |m(p)|. \quad (930)$$

So the general theorem reduces exactly to

$$\boxed{\delta_{\text{non-enr},\rho}^{(1)} > \eta_{\text{non-enr},\rho}^{(1)} \implies \Delta_{\text{non-enr},\rho}^{(1)} > 0}, \quad (931)$$

which is the same as the uniform Gershgorin condition (927).

CCXIX. SMALL- p EXPANSION

Write

$$d_{\pm}(p) = d_{\pm}^{(0)} + p_i a_{\pm,i} + O(|p|^2), \quad m(p) = m_0 + p_i b_i + O(|p|^2). \quad (932)$$

Then a sufficient zero-momentum closure condition is

$$\boxed{\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|}. \quad (933)$$

If, in addition, the linear correction obeys

$$\sup_{|p| \leq \rho} \max\{|p_i a_{+,i}|, |p_i a_{-,i}|, |p_i b_i|\} \leq \varepsilon_\rho \quad (934)$$

with

$$\varepsilon_\rho < \min\{|d_+^{(0)}|, |d_-^{(0)}|\} - |m_0|, \quad (935)$$

then

$$\boxed{\Delta_{\text{non-enr},\rho}^{(1)} > 0}. \quad (936)$$

Thus the $m = 1$ slot is fully reduced to the three finite quantities

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0,$$

plus one small- p perturbation budget.

CCXX. INSERTION INTO THE MASTER STEP-IV BENCHMARK

Replacing the non-enriched slot in the master benchmark, the working Step-IV gap becomes

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}, q_* \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\}. \quad (937)$$

Therefore the final Step-IV condition is still

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}}. \quad (938)$$

The difference is that the last non-automatic slot is now an explicit 2×2 inequality problem, not an abstract block.

CCXXI. FINAL SUMMARY

In the minimal same-shell non-enriched truncation $m = 1$, the block is

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}.$$

Its exact eigenvalues are

$$\lambda_{\pm}(p) = \frac{d_+(p) + d_-(p)}{2} \pm \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}.$$

Hence

$$\Delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|\lambda_+(p)|, |\lambda_-(p)|\}.$$

A clean closed sufficient condition is the uniform Gershgorin/coercive inequality

$$\inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\} > \sup_{|p| \leq \rho} |m(p)|.$$

Under this condition,

$$\Delta_{\text{non-enr},\rho}^{(1)} > 0.$$

Equivalently, in the coercive split

$$D_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & 0 \\ 0 & d_-(p) \end{pmatrix}, \quad R_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} 0 & m(p) \\ m(p) & 0 \end{pmatrix},$$

one has

$$\delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\}, \quad \eta_{\text{non-enr},\rho}^{(1)} = \sup_{|p| \leq \rho} |m(p)|,$$

and therefore

$$\delta_{\text{non-enr},\rho}^{(1)} > \eta_{\text{non-enr},\rho}^{(1)} \implies \Delta_{\text{non-enr},\rho}^{(1)} > 0.$$

At $p = 0$, a still simpler sufficient condition is

$$\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|.$$

With a small enough p -dependent perturbation budget, this extends to $|p| \leq \rho$.

Thus the last non-automatic same-shell slot is now reduced to an explicit 2×2 Hermitian inequality problem.

CCXXII. WORKING PDE AND ITS REDUCED DIRAC FORM

The current working microscopic PDE is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi].$$

It is linearized around the locked 12-mode BCC background

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad P_0 = z_0 z_0^\dagger,$$

and treated in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p.$$

After Bloch reduction and heavy-mediator elimination,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k).$$

The shell-projected infrared principal symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

In the explicit concrete Class II local truncation,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

Thus the working PDE is summarized as:

$$\text{microscopically: a refined locked TECT triplet PDE coupled to a heavy Class II mediator,}$$

$$\text{infraredly: a graded } 2 + 2 \text{ Dirac-type first-order PDE with S2 symbol } A_1(p).$$

CCXXIII. FINAL ORGANIZED FORM OF THE CURRENT PDE

CCXXIV. MICROSCOPIC WORKING PDE

The strongest explicit microscopic branch currently available is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (939)$$

This is not yet claimed to be the final complete TECT action in full generality. Rather, it is the strongest explicit branch in which the full-PDE derivation can be pushed concretely.

CCXXV. PERIODIC BACKGROUND AND BLOCH REDUCTION

We linearize around the locked 12-mode BCC background

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad P_0 = z_0 z_0^{\dagger}. \quad (940)$$

Using the folded Bragg convention,

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|, \quad (941)$$

the actual operator-construction chain is finite:

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta). \quad (942)$$

CCXXVI. REDUCED INFRARED PDE

The shell-projected infrared principal symbol is the S2 branch

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (943)$$

Its nondegeneracy condition is exactly

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (944)$$

CCXXVII. CONCRETE CLASS II COEFFICIENT FORM

Inside the exact concrete Class II local branch,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}. \quad (945)$$

Therefore:

$$\text{transverse sector starts at } I_{\theta}\text{-order,} \quad \text{longitudinal sector starts at primitive } I_{\theta'^3}\text{-order.} \quad (946)$$

Hence the concrete Class II nondegeneracy test is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (947)$$

CCXXVIII. REMOTE-SECTOR COMPLETION CONDITIONS

For this reduced Dirac PDE to be the actual low-energy effective PDE, three further conditions are required.

A. Zero constant mixing

At $p = 0$,

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D). \quad (948)$$

Hence

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0. \quad (949)$$

B. Positive remote gap

The remote block must satisfy a uniform gap condition

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho} \implies \Delta_{\rho} > 0. \quad (950)$$

A current explicit benchmark is

$$\Delta_{\text{bench}, \rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_{\sigma}^2 + c_{\sigma}^2 (q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \right|, \Delta_{\text{non-enr}, \rho}^{(N)} - \varepsilon_{\text{non-enr}, \rho}^{(N)} \right\}. \quad (951)$$

C. Mass protection

Finally, residual symmetry must forbid constant Dirac masses:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (952)$$

CCXXIX. FINAL ORGANIZED STATEMENT

Thus the current PDE is organized in the following two-layer form:

$$\boxed{\text{Microscopic level: refined locked TECT triplet PDE coupled to a heavy Class II mediator.}} \quad (953)$$

$$\boxed{\text{Infrared level: a graded } 2 + 2 \text{ Dirac-type first-order PDE with principal symbol } A_1(p).} \quad (954)$$

What is already closed:

1. the working microscopic branch (939);
2. the Bloch/shell reduction pipeline (942);
3. the reduced S2 infrared symbol (943);
4. the exact concrete Class II coefficient dictionary (945);
5. the finite completion criteria $V(0) = 0$, $\Delta_\rho > 0$, and mass protection.

What remains honestly open:

1. derive the actual transport kernel from the full complete TECT action rather than from the strongest explicit branch;
2. compute the actual microscopic values of g_c, g_E, g_O (equivalently $\lambda_c, \Lambda_E, \Lambda_O$);
3. decide whether complete TECT realizes a constant-leading longitudinal branch or only the primitive I_{θ^3} branch;
4. evaluate the final same-shell non-enriched benchmark and remote mixing budget in the chosen truncation.

CCXXX. FINAL SUMMARY

The current PDE is organized as follows.

Microscopic working PDE:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi].$$

After Bloch reduction around the locked 12-mode BCC background and shell projection at

$$k_* = \frac{G_*}{2}, \quad k = k_* + p,$$

the infrared principal symbol becomes

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

In the exact concrete Class II branch,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

so the reduced nondegeneracy condition is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Thus the current PDE has the following two-layer interpretation:

$$\boxed{\text{microscopically: refined locked TECT + heavy Class II mediator;}}$$

$$\boxed{\text{infraredly: graded } 2 + 2 \text{ Dirac-type first-order PDE.}}$$

To make this the full low-energy effective PDE, one still needs:

$$V(0) = 0, \quad \Delta_\rho > 0, \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

So the reduced Dirac structure is already fixed, while the final complete microscopic coefficient extraction remains open.

CCXXXI. CURRENT PDE STATEMENT OF TECT

CCXXXII. MICROSCOPIC WORKING PDE

The strongest explicit microscopic branch currently available is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (955)$$

This is not yet the final complete TECT action in full generality. It is the strongest explicit branch on which the full PDE derivation can currently be pushed concretely.

CCXXXIII. BACKGROUND AND BLOCH REDUCTION

We linearize around the locked 12-mode BCC background

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad P_0 = z_0 z_0^\dagger. \quad (956)$$

Using the folded Bragg convention,

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|, \quad (957)$$

the actual operator-construction chain is finite:

$$H_{\text{eff}}(k) \rightarrow H_{\text{shell}}(p) \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta). \quad (958)$$

CCXXXIV. INFRARED REDUCED PDE

The shell-projected infrared principal symbol is the natural S2 branch

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (959)$$

Its exact nondegeneracy condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (960)$$

Equivalently, the finite nondegeneracy matrix is positive iff

$$G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (961)$$

CCXXXV. EXACT CONCRETE CLASS II COEFFICIENT FORM

Inside the exact concrete Class II local branch, the reduced coefficients are

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}. \quad (962)$$

Hence:

$$\text{transverse sector starts at } I_{\theta}\text{-order,} \quad \text{longitudinal sector starts at primitive } I_{\theta'^3}\text{-order.} \quad (963)$$

Therefore the concrete Class II nondegeneracy test is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (964)$$

CCXXXVI. COMPLETION CONDITIONS FOR AN ACTUAL LOW-ENERGY DIRAC PDE

To conclude that the reduced symbol (959) is the actual low-energy effective PDE, three additional conditions are required.

First, zero constant remote mixing:

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.} \quad (965)$$

Second, a positive remote spectral gap:

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho} \implies \Delta_\rho > 0.} \quad (966)$$

A current explicit benchmark is

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}. \quad (967)$$

Third, mass protection:

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (968)$$

CCXXXVII. FINAL ORGANIZED STATEMENT

Thus the current PDE is organized in the following two-layer form:

$$\boxed{\text{Microscopic level: refined locked TECT triplet PDE coupled to a heavy Class II mediator.}} \quad (969)$$

$$\boxed{\text{Infrared level: a graded } 2 + 2 \text{ Dirac-type first-order PDE with principal symbol } A_1(p).} \quad (970)$$

What is already closed:

1. the working microscopic branch (955);
2. the finite Bloch/shell operator chain (958);
3. the reduced S2 principal symbol (959);
4. the exact concrete Class II coefficient dictionary (962);
5. the finite completion criteria $V(0) = 0$, $\Delta_\rho > 0$, and mass protection.

What remains honestly open:

1. derive the actual transport kernel from the full complete TECT action rather than from the strongest explicit branch;
2. compute the actual microscopic coefficient point (g_c, g_E, g_O) , equivalently $(\lambda_c, \Lambda_E, \Lambda_O)$;
3. determine whether complete TECT realizes a constant-leading longitudinal branch or only the primitive I_{θ^3} branch;
4. evaluate the final same-shell non-enriched benchmark and the remote mixing budget in the chosen truncation.

CCXXXVIII. FINAL SUMMARY

The current TECT PDE is organized as follows.
Microscopic working PDE:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi].$$

After linearization around the locked 12-mode BCC background and Bloch reduction at

$$k_* = \frac{G_*}{2}, \quad k = k_* + p,$$

the infrared principal symbol becomes

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

In the exact concrete Class II branch,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

Hence the reduced nondegeneracy condition is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

So the current PDE has the two-layer interpretation:

microscopically: refined locked TECT + heavy Class II mediator;

infraredly: graded 2 + 2 Dirac-type first-order PDE.

To make this the actual low-energy effective PDE, one still needs

$$V(0) = 0, \quad \Delta_{\rho} > 0, \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Thus the reduced Dirac structure is already fixed, while the final complete microscopic coefficient extraction remains open.